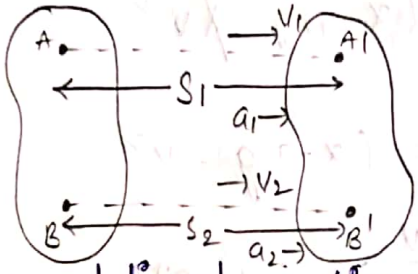
Rigid Body Motion :- Rotational Mechanics

Rigid body :

The seperation b/w any two particles of the body does not change with time then it is called rigid body.

① Translational Motion :



* In translational motion, all the particles have same linear displacements i.e, $S_1 = S_2 = \dots$, all particles have same velocities i.e, $V_1 = V_2 = \dots$, all particles have same accelerations i.e, $a_1 = a_2 = \dots$.

Eqn of motion :

① a = const :

(a) $v = u + at$

(b) $s = ut + \frac{1}{2} a t^2$

$\downarrow$

$x_f - x_i$ = net displacement.

(c) $v^2 - u^2 = 2as$

(d) $s = \left(\frac{v+u}{2}\right) t$

(e) $S_n = u + a\left(n - \frac{1}{2}\right)$

② a is variable :

i.e, $a = f(t)$

$a = f(s)$

$a = f(v)$

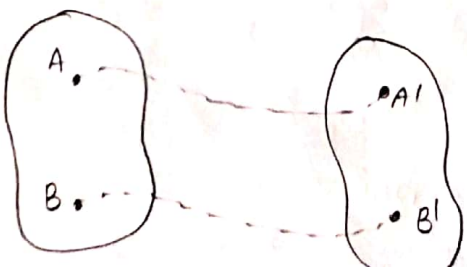
* use calculas method.

(a) $a = \frac{dv}{dt} \Rightarrow \int dv = \int a \cdot dt$

(b) $a = v \cdot \frac{dv}{dx} \Rightarrow \int v \cdot dv = \int a \cdot dx$

(c) $v = \frac{dx}{dt} \Rightarrow \int dx = \int v \cdot dt$

② Curvilinear Motion :



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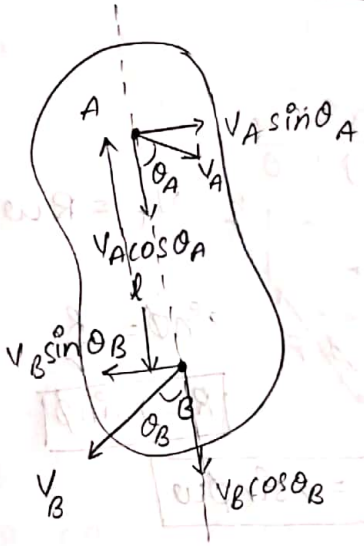
③ Rotational Motion:



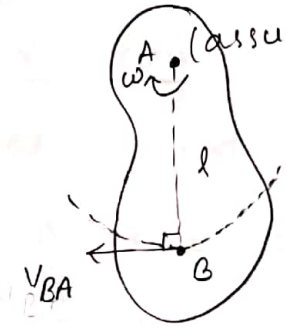
l = const

$$\frac{dl}{dt} = 0$$

we have,

$$\frac{dl}{dt} = V_B \cos\theta_B - V_A \cos\theta_A = 0$$

$$\Rightarrow \boxed{V_B \cos\theta_B = V_A \cos\theta_A}$$

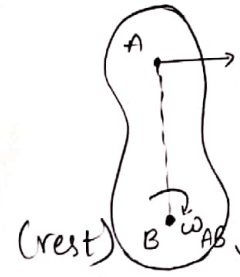


now, $\bar{V}_{BA} = \bar{V}_B - \bar{V}_A$

$= V_B \sin\theta_B - (-V_A \sin\theta_A)$

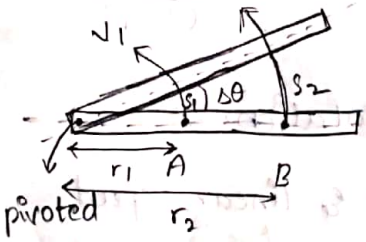
$= V_A \sin\theta_A + V_B \sin\theta_B$.

$$\boxed{\omega_{BA} = \frac{V_{BA}}{l}}$$



$$\boxed{\omega_{AB} = \frac{V_{AB}}{l}}$$

clearly $\boxed{\omega_{AB} = \omega_{BA}}$.



$S_2 > S_1$

$\Rightarrow V_2 > V_1$

In Rigid body motion, all particles have

① same angular displacements

② same angular velocities

③ same angular accelerations.

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i.e, $\boxed{\Delta\theta = \frac{s_1}{r_1} = \frac{s_2}{r_2}}$

$\boxed{\theta = 2n\pi}$ — n: no. of revolutions

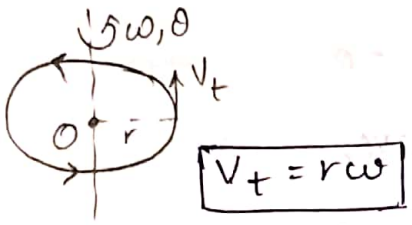
Angular velocity (ω): [rad/sec]

* $\omega_{avg} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_2 - \theta_1}{t_2 - t_1}$

(or) $\omega_{avg} = \frac{\int_{t_1}^{t_2} \omega \cdot dt}{\int_{t_1}^{t_2} dt}$ $[\omega = f(t)]$

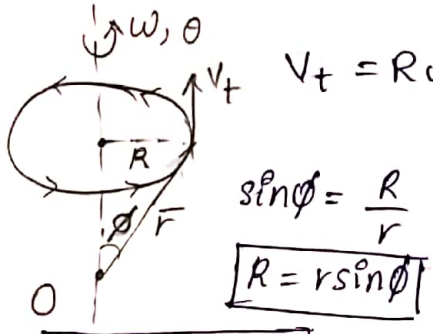
* $\omega_{inst} = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$

*



$\boxed{V_t = r\omega}$

*



$V_t = R\omega$

$\sin\phi = \frac{R}{r}$

$\boxed{R = r\sin\phi}$

$\Rightarrow \boxed{V_t = r\sin\phi\,\omega}$

* $\boxed{\vec{V_t} = \vec{\omega} \times \vec{r}}$ *

Angular Acceleration (α):

$\boxed{\alpha = \frac{d\omega}{dt}}$ [rad/sec^2]

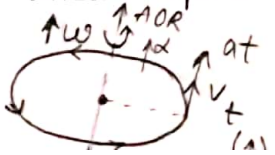
* $\alpha_{avg} = \frac{\omega_2 - \omega_1}{t_2 - t_1}$

(or) $\alpha_{avg} = \frac{\int_{t_1}^{t_2} \alpha \cdot dt}{\int_{t_1}^{t_2} dt}$ $[\alpha = f(t)]$

* $\alpha_{inst} = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$

* $\vec{\alpha}$ also acting along axis of rotation.

* If $\alpha > 0$ then "ω" increases & linear speed $\omega > 0$ also increases.



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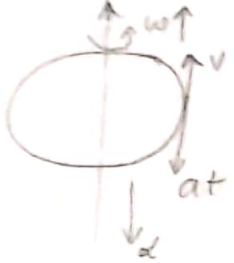

$\omega > 0$, $\alpha < 0$

$\omega(\downarrow)$ & $v(\downarrow)$.

$$\bar{v} = \bar{\omega} \times \bar{r}$$

$$a = \frac{dv}{dt} = \frac{d}{dt}(\bar{\omega} \times \bar{r}) = \bar{\omega} \times \frac{d\bar{r}}{dt} + \frac{d\bar{\omega}}{dt} \times \bar{r}$$

$$\boxed{\bar{a} = \bar{\omega} \times \bar{v} + \bar{\alpha} \times \bar{r}}$$

$$\bar{a} = \bar{a}_c + \bar{a}_t$$

$$\bar{a}_t = \bar{\alpha} \times \bar{r}$$

$$\boxed{a_t = r\alpha} \ *$$

Rotational Kinematics About fixed axis :-

① α = const :

② $\omega = \omega_0 + \alpha t$

③ $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

└ total angular displacement in time t .

$$\boxed{\theta = 2\pi N}$$

④ $\omega^2 - \omega_0^2 = 2\alpha\theta$

⑤ $\theta = \left(\frac{\omega_0 + \omega}{2}\right) t$

) α = variable : α — $f(t)$, $f(\theta)$, $f(\omega)$

ⓐ $\alpha = \frac{d\omega}{dt}$ =) $\int d\omega = \int \alpha \cdot dt$

ⓑ $\alpha = \omega \cdot \frac{d\omega}{d\theta}$ =) $\int \omega \cdot d\omega = \int \alpha \cdot d\theta$

ⓒ $\omega = \frac{d\theta}{dt}$ =) $\int d\theta = \int \omega \cdot dt$

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→ A fan is rotating with an angular velocity ω_0. Now power supply to the fan is switched off. Angular velocity of fan now decreases with time as $\omega = \omega_0 \cdot e^{-t/T}$ where T = const. Find

① Time at which $\omega = \omega_0/2$.

② Intial angular acceleration.

③ Avg angular accelesation b/w the instants when $\omega = \omega_0$ & $\omega = \omega_0/2$.

④ Total no. of revolutions during the time b/w $\omega = \omega_0$ & $\omega = \omega_0/2$.

① $\omega = \omega_0 \cdot e^{-t/T}$

$$\frac{\omega}{\omega_0} = e^{-t/T}$$

$$\ln \frac{\omega}{\omega_0} = \frac{-t}{T}$$

$$\frac{t}{T} = \ln \frac{\omega_0}{\omega}$$

$$\boxed{t = T \ln \frac{\omega_0}{\omega}}$$ general expression.

for $\omega = \omega_0/2$,

$$\boxed{t = T \ln 2}$$

② $\alpha = \frac{d\omega}{dt} = \frac{d}{dt}\omega_0 (e^{-t/T}) = \omega_0 e^{-t/T} \cdot \left(-\frac{1}{T}\right)$

$$\boxed{\alpha = -\frac{\omega_0}{T} e^{-t/T}}$$ at $t = 0$ $\boxed{\alpha = -\omega_0/T}$

③ $\alpha_{avg} = \frac{\int_{t_1}^{t_2} \alpha \cdot dt}{\int_{t_1}^{t_2} dt}$ or $\alpha_{avg} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\frac{\omega_0}{2} - \omega_0}{t - 0}$

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$$\alpha_{avg} = \frac{-\omega_0}{t}$$

$$\boxed{\alpha_{avg} = \frac{-\omega_0}{T \ln 2}}$$

(4) $\omega = \frac{d\theta}{dt} \Rightarrow d\theta = \omega \cdot dt$

$$\int d\theta = \int \omega \cdot dt$$

$$\theta = \int_0^{T\ln 2} \omega_0 e^{-t/T} \cdot dt$$

$$\theta = \left[\frac{\omega_0 e^{-t/T}}{-1/T}\right]_0^{T\ln 2}$$

$$\theta = -\omega_0 T \left[e^{-t/T}\right]_0^{T\ln 2}$$

$$\theta = -\omega_0 T \left[e^{-\ln 2} - e^0\right]$$

$$= -\omega_0 T \left[\frac{1}{2} - 1\right]$$

$$= -\omega_0 T \left(-\frac{1}{2}\right)$$

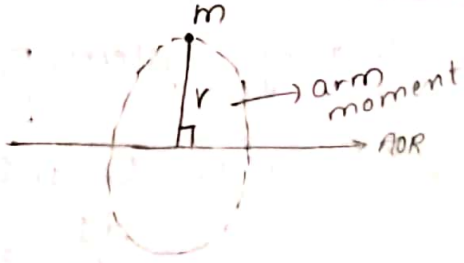
$$\boxed{\theta = \frac{\omega_0 T}{2}}$$

$$N = \frac{\theta}{2\pi} = \frac{\omega_0 T}{2(2\pi)} = \frac{\omega_0 T}{4\pi}$$

∴ No. of revolutions $= \omega_0 T / 4\pi$

Moment of Inertia :

The property of inability of a body to change its state of rest or uniform rotational motion about axis of rotation by itself is called rotational inertia or moment of inertia (I).



$$\boxed{I = mr^2}$$

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* Units: $kg\,m^2$ — S.I unit.

$g\,cm^2$ — C.G.S unit.

* It depends on → mass of body
→ distribution of mass of body (particles)
→ position of axis of rotation.

* Moment of inertia is a Tensor quantity i.e, scalar of vector.

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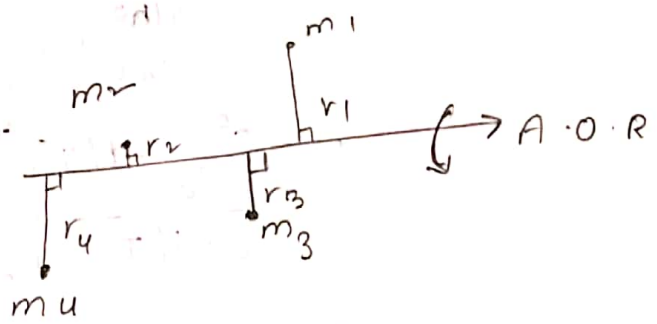

Moment of inertia of particle system,

$$I = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2.$$

In general, $\boxed{I = \sum_{i=1}^{n} m_i r_i^2}$

→ representing:

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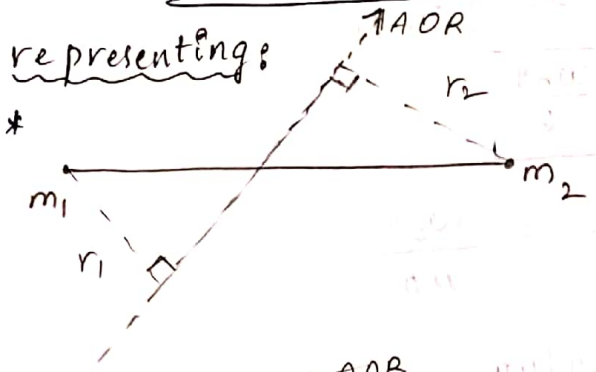

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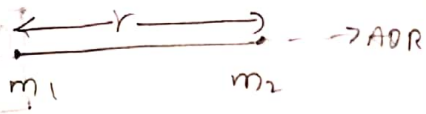

$I = m_1(0)^2 + m_2(0)^2$

$= 0.$

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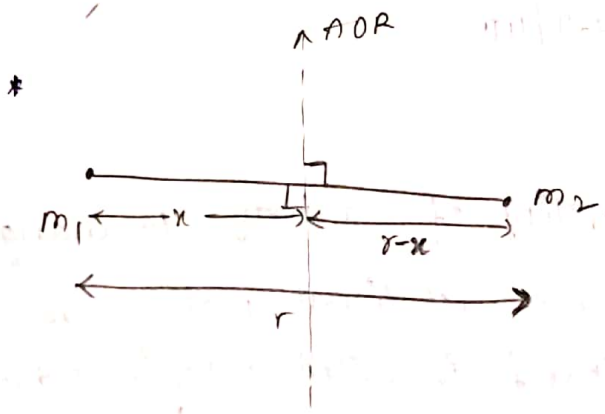

M.I of system, $\boxed{I = m_1 x^2 + m_2 (r-x)^2}$

* for I to be minimum,

$$\frac{dI}{dx} = 0$$

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$$m_1(2x) + m_2(2(r-x)(-1)) = 0$$

$$2m_1x = 2m_2r - 2m_2x$$

$$2x(m_1+m_2) = 2m_2r$$

$$\boxed{x = \frac{m_2 r}{m_1+m_2}}$$

clearly distance of CM from m_1 is $\boxed{x = \frac{m_2 r}{m_1+m_2}}$ *

* M.I of system is minimum when A.O.R is passing through C.M of system.

$$I = m_1\left(\frac{m_2 r}{m_1+m_2}\right)^2 + m_2\left(r - \frac{m_2 r}{m_1+m_2}\right)^2$$

$$= \frac{m_1 m_2^2 r^2}{(m_1+m_2)^2} + \frac{m_1^2 m_2 r^2}{(m_1+m_2)^2}$$

$$I = \frac{m_1 m_2 r^2 (m_1+m_2)}{(m_1+m_2)^2}$$

$$\boxed{I = \frac{m_1 m_2}{m_1+m_2} \times r^2}$$

* $\boxed{I = \mu r^2} \Rightarrow \mu = \frac{m_1 m_2}{m_1+m_2}$ = reduced mass.

Radius of gyration (K):

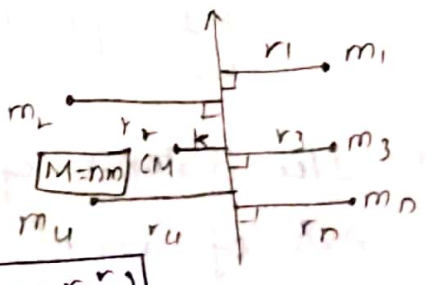
* It is the effective distance of all the particles of the system from axis of rotation to the pt where the total mass of the system appears to be concentrated.

* For n-particle system,

$$I = m_1 r_1^2 + m_2 r_2^2 + \cdots + m_n r_n^2$$

if $m_1 = m_2 = m_3 = \cdots = m_n = m$

$$\boxed{I = m(r_1^2 + r_2^2 + \cdots r_n^2)}$$



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$M = nm$

Clearly, $\boxed{I = MK^2}$ (CM)

— (1)

$$I = \frac{mn(r_1^2 + r_2^2 + \dots)}{n}$$

$$MK^2 = \frac{M}{n}(r_1^2 + r_2^2 + \dots r_n^2)$$

$$\boxed{K = \sqrt{\frac{r_1^2 + r_2^2 + r_3^2 + \dots + r_n^2}{n}}}$$

(or) $I = MK^2 \Rightarrow \boxed{K = \sqrt{\frac{I}{M}}}$

Units:

m — S.I unit

cm — C.G.S unit

* It depends on → position of axis of rotation
→ mass distribution.

→ $I = 12x^2 - 3x + 4$. Find CM of system.

$$\frac{dI}{dx} = 0$$

$$24x - 3 = 0$$

$$x = \frac{1}{8} \text{ units.}$$

Moment of Inertia of Extended bodies:

Formula:

$$\boxed{I = \int r^2 . dm}$$

r = ⊥ler distance of small element of mass 'dm' from Axis of rotation.

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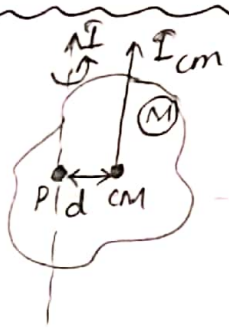
Theorems of Moment of Inertia:

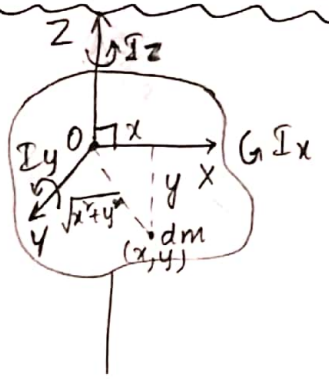
1. Parallel axis theorem.
2. Perpendicular axis theorem.

Parallel axis Theorem:



$$I = I_{cm} + Md^2 \quad *$$

Perpendicular Axis Theorem:



$$I_z = I_x + I_y \quad *$$

$$I_x = \int dm\, y^2$$

$$I_y = \int dm\, x^2$$

$$I_z = \int dm(x^2+y^2) = \int dm\, x^2 + \int dm\, y^2.$$

Mass densities:

1) Linear mass density:

$$\lambda = \frac{dm}{dx} \longrightarrow \text{wire, ring, thin rods.}$$

2) Areal mass density:

$$\sigma = \frac{dm}{dA} \longrightarrow \text{disc, rectangular plate, hollow sphere, hollow cylinder.}$$

3) Volume mass density:

$$\rho = \frac{dm}{dv} \longrightarrow \text{solid sphere, solid cone, solid cylinder, solid cube etc...}$$

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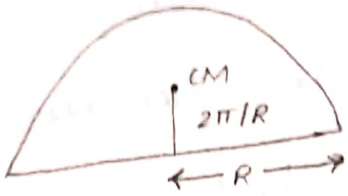
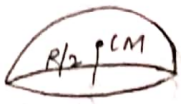
Centre of mass of some objects:-

1) Half ring (or) Hollow hemi cylinder:

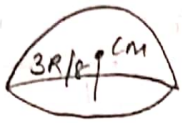


2) Semicircular disc/solid hemi cylinder:

3) Hollow hemi sphere:



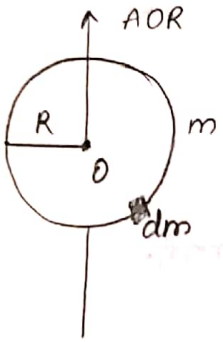
4) Solid hemi sphere:



Applications:

① M.I of ring:

ⓐ About it's natural axis:

* 

$dI = (dm)R^2$

∴ To total M.I of ring:

$I_0 = \int dm R^2$

$= R^2 \int dm$

$\boxed{I_0 = MR^2}$

we have,

$\boxed{I_0 = Mk_0^2}$ $\quad Mk_0^2 = MR^2 \Rightarrow \boxed{k_0 = R}$

*

m = remaining mass

$= \frac{M}{2\pi}(2\pi - \theta)$

$\boxed{I = MR^2}$ M = total mass of ring.

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AOR is tangent to ring & ⊥ar to it's plane.

$I = I_0 + md^2$

$I = MR^2 + MR^2$

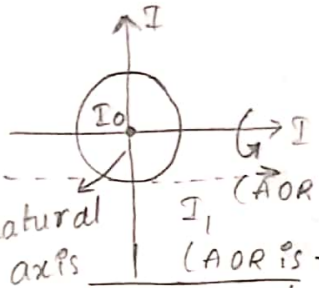
$\boxed{I = 2MR^2}$

we have,

$I = mK^2$

$MK^2 = 2MR^2$

$\boxed{K = \pm\sqrt{2}R}$



(AOR is a diametrical axis)

(AOR is tangent to ring & in it's plane)

$\boxed{I_0 = MR^2}$

perpendicular axis theorem,

$I_0 = I + I$

$I_0 = 2I$

$MR^2 = 2I$

$\boxed{I = \frac{MR^2}{2}}$

we have,

$I = MK^2$

$\boxed{K = \pm R/\sqrt{2}}$

we have,

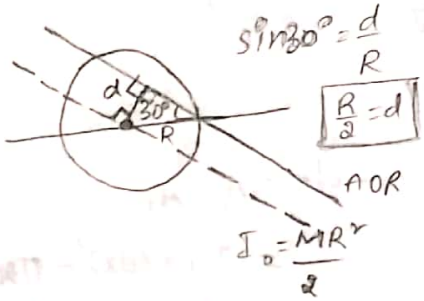
$I_1 = I + md^2$

$= \frac{MR^2}{2} + MR^2$

$\boxed{I_1 = \frac{3MR^2}{2}}$

and $K = \pm\sqrt{\frac{3}{2}R^2}$

$\boxed{K = \pm\sqrt{\frac{3}{2}}R}$



$\sin 30° = \frac{d}{R}$

$\boxed{\frac{R}{2} = d}$

$R = 2d$

$\boxed{d = R/2}$

$I = I_0 + Md^2$

$I = \frac{MR^2}{2} + \frac{MR^2}{4} = \frac{3MR^2}{4}$

$\boxed{I = 3MR^2/4}$

$\Rightarrow \boxed{K = \pm\frac{R}{2}\sqrt{3}}$

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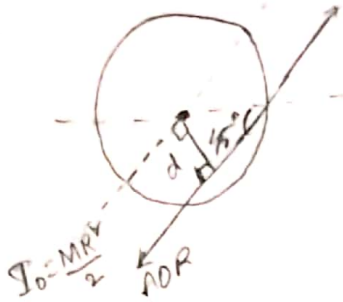


$\sin 45° = \frac{d}{R}$

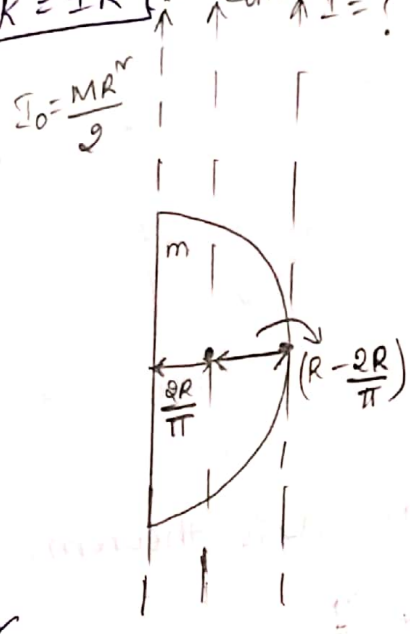
$\frac{R}{\sqrt{2}} = d$

$I = I_0 + Md^2$

$= \frac{MR^2}{2} + \frac{MR^2}{2}$

$\boxed{I = MR^2}$

$\Rightarrow \boxed{K = \pm R}$



* $I_0 = I_{cm} + m\left(\frac{2R}{\pi}\right)^2$

$\frac{mR^2}{2} = I_{cm} + m\frac{4R^2}{\pi^2}$

$I_{cm} = mR^2\left(\frac{1}{2} - \frac{4}{\pi^2}\right)$

and

$I = I_{cm} + m\left(R - \frac{2R}{\pi}\right)^2$

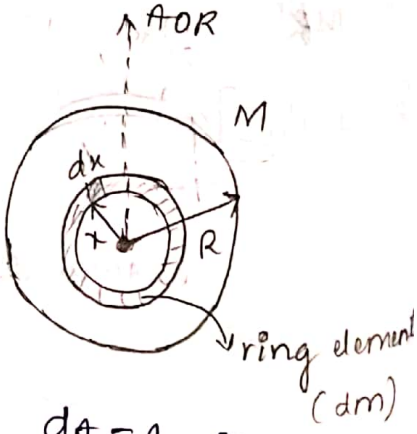
$= mR^2\left(\frac{1}{2} - \frac{4}{\pi^2}\right) + MR^2\left(1 + \frac{4}{\pi^2} - \frac{4}{\pi}\right)$

$\boxed{I = mR^2\left(\frac{3}{2} - \frac{4}{\pi}\right)}$

② M.I of Disc:

ⓐ About natural axis:

AOR is ⊥ler to plane & passing through centre.



$\sigma = \frac{M}{\pi R^2} = \text{const.}$

* mass of ring element,

$dm = \sigma\, dA$

$= \frac{m}{\pi R^2}(2\pi x\, dx) = \frac{2mx\, dx}{R^2}$

$dA = A_2 - A_1$

$= \pi(x+dx)^2 - \pi x^2$

$= 2\pi x\, dx$

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. M.I of ring element,

$$dI = (dm)x^2$$

∴ Total M.I,

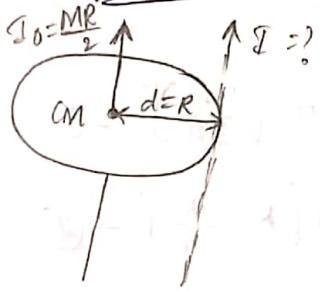
$$I = \int dm = \int_0^R \left(\frac{2m}{R^2} x \cdot dx\right) \cdot x^2$$

$$I = \frac{2m}{R^2} \int_0^R x^3 \cdot dx$$

$$= \frac{2m}{R^2} \times \frac{R^4}{4}$$

$$I = \frac{2mR^2}{2}$$

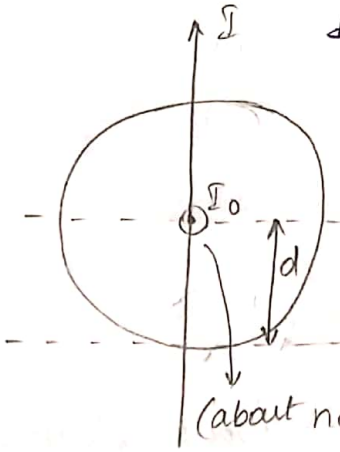
$$\boxed{I = \frac{mR^2}{2}}$$



$$I = I_0 + Md^2$$

$$= \frac{MR^2}{2} + MR^2$$

$$\boxed{I = \frac{3MR^2}{2}}$$



$\rightarrow I$ (AOR is diametrical axis)

$\rightarrow I_1 = ?$

$$I_0 = I + I$$

$$I_0 = 2I$$

$$\boxed{I = \frac{MR^2}{4}}$$

parallel axis theorem,

$$I_1 = I + Md^2$$

$$I_1 = \frac{MR^2}{4} + MR^2 = \frac{5MR^2}{4}$$

$$\boxed{I_1 = \frac{5MR^2}{4}}$$

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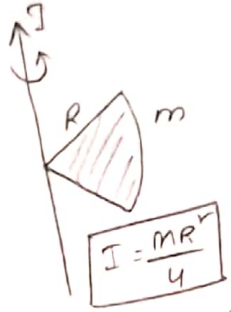
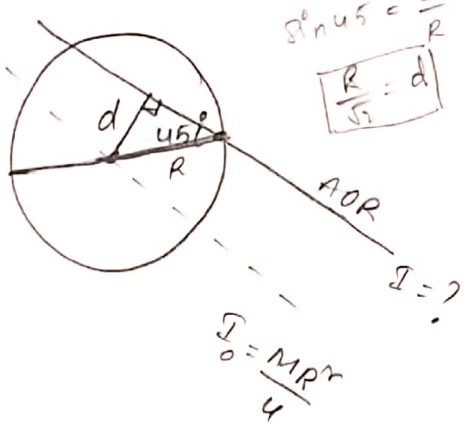
*



$$I = \frac{MR^2}{4}$$

*



$\sin 45^\circ = \frac{d}{R}$

$$d = \frac{R}{\sqrt{2}}$$

$$I = I_{cm} + Md^2$$

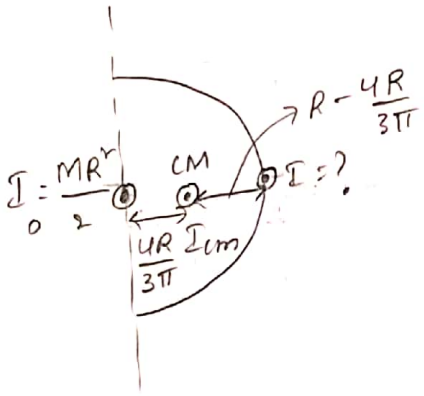
$$= I_0 + Md^2$$

$$= \frac{MR^2}{4} + M\left(\frac{R}{\sqrt{2}}\right)^2$$

$$= \frac{MR^2}{4} + \frac{MR^2}{2}$$

$$I = \frac{3MR^2}{4}$$

*



$$I_0 = I_{cm} + m\left(\frac{4R}{3\pi}\right)^2 \quad -①$$

$$I = I_{cm} + m\left(R - \frac{4R}{3\pi}\right)^2 \quad -②$$

$$I - I_0 = m\left(R - \frac{4R}{3\pi}\right)^2 - m\left(\frac{4R}{3\pi}\right)^2$$

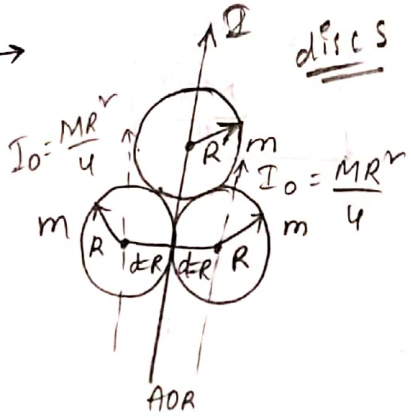
$$I - \frac{MR^2}{2} = m\left(R\right)\left(R - \frac{8R}{3\pi}\right)$$

$$I = MR^2\left(\frac{3\pi - 8}{3\pi}\right) + \frac{MR^2}{2}$$

$$I = MR^2\left(\frac{6\pi - 16 + 3\pi}{6\pi}\right)$$

$$I = MR^2\left(\frac{9\pi - 16}{6\pi}\right)$$

→ *



M.I of system about given AOR

$$I = I_1 + I_2 + I_3$$

$$I = \left(\frac{MR^2}{4} + MR^2\right) + \left(\frac{MR^2}{4} + MR^2\right) + \frac{MR^2}{4}$$

$$= \frac{3MR^2}{4} + 2MR^2 = \frac{11MR^2}{4}$$

we have, $I = M_{total}k^2 \Rightarrow 3M(k^2) = \frac{11MR^2}{4}$

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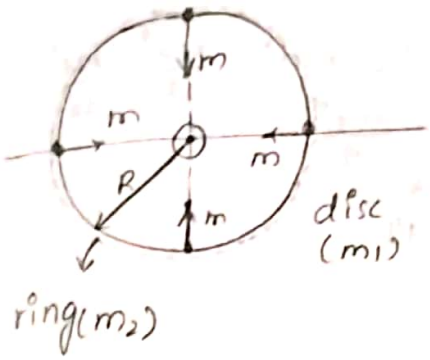
$$K = \pm \frac{R}{2}\sqrt{\frac{11}{3}}$$

→ 

Find M·I of system about an AOR ⊥ler to plane and passing through centre.

$$I = I_{ring} + I_{disc} + I_{particles}$$

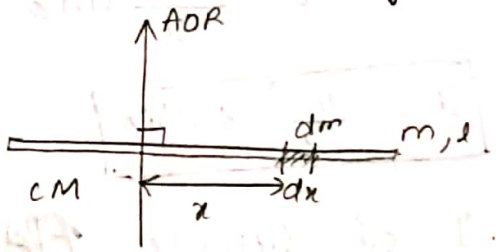
$$= m_2 R^2 + m_1 \frac{R^2}{2} + 4\left(m\left(\frac{R}{2}\right)^2\right)$$

$$= R^2\left(m_2 + \frac{m_1}{2} + m\right)$$

$$\boxed{I = \frac{R^2}{2}(2m_2 + 2m + m_1)}$$

③ M·I of thin rod/wire :

ⓐ AOR is passing through centre & ⊥ler to it's length.



∴ mass of small element,

$$dm = \lambda \cdot dx = \frac{m}{l} dx \quad -①$$

M·I of small element about AOR,

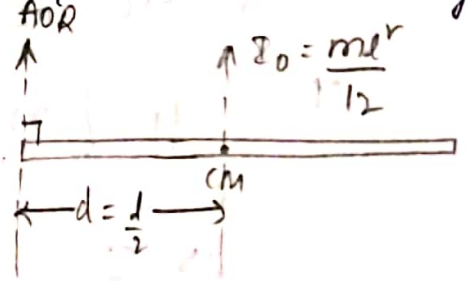
$$dI = (dm)\, x^2$$

$$= \frac{m}{l} dx\,(x^2)$$

∴ Total M·I,

$$I = \int_{x=-l/2}^{l/2} dI = \frac{m}{l}\frac{[x^3]_{-l/2}^{l/2}}{3} = \frac{m}{3l}\left(\frac{l^3}{4} + \frac{l^3}{4}\right) = \frac{ml^2}{12}$$

ⓑ AOR is passing through one end & ⊥ler to it's length



① $$I = \int_{x=0}^{x=l} dI = \frac{m}{l}\frac{l^3}{3} = \frac{ml^2}{8}.$$

(OR)

② $\Rightarrow I = I_0 + md^2$.

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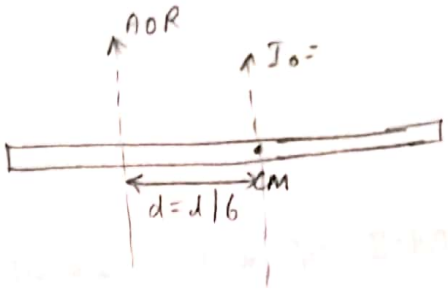
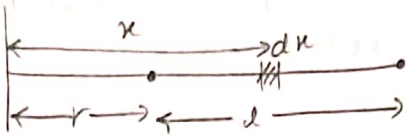


(1) $I = \int_{x=-l/3}^{2l/3} dI$

(2) $I = I_0 + md^2 = \frac{ml^2}{12} + \frac{ml^2}{36}$

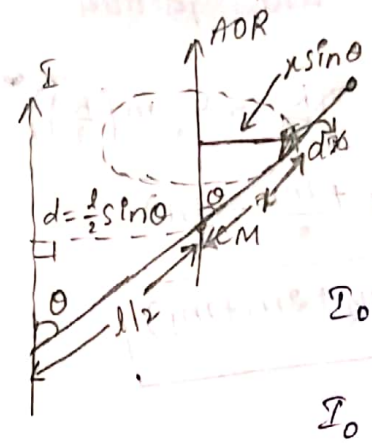
$= \frac{ml^2}{9}$.

*



$I = \int_{x'=r}^{x=r+l} dI$

*



$dI = (dm)(x\sin\theta)^2$

$I = \int_{x=-l/2}^{x=l/2} dI$

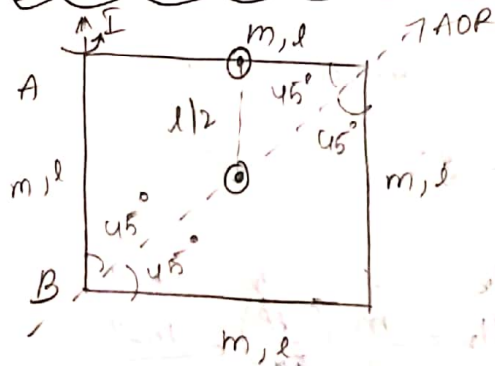
$I_0 = \int_{x=-l/2}^{x=l/2} \left(\frac{m}{l}\right) \cdot dx \cdot x^2 \sin^2\theta$

$I_0 = \frac{ml^2}{12} \sin^2\theta$

$I = I_0 + \frac{ml^2}{4}\sin^2\theta = \frac{ml^2}{12}\sin^2\theta + \frac{ml^2}{4}\sin^2\theta$

$$\boxed{I = \frac{ml^2}{3}\sin^2\theta}$$

M.I of square FRAME:



(a) M.I of system about an AOR passing through center and ⊥ar to its plane.

Sol: $I = 4[I_{wire}]$

$= 4\left[\frac{ml^2}{12} + m(l/2)^2\right]$

$I = 4\left[\frac{ml^2}{12} + \frac{ml^2}{4}\right]$

$\Rightarrow I = \frac{4ml^2}{3}$

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(b) M.I about an AOR AB:

$$I = m(0)^2 + m\frac{l^2}{3} + m\frac{l^2}{3} + ml^2$$

$$= 5\frac{ml^2}{3}$$

(c) M.I about AOR BC:

$$I = \left(\frac{ml^2}{3}\sin^2 45\right)\times 4$$

$$= \frac{ml^2}{3}\times\frac{1}{2}\times 4$$

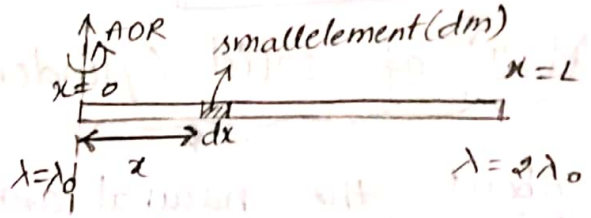
$$= 2\frac{ml^2}{3}$$

$$\therefore (Ki^2 m)k^2 = 2\frac{ml^2}{3}$$

$$k^2 = \frac{l^2}{6}$$

$$k = \pm\frac{l}{\sqrt{6}}$$

→ Linear mass density of rod increases linearly from λ_0 to $2\lambda_0$ as shown. Find the M.I of the rod about an axis passing through one end and ⊥ler to it's length.



(variable mass density)

$\lambda = a + bx$

$x = 0,\ \lambda = \lambda_0$

$\boxed{\lambda_0 = a}$

& at $x = L,\ \lambda = 2\lambda_0$

$2\lambda_0 = \lambda_0 + bL$

$\boxed{\frac{\lambda_0}{L} = b} \Rightarrow \boxed{\lambda_0 = bL}$

$\boxed{\lambda = \lambda_0 + \frac{\lambda_0}{L}x}$

∴ mass of element,

$$dm = \lambda_0\left(1 + \frac{x}{L}\right)\cdot dx$$

now, $dI = (dm)x^2$

∴ total M.I,

$$I = \int_{x=0}^{x=L} dI = \int_{x=0}^{x=L} \lambda_0\left(1+\frac{x}{L}\right)x^2\,dx$$

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$$I = \lambda_0 \int_0^L x^2 \, dx + \frac{\lambda_0}{L} \int_0^L x^3 \cdot dx$$

$$= \lambda_0 \frac{[x^3]_0^L}{3} + \frac{\lambda_0}{L} \frac{[x^4]_0^L}{4}$$

$$= \frac{\lambda_0 L^3}{3} + \frac{\lambda L^3}{4} = \frac{7\lambda L^3}{12}$$

$$\boxed{I = \frac{7\lambda L^3}{12}}$$

mass of element, $dm = \lambda dx$

mass of rod,

$$M = \int_{x=0}^{L} dm = \frac{3\lambda_0}{2} \ell$$

Extra:

centre of Mass of rod:

$$x_{cm} = \frac{\int x \, dm}{\int dm}$$

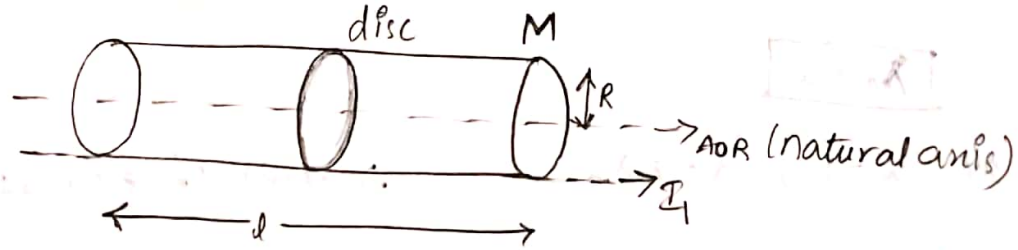
$$I = \frac{7 dx^2}{12_6} \times \frac{3M}{3x}$$

$$= \frac{7 M \ell^2}{18}$$

$$\boxed{I = \frac{7M\ell^2}{18}}$$

(4) M.I of solid Cylinder:

(a) About the natural axis:



M.I of disc element,

about AOR,

$$dI = \frac{(dm)R^2}{2}$$

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Total M.I,

$$I = \int \frac{dm R^2}{2}$$

$$= \frac{R^2}{2} \int dm$$

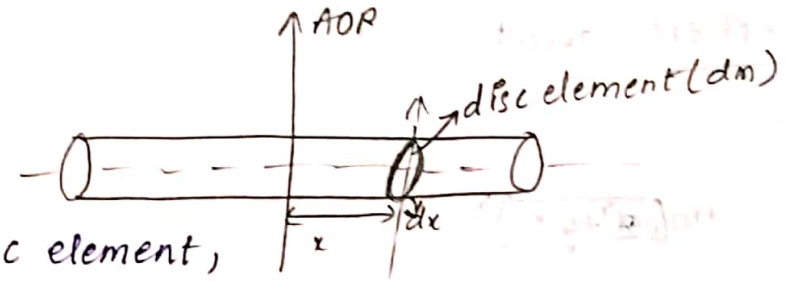
$$\boxed{I = \frac{MR^2}{2}}$$

(b) M.I about AOR ~~about~~ tangent to the cylinder & ||el to ~~the~~ it's length,

$$I_1 = I_0 + MR^2$$

$$I_1 = \frac{MR^2}{2} + MR^2 = \frac{3MR^2}{2}.$$

(c) M.I about AOR passing through centre & ⊥er to it's length,



mass of disc element,

$$dm = \frac{M}{\pi R^2 l} (\pi R^2 . dx)$$

$$\boxed{dm = \frac{M}{L} . dx}$$

M.I of disc element about given AOR,

$$dI = \frac{(dm)R^2}{4} + dm(x^2)$$

$$= \frac{MR^2}{4L} dx + \frac{M}{L} x^2 . dx$$

∴ Total M.I;

$$I = \int_{x=-l/2}^{l/2} dI \Rightarrow \boxed{I = \frac{MR^2}{4} + \frac{ML^2}{12}}$$

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⑤ M.I of hollow cylinder:

ⓐ Total M.I = MR^2 — about it's natural axis.

ⓑ M.I about an AOR tangent to cylinder & ⊥el to it's length:

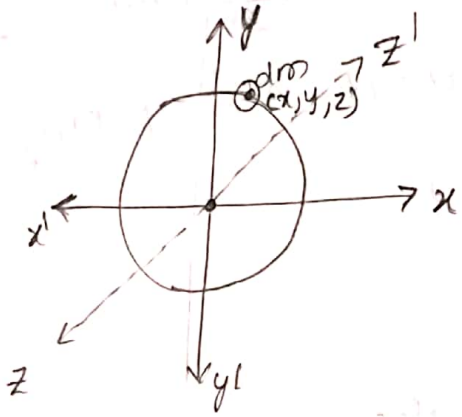
$$I_1 = MR^2 + MR^2 = 2MR^2.$$

ⓒ M.I about an AOR passing through centre & ⊥er to it's length,

$$\boxed{I_1 = \frac{MR^2}{2} + \frac{ML^2}{12}}$$

⑥ M.I of hollow sphere:

$$R^2 = \sqrt{x^2+y^2+z^2}$$



* M.I of sphere about x-axis:

$$I_{xx'} = \int dm(\sqrt{x^2+y^2})^2$$

$$= \int (y^2+z^2).dm$$

(ly M.I about y-axis:

$$I_{yy'} = \int (x^2+z^2).dm$$

M.I about z-axis:

$$I_{zz'} = \int (x^2+y^2).dm$$

now, $I_{xx'} + I_{yy'} + I_{zz'} = \int (2x^2+2y^2+2z^2).dm$

$$3I_{xx'} = 2R^2 \int dm$$

$$\boxed{I_{xx'} = \frac{2}{3}MR^2}$$

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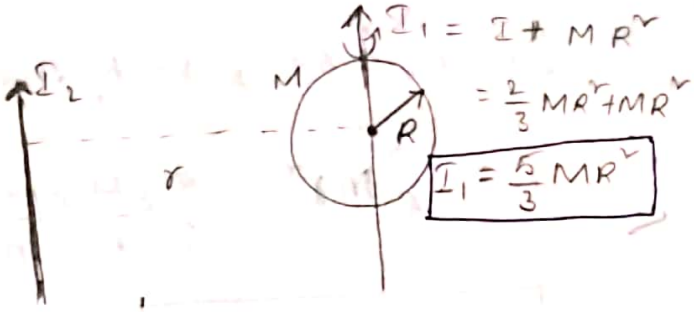
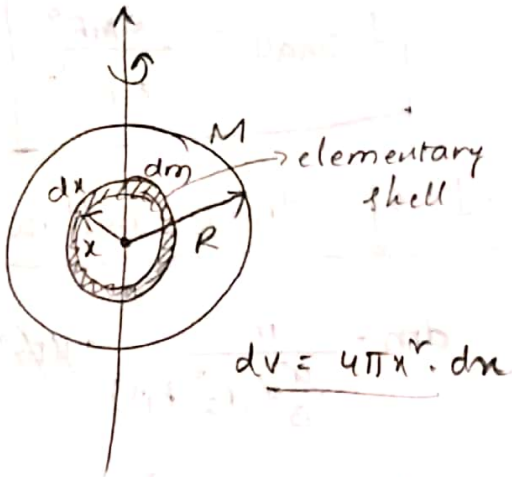
$I_2 = I + Mr^2$

$$I_2 = \frac{2}{3}MR^2 + Mr^2$$



M.I of solid sphere:



* $dm = \frac{m}{\frac{4}{3}\pi R^3} \times 4\pi x^2 . dx$

$$dm = \frac{3m}{R^3} x^2 . dx$$

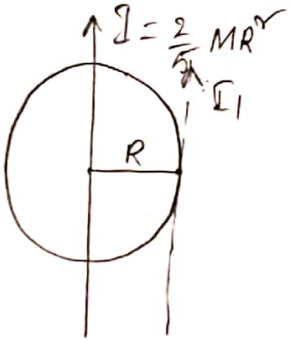
$dV = 4\pi x^2 . dx$

M.I of small element,

$$dI = \frac{2}{3}(dm)x^2$$

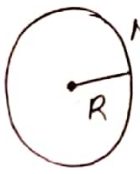
Total M.I, $I = \int dI = \frac{2}{3} \times \frac{3}{R^3} m \int x^4 . dx$

$$I = \frac{2}{5} MR^2$$



* $I = I + MR^2 = \frac{2}{5}MR^2 + MR^2$

$$I = \frac{7}{5}MR^2 \Rightarrow K = \pm R\sqrt{\frac{7}{5}}$$

→ 

M It melts and recasts into n identical solid spheres. ○ ○ ○ $m = \frac{M}{n}$

Find M.I of → ○ ○ ○ ○ ○

Here, Volume = const

$$\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3$$

$$r^3 = R^3/n \Rightarrow \boxed{R^3 = nr^3}$$

for $n = 8$, $r = R/2$

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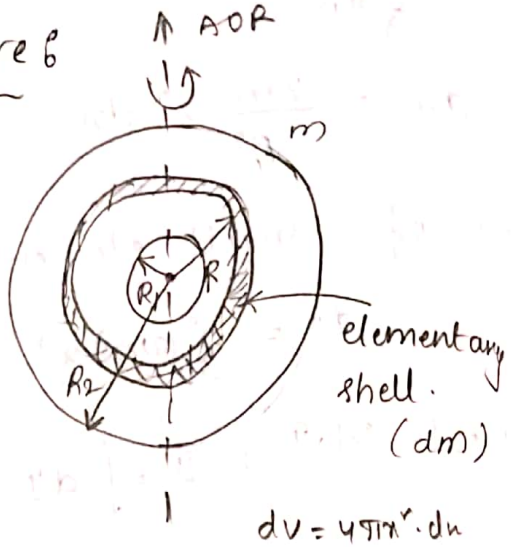
for $n = 64$, $r = \frac{R}{4}$.

$$I_{small} = \frac{2}{5} m r^2 = \frac{2}{5} \times \frac{M}{n} \times \frac{R^2}{n^{2/3}} = \frac{2MR^2}{5n^{5/3}}$$

$$\boxed{I_{small} = \frac{2MR^2}{5n^{5/3}}}$$

M.I of thick hollow sphere



$$dm = \frac{m}{\frac{4}{3}\pi(R_2^3 - R_1^3)} (4\pi x^2 . dx)$$

$$dm = \frac{3m x^2 . dx}{R_2^3 - R_1^3}$$

$dV = 4\pi x^2 . dx$

* $$dI = \frac{2}{3} dm\, x^2$$

$$dI = \frac{2}{3} \times \frac{3m}{R_2^3 - R_1^3} x^4 . dx.$$

$$I = \frac{2m}{R_2^3 - R_1^3} \int_{R_1}^{R_2} x^4 . dx$$

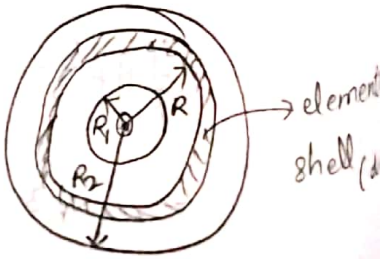
$$= \frac{2m}{5(R_2^3 - R_1^3)} (R_2^5 - R_1^5)$$

$$\boxed{I = \frac{2m(R_2^5 - R_1^5)}{5(R_2^3 - R_1^3)}}$$

M.I of thick rings

* $$dm = \frac{m}{\pi(R_2^2 - R_1^2)} (2\pi x . dx)$$

$$dm = \frac{2m x . dx}{R_2^2 - R_1^2}$$



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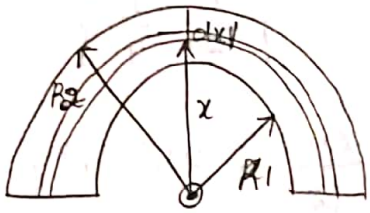
now, $I = \int (dm)x^2 = \frac{2m}{R_2^2 - R_1^2} \int x^3 . dx = \frac{2m}{\pi(R_2^2 - R_1^2)} (R_2^4 - R_1^4)$

$$\boxed{I = \frac{m(R_2^2 + R_1^2)}{2}}$$

* $dm = \frac{m}{\frac{\pi(R_2^2 - R_1^2)}{2}} \times \pi x . dx$

$= \frac{2mx . dx}{R_2^2 - R_1^2}$



(AOR)

$I = \int dm . x^2 = \frac{2m \times 1}{\pi(R_2^2 - R_1^2)} \int x^3 . dx .$

$$\boxed{I = \frac{m(R_2^2 + R_1^2)}{2}}$$

M.I of Triangular plate

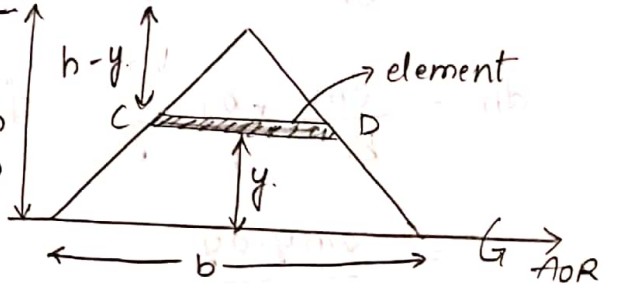
↳ uniform mass distribution

$dm = \frac{M}{\frac{1}{2}(b)(h)} \times (CD) dy$

$= \frac{2m}{bh} (CD) . dy$

$= \frac{2m}{bh} \left(\frac{b}{h}\right)(h-y) . dy$

$dm = \frac{2m(h-y) . dy}{h^2}$



from similar Δle concept

$\frac{h-y}{CD} = \frac{h}{b}$

$CD = \frac{b}{h}(h-y)$

now, $dI = (dm) y^2$

$I = \int_0^h \frac{2m(h-y)}{h^2} . y^2 . dy$

$= \frac{2m}{h^2} \left\{ \int_0^h hy^2 . dy - \int_0^h y^3 . dy \right\}$

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$$I = \frac{2m}{h^2}\left\{ h\int_0^h y^2\,dy - \int_0^h y^3\,dy \right\}$$

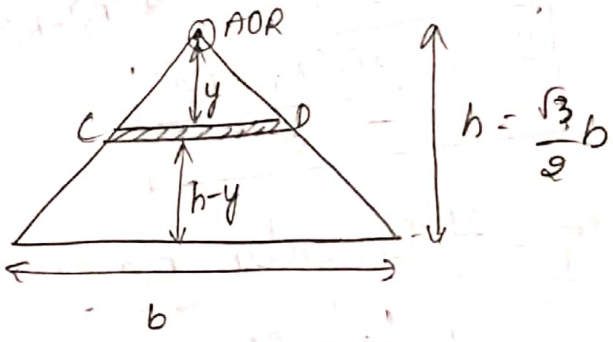
$$= \frac{2m}{h^2}\left\{ h\left(\frac{h^3}{3}\right) - \frac{h^4}{4} \right\}$$

$$= 2m\left\{ \frac{h^2}{3} - \frac{h^2}{4} \right\}$$

$$= \frac{2m}{12}\left\{ 4h^2 - 3h^2 \right\}$$

$$\boxed{I = \frac{mh^2}{6}}$$

Equilateral Sle :



$$dm = \frac{m}{\frac{1}{2}bh} \times (CD).dy$$

$$dm = \frac{2m}{bh}(CD).dy$$

$$dm = \frac{2m}{bh} \times \frac{by}{h}\,dy$$

$$= \frac{2my.dy}{h^2}$$

from similar Sles concept,

$$\frac{y}{CO} = \frac{h}{b}$$

$$\boxed{CD = \frac{by}{h}}$$

$$dI = \frac{dm(CD)^2}{12} + (dm)y^2$$

$$= \frac{2my}{6h^2}\left(\frac{b^2y^2}{h^2}\right).dy + \frac{2my.dy}{h^2}y^2$$

$$dI = \frac{my^3b^2}{6h^4}.dy + \frac{2my^3.dy}{h^2}$$

$$dI = \left(\frac{mb^2}{6h^4} + \frac{2m}{h^2}\right)(y^3).dy$$

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$$I = \left(\int_0^b y^2 \cdot dy\right)\left(\frac{mb^2}{6h^2} + \frac{2m}{h^2}\right)$$

$$I = \left(\frac{mb^2}{6h^2} + \frac{2m}{h^2}\right)\left(\frac{h^3}{4}\right)$$

$$\boxed{I = \frac{mb^2}{24} + \frac{mh^2}{2}}$$

$$I = \frac{mb^2}{24} + \frac{m\left(\frac{3}{4}b^2\right)}{2}$$

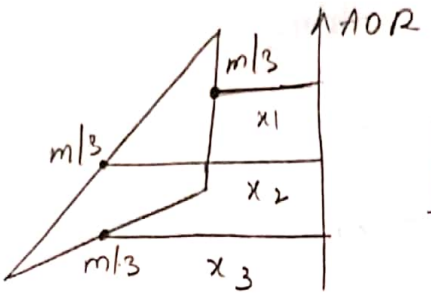
$$= \frac{mb^2}{24} + \frac{9mb^2}{24}$$

$$\boxed{I = \frac{5mb^2}{12}}$$

short-cut:

↳ uniform mass distribution.

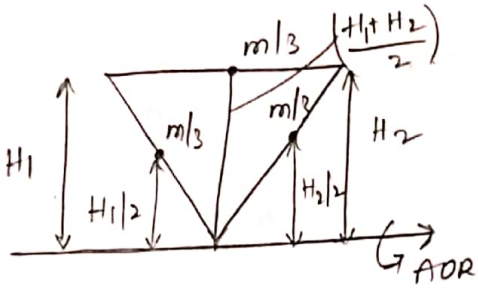
$$I = \sum m_i r_i^2$$



$$I = \frac{m}{3}x_1^2 + \frac{m}{3}x_2^2 + \frac{m}{3}x_3^2$$

$$\boxed{I = \frac{m}{3}(x_1^2 + x_2^2 + x_3^2)}$$

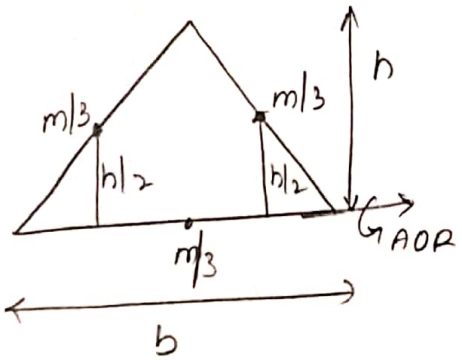
→



$$I = \frac{m}{3}\left(\frac{H_1^2}{4} + \frac{H_2^2}{4} + \frac{H_3^2}{4}\right)$$

$$= \frac{m}{12}(H_1^2 + H_2^2 + H_3^2)$$

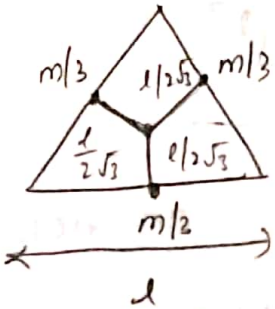
→



$$I = \frac{m}{3}(h^2/4 + h^2/4 + 0)$$

$$\boxed{I = \frac{mh^2}{6}}$$

Equilateral Δle,



$$I = \frac{m}{3}\left(\frac{l}{2\sqrt{3}}\right)^2 \times 3$$

$$\boxed{I = \frac{ml^2}{12}}$$

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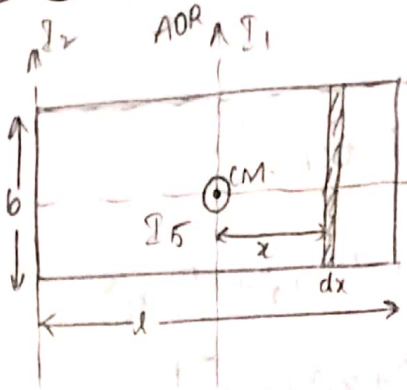
→

M.I of rectangular Lamina:



$\rightarrow I_4 = I_3 + m\left(\frac{b}{2}\right)^2 = \frac{mb^2}{12} + \frac{mb^2}{4} = \frac{mb^2}{3}$

$\rightarrow I_3 = \frac{mb^2}{12}$

* $dm = \frac{m}{l\cancel{b}} \times \cancel{b} \times dx = \frac{m}{l} \cdot dx$

& $dI = (dm)\,x^2$

$$I_1 = \int_{x=-l/2}^{l/2} dI = \int_{x=-l/2}^{l/2} \frac{m}{l} x^2 \cdot dx$$

$$\boxed{I_1 = \frac{ml^2}{12}}$$

$$I_2 = I_1 + m\left(\frac{l}{2}\right)^2$$

$$= \frac{ml^2}{12} + \frac{ml^2}{4}$$

$$\boxed{I_2 = \frac{ml^2}{3}}$$

* M.I about an AOR passing through centre & ⊥er to it's plane,

$$I_5 = I_3 + I_1 \Rightarrow \boxed{\frac{m(l^2+b^2)}{12} = I_5}$$

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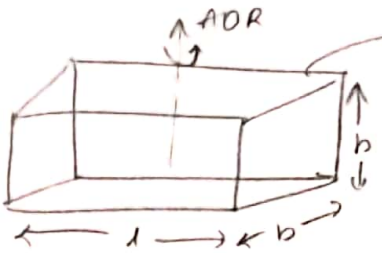
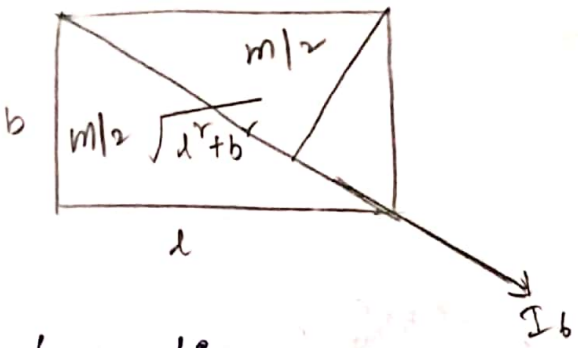


rectangular parallelopiped,

$$I = \frac{m(l^2+b^2)}{12}$$

*



$$I_b = 2\left(\frac{\frac{m}{2}b^2}{6}\right)$$

$$I_b = \frac{mb^2}{6}$$

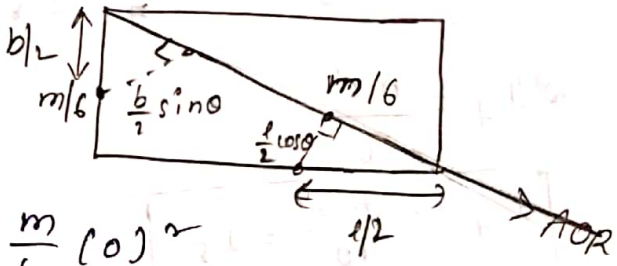
$$I_b = \frac{ml^2b^2}{6(l^2+b^2)}$$

from figure,

$$lb = 2\left(\frac{1}{2}(\sqrt{l^2+b^2})h\right)$$

$$\frac{l^2b^2}{l^2+b^2} = h^2$$

short cut :



$$I_1 = \frac{m}{6}\left(\frac{b}{2}\sin\theta\right)^2 + \frac{m}{6}\left(\frac{l}{2}\cos\theta\right)^2 + \frac{m}{6}(0)^2$$

$$= \frac{m}{24}(b^2\sin^2\theta + l^2\cos^2\theta)$$

$$= \frac{m}{24}\left(b^2\left(\frac{l^2}{b^2+l^2}\right) + l^2\left(\frac{b^2}{l^2+b^2}\right)\right)$$

$$= \frac{m}{24}\,\frac{(2l^2b^2)}{b^2+l^2}$$

$$I_1 = \frac{m(l^2b^2)}{12(b^2+l^2)}$$

$$I = 2I_1$$

$$I = \frac{m}{6}\left(\frac{l^2b^2}{l^2+b^2}\right)$$

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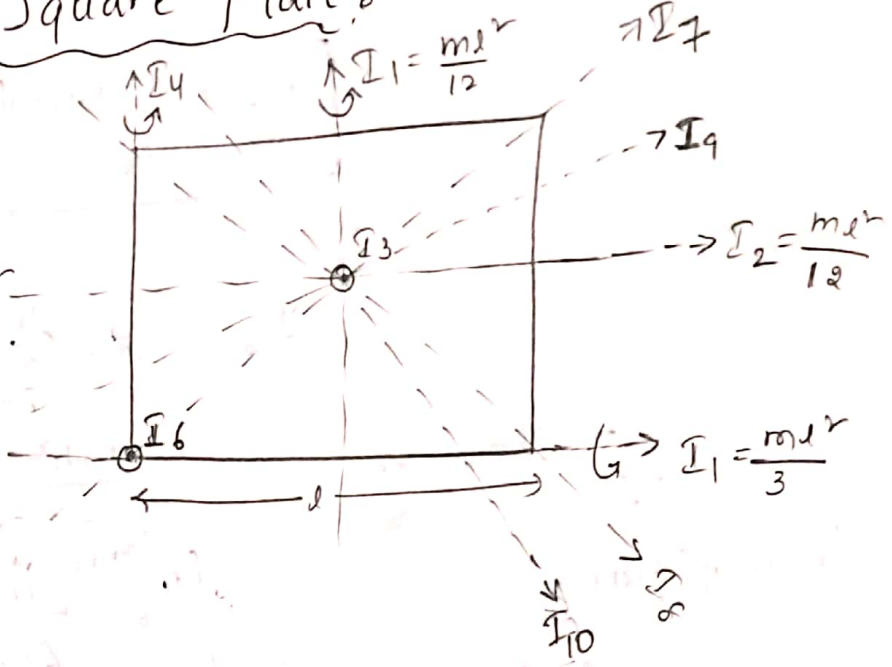
M.I of Square Plate:



$$I_3 = I_1 + I_2$$

$$= \frac{ml^2}{12} + \frac{ml^2}{12}$$

$$\boxed{I_3 = \frac{ml^2}{6}}$$

$$\boxed{I_2 = I_3/2}$$

$$I_4 = I_1 + m\left(\frac{l}{2}\right)^2$$

$$\boxed{I_4 = \frac{ml^2}{3}}$$

$$I_6 = I_5 = \frac{ml^2}{3} + \frac{ml^2}{3} = \frac{2ml^2}{3}$$

$$\boxed{I_5 = I_6 = \frac{2ml^2}{3}}$$

$$I_7 + I_8 = I_3 \quad \& \quad I_7 = I_8$$

$$2I_7 = \frac{ml^2}{6}$$

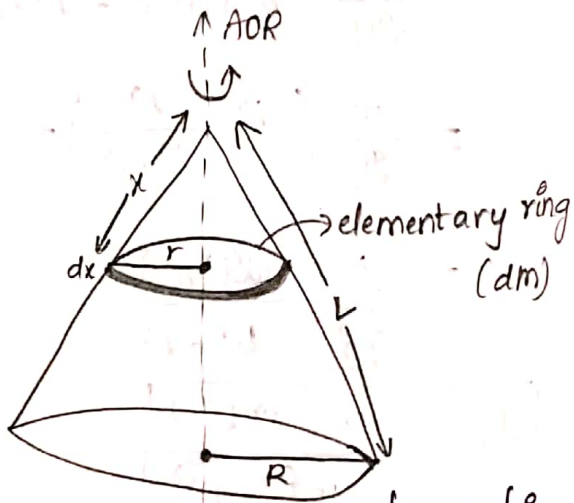
$$\boxed{I_7 = \frac{ml^2}{12} = I_8}$$

$$I_9 = I_{10}$$

$$I_9 + I_{10} = I_3$$

$$\boxed{I_9 = \frac{ml^2}{12} = I_{10}}$$

M.I of hollow cone:



* $dm = \frac{m}{\pi RL}(2\pi r)\cdot dx$

$$dm = \frac{2mr}{RL}\cdot dx$$

$$dI = (dm)\cdot r^2$$

$$= \frac{2mr^3}{RL}\cdot dx$$

$$dI = \frac{2m}{RL}(r^3\cdot dx) = \frac{2mR^3x^3}{L^4}\cdot dx$$

from fig,

$$\frac{R}{L} = \frac{r}{x}$$

$$r = \frac{Rx}{L}$$

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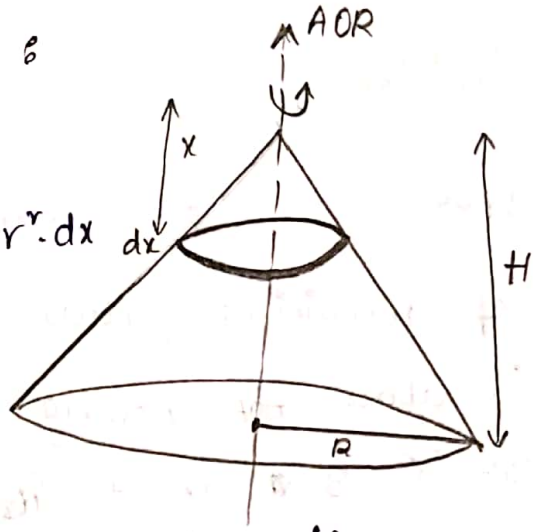
$$I = \int_0^L dI = \frac{2mR^2}{L^4} \int_0^L x^3 \cdot dx = \frac{2mR^2}{L^4}\left(\frac{x^4}{4}\right)_0^L$$

$$\boxed{I = \frac{mR^2}{2}}$$

M.I of solid cone :



$$* \quad dm = \frac{m}{\frac{1}{3}\pi R^2 H} \times \pi r^2 \cdot dx$$

$$= \frac{3mr^2 \cdot dx}{R^2 H}$$

$$= \frac{3mR^2x^2}{R^2H^3} \cdot dx$$

$$dm = \frac{3mx^2 \cdot dx}{H^3}$$

from fig,

$$\frac{R}{H} = \frac{x}{x}$$

$$\Rightarrow r = \frac{Rx}{H}$$

$$dI = \frac{dm(r^2)}{2}$$

$$dI = \frac{dm}{2}\left(\frac{R^2x^2}{H^2}\right)$$

$$dI = \frac{3mR^2x^4}{2H^5} \cdot dx$$

$$I = \frac{3mR^2}{2H^5}\left[\frac{x^5}{5}\right]_0^H$$

$$I = \frac{3mR^2}{2\times 5} \Rightarrow \boxed{I = \frac{3mR^2}{10}}$$

→ There are 4 spheres at the corners of a weightless square frame. Find the moment of inertia of system about it's Axis of rotation.

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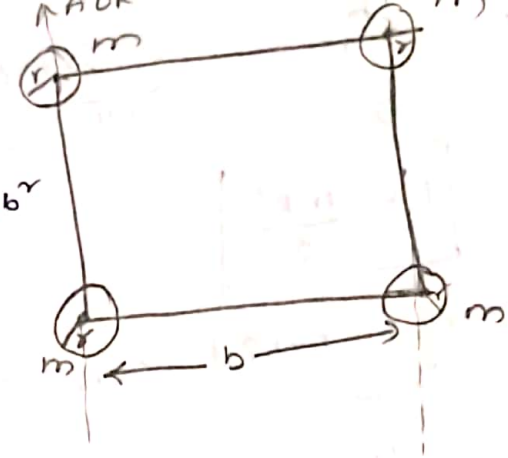
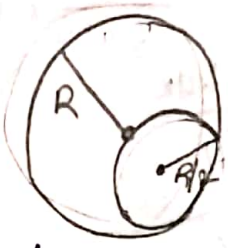
$$I = \frac{2}{5}mr^2 + \frac{2}{5}mr^2 + \frac{2}{5}mr^2 + mb^2 + \frac{2}{5}mr^2 + mb^2$$

$$= \frac{8}{5}mr^2 + 2mb^2$$



Cavity Problems:

disc problems:



* M.I of remaining portion of disc about AOR passing through 'O' & ⊥ler to plane.

Case (1):-

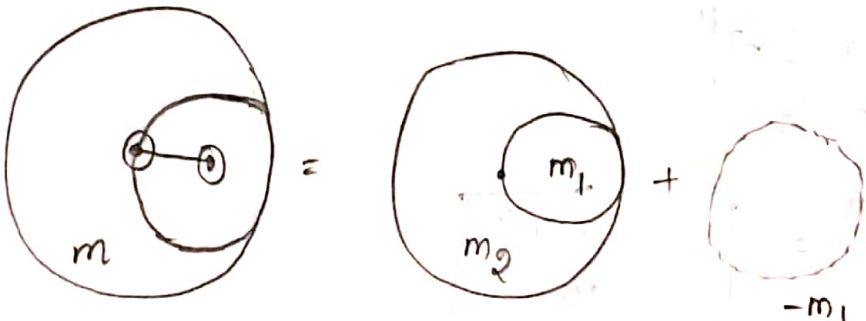
m = total mass of disc (without cavity)

* removed mass,

$$m_1 = \frac{m}{\pi R^2} \times \pi\left(\frac{R}{2}\right)^2$$

$$= \frac{m}{R^2} \times \frac{R^2}{4} = \frac{m}{4}$$

$\therefore$ remaining mass $(m_2) = m - m/4 = \frac{3m}{4}$.



$$I = \frac{mR^2}{2} + \left[-\frac{m_1}{2}\left(\frac{R}{2}\right)^2 + \frac{(-m_1)}{1}\left(\frac{R}{2}\right)^2\right]$$

$$= \frac{mR^2}{2} - \left[\frac{m_1R^2}{8} + \frac{m_1R^2}{4}\right]$$

$$= \frac{mR^2}{2} - \left[\frac{3m_1R^2}{8}\right] = \frac{mR^2}{2} - \frac{3mR^2}{32}$$

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$$\boxed{I = \frac{13mR^2}{32}}$$

$I_{remaining} = I_{total} - I_{removed\ mass}$

case (2):-

m = mass of disc with cavity.

Area of removed portion $= \frac{\pi R^2}{4}$

Area of remaining portion $= \frac{3\pi R^2}{4}$.

* mass per unit area $= \frac{m}{\frac{3\pi R^2}{4}} = \frac{4m}{3\pi R^2}$

* removed mass, $m_1 = \frac{4m}{3\pi R^2} \times \frac{\pi R^2}{4} = \frac{m}{3}$.

* Total mass, $m_2 = \frac{4m}{3\pi R^2} \times \pi R^2 = \frac{4m}{3}$.

$I_{remaing} = I_{total} - I_{removed}$

$$= \frac{m_2 R^2}{2} - \left[\frac{m_1\left(\frac{R}{2}\right)^2}{2} + m_1\left(\frac{R}{2}\right)^2\right]$$

$$= \frac{m_2 R^2}{2} - \left[\frac{m_1 R^2}{8} + \frac{m_1 R^2}{4}\right]$$

$$= \frac{m_2 R^2}{2} - \frac{3m_1 R^2}{8}$$

$$= \frac{4mR^2}{6} - \frac{mR^2}{8} = \frac{32mR^2 - 6mR^2}{48} = \frac{13}{24}mR^2$$

$$\boxed{I = \frac{13mR^2}{24}}$$

→ Find moment of Inertia of remaining portion in the given figures.

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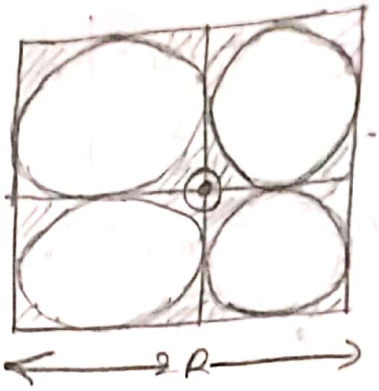
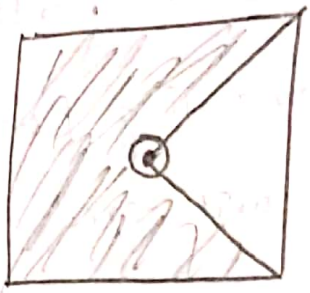
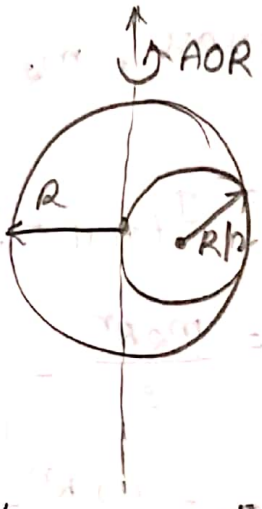
(i)



(ii)



* For sphere :



Case - (1) :-

m = Total mass of disc (without cavity)

removed mass $(m_1) = \frac{M}{\frac{4}{3}\pi R^3}\left(\frac{4}{3}\pi\left(\frac{R}{2}\right)^3\right) = \frac{M}{8}$

remaining mass $(m_2) = M - \frac{M}{8} = \frac{7M}{8}$

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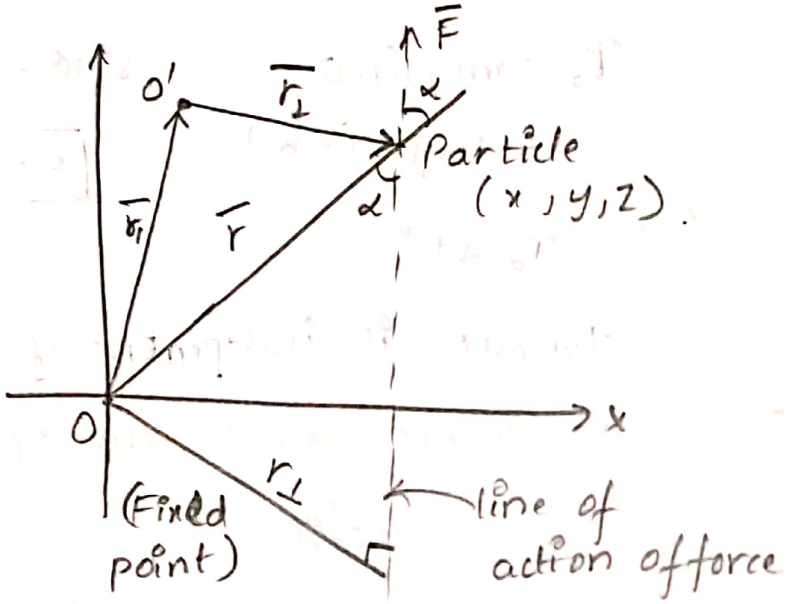
Torque (τ) :

* It is the turning effect of force about a "fixed pt / fixed axes."

* Torque is the moment of force acting on the body about a fixed point.

* τ = Force x ⊥er distance from fixed pt on to the line of action of force.

$$\boxed{\tau = r_\perp F}$$



* $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\bar{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

* Torque about 'O'

$\tau_o = r_\perp F$

$$\boxed{\tau_o = rF\sin\alpha}$$

from $\sin\alpha = \frac{r_\perp}{r}$

$r_\perp = r\sin\alpha$

In vector form, $\boxed{\bar{\tau} = \bar{r} \times \bar{F}}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

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* It is an axial vector. It's direction is always along AOR and is decided by Right hand thumb rule.

* Torque about O',

$\bar{\tau}_{O'} = \bar{r}_2 \times \bar{F}$

$\boxed{\bar{\tau}_{O'} = (\bar{r} - \bar{r}_1) \times \bar{F}}$

$\bar{r}_1 = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

addn of vectors,

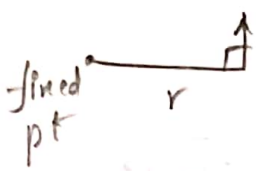
$\bar{r}_1 + \bar{r}_2 = \bar{r}$

$\boxed{\bar{r}_2 = \bar{r}_0 - \bar{r}_1}$

* If "line of action of force" will pass through fixed point. Then $\bar{r} \parallel \bar{F}$ (or) $\bar{r} \parallel -\bar{F}$ $\boxed{\tau = 0}$

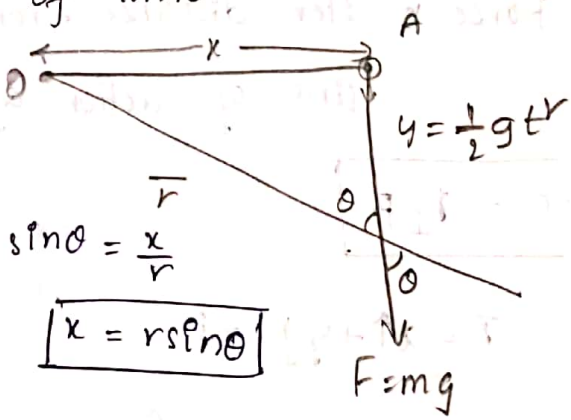
* Torque is max when $\sin\alpha = 1$

$\Rightarrow \boxed{\alpha = 90^\circ}$



$\boxed{\tau = rF}$ max.

→ A particle is dropped freely from A. Find the torque due to gravity acting on the particle about O as a fuction of time.



$\tau_o = (\bar{r} \times \bar{F})$

$\tau_o = mg\sin\theta$

$= mgx(-\hat{k})$

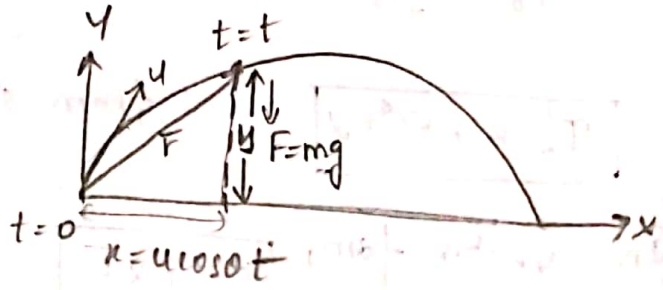
$\sin\theta = \frac{x}{r}$

$\boxed{x = r\sin\theta}$

i.e, $\tau_o \propto t^0$

⇒ Torque is independent of time.

→ Find torque due to gravity on the projectile about pt of projection.



$\bar{r} = x\hat{i} + y\hat{j}$

$\bar{F} = mg(-\hat{j})$

$\tau_o = \bar{r} \times \bar{F}$

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$$\tau_o = (x\hat{\imath} + y\hat{\jmath}) \times mg(-\hat{\jmath})$$

$$\tau_o = mgx(-\hat{k})$$

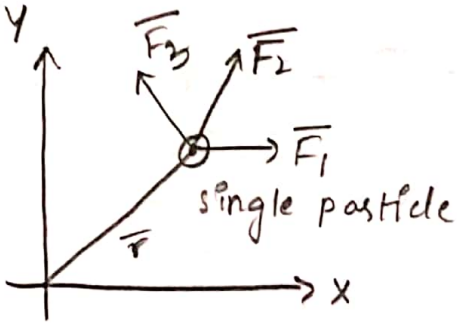
$$\boxed{\tau_o = mgu\cos\theta t(-\hat{k})} \quad \text{i.e, } (\tau_o \propto t)$$

* Avg torque,

$$\tau_{avg} = \frac{\int \tau_o dt}{\int dt} = \frac{\int_0^t mgu\cos\theta t \cdot dt}{t} = \frac{mgu\cos\theta t}{2}$$

$$\boxed{\tau_{avg} = \frac{1}{2} mgu\cos\theta t}$$

* Net torque,



$$\bar{\tau} = \bar{r}_1 \times \bar{F}_1 + \bar{r} \times \bar{F}_2 + \bar{r} \times \bar{F}_3$$

$$\bar{\tau} = \bar{r} \times (\bar{F}_1 + \bar{F}_2 + \bar{F}_3)$$

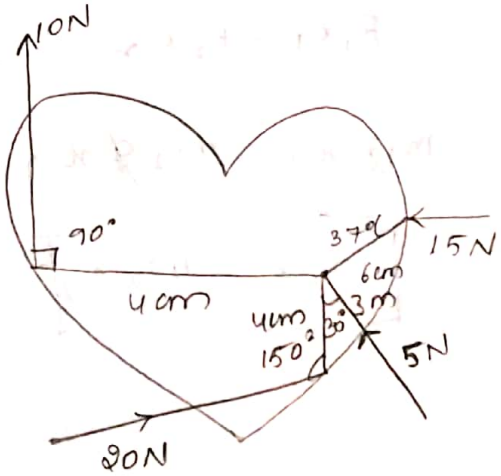
$$\boxed{\bar{\tau} = \bar{r} \times \bar{F}_{net}}$$

Equilibrium :

① $\bar{F}_{net} = 0$ (translational eqm)

② $\bar{\tau}_{net} = 0$ (rotational eqm)

→ Find net torque about O ?



$$\tau = rF\sin\theta$$

$$= 15 \times 6 \sin(180 - 37°) + (-10 \times 4) + 20 \times 4 \times \sin 150°$$

$$= 90 \sin 37° - 40 + 80 \sin 150°$$

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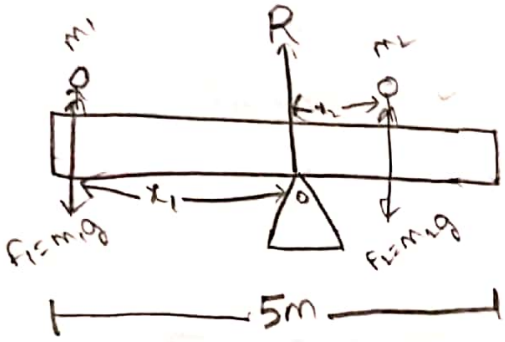
$$= \frac{\cancel{90}^{18} \times 3}{\cancel{5}} + 80 \times \frac{1}{2} - 40$$

$$= 54 + 40 - 40$$

$$= 54 \text{ N/cm}$$

$$\boxed{\tau = 0.54 \text{ N/m}}$$

Eqm (See-Saw Problem):



① $R = F_1 + F_2$

② $\tau_0 = 0$

$$F_1 x_1 - F_2 x_2 = 0$$

$$F_1 x_1 = F_2 x_2$$

$$m_1 g x_1 = m_2 g x_2$$

$$\boxed{m_1 x_1 = m_2 x_2}$$

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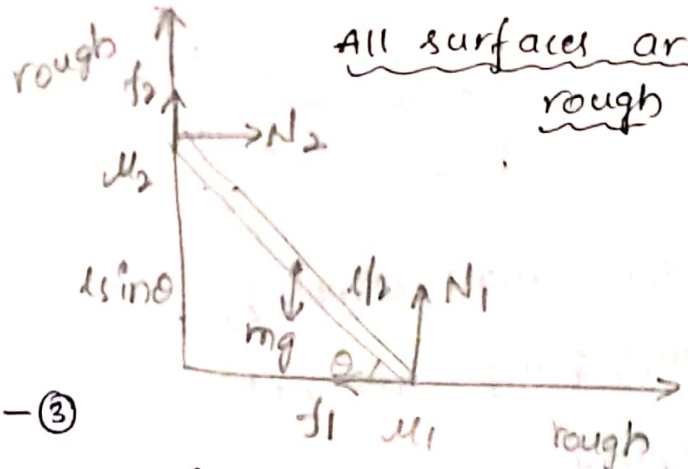
Ladder Problems:

All surfaces are rough



d = length of ladder

m = mass of ladder

In limiting eqm,

$\Sigma F_H = 0 \Rightarrow N_2 = f_1$ —③

$\Sigma F_V = 0 \Rightarrow N_1 = mg$ —④

$f_1 = \mu_1 N_1$ —①

$f_2 = \mu_2 N_2$ —②

* $\Sigma \tau_A = 0$

$$mg\frac{l}{2}\cos\theta - l\sin\theta \cdot N_2 - f_2 l\cos\theta = 0$$

$$\left(\frac{mg}{2} - f_2\right)\cos\theta - \sin\theta\, N_2 = 0 \quad —⑤$$

$N_1 + \mu_2 N_2 = mg$

$N_1 + \mu_2 f_1 = mg$

$N_1 + \mu_2 \mu_1 N_1 = mg$

$$\boxed{N_1 = \frac{mg}{1+\mu_1\mu_2}}$$

& $N_2 = f_1 = \mu_1 N_1$

$$\boxed{N_2 = \frac{\mu_1 mg}{1+\mu_1\mu_2}}$$

& $\frac{mg\cos\theta}{2}$

$$= \frac{\mu_1 mg\sin\theta}{1+\mu_1\mu_2} + \frac{\mu_1\mu_2 mg\cos\theta}{1+\mu_1\mu_2}$$

$$\frac{\cot\theta}{2} = \frac{\mu_1}{1+\mu_1\mu_2} + \frac{\mu_1\mu_2\cot\theta}{1+\mu_1\mu_2}$$

$$\cot\theta\left[\frac{1}{2} - \frac{\mu_1\mu_2}{1+\mu_1\mu_2}\right] = \frac{\mu_1}{1+\mu_1\mu_2}$$

$$\cot\theta\left[\frac{1+\mu_1\mu_2-2\mu_1\mu_2}{2(1+\mu_1\mu_2)}\right] = \frac{\mu_1}{1+\mu_1\mu_2}$$

$\Rightarrow \boxed{\cot\theta = \frac{2\mu_1}{1-\mu_1\mu_2}}$ or $\boxed{\tan\theta = \frac{1-\mu_1\mu_2}{2\mu_1}}$

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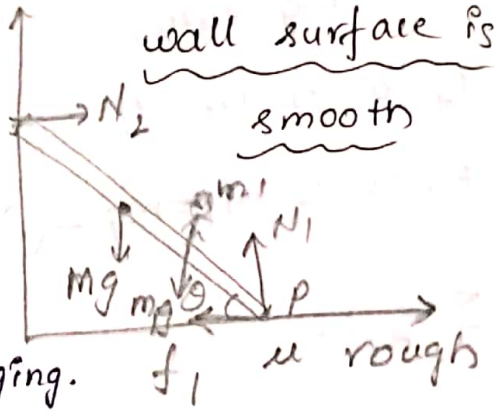
* If wall is smooth, $(\mu_2 = 0)$

$$\boxed{\tan\theta = \frac{1}{2\mu_1}}$$

* As the man climbing, $m_1 g$ & mg are not changing,

$\Sigma F_V = 0 \Rightarrow N_1 = m_1 g + Mg$ is also not changing.



* $\tau_p = 0$

$$[Mg\frac{l}{2}\cos\theta + m_1 g x\cos\theta](\circlearrowleft) = N_2 l\sin\theta \ (\circlearrowright)$$

Here as $x(\uparrow) \Rightarrow N_2(\uparrow)$

we have $\Sigma F_H = 0$

$\Rightarrow \boxed{N_2 = f_1}$ As $N_2 \uparrow \Rightarrow f_1(\uparrow)$

when $\boxed{f_1 = \mu N_1}$ $\Rightarrow$ ladder in limiting eqm.

* $\Sigma F_H = 0$

$N_2 = f_1$ — (1)

$\Sigma F_V = 0$

$N_1 = (m_1 + M)g$ — (2)

$\Rightarrow \Sigma \tau_p = 0$

$mg x\cos\theta + Mg\frac{l}{2}\cos\theta = N_2 l\sin\theta$ — (3)

& $f_1 = \mu(m_1 + M)g$ — (4)

from this 4 eqns we get value of x.

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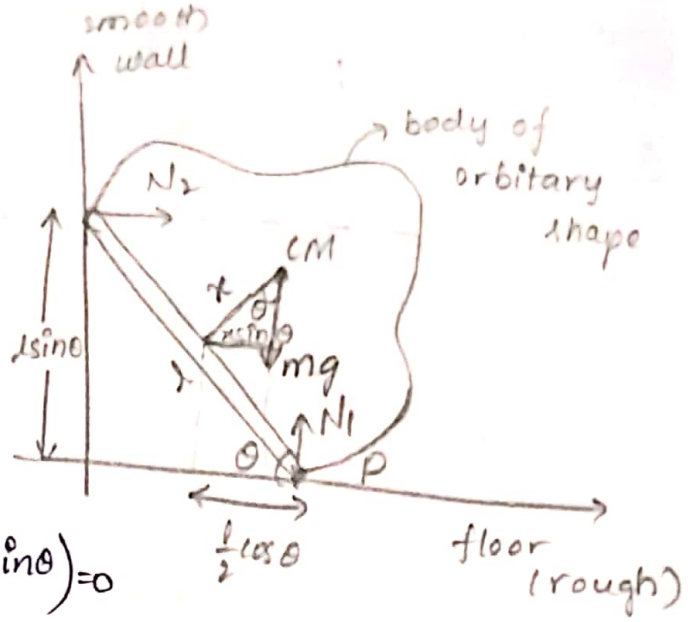
* For General Case:-



In eqm,

$\Sigma F_H = 0 \Rightarrow N_2 = f_1 \quad -①$

$\Sigma F_V = 0 \Rightarrow N_1 = mg \quad -②$

$\Sigma \tau_P = 0$

$$-N_2 l\sin\theta + mg\left(\frac{l}{2}\cos\theta - x\sin\theta\right) = 0$$

$$N_2 l\sin\theta = mg\frac{l}{2}\cos\theta - x\sin\theta$$

$$\boxed{f_1 = mg\frac{\cot\theta}{2} - \frac{mgx}{l}}$$

we have,

$f_1 \le \mu N_1$ (In eqm)

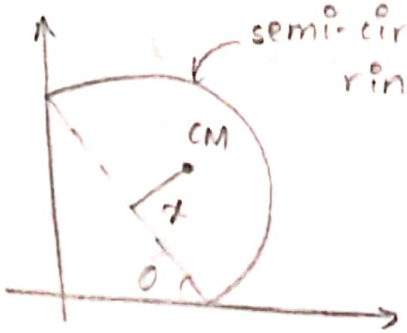
$$\frac{mg\cot\theta}{2} - \frac{mgx}{l} \le \mu mg$$

$$\frac{\cot\theta}{2} - \frac{x}{l} \le \mu$$

$$\boxed{\mu_{min} = \frac{\cot\theta}{2} - \frac{x}{l}}$$ ← min coefficient of friction required.

Special Cases :-

①



$l = 2R$, $x = \frac{2R}{\pi}$

$$\mu_{min} = \frac{\cot\theta}{2} - \frac{2R}{\pi(2R)}$$

$$\boxed{\mu_{min} = \frac{\cot\theta}{2} - \frac{1}{\pi}}$$

(or) $\mu + \frac{1}{\pi} = \frac{\cot\theta}{2} \Rightarrow \cot\theta = 2\left(\mu + \frac{1}{\pi}\right)$

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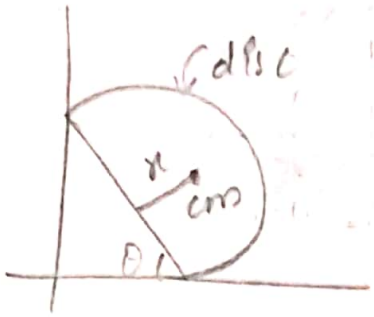
$\Rightarrow \boxed{\theta = \cot^{-1}\left(2\left(\mu + \frac{1}{\pi}\right)\right)}$

(2)

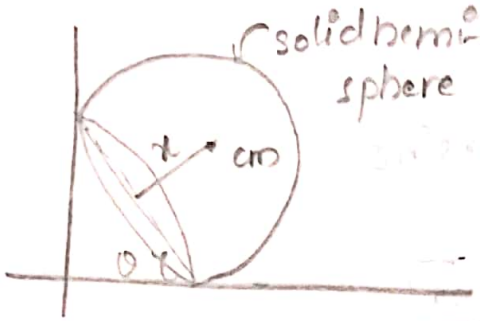


$l = 2R, \quad x = \frac{4R}{3\pi}$

$\mu_{min} = \frac{\cot\theta}{2} - \frac{4R}{(3\pi)(2R)}$

$\boxed{\mu_{min} = \frac{\cot\theta}{2} - \frac{2}{3\pi}}$

(3)

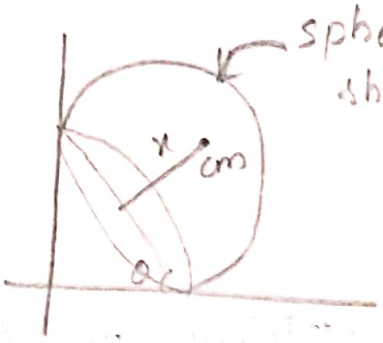


$l = 2R, \quad x = \frac{3R}{8}$

$\mu_{min} = \frac{\cot\theta}{2} - \frac{3R}{8(2R)}$

$\boxed{\mu_{min} = \frac{\cot\theta}{2} - \frac{3}{16}}$

(4)

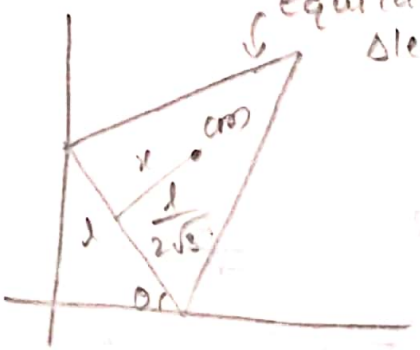


$l = 2R, \quad x = \frac{R}{2}$

$\mu_{min} = \frac{\cot\theta}{2} - \frac{R}{2(2R)}$

$\boxed{\mu_{min} = \frac{\cot\theta}{2} - \frac{R}{4}}$

(5)



$l = l, \quad x = \frac{l}{2\sqrt{3}}$

$\mu_{min} = \frac{\cot\theta}{2} - \frac{l}{2\sqrt{3}(l)}$

$\boxed{\mu_{min} = \frac{\cot\theta}{2} - \frac{1}{2\sqrt{3}}}$

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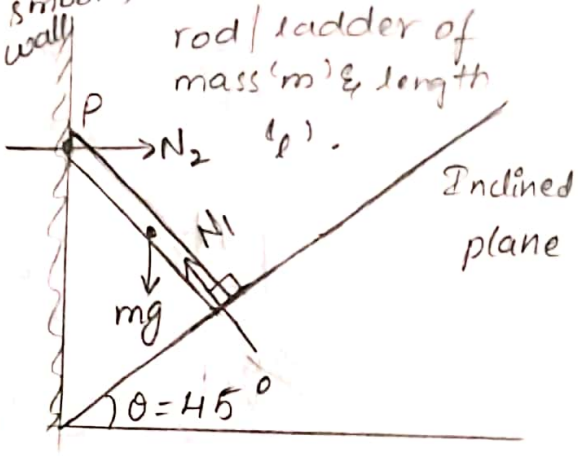
* If inclined plane is smooth,



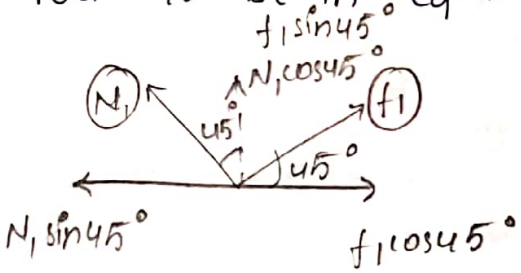
then $\tau_P = mg\frac{l}{2}\cos 45^\circ$

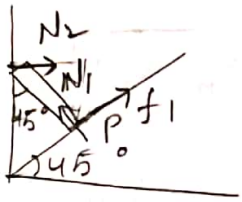
$= \frac{mgl}{2\sqrt{2}}(\circlearrowright) \neq 0$

i.e, rod is not at eqm.

∴ rod can slip down the inclined plane.

* If inclined plane is rough, then μ_{min} for the rod to be in eqm is.





$f_1 = \mu N_1$

$\Sigma F_H = 0$

$N_2 + f_1\cos 45^\circ = N_1\sin 45^\circ$ — ①

$N_1\cos 45^\circ + f_1\sin 45^\circ = mg$ — ②

$\tau_P = 0$

$mg\frac{l}{2}\cos 45^\circ - N_2 l\sin 45^\circ = 0$ — ③

① ⇒ $N_2 + \frac{\mu N_1}{\sqrt{2}} = \frac{N_1}{\sqrt{2}}$ ② ⇒ $\frac{N_1}{\sqrt{2}} + \frac{\mu N_1}{\sqrt{2}} = mg$.

$N_2 = \frac{N_1}{\sqrt{2}}(1-\mu)$ $\frac{N_1}{\sqrt{2}}(1+\mu) = mg$

$$\boxed{N_2 = \frac{mg(1-\mu)}{1+\mu}}$$ $$\boxed{N_1 = \frac{mg\sqrt{2}}{1+\mu}}$$

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③ $\Rightarrow \frac{mg}{2\sqrt{5}} = \frac{N_2}{\sqrt{2}}$

$$\boxed{N_2 = \frac{mg}{2}}$$

$$\frac{mg(1-\mu)}{1+\mu} = \frac{mg}{2}$$

$$2 - 2\mu = 1 + \mu$$

$$1 = 3\mu \Rightarrow \boxed{\mu = 1/3}$$

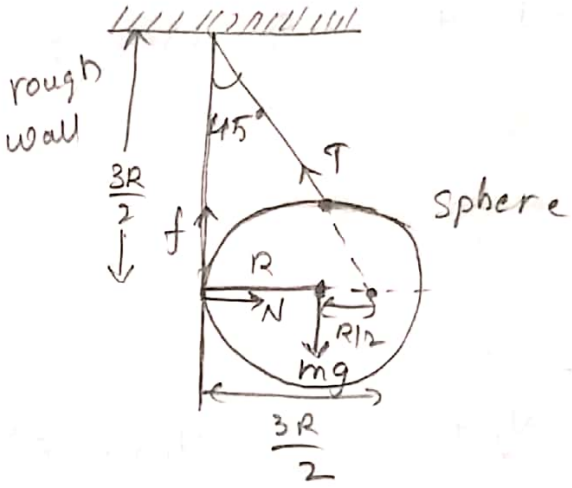
→

Since sphere is in eqm,

$$\boxed{\Sigma F_H = 0}$$

$\Rightarrow N = T\sin 45^\circ$

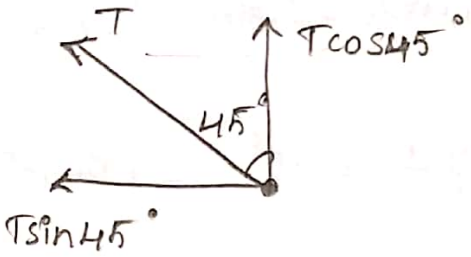
$$\boxed{N = \frac{T}{\sqrt{2}}} \quad -①$$



$\Sigma F_V = 0$

$$f + T\cos 45^\circ = mg$$

$$f + \frac{T}{\sqrt{2}} = mg \quad -②$$



we have,

$f \le \mu N$

$$\frac{mg}{3} \le \mu\left(\frac{2mg}{3}\right)$$

$$\boxed{\mu_{min} = \frac{1}{2}}$$

$$\boxed{\frac{T}{\sqrt{2}} = \frac{2mg}{3} = N}$$

$\Sigma \tau_A = 0$

$$-f\left(\frac{3R}{2}\right) + \frac{mgR}{2} = 0$$

$$\boxed{f = \frac{mg}{3}} \quad -③$$

$$\frac{mg}{3} + \frac{T}{\sqrt{2}} = mg$$

$$\frac{T}{\sqrt{2}} = \frac{2mg}{3} \Rightarrow T = \frac{2\sqrt{2}mg}{3}$$

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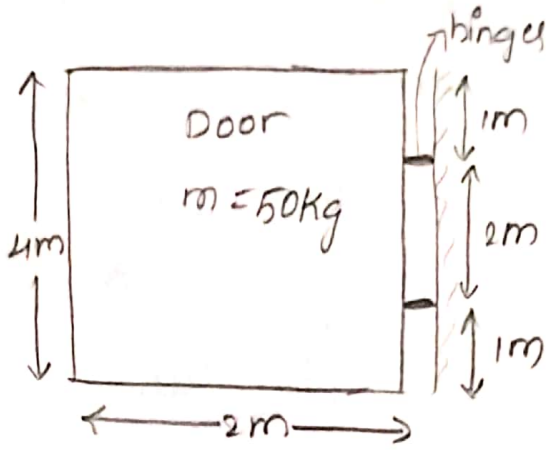
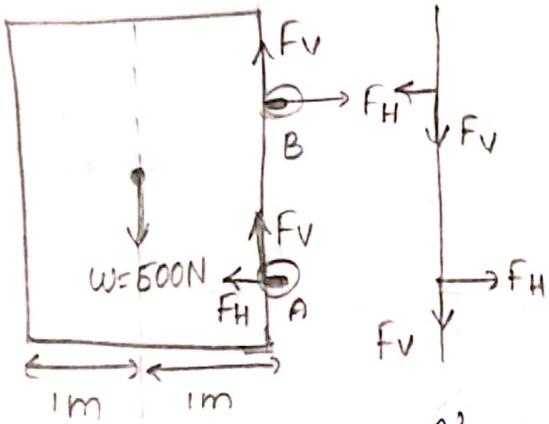
Door Problem:

→ If magnitude of force of each hinge on the door is same. Find that Force.





For door:

$\Sigma F_H = 0$

$\Sigma F_V = 0 \Rightarrow 2F_V = \omega$

$F_V = \frac{\omega}{2} = 250N$

$\boxed{F_V = 250N}$

$\Sigma \tau_A = 0$

$F_H(2) = \omega(1)$

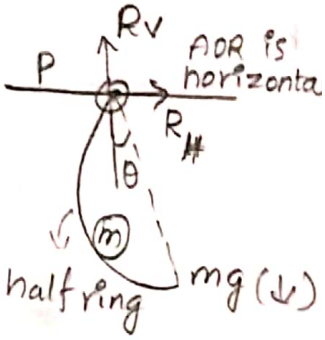
$F_H = \frac{500}{2}$

$\boxed{F_H = 250N}$

$F = \sqrt{F_V^2 + F_H^2} = 250\sqrt{2}$

$\boxed{F_{net} = 500N}$

→ Find 'θ' so that ring is in eqm.



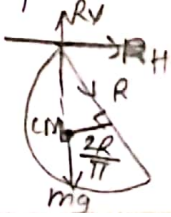
$\Sigma F_H = 0$, $\Sigma F_V = 0$

$\Rightarrow \boxed{R_H = 0}$ $\Rightarrow \boxed{R_V = mg}$

$\Sigma \tau_P = 0$

here line of axis of ring should pass through fixed point.

$\tan\theta = \frac{2R}{\pi R} \Rightarrow \boxed{\theta = \tan^{-1}(R/\pi)}$



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→ Find 'θ' so that disc / solid hemisphere / hollow hemi sphere is in eqm.

$\tan\theta = \frac{4R}{3\pi(R)}$

$\theta = \tan^{-1}(4/3\pi)$

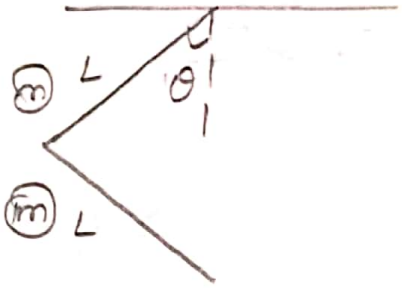
$\tan\theta = \frac{3R}{8R}$

$\theta = \tan^{-1}(3/8)$

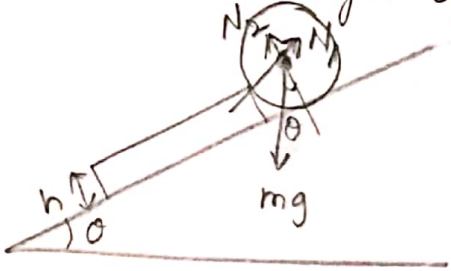
$\tan\theta = \frac{R}{2R}$

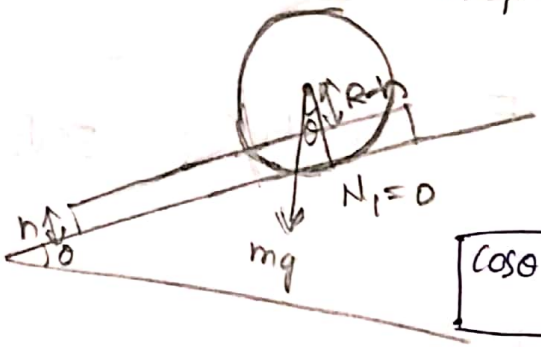
$\theta = \tan^{-1}(1/2)$

→ Find 'θ' to keep the system at eqm as shown in figure.



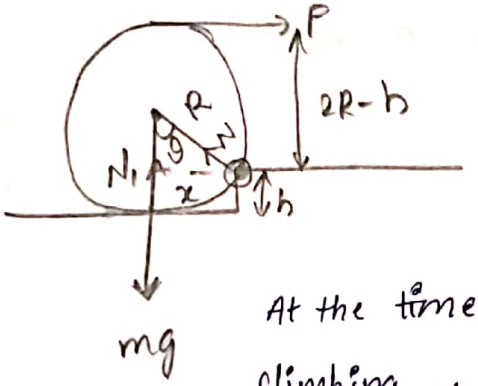
→ For what angle of θ ball climbs the step.

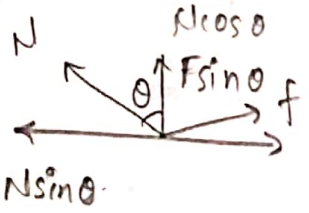




$\cos\theta = \frac{R-h}{R}$

*





At the time of climbing step, $[N_1 = 0]$

$\Sigma F_H = 0$

$\Rightarrow F + F\cos\theta = N\sin\theta$ — ①

$\Sigma F_V = 0$

$\Rightarrow N\cos\theta + f\sin\theta = mg$ — ②

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$\tau_p = 0$, $f(2R-h) = mgx$ — (3)

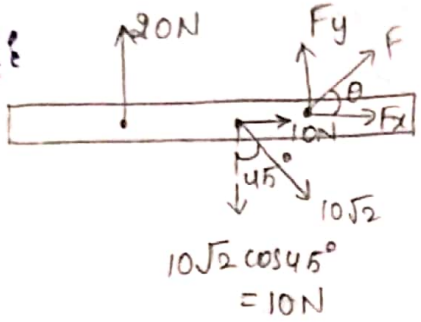
from figure $R^2 = (R-h)^2 + x^2$

$$\boxed{x = \sqrt{R^2 - (R-h)^2}}$$

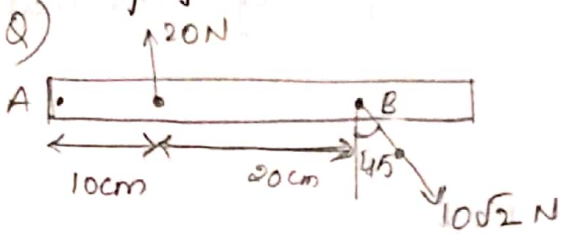
Point of application of force :

→ Find the pt of application of force about A.

Sol:



Q)



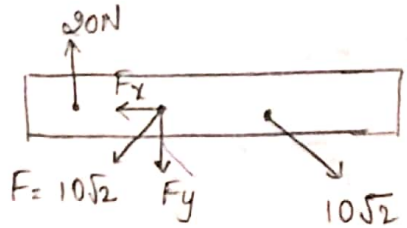
since rod is in eqm,

$$\Sigma F_x = 0$$

$$F_x + 10 = 0 \Rightarrow \boxed{F_x = -10N}$$

$$\Sigma F_y = 0$$

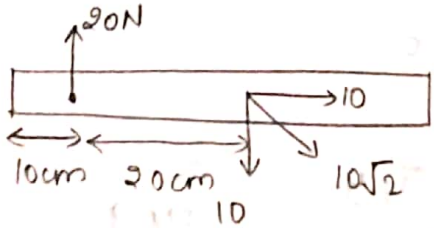
$$F_y + 20 = 0 \Rightarrow \boxed{F_y = -20N}$$



$\boxed{\theta = 45°}$

with out F,

Torque on the rod about 'A'.



$\tau_A = 200 - (-10)(30) = -100N\text{-}m = 100N\text{-}m$ (↻)

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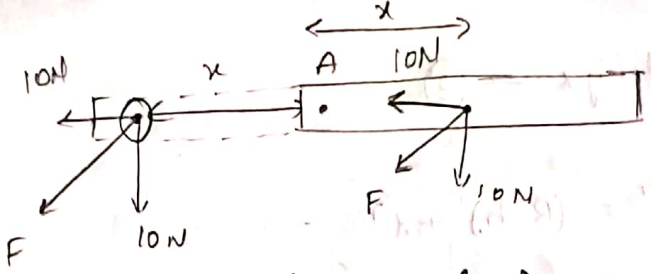
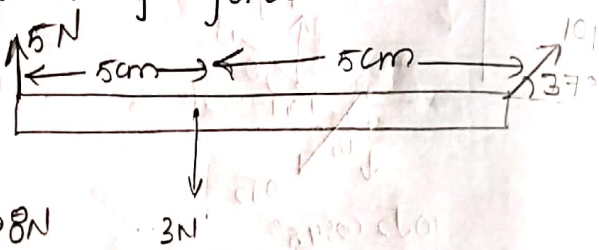


$$\tau_A = 10x \quad (2)$$

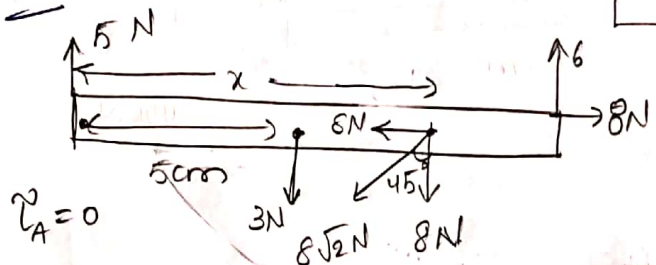
$$10x = 100$$

$$\boxed{x = 10\text{ cm}}$$

→ Find pt of application of force.



Sol:



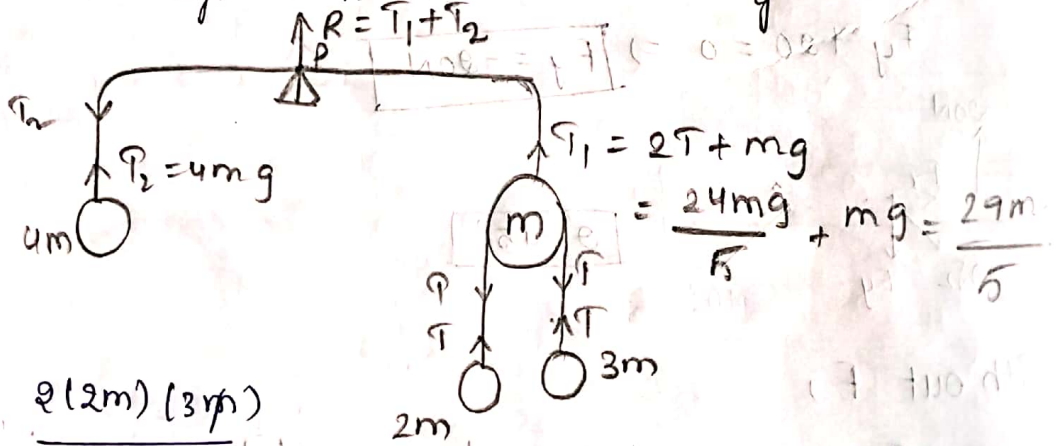
$\tau_A = 0$

$$-5\times3 - 8x + 6\times10 = 0$$

$$8x = 45$$

$$\boxed{x = 5.6\text{ cm}}$$

→ If the system is in eq^m. Find $\frac{x}{y}$.



$$T = \frac{2(2m)(3m)}{5m}$$

$$\boxed{T = \frac{12mg}{5}}$$

$$\tau_P = 0$$

$$T_2 x = T_1 y$$

$$\frac{x}{y} = \frac{T_1}{T_2} = \frac{\frac{12mg}{5}}{4mg} = \frac{3}{5}.$$

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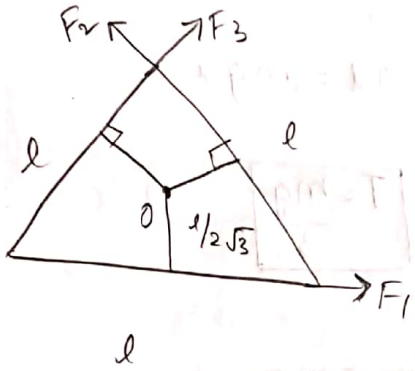
→ Along the sides of equilateral Δle. 2f system is in eqm about 'O'.

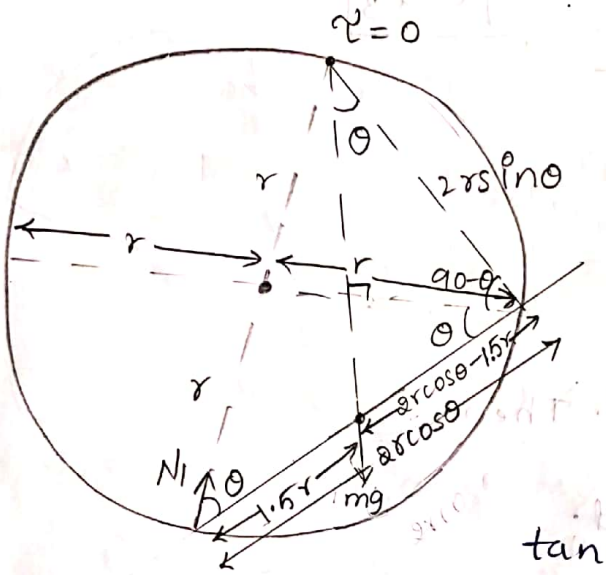


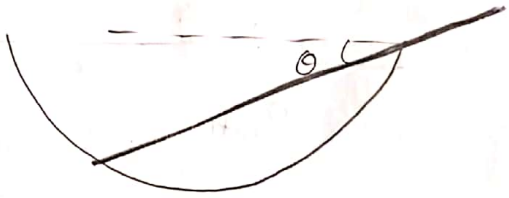
$\tau_o = 0$

$$F_2 \frac{l}{2\sqrt{3}} + F_1 \frac{l}{2\sqrt{3}} - F_3 \frac{l}{2\sqrt{3}} = 0$$

$$\boxed{F_3 = F_1 + F_2}$$

→ Contact surfaces are smooth and rod of length 3r. Find 'θ' for which rod is in eqm.



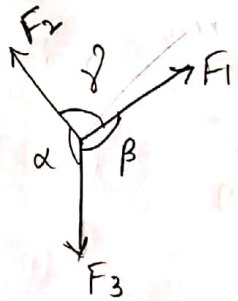


N_1, mg & N_2 are concurrent forces.

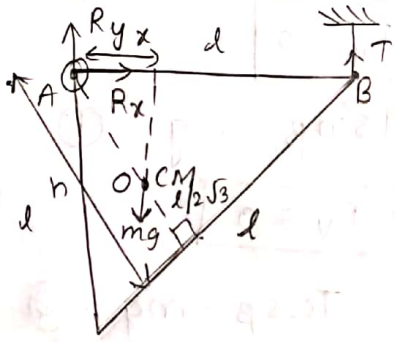
$$\tan\theta = \frac{2r\cos\theta - 1.5r}{2r\sin\theta} = \frac{\sin\theta}{\cos\theta}$$

$$\theta = \cos^{-1}(0.92)$$

$$\boxed{\theta = 23.07^\circ}$$



→ Triangular plate is in eqm



In eqm,

$\Sigma F_x = 0 \Rightarrow R_x = 0$

$\Sigma F_y = 0 \Rightarrow R_y + T = mg$ —②

$\tau_A = 0$

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$T_d - mgx = 0$

$T_d = mgx$

$$\boxed{T = \frac{mg}{3}}$$

$\therefore R_y = mg - T$

$= mg - \frac{mg}{3}$

$$\boxed{R_y = \frac{2mg}{3}}$$

$\cos 45^\circ = \frac{h}{d}$

$h = d/\sqrt{2}$

CM from base $= \frac{h}{3} = \frac{d}{3\sqrt{2}}$

$OA = h - \frac{d}{3\sqrt{2}}$

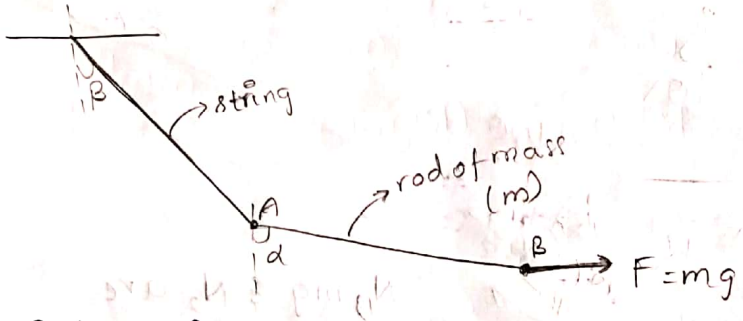
$= \frac{d}{\sqrt{2}} - \frac{d}{3\sqrt{2}} = \frac{2d}{3\sqrt{2}} = \frac{\sqrt{2}d}{3}$

$\cos 45^\circ = \frac{x}{OA}$

$x = OA \cos 45^\circ = \frac{\sqrt{2}d}{3} \times \frac{1}{\sqrt{2}} = \frac{d}{3}$

$$\boxed{x = d/3}$$

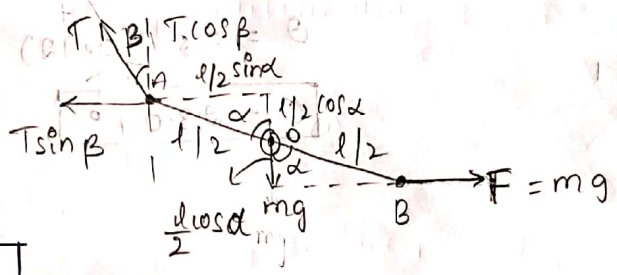
K



System is at eqm. Then,

let L is length of rod.

unknowns → α, β, T.



$$\boxed{\Sigma F_H = 0}$$

$T \sin\beta = mg$ — (1)

$$\boxed{\Sigma F_V = 0}$$

$T \cos\beta = mg$ — (2)

from eqns (1) & (2) $\Rightarrow$ $\boxed{\beta = 45^\circ}$

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$Td - mgx = 0$

$Td = mgx$

$$\boxed{T = \frac{mg}{3}}$$

$\therefore R_y = mg - T$

$= mg - \frac{mg}{3}$

$$\boxed{R_y = \frac{2mg}{3}}$$

$\cos 45^\circ = \frac{h}{l}$

$h = l/\sqrt{2}$

CM from base $= \frac{h}{3} = \frac{l}{3\sqrt{2}}$

$OA = h - \frac{l}{3\sqrt{2}}$

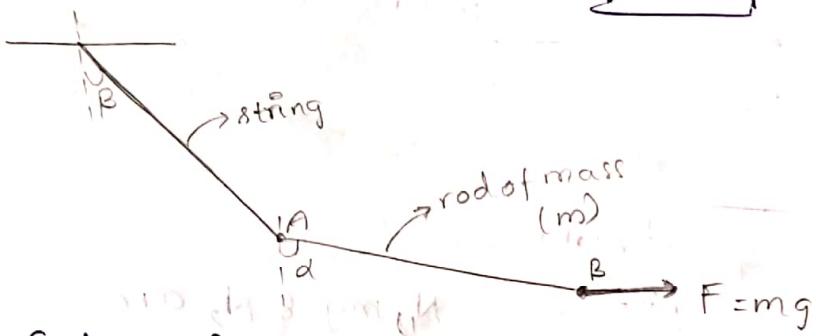
$= \frac{l}{\sqrt{2}} - \frac{l}{3\sqrt{2}} = \frac{2l}{3\sqrt{2}} = \frac{\sqrt{2}l}{3}$

$\cos 45^\circ = \frac{x}{OA}$

$x = OA \cos 45^\circ = \frac{\sqrt{2}l}{3} \times \frac{1}{\sqrt{2}} = \frac{l}{3}$

$$\boxed{x = l/3}$$

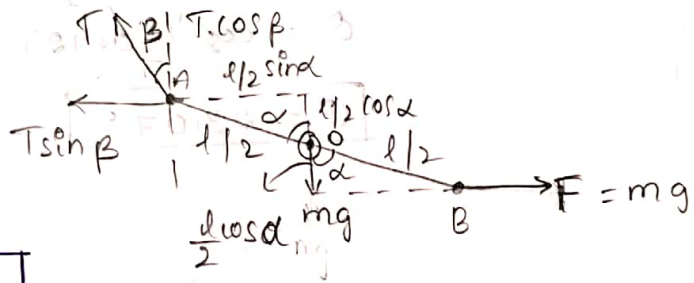
*



System is at eqm. Then,

let L is length of rod.

unknowns → α, β, T.



$$\boxed{\Sigma F_H = 0}$$

$T \sin\beta = mg$ — (1)

$$\boxed{\Sigma F_V = 0}$$

$T \cos\beta = mg$ — (2)

from eqns (1) & (2) $\Rightarrow$ $\boxed{\beta = 45^\circ}$

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$\therefore T\cos 45^\circ = mg \Rightarrow \boxed{T = \sqrt{2}mg}$

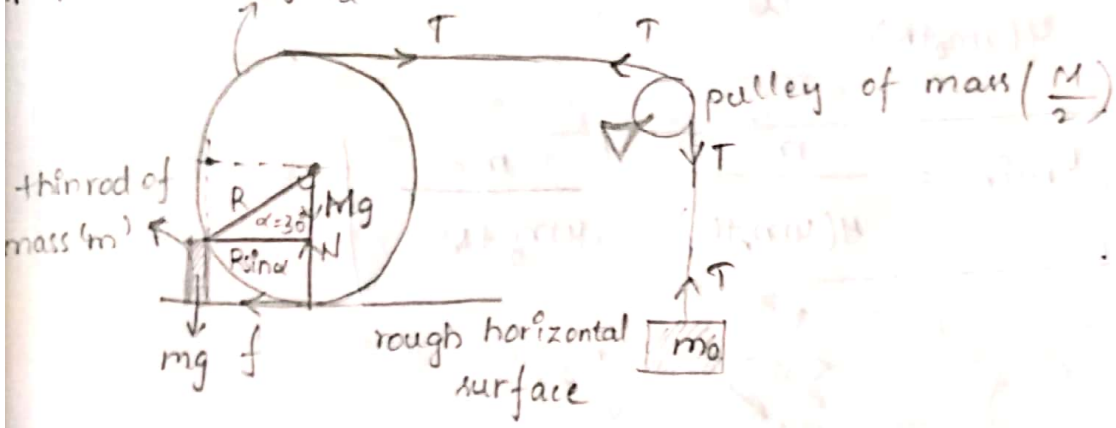
$\boxed{\tau_0 = 0}$

$$mg\frac{\ell}{2}\cos\alpha + T\cos\beta\left(\frac{\ell}{2}\cos\alpha\right) - T\sin\beta\left(\frac{\ell}{2}\cos\alpha\right) - T\cos\beta\frac{\ell}{2}\sin\alpha = 0$$

$$mg\cos\alpha + mg\cos\alpha - mg\sin\alpha - mg\cos\alpha = 0$$

$$\cos\alpha = \sin\alpha$$

$$\boxed{\tan\alpha = 1}$$



For the system is in eqm. Find mass of rod m.

* Since cylinder is in eqm, friction acts on the cylinder to "left".

$$T = m_0 g \quad -(1)$$

* $\Sigma F_V = 0 \Rightarrow N = (m+M)g \quad -(2)$

$\Sigma F_H = 0 \Rightarrow \boxed{T = f} \quad -(3)$

* Torque about centre of cylinder = 0.

$$-Tk - fk + mg\frac{k}{2}\sin\alpha = 0$$

$$T + f = \frac{mg}{2}\sin\alpha \quad -(4)$$

$$2T = \frac{mg}{2} \Rightarrow \boxed{T = \frac{mg}{4}} \quad T = f \Rightarrow \frac{mg}{4} = m_0 g$$

$$\boxed{m_0 = m/4}$$

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$m = 4m_0$

finally, $\boxed{N = (4m_0 + M)g}$

we have,

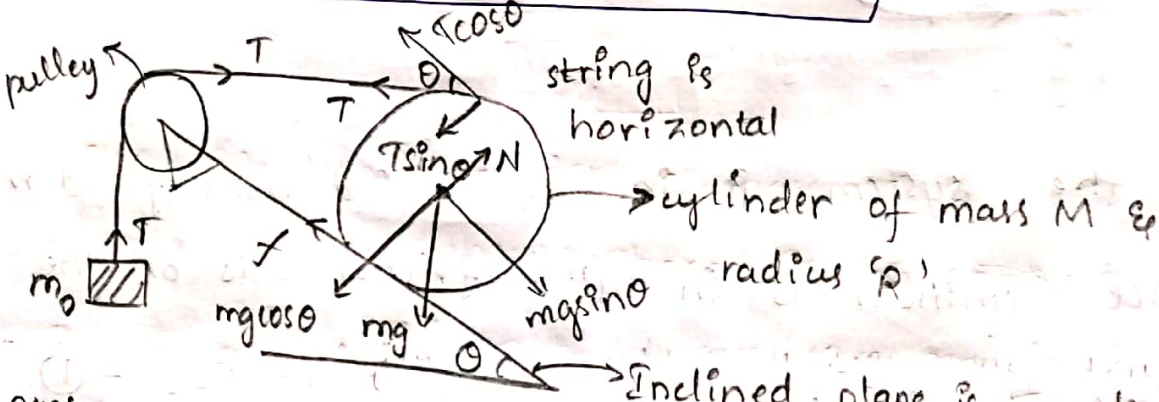
$f \leq \mu N$

$\frac{mg}{4} \leq \mu (4m_0 + M)g$

$\frac{m}{4(4m_0+M)} \leq \mu$

$$\boxed{\mu_{min} = \frac{m}{4(4m_0+M)} = \frac{m_0}{4m_0+M}}$$

→



System is in equilibrium. Find 'm_0'

In eqm, $T = mg$ — ①

for cylinder, along incline,

$T\cos\theta + f = Mg\sin\theta$ — ②

Normal to incline,

$N = T\sin\theta + Mg\cos\theta$ — ③

Torque about centre of cylinder = 0.

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$fR = TR$

$\boxed{f = T}$ — ④

④ ⇒ $m_0 f\cos\theta + m_0 g = Mg\sin\theta$

$m_0(\cos\theta + 1) = M\sin\theta \Rightarrow \boxed{m_0 = \frac{M\sin\theta}{1+\cos\theta}}$

③ ⇒ $N = m_0 g\sin\theta + Mg\cos\theta$

finally, $f \le \mu N$

$\mu_{min} = \frac{f}{N} = \frac{m_0}{m_0\sin\theta + M\cos\theta}$

$\mu = \frac{M\sin\theta}{M\sin^2\theta + M\cos\theta + M\cos^2\theta}$

$= \frac{M\sin\theta}{M + M\cos\theta}$

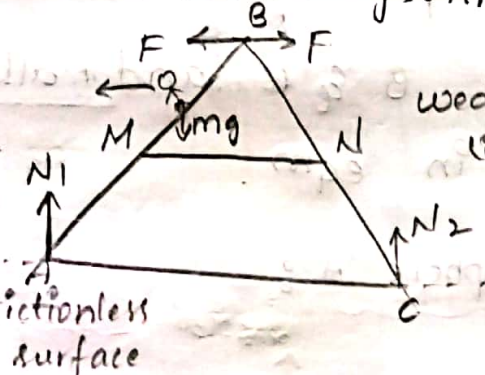
$= \frac{\sin\theta}{1+\cos\theta}$

$\boxed{\mu_{min} = \frac{m_0}{m_0\sin\theta + M\cos\theta}}$

$\boxed{\mu_{min} = \frac{\sin\theta}{1+\cos\theta}}$

→ Given $MB = MN = BN$ & 'M' & 'N' are the mid pts of AB & BC. Find Tension in the connecting string?

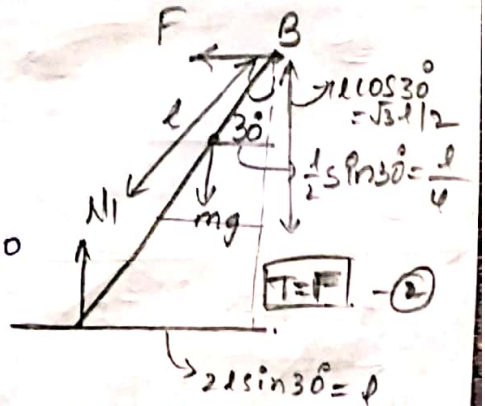
man of mass 'm' is standing at mid pt of MB



for a complete system,

$N_1 + N_2 = mg$ — ①

let $AB = 2l = BC \Rightarrow \boxed{MB = NB = l}$



Torque about B on left arm AB = 0

$mg\frac{l}{4} + T\frac{\sqrt{3}l}{2} - N_1 l = 0$ — ③

$\boxed{\frac{mg + 2\sqrt{3}T}{4} = N_1}$

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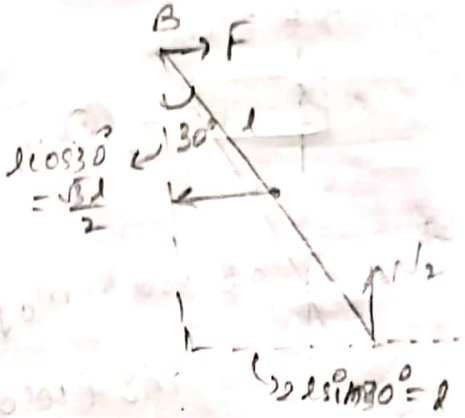
for rod BC,

$\tau_B = 0$

$-T\frac{\sqrt{3}}{2}\ell + \ell N_2 = 0$

$$\boxed{N_2 = \frac{\sqrt{3}T}{2}}$$



(1) $\Rightarrow$ $N_1 + N_2 = mg$

$$\frac{mg + 2\sqrt{3}T}{4} + \frac{\sqrt{3}T}{2} = mg$$

$$mg + 4\sqrt{3}T = 4mg$$

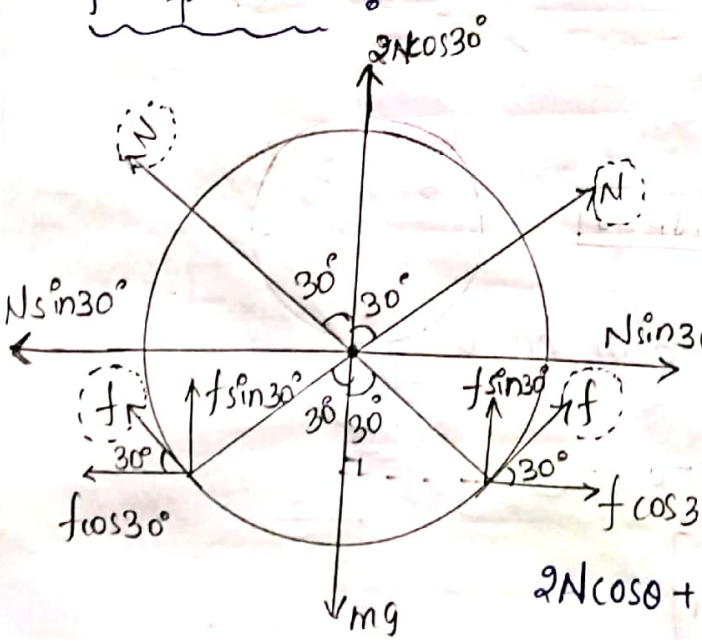
$$4\sqrt{3}T = 3mg$$

$$\boxed{T = \frac{\sqrt{3}mg}{4}}$$

→ Three identical cylinders of mass M & radius R are placed as shown. There is no contact force b/w B & C and all surfaces are rough. System is in eqm.

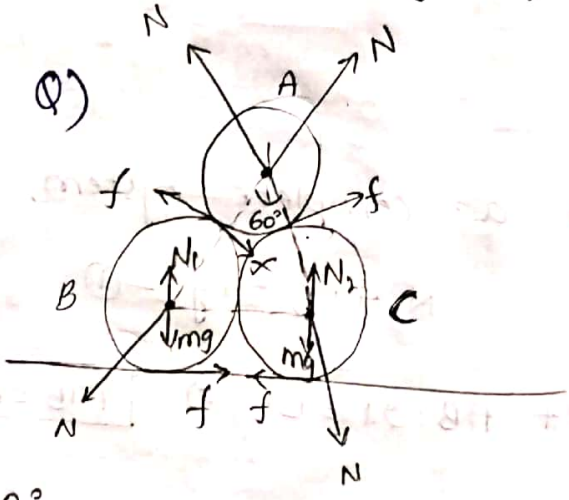
Sol:

for sphere 'A':



$2N\cos\theta + 2f\sin\theta = mg$ —(1)

Q)



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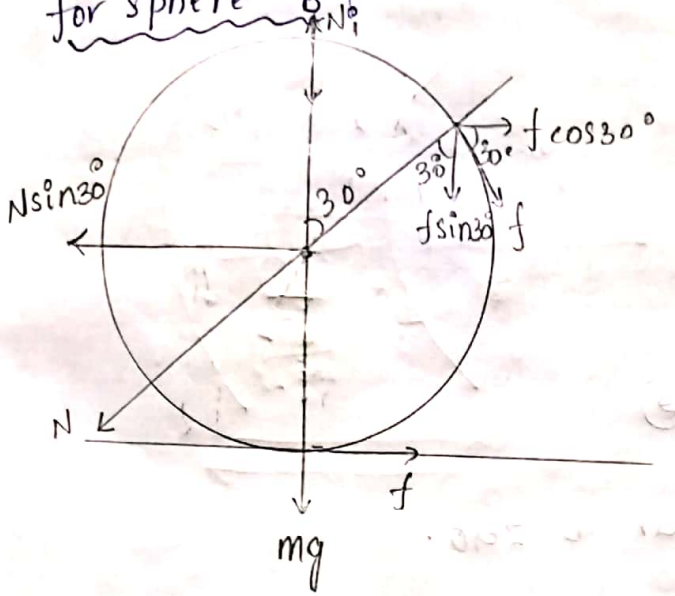
for sphere 'B':



$$N_1 = mg + N\cos 30° + f\sin 30° \quad \text{—②}$$

& $$N\sin 30° = f + f\cos 30° \quad \text{—③}$$

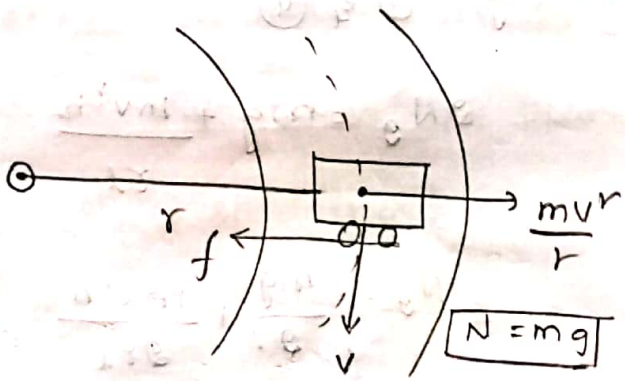
$$N = \frac{mg}{2}$$

Overturning of a vehicle (while taking a circular turn):

Case (i):

for no skidding,

$$f = \frac{mv^2}{r} \quad \text{—①}$$



we have, $f \leq \mu_s N$

$$\frac{mv^2}{r} \leq \mu_s mg$$

$$\boxed{v \leq \sqrt{\mu_s rg}}$$

max safe speed of vehicle without skidding.

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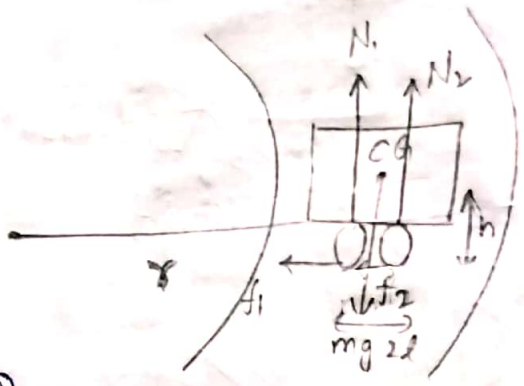
case-(2) :



In eqm,

$$N_1 + N_2 = mg \quad -(1)$$

& $$f_1 + f_2 = \frac{mv^2}{r} \quad -(2)$$

Torque about CG of car is zero.

$$-f_1 h - f_2 h - N_1 l + N_2 l = 0$$

$$(N_2 - N_1) l = (f_1 + f_2) h$$

$$N_2 - N_1 = (f_1 + f_2) \frac{h}{l} \quad -(3)$$

from eqn (2) & (3)

$$N_2 - N_1 = \frac{mv^2}{r}\left(\frac{h}{l}\right) \quad -(4)$$

adding (1) & (4)

$$2N_2 = mg + \frac{mv^2 h}{rl}$$

$$N_2 = \frac{mg}{2} + \frac{mv^2 h}{2rl}$$

$$\boxed{N_2 = \frac{m}{2}\left[g + \frac{v^2 h}{rl}\right]} \quad -(5)$$

└ always +ve.

& $N_1 = mg - N_2$

$$\boxed{N_1 = \frac{m}{2}\left(g - \frac{v^2 h}{rl}\right)} \quad -(6)$$

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when (v) increases to $\sqrt{\frac{grd}{h}}$, inner wheel looses contact with ground. $\boxed{N_1 = 0}$

* vehicle takes 'over turning'

* maximum safe speed of vehicle, for not overturning,

$$\boxed{v = \sqrt{\frac{grd}{h}}}$$

Here if $d(\uparrow)$ or $h(\downarrow)$ then $v(\uparrow)$.

Theoritical values:

$v_1 = \sqrt{\mu s r g}$ (without skidding)

$v_2 = \sqrt{\frac{grd}{h}}$ (without overturning)

* If $v_1 < v_2$ then skidding takes place first.
* If $v_1 > v_2$ then overturning takes place first.
* If $v_1 = v_2$ then both skidding & overturning takes place simultaneously.

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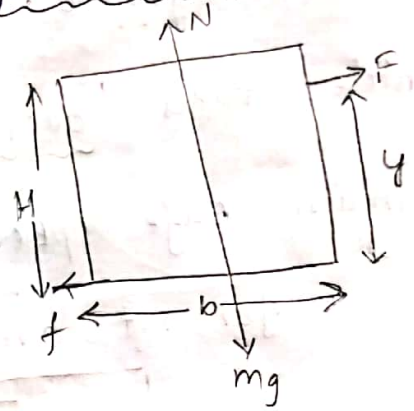
Toppling:

Case:-1: Body placed on rough horizontal surface:



* In eqm,

$$F = f \quad -(1)$$

$$N = mg \quad -(2)$$

Torque about 'p', $\tau_p = 0$

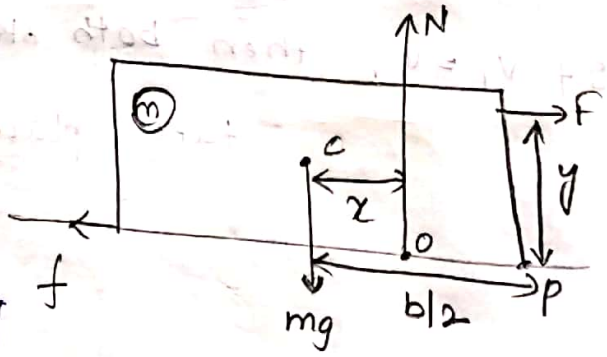
$$-F(y) + mg\left(\frac{b}{2}\right) - N\left(\frac{b}{2}\right) = 0$$

$-F(y) \neq 0$ i.e, block is not in eqm (for any value of F & y).

* practically which is not correct*

* **shifting of normal rxn:**

Bcz to keep body in rotational eqm shifing of normal rxn takes place towards the force end.



* In eqm,

$F = F$, $\tau_o = 0$

$N = mg$

$mgx = Fy \Rightarrow \boxed{x = \frac{Fy}{mg}}$

* As $F \uparrow$ or $y \uparrow \Rightarrow x \uparrow$.

when $x = b/2$, body is just begin to topple about 'p'.

* Minforce required to slide the body, $\boxed{F_{min} = f_L = \mu_s mg}$

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* Min force required to topple the body, ~~$y = H$~~

$$F_{min} = \frac{mgb}{2y}$$

$x = \frac{b}{2}$

* Least force required to topple the body, $y = H$

$$F_{least} = \frac{mgb}{2H}$$

$\tau_c = 0$; $f y_c + F(y - y_c) = Nx$

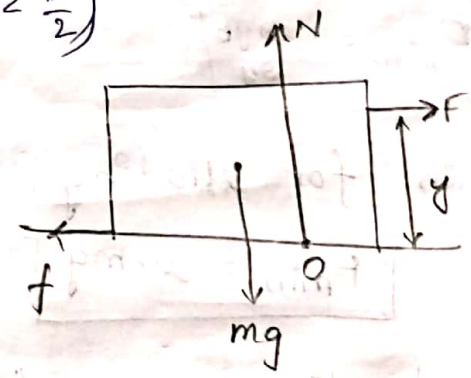
$f y_c - F y_c + Fy = Mgx$

$$F = \frac{Mgx}{y}$$

Important pts:

a) Sliding before toppling: $(x < \frac{b}{2})$

'F' is increased gradually from zero.



* for skidding,

$F > f_s$

$$F > \mu_s mg \quad \text{—①}$$

min force for skidding,

$$F_{min} = \mu_s mg$$

Here $\tau_0 = 0$

$mgx = Fy$

$$x = \frac{Fy}{mg}$$ *

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* min force required for toppling,

$$F = \frac{mgx}{y} \quad \text{Here } x = \frac{b}{2}$$

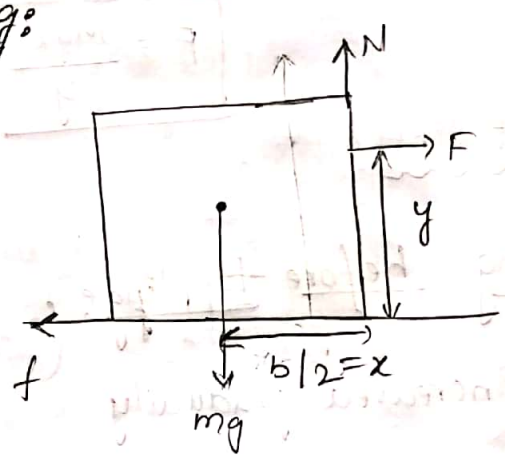
$$\boxed{F = \frac{mgb}{2y}}$$

$$\frac{mgb}{2y} > \mu_s mg$$

$$\boxed{\mu_s < \frac{b}{2y}} \; ** \quad \text{—This is condn for slidding occurs at first.}$$

b) Topplying before slidding:

$$\boxed{x = \frac{b}{2}}$$



* $Fy = mgb/2$

$$\boxed{F_{min} = \frac{mgb}{2y}}$$

* F_{min} for skidding,

$$\boxed{F_{min} = \mu_s mg}$$

* for toppling occurs first,

$$\mu_s mg > \frac{mgb}{2y}$$

$$\boxed{\mu_s > \frac{b}{2y}} \; **$$

Here, $\boxed{F_{least} = \frac{mgb}{2H}}$

c) Both occurs simultaneously:

when $\mu_s mg = \frac{mgb}{2y} \Rightarrow \boxed{\mu_s = \frac{b}{2y}}$

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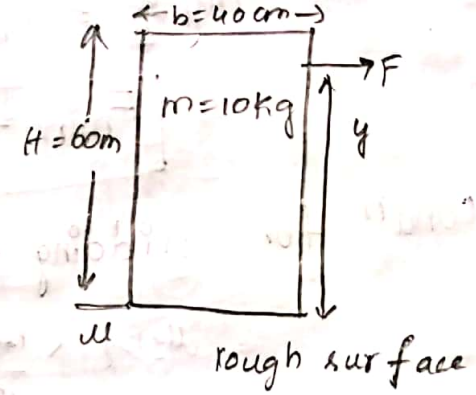
Here $\boxed{F_{min} = \mu_s mg = \frac{mgb}{2y}}$

→ Decide which occurs first whether skidding or toppling for given μ & y values. Also find F_{min} & shifting in normal rxn in each case.

a) $y = 30$, $\mu = 1/3$

b) $y = 50$, $\mu = 2/3$



a) $* \frac{b}{2y} = \frac{40}{2 \times 30} = \frac{2}{3}$.

$\mu < \frac{b}{2y}$ → skidding occurs first.

$* F_{min} = \mu mg = \frac{1}{3}(10)(10) = \frac{100}{3} N$

where as f_{min} for toppling is $F_{min} = \frac{200}{3} N$

* shift in normal rxn,

$$x = \frac{Fy}{mg} = \frac{\frac{100}{2} \times 30}{100} = 30 cm.$$

b) $* \frac{b}{2y} = \frac{40}{2 \times 50} = \frac{2}{5}$

$\mu > \frac{b}{2y}$ → toppling occurs first.

$* F_{min} = \frac{mgb}{2y} = \frac{100 \times 40}{2 \times 50} = 40N$

* least force for toppling,

$$F_{least} = \frac{mgb}{2h} = \frac{100 \times 40}{2 \times 60} = \frac{100}{3} N$$

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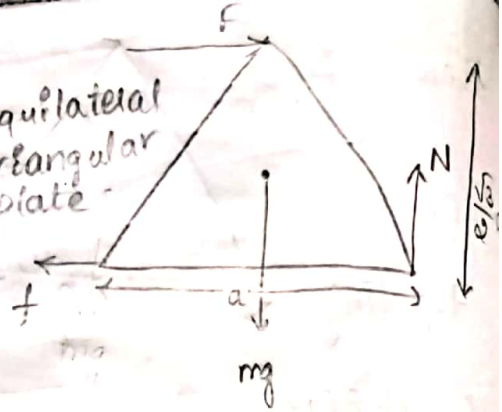
→ cond^n for skidding,

$F > \mu_s mg$



$F_{min} = \mu_s mg$ — ①

cond^n for toppling,

$mg\frac{a}{2} = F_{min}\left(\frac{\sqrt{3}}{2}a\right)$

$$\boxed{F_{min} = \frac{mg}{\sqrt{3}}}$$

cond^n for slidding first,

$\mu_s mg < \frac{mg}{\sqrt{3}}$

$$\boxed{\mu_s < \frac{1}{\sqrt{3}}}$$

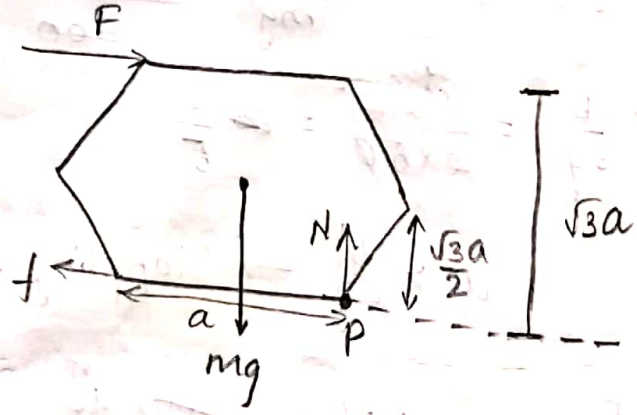
cond^n for toppling first,

$$\boxed{\mu_s > \frac{1}{\sqrt{3}}}$$

both occurs,

$$\boxed{\mu_s = 1/\sqrt{3}}$$

→ cond^n for slidding,



$F_{min} = \mu_s mg$ — ①

cond^n for toppling,

$$\boxed{\tau_p = 0}$$

$mg\left(\frac{a}{2}\right) = F_{min}\left(\frac{\sqrt{3}}{2}a\right)$

$$\boxed{F_{min} = mg/\sqrt{3}}$$

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* both occurs simultaneously,

$$\mu_s mg = \frac{mg}{2\sqrt{3}}$$

$$\boxed{\mu_s = \frac{1}{2\sqrt{3}}}$$

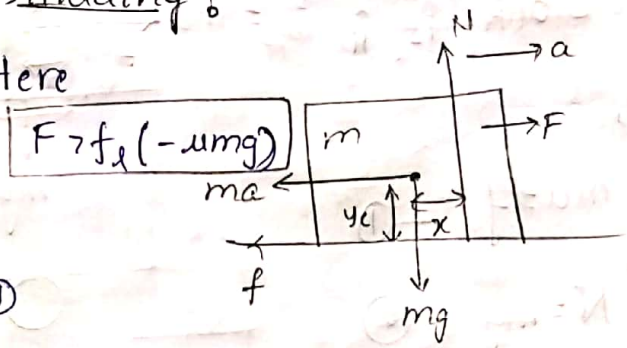
d) Toppling during skidding:

* we have,

$$F - f_k = ma$$

$$\Rightarrow F - \mu_k mg = ma \quad \text{—①}$$

Here $F > f_k (= \mu mg)$



* Torque about 'O' = 0 (w.r.to block)

$$\Rightarrow mgx + ma y_c = Fy \quad \text{—②}$$

solving ① & ②, we get x & a. [for given F]

$$mgx + (F - \mu_k mg) y_c = Fy$$

$$mgx = Fy - y_c (F - \mu_k mg)$$

$$x = \frac{Fy - Fy_c + \mu_k mg y_c}{mg}$$

$$\boxed{x = \frac{F}{mg}(y - y_c) + \mu_k y_c}$$

$$\boxed{a = \frac{F}{m} - \mu_k g}$$

* To find min force required for toppling,

put $x = \frac{b}{2}$ in eqn ②

$$\boxed{mg\frac{b}{2} + may_c = F_{min} y}$$

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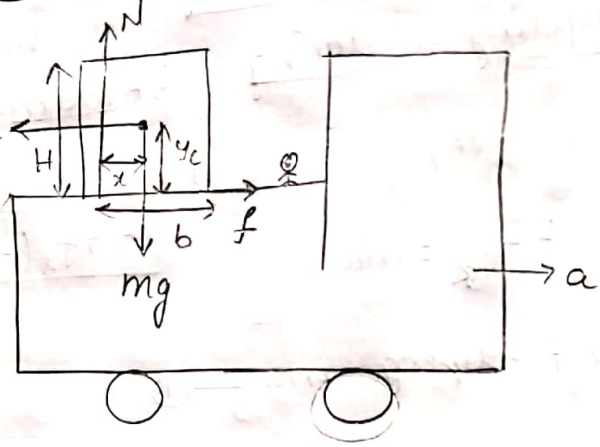
* If $\boxed{y = H}$ then,

$$\boxed{F_{min} = F_{least}}$$

Toppling of body in accelerating frame :-

w.r.to truck,

a) if block is in eqm,



$$ma = f \quad -①$$

$$N = mg \quad -②$$

∴ we have,

$$f \leq \mu N$$

$$ma \leq \mu mg$$

$$a \leq \mu g$$

$\boxed{a_{max} = \mu g}$ — upto this, there is no skidding.

Again,

$$\tau_0 = 0$$

$$may_c = mgx$$

$$\boxed{a = \frac{gx}{y_c}}$$

toppling occurs when $x = b/2$,

$$\therefore \boxed{a = \frac{gb}{2y_c}}$$ — upto this acceleration, toppling doesn't occurs.

* If acceleration is gradually increased from zero.

when $\mu_s g < \frac{gb}{2y_c} \Rightarrow \boxed{\mu_s < b/2y_c}$

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* skidding occurs first for $\mu_s < \frac{b}{2y_c}$.

* Eq min force for skidding,

$$\boxed{F_{min} = \mu_s mg}$$

* Eq shift in normal rxn, $\boxed{x = \frac{ay_c}{g}}$

* If $\mu_s g > \frac{gb}{2y_c} \Rightarrow \boxed{\mu_s > \frac{b}{2y_c}}$

* Toppling occurs at first.

shift; $\boxed{x = b/2}$

* min force for toppling;

$$F_{min} = ma = \frac{mbg}{2y_c}$$

$$\boxed{f_{min} = \frac{mbg}{2y_c}}$$

* If $\mu_s g = \frac{gb}{2y_c}$ i.e, $\boxed{\mu_s = \frac{b}{2y_c}}$

* both occurs simultaneously.

* min force,

$$\boxed{f_{min} = \mu_s mg = \frac{mgb}{2y_c}}$$

Toppling of a body in a rotating frame:-

* w.r.to rotating frame,

if body is in eqm,

$mr\omega^2 = f$ & $N = mg$.

* we have,

$f \leq \mu_s N$

$mr\omega^2 \leq \mu_s mg$

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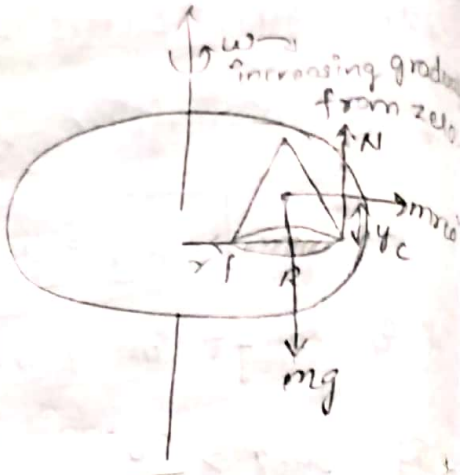
$$\omega^2 \leq \frac{\mu_s g}{r}$$

$$\boxed{\omega \leq \sqrt{\frac{\mu_s g}{r}}}$$

$$\boxed{\omega_{min} = \sqrt{\frac{\mu_s g}{r}}} \quad \text{for skidding.}$$



* shift in Normal rxn,

$$mg x = mr\omega^2 y_c$$

$$\boxed{\omega = \sqrt{\frac{gx}{ry_c}}}$$

when $x = b/2$ $\quad [b = 2R]$

$$\boxed{\omega_{min} = \sqrt{\frac{gb}{2ry_c}}} \quad \text{for toppling.}$$

a) $\frac{\mu_s g}{r} < \frac{gb}{2ry_c}$

$$\boxed{\mu_s < \frac{b}{2y_c}}$$

$$\boxed{\mu_s < \frac{b}{2y_c}} \quad \text{skidding occurs first.}$$

* min force required for skidding.

$$F_{min} = \mu_s mg$$

& $mr\omega^2 = \mu_s mg$

$$\boxed{\omega_{min} = \sqrt{\frac{\mu_s g}{r}}}$$

b) If $\frac{\mu_s g}{r} > \frac{gb}{2y_c r} \Rightarrow \boxed{\mu_s > \frac{b}{2y_c}}$

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$\mu_s > \frac{b}{2y}$ toppling occurs first.

* min force for toppling,

$$F = mr\omega^2_{min}$$

$$= m\cancel{r} \times \frac{gb}{2\cancel{r}y_c}$$

$$F_{min} = \frac{mgb}{2y_c}$$

& $\omega_{min} = \sqrt{\frac{gb}{2ry_c}}$

c) If $\frac{\mu_s \cancel{r}}{\cancel{r}} = \frac{\cancel{g}b}{2y_c\cancel{r}} \Rightarrow \mu_s = \frac{b}{2y_c}$ → both occurs simultaneously.

$$F_{min} = \mu_s mg = \frac{mgb}{2y_c}$$

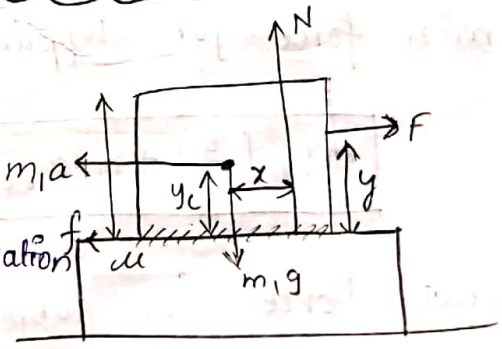
$$F_{min} = \frac{mgb}{2y_c}$$

at $\omega_{min} = \sqrt{\frac{\mu_s g}{r}} = \sqrt{\frac{gb}{2ry_c}}$

Toppling incase of block on block problems:

Case-(i):

* let both blocks are moving with common acceleration



$$a_c = \frac{F}{m_1+m_2}$$

w.r.to lower block, upper block is in eqm.

$$F = m_1a + f$$

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$$F = m_1\left(\frac{F}{m_1+m_2}\right) + f$$

$$f = F - \frac{m_1 F}{m_1+m_2}$$

$$\boxed{f = \frac{m_2 F}{m_1+m_2}}$$ static friction.

we have, $f \leq \mu_s N$

$$\frac{m_2 F}{m_1+m_2} \leq \mu_s m_1 g$$

$$\boxed{F \leq \mu_s \frac{m_1}{m_2} g (m_1+m_2)}$$

* min force required for skidding,

$$\boxed{F_{min} = \mu_s \frac{m_1}{m_2}(m_1+m_2)g} \quad -①$$

* torque about $O_1 = 0$

$$\tau_0 = 0$$

$$m_1 g x + m_1 a y_c = Fy$$

* min force for toppling, $\boxed{x = \frac{b}{2}}$

$$\boxed{m_1 g\left(\frac{b}{2}\right) + m_1 a y_c = F_{min} y}$$

* least force for toppling, $\boxed{y = H}$

$$\boxed{m_1 g\left(\frac{b}{2}\right) + m_1 a y_c = F_{least} \cdot H}$$

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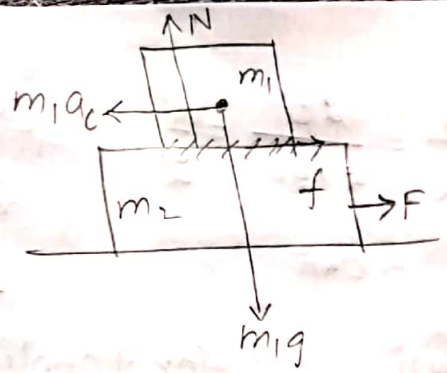
case-(ii):

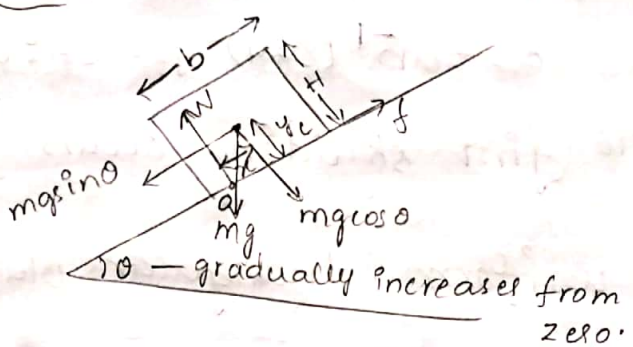


* common acceleration,

$$a_c = \frac{F}{m_1 + m_2}$$

* similar to that of block place in accelerating truck.

Toppling On inclined plane:



* since block is in eqm,

$mg\sin\theta = f$

$mg\cos\theta = N$

we have, $f \leq \mu_s N$

$mg\sin\theta \leq \mu_s (mg\cos\theta)$

$$\boxed{\tan\theta \leq \mu_s}$$

critical angle for skidding,

$$\boxed{\theta = \tan^{-1}(\mu_s)}$$

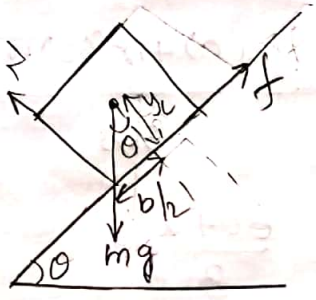
In this case,

shift in normal rxn,

$x_0 = 0$

$mg\sin\theta \cdot y_c = mg\cos\theta\,(x)$

$$\boxed{\tan\theta = \frac{x}{y_c}}$$



Just for toppling, $x = b/2$ $\Rightarrow$ $\boxed{\tan\theta = \frac{b}{2y_c}}$

$\therefore$ critical angle for toppling, $\boxed{\theta = \tan^{-1}\left(\frac{b}{2y_c}\right)}$

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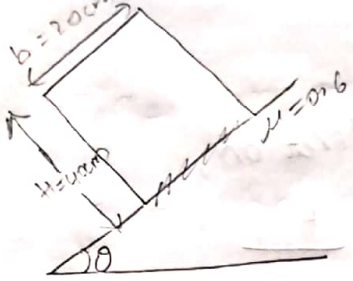
→

$$\frac{b}{2y_c} = \frac{20}{2\times 2b} = \frac{1}{2}$$

$$= 0.5$$



$\theta = \tan^{-1}(0.5)$ for toppling.

$\theta = \tan^{-1}(0.6)$ for skidding.

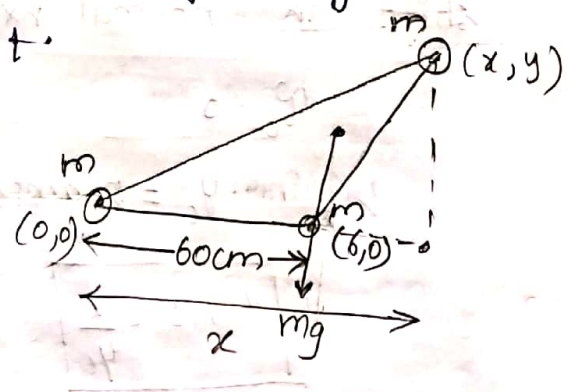
* If $\mu_s = 0.4$

$\theta = \tan^{-1}(0.4)$ for skidding.

here first skidding occurs.

→ A uniform triangular plate of base length 60cm, area $1\,m^2$, & thickness 10mm is lying vertically on the smooth ground as shown. Find the max value of $\cot\theta$ for which it doesn't topple.

In critical state line of action of weight should pass through tipping pt.



$$x_{cm} = \frac{m(0) + m(60) + m(x)}{3m}$$

$$60 = \frac{60+x}{3}$$

$$180 = 60 + x \Rightarrow x = 120\,cm$$

$$\boxed{x = 1.2m}$$

& area of Δle,

$$A = \frac{1}{2} \times b \times h$$

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$$1 = \frac{1}{2} \times \frac{36}{100} \times y$$

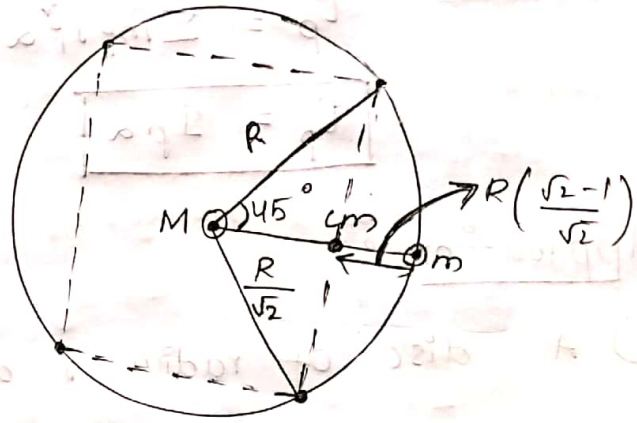
$$\boxed{y = 10/3 m}$$

From figure, $\cot\theta = \frac{x}{y} = \frac{1.2}{\frac{10}{3}} = \frac{3.6}{10}$

$$\boxed{\cot\theta = 0.36}$$

→ A turn table has heavy circular top of radius 1m & mass 1kg and is placed symmetrically on circumference. The max mass that can be kept any where on circumference of turntable without topping.

$\tau_{cm} = 0$



$$mg\frac{R}{\sqrt{2}} = Mg\frac{R}{\sqrt{2}}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$m = M$$

$$Mg\frac{R}{\sqrt{2}} = mgR\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$m = \frac{M}{\sqrt{2}-1}$$

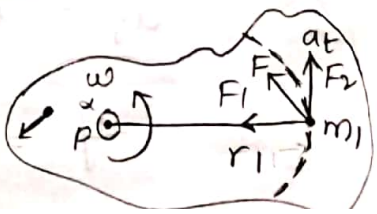
$$\boxed{m = M(\sqrt{2}+1)}$$

Area of table without toppling,

= Area of square of side $\sqrt{2}R$

$= (\sqrt{2}R)^2 = 2R^2$.

Newton's 2nd Law for Rotation:



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And $F_1 = m_i r_i \omega^2$, $F_2 = m_i a_t = m_i r_i \alpha$

* torque acting on the body about AOR,

$$d\tau_p = r_i F_2 = (m_i r_i \alpha) r_i$$

$$d\tau_p = m_i r_i^2 \alpha$$

here ω & α-values are same for all particles of rigid body.

* net torque acting on the body about AOR,

$$\tau_p = \Sigma m_i r_i^2 \alpha$$

$$\boxed{\tau_p = I_p \alpha}$$

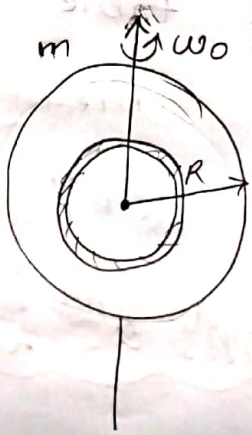
$$\boxed{\begin{aligned} F_{net} &= ma \\ \tau_{net} &= I\alpha \end{aligned}}$$

Applications:

① A disc of radius r and mass m is first spinned about its axis to the angular velocity ω_0 and then carefully placed with its flat face on the horizontal surface.

i) How long will the disc be rotating on the surface if friction coefficient is μ.

* Kinetic friction on ring element,



$$df = \mu (dm) g$$

$$df = \mu \frac{m}{\pi R^2} (2\pi x \cdot dx) g$$

$$df = \frac{2\mu mg}{R^2} x \cdot dx$$

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* opposing torque on ring element due to friction,

$$d\tau = (x) \cdot df$$

$$d\tau = \frac{2\mu mg}{R^2} x^2 \cdot dx$$

∴ net opposing torque on disc due to friction,

$$\tau = \int_{x=0}^{x=R} d\tau$$

$$\tau = \int_{x=0}^{x=R} \frac{2\mu mg}{R^2} \cdot x^2 \cdot dx$$

$$\tau = \frac{2\mu mg}{R^2} \int_{x=0}^{x=R} x^2 \cdot dx$$

$$= \frac{2\mu mg R^3}{3R^2}$$

$$\boxed{\tau = \frac{2\mu mgR}{3}}$$

* we have, $\boxed{\tau = I\alpha}$

∴ Angular acceleration,

$$\alpha = -\frac{\tau}{I_{cm}} = \frac{-2\mu mgR}{3\left(\frac{mR^2}{2}\right)} = -\frac{4\mu g}{3R} = \text{const.}$$

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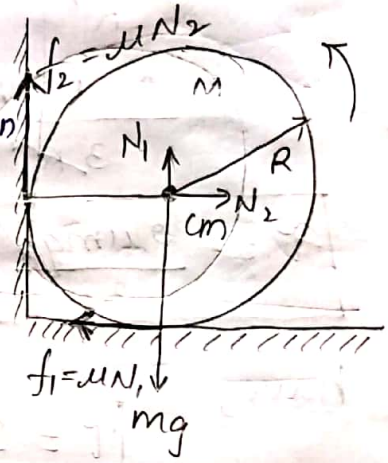
(2) A cylinder of mass m and Radius R is spinned about its axis to the angular velocity ω_0 and then placed into a corner as shown. The coeff of friction b/w the corner wall, floor & cylinder is μ. How many turns are made by the cylinder before it stops.



since cylinder has no translation motion,

$$\Sigma F_x = 0 \Rightarrow N_2 = f_1$$

$$\Rightarrow N_2 = \mu N_1 \quad \text{—①}$$

$$\Sigma F_y = 0 \Rightarrow N_1 + f_2 = mg \quad \text{—②}$$

by solving ① & ②,

$$N_1 + \mu N_2 = mg$$

$$N_1 + \mu^2 N_1 = mg$$

$$\boxed{N_1 = \frac{mg}{1+\mu^2}}$$

& $$\boxed{N_2 = \frac{\mu mg}{1+\mu^2}}$$

now, opposing torque,

$$\tau_{cm} = Rf_2 + Rf_1 = \mu(N_1 + N_2)R$$

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$$= \mu R \left[\frac{mg}{1+\mu^2} + \frac{\mu mg}{1+\mu^2} \right]$$

$$= \frac{mgR(\mu)(1+\mu)}{1+\mu^2}$$

$$\boxed{\tau_{cm} = \frac{mg\mu R(1+\mu)}{1+\mu^2}}$$

* Angular accelesation,

$$\alpha = -\frac{\tau_{cm}}{I_{cm}} = \frac{-mg\mu R(1+\mu)}{(1+\mu^2)\left(\frac{mR^2}{2}\right)}$$

$$\alpha = \frac{-2\mu(1+\mu)g}{(1+\mu^2)R} = \text{const.}$$

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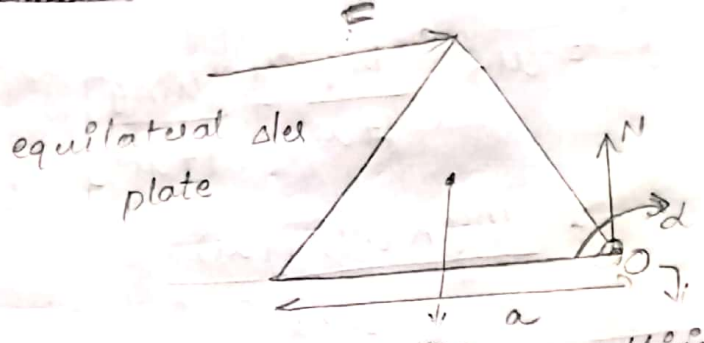


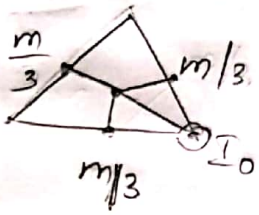
* min force for toppling,

$$F\left(\frac{\sqrt{3}}{2}a\right) = mg\left(\frac{a}{2}\right)$$

$$\boxed{F = \frac{mg}{\sqrt{3}}}$$

* if $F = \frac{2mg}{\sqrt{3}}$ then angular acceleration of plate about 'O' is

$$\tau_{net} = I\alpha$$



$$F\left(\frac{\sqrt{3}}{2}a\right) - mg\frac{a}{2} = I_O\alpha$$

$$\frac{2mg}{\sqrt{3}} \times \frac{\sqrt{3}}{2}a - mg\frac{a}{2} = I_O\alpha$$

$$\frac{mga}{2} = I_O\alpha$$

$$\alpha = \frac{mga}{2I_O}$$

$$\alpha = \frac{mga}{2 \times \frac{5ma^2}{12}}$$

$$\boxed{\alpha = \frac{6g}{5a}}$$

$$I_{cm} = \frac{m}{3} \times \left(\frac{a}{2\sqrt{3}}\right)^2 \times 3$$

$$= \frac{ma^2}{12}$$

$$I_O = I_{cm} + m\left(\frac{a}{\sqrt{3}}\right)^2$$

$$= \frac{ma^2}{12} + \frac{ma^2}{3}$$

$$\boxed{I_O = \frac{5ma^2}{12}}$$

Rope and pulley contraints for rotation

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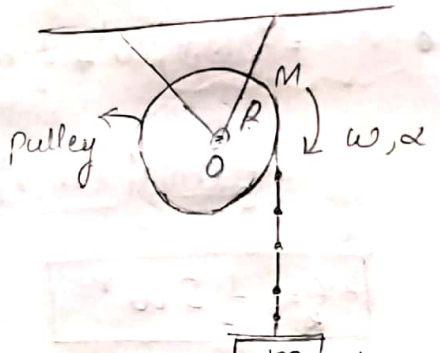
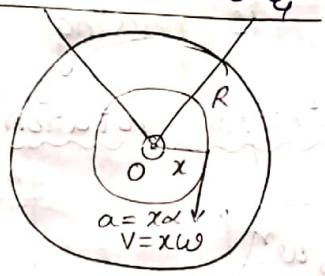
* There is friction b/w string & pulley & no slipping b/w string and pulley.



* All particles of pulley have same ω & α-values.



constraint eqn,

$V_{string} = V_{particle\ on\ the\ surface\ of\ pulley}$.

$V_{string\ particle}$

$\Rightarrow \boxed{V = R\omega}$, $\boxed{a = R\alpha}$ — for fixed axis of rotation.

Steps:

① Identify the motion of the bodies in the system.

combined motion.
pure translation, pure rotation

② Identify the force acting on the bodies in the system.

T, Fg, f, N, hinge force

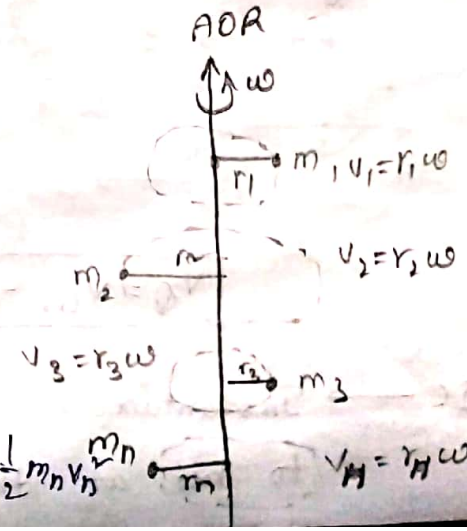
③ Identify the unknows.

④ use eqns $\Sigma F_x = ma_x$, $\Sigma F_y = ma_y$, $\Sigma \tau = I\alpha$, constraint eqns.

Rotational Kinetic Energy:

* K.E of rotating body about AOR is,

$$K_R = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + \dots + \frac{1}{2} m_n v_n^2$$



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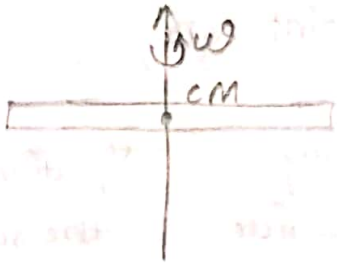
$$= \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \cdots + \frac{1}{2} m_n r_n^2 \omega^2$$

$$= \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2 + \cdots + m_n r_n^2) \omega^2$$

$$\boxed{K.E_R = \frac{1}{2} I \omega^2}$$

* If A.O.R is passing through COM of body - it is called "centroidal rotation".

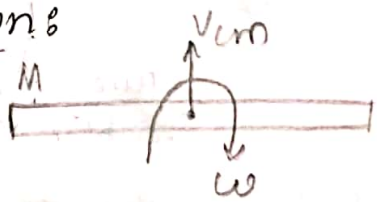


$$\boxed{K.E_R = \frac{1}{2} I_{cm} \omega^2}$$

* If AOR is not passing through COM of body then it is called "Non-centroidal rotation"

$$\boxed{K.E_R = \frac{1}{2} I_P \omega^2}$$

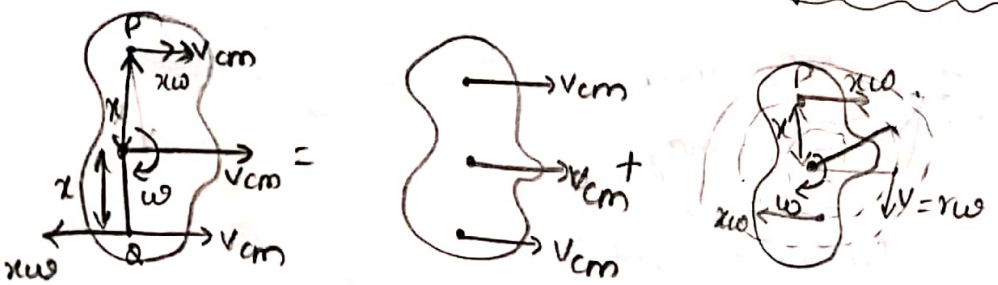
* K.E of body in combined motion:



$$K.E = K.E_{\text{of cm}} + K.E_{\text{about cm}}$$

(translate) (pure rotation)

Velocity of a pt in the body during combined motion:



Combined motion — pure translation — pure rotation about cm.

$$\boxed{V_Q = V_{cm} - x\omega}$$

IC ⟶ Instantaneous centre is the pt when particle comes to rest at that moment.

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In general, $\boxed{\bar{V} = \bar{V}_{cm} + \bar{\omega} \times \bar{x}}$

* Similarly, acceleration at any pt,

$$\boxed{\bar{a} = \bar{a}_{cm} + \bar{\alpha} \times \bar{r} + \bar{a}_n}$$

Work-Energy Theorem:

* work done by rotating torque,

$$dw = \bar{\tau} \cdot d\bar{\theta}$$

$$\boxed{w = \int \bar{\tau} \cdot d\bar{\theta}}$$

τ = const
$\tau = f(\theta)$
$\tau = f(\omega)$
$\tau = f(t)$

we use,

$\omega = \frac{d\theta}{dt}$

$\alpha = \frac{d\omega}{dt} = \omega \cdot \frac{d\omega}{d\theta}$

$\tau = I\alpha$

* ω-ϵ-Theorem,

$$w_c + w_{nc} + w_{ext} = \Delta K_{rot}$$

i.e, $\boxed{w_{net} = \int \bar{\tau} \cdot d\bar{\theta} = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2}$

Law of conservation of Energy:

If only internal forces are doing work. Then they are conservative. then $\boxed{T \cdot \epsilon = \text{const}}$

$$\boxed{U_1 + K_1 = U_2 + K_2}$$

k < Translational
Rotational

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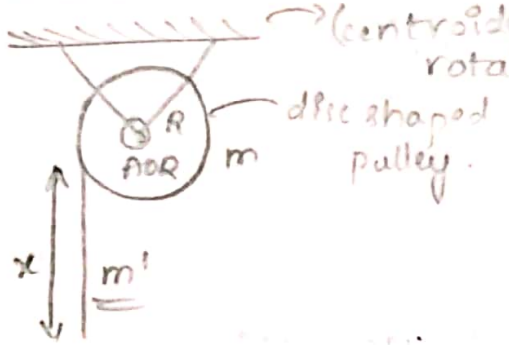
Rotating Power:

* $P_{avg} = \frac{\Delta W}{\Delta t} = \frac{\frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2}{\Delta t} = \frac{\int P_{inst} \cdot dt}{\int dt}$

* $P_{inst} = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt} = \vec{\tau} \cdot \frac{d\vec{\theta}}{dt} = \vec{\tau} \cdot \vec{\omega}$.

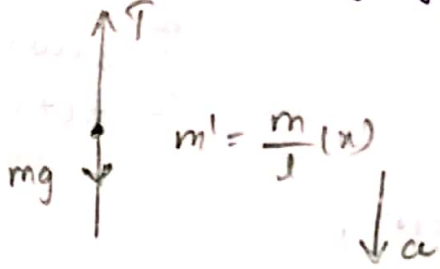
Applications:

(1)



A cord is of mass 'm' & initially wound by one turn. Find the angular acceleration of pulley as a function of 'x' of hanging part of chord.

* For hanging cord (rope),

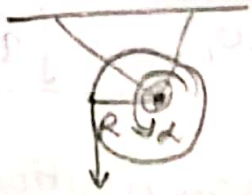


$\therefore m'g - T = m'a$

$\frac{m}{l} x g - T = \frac{m}{l} x a \quad -(1)$

* for pulley,

$TR = \left[\frac{MR^2}{2} + \frac{m}{l}(l-x)R^2\right]\alpha \quad -(2)$

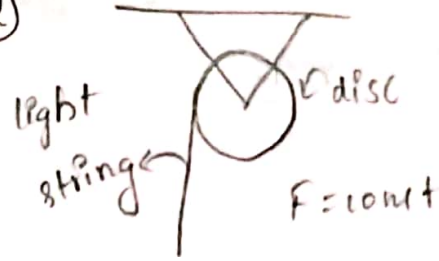


and constraint eqn,

$a = R\alpha \quad -(3)$

unknowns $\longrightarrow a, \alpha$ & T.

(2)



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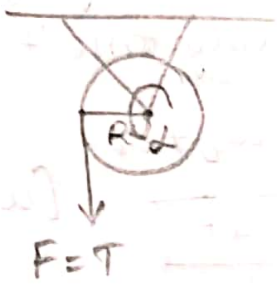
* for hanging part,

$F - T = m\cancel{a}^{0}$

$m = 0$ $\quad \boxed{F = T}$

here $F_{net} = 0$ but $\boxed{a \neq 0}$.

* for pulley,



$TR = I_{cm}\alpha$

$F\cancel{R} = \frac{MR^{\cancel{2}}}{2}(\alpha)$

$\boxed{\alpha = \frac{2F}{RM}}$ & $a = R\alpha \Rightarrow \boxed{\frac{2F}{m} = a}$

* Angular velocity of pulley when it is rotated through an angle θ is

$$\int \tau \cdot d\theta = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \cancel{\omega_i^2}^{0}$$

$$\int FR \cdot d\theta = \frac{1}{2} I \omega_f^2$$

$$FR\theta = \frac{1}{2} I \omega_f^2$$

$$F\cancel{R}\theta = \frac{1}{2}\left(\frac{MR^{\cancel{2}}}{2}\right)\omega_f^2$$

$$\omega_f^2 = \frac{4F\theta}{MR}$$

$$\boxed{\omega_f = 2\sqrt{\frac{F\theta}{MR}}}$$ $\theta \rightarrow$ radians.

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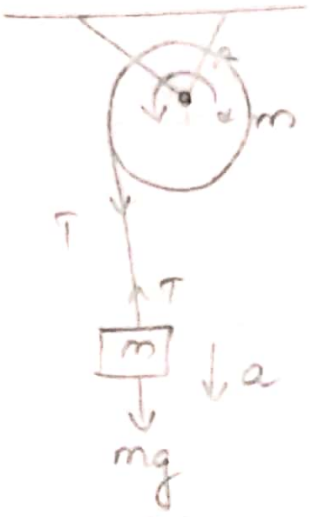
(3)



unknowns

a, α & T

M-I:

* for block,

$$mg - T = ma$$

$$T = m(g-a) \quad -(1)$$

* for pulley,

$$TR = \frac{mR^2}{2}\alpha$$

$$m(g-a) = \frac{mR\alpha}{2}$$

$$g - a = \frac{R\alpha}{2} \qquad (a = R\alpha)$$

$$g - R\alpha = \frac{R\alpha}{2}$$

$$g = \frac{3R\alpha}{2} \Rightarrow \boxed{\alpha = \frac{2g}{3R}}$$

$$\boxed{a = 2g/3}$$

☆ points:

* Velocity of block at any time 't',

$$v = u + at$$

$$\boxed{v = at} \quad [u = 0]$$

* Angular velocity of pulley at any time 't',

$$\omega = \omega_0 + \alpha t$$

$$\boxed{\omega = \alpha t} \quad [\omega_0 = 0]$$

* distance travelled by block in a time 't',

$$\boxed{h = \frac{1}{2}at^2} \quad [u = 0]$$

* Angala turned by pulley in a time 't',

$$\boxed{\theta = \frac{1}{2}\alpha t^2}$$

* No. of rotations,

$$\boxed{N = \frac{\theta}{2\pi}}$$

M-II: L.C.E:

Decrease in P.E of block = Increase in K.E of syste

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$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2} \times \frac{MR^2}{2} \times \omega^2$$

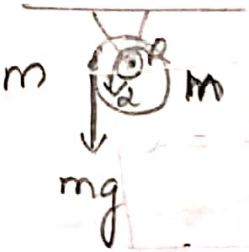
$$4gh = 2v^2 + R^2\omega^2$$

$$4gh = 3R^2\omega^2$$

$$\boxed{h = \frac{3R^2\omega^2}{4g}} \quad \text{or} \quad \boxed{\omega = \frac{2}{R}\sqrt{\frac{gh}{3}}}$$

$$\boxed{v = 2\sqrt{\frac{gh}{3}}}$$

short cut:



$\tau = I\alpha \Rightarrow mgR = I\alpha$

$mgR = \frac{MR^2}{2}\alpha$

α

$\tau_0 \Rightarrow mgR = I\alpha$

$$mgR = \left(\frac{MR^2}{2} + mR^2\right)\alpha$$

$$\boxed{\alpha = \frac{2mg}{R(M+2m)}} \quad \& \quad \boxed{a = \frac{2mg}{M+2m}}$$

(3) Atwood Machine:

M-I:

* for m_1:

$$m_1g - T_1 = m_1a \quad -(1)$$

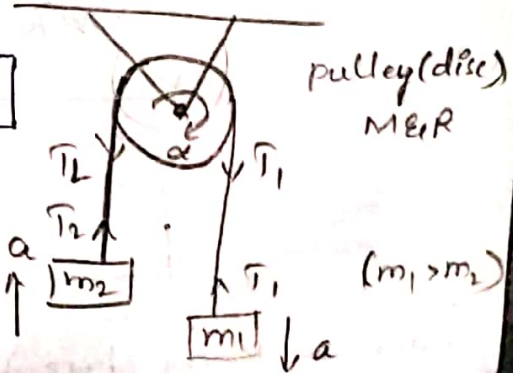
* for m_2:

$$T_2 - m_2g = m_2a \quad -(2)$$

$\boxed{T_1 = T_2 e^{\mu\theta}}$

unknowns

a, α, T_1, T_2.



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* for pulley,

$$T_1 R - T_2 R = \frac{MR^2}{2}\alpha$$

$$T_1 - T_2 = \frac{MR\alpha}{2} \quad -(3)$$

* constraint eqn,

$$a = R\alpha \quad -(4)$$

(1) $\Rightarrow$ $m_1(g-a) = T_1$

(2) $\Rightarrow$ $m_2(g+a) T_2$

(3) $m_1(g-a) - m_2(g+a) = \frac{MR\alpha}{2}$

$$\frac{2(m_1-m_2)g - 2(m_1+m_2)a}{MR} = \alpha$$

$$\frac{2(m_1-m_2)g - 2(m_1+m_2)(R\alpha)}{MR} = \alpha$$

$$\frac{2(m_1-m_2)g}{MR} - \frac{2(m_1+m_2)}{M}\alpha = \alpha$$

$$\frac{2(m_1-m_2)g}{MR} = \alpha\left(\frac{M+2(m_1+m_2)}{M}\right)$$

$$\alpha = \frac{2(m_1-m_2)g}{(M+2(m_1+m_2))R}$$

(Same ✱ points here) $\rightarrow$ on the back page.

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M-II &

L.C.E &

Decrease in P.E of blocks = Increase in K.E of system.

$\Rightarrow m_1gh - m_2gh = \frac{1}{2} m_1v^2 + \frac{1}{2} m_2v^2 + \frac{1}{2} I_{cm}\omega^2$ & $V = R\omega$

we get v & ω - values.

Important points:

for mass less pulley,

$T_1 = T_2$ but $\alpha \neq 0$

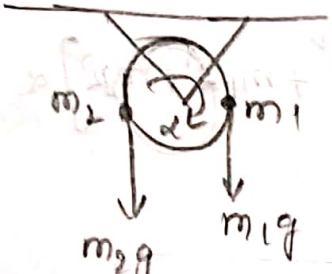
$M \longrightarrow 0$

$I \longrightarrow 0$

$T_1R - T_2R = 0$

$\boxed{T_1 = T_2}$

short cut:

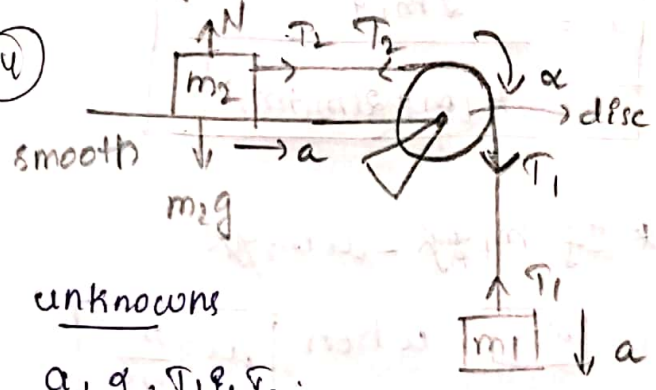


$m_1gR - m_2gR = \left[\frac{MR^2}{2} + m_1R^2 + m_2R^2\right]\alpha$

& $\boxed{a = R\alpha}$

$$\boxed{\alpha = \frac{2(m_1 - m_2)g}{[M + 2(m_1 + m_2)]R}}$$

(4)



unknowns

a, α, T_1 & T_2.

M-I &

* for m_1 &

$m_1g - T_1 = m_1a$

$T_1 = m_1(g - a)$ — (1)

* for m_2 &

$T_2 = m_2a$ — (2)

* for pulley,

$T_1 - T_2 = \frac{MR\alpha}{2}$ — (3)

* constraint eqn,

$a = R\alpha$ — (4)

from (1) & (2), $m_1(g - a) - m_2a = \frac{MR\alpha}{2}$

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$$m_1 g - a(m_1+m_2) = \frac{MR\alpha}{2}$$

$$m_1 g = \left[\frac{MR}{2} + R(m_1+m_2)\right]\alpha$$

$$\boxed{\alpha = \frac{2m_1 g}{R\left[M+2(m_1+m_2)\right]}}$$

(same ✱ pts here)

M-II &

L.C.E &

$$m_1 gh = \frac{1}{2} m_1 v^2 + \frac{1}{2} m_2 v^2 + \frac{1}{2} I_{cm} \omega^2$$

& $v = R\omega$

⇒ we get v & ω values.

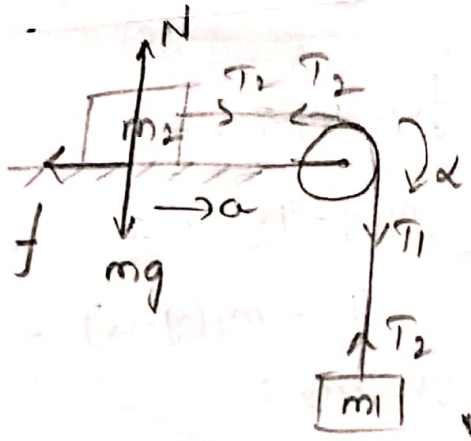
short cut:

$$m_1 gR = \left[\frac{MR^2}{2} + m_1 R^2 + m_2 R^2\right]\alpha$$

& $a = R\alpha$

$$\Rightarrow \boxed{\alpha = \frac{2m_1 g}{R\left[M+2(m_1+m_2)\right]}}$$

(5)



* If $m_1 g R \geq \mu m_2 g R$

$\boxed{\alpha = 0}$ when $\boxed{\mu > \frac{m_1}{m_2}}$

In this case,

$$\boxed{T_1 = T_2 = mg = f}$$

* If $\mu < \frac{m_1}{m_2}$ then $\boxed{\alpha \neq 0}$.

unknowns, a, α, T_1, T_2.

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* for m_1 :

$$m_1 g - T_1 = m_1 a \quad \text{---(2)}$$

$$T_1 = m_1 (g - a) \quad \text{---(1)}$$

* for m_2 :

$$T_2 - \mu m_2 g = m_2 a \quad \text{---(2)}$$

$$T_2 = m_2 (a + \mu g) \quad \text{---(2)}$$

* for pulley :

$$T_1 - T_2 = \frac{MR\alpha}{2}$$

$$m_1 (g - a) - m_2 (a + \mu g) = \frac{M \alpha R}{2}$$

$$(m_1 - \mu m_2) g - (m_1 + m_2)(R\alpha) = \frac{MR\alpha}{2}$$

$$\frac{(m_1 - \mu m_2) g}{R} = \left[\frac{M + 2(m_1 + m_2)}{2}\right] \alpha$$

$$\boxed{\alpha = \frac{2 (m_1 - \mu m_2) g}{R [M + 2(m_1 + m_2)]}}$$

Short cute

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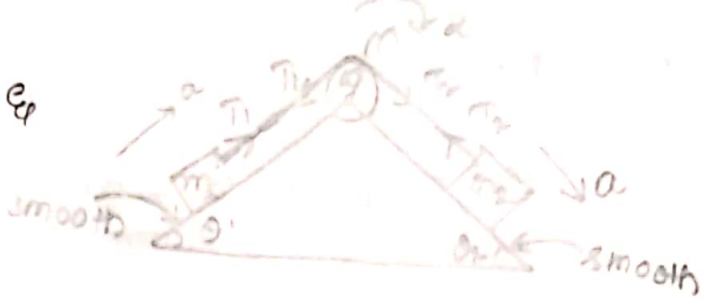
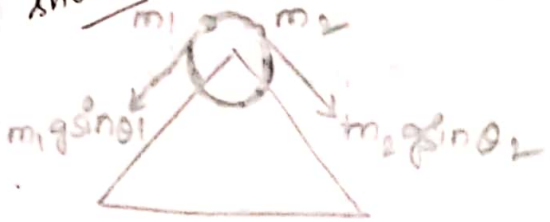
→ Find 'α' of pulley & 'a' of blocks.



short cut:



$$\tau_{cm} \Rightarrow m_2 g\sin\theta_2 R - m_1 g\sin\theta_1 R = \left[\frac{MR^2}{2} + m_1R^2 + m_2R^2\right]\alpha$$

$$\alpha = \frac{2[m_2\sin\theta_2 - m_1\sin\theta_1]g}{[M + 2(m_1+m_2)]R}$$

$$\boxed{\alpha = \frac{2g(m_2\sin\theta_2 - m_1\sin\theta_1)}{R(M+2(m_1+m_2))}}$$

standard:

* $m_2 g\sin\theta_2 - T_2 = m_2 a$ —①

$$T_2 = m_2(g\sin\theta_2 - a) \quad -①$$

* $T_1 - m_1 g\sin\theta_1 = m_1 a$

$$T = m_1(g\sin\theta_1 + a) \quad -②$$

* for pulley,

$$T_2R - T_1R = \frac{mR^2}{2}\alpha \quad -③$$

* constraint eqn,

$$a = R\alpha \quad -④ \Rightarrow \text{By solving we get } \alpha.$$

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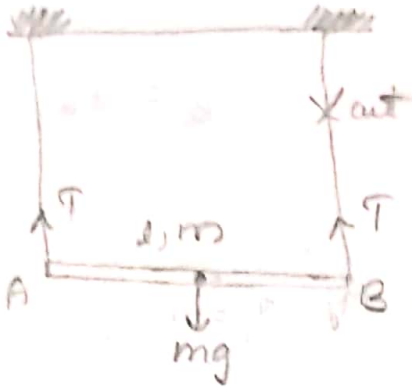
Strings Concept :-

M-I

* $2T = mg$

Two identical strings

$$T = \frac{mg}{2} \text{ [when string snaps]}$$



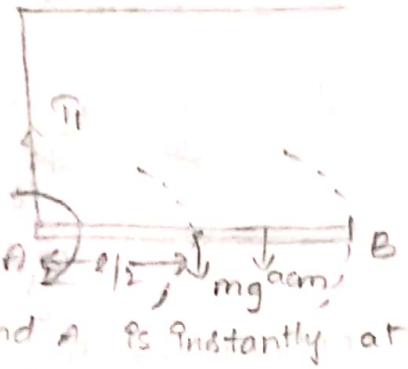
* when right string snaps

* $mg - T_1 = ma_{cm}$ — ①

& $\tau_A = I_A \alpha$

$$\frac{mgl}{2} = \frac{ml^2}{3}\alpha$$

$$\boxed{\alpha = \frac{3g}{2l}} \quad -②$$



end A is instantly at rest.

* Constraint eqn, $a_{cm} = \frac{l}{2}\alpha$ — ③

$$mg - T_1 = \frac{ml}{2}\alpha$$

& from eqn ②, $\boxed{\alpha = \frac{3g}{2l}}$

$$T_1 = mg - \frac{3mg}{4} = \frac{mg}{4} \Rightarrow \boxed{T_1 = \frac{mg}{4}}$$

* % of change in tension in left string

$$= \frac{\frac{mg}{4} - \frac{mg}{2}}{\frac{mg}{2}} \times 100$$

$$= -50\%$$

* acceleration of centre of mass of rod,

$$a_{cm} = \frac{l}{2}\alpha = \frac{l}{2} \times \frac{3g}{2l} \Rightarrow \boxed{a = 3g/4}$$

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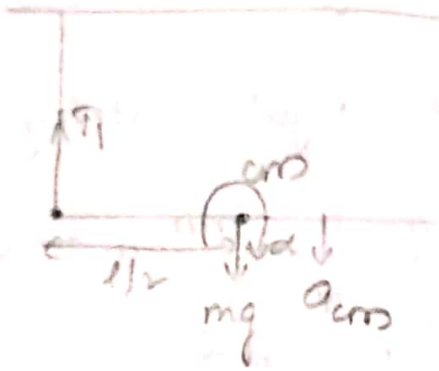
→ * acceleration of end B,

$$a_B = l\alpha = l \times \frac{3g}{2l} \Rightarrow \boxed{a_B = \frac{3g}{2}}$$

SIM-I:

* $mg - T_1 = m a_{cm} \quad -①$



* $\tau_{cm} = I_{cm}\alpha$

$$T_1 \frac{l}{2} = \frac{ml^2}{12} \times \alpha$$

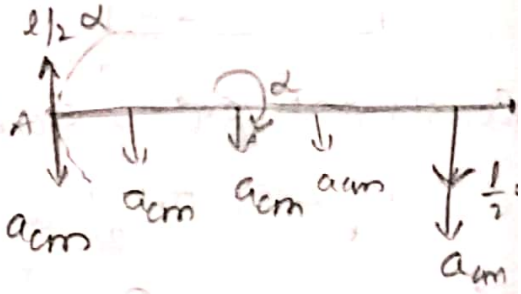
$$T_1 = \frac{ml}{6}\alpha \quad -②$$

* net acceleration of end 'A' along the string = 0

[because $\omega = 0$]

* constraint eqn,

$$\therefore \frac{l}{2}\alpha = a_{cm} \quad -③$$



$$mg - \frac{ml}{6}\alpha = m\frac{l}{2}\alpha$$

$$\boxed{\alpha = \frac{3g}{2l}}$$

$$a_{cm} = \frac{l}{2}\alpha = \frac{l}{2} \times \frac{3g}{2l} = \frac{3g}{4}$$

$$\boxed{a_{cm} = \frac{3g}{4}}$$

$$a_B = a_{cm} + \frac{l}{2}\alpha$$

$$= \frac{3g}{4} + \frac{3g}{4}$$

$$\boxed{a_B = 3g/2}$$

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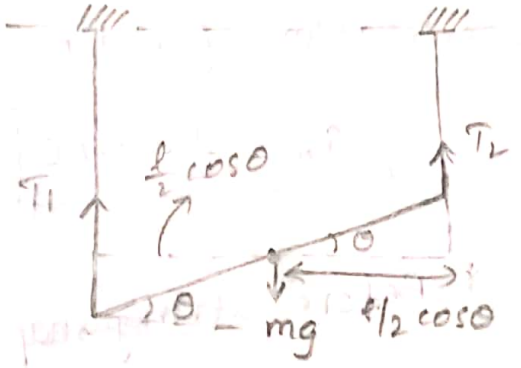
Final, $mg - T_1 = m\left(\frac{3g}{4}\right)$

$$\boxed{T_1 = \frac{mg}{4}}$$

→

* before string snaps,



$$T_1 + T_2 = mg$$

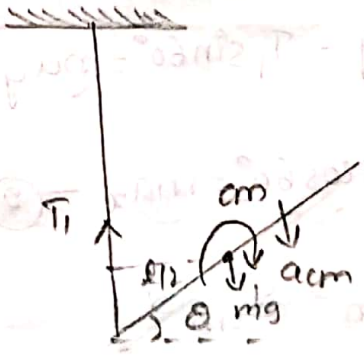
$$\tau_{cm} = 0 \quad [\text{In eq m}]$$

$$T_2\left(\frac{l}{2}\cos\theta\right) = T_1\left(\frac{l}{2}\cos\theta\right)$$

$$\Rightarrow \boxed{T_1 = T_2}$$

$$\therefore \boxed{T_1 = T_2 = \frac{mg}{2}}$$

* Just after right string snaps,



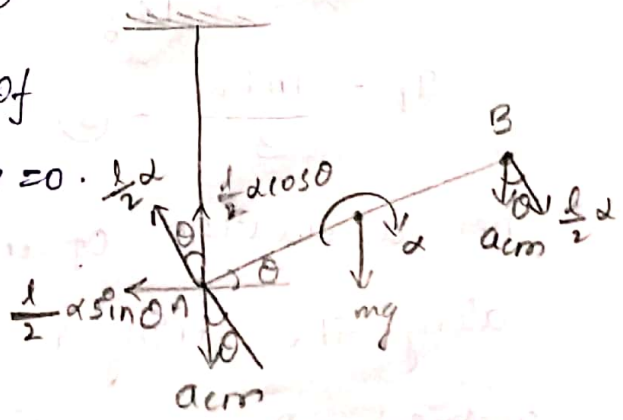
$$mg - T_1 = m a_{cm} \quad \text{—①}$$

$$\tau_{cm} = I_{cm}\alpha$$

$$T_1 \frac{l}{2}\cos\theta = \frac{ml^2}{6}(\alpha)$$

$$T_1 \cos\theta = \frac{ml\alpha}{6} \quad \text{—②}$$

* since net acceleration of end A along the string = 0.



$$\frac{l}{2}\alpha\cos\theta = a_{cm} \quad \text{—③}$$

solving eqns ①, ②, ③ we get T_1, α & a_{cm}.

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Finally,

acceleration of end B,

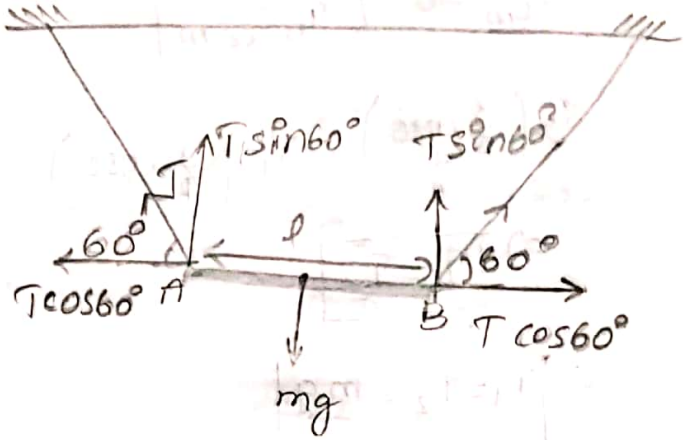
$$a_B = \sqrt{a_{cm}^2 + \left(\frac{l}{2}\alpha\right)^2 + 2a_{cm}\frac{l}{2}\cos\theta(\alpha)}$$

* acceleration of end A,

$$\boxed{a_A = \frac{l}{2}\alpha\sin\theta}$$

→

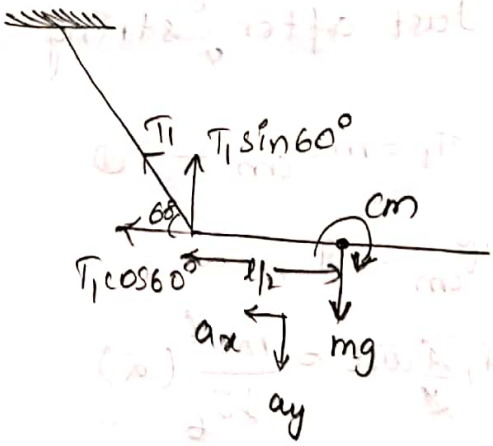
* Before string snaps,



$$2T\sin60° = mg$$

$$\boxed{T = \frac{mg}{\sqrt{3}}}$$

* Just after right string snaps,



$$mg - T_1\sin60° = ma_y \quad -①$$

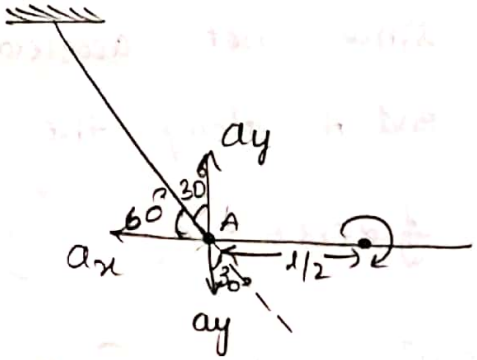
$$T_1\cos60° = ma_x \quad -②$$

$$\tau_{cm} = I_{cm}\alpha$$

$$T_1\cos60°\left(\frac{l}{2}\right) = \frac{ml^2}{12}(\alpha)$$

$$T_1 = \frac{ml\alpha}{3} \quad -③$$

* net acceleration of end A along the string is 0.



constraint eqn, $a_x\cos60° + \frac{l}{2}\alpha\cos30° = a_y\cos30°$ —④

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① ⇒

$$mg - \frac{ml\alpha}{3}\sin 60^\circ = may$$

$$g - \frac{l\alpha}{3} \times \frac{\sqrt{3}}{2} = a_y$$

$$a_y = \frac{6g - l\alpha\sqrt{3}}{6}$$

② ⇒

$$\frac{ml\alpha}{3} \times \frac{1}{2} = m\, a_x$$

$$a_x = \frac{l\alpha}{6}$$

④ ⇒

$$\frac{l\alpha}{6} \times \frac{1}{2} + \frac{l}{2}\alpha \times \frac{\sqrt{3}}{2} = \left(\frac{6g - l\alpha\sqrt{3}}{6}\right)\left(\frac{\sqrt{3}}{2}\right)$$

[Calculation wrong]

$$l\alpha\left(\frac{1}{6} + \frac{\sqrt{3}}{2}\right) = \sqrt{3}g - \frac{l\alpha\sqrt{3}}{6} \times \sqrt{3}$$

$$l\alpha\left(\frac{1}{6} + \frac{\sqrt{3}}{2}\right) = \sqrt{3}g - \frac{l\alpha(3)}{6\,2}$$

$$l\alpha\left(\frac{1}{6} + \frac{\sqrt{3}}{2} + \frac{1}{2}\right) = \sqrt{3}\,g$$

$$l\alpha\left(\frac{1+3+3\sqrt{3}}{6}\right) = \sqrt{3}\,g$$

$$l\alpha = \frac{\sqrt{3}\,g \times 6}{4+3\sqrt{3}}$$

$$T_1 = \frac{mg}{2} \times \frac{\sqrt{3} \times 6\,2}{4+3\sqrt{3}} = \frac{2mg\sqrt{3}}{13}$$

$$\boxed{T_1 = \frac{2mg\sqrt{3}}{13}}$$ ##

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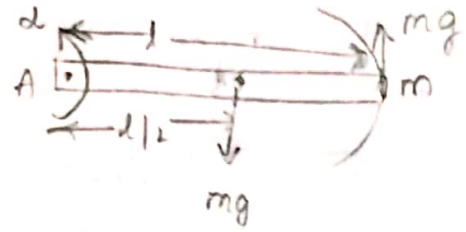
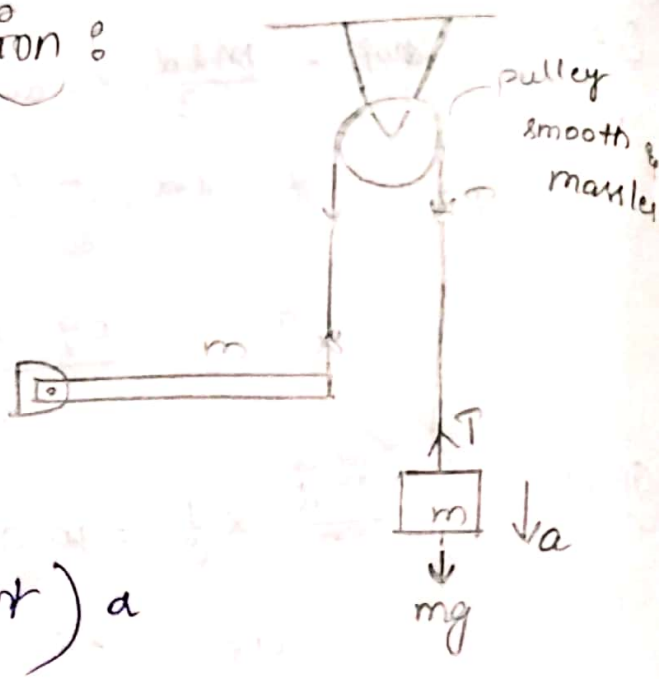
→ Non-Centroidal Rotation :





$$\tau_A = I_A \alpha$$

$$mgl - mg\frac{l}{2} = \left(\frac{ml^2}{3} + ml^2\right)\alpha$$

$$g - g/2 = \frac{4l}{3}\alpha$$

$$\boxed{\frac{3g}{8l} = \alpha}$$

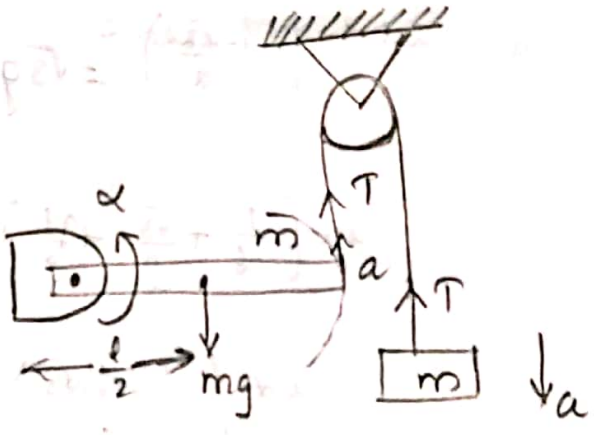
and $\omega = l\alpha = \frac{3g}{8}$

$$mg - T = ma$$

$$mg - T = mg\left(\frac{3}{8}\right)$$

$$\boxed{T = \frac{5mg}{3}}$$

general method :



$mg - T = ma$ — ①

[for block]

for rod, $Tl - mg\frac{l}{2} = \frac{ml^2}{3}\alpha$ — ②

constraint eqn,

$a = l\alpha$ — ③

by solving ①, ②, ③ we get required values.

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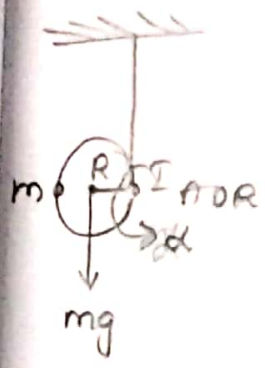
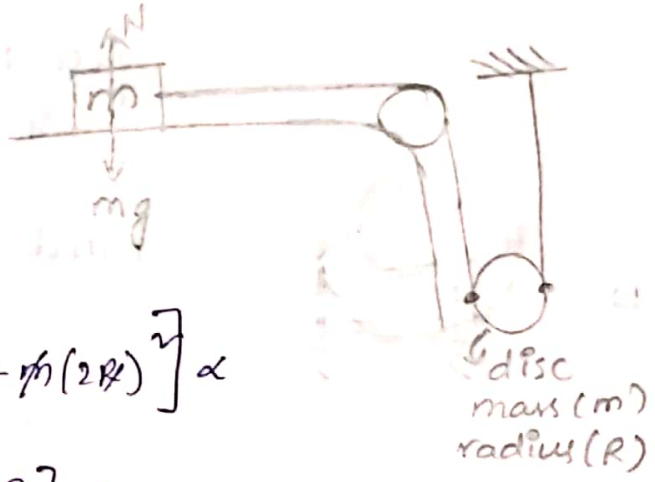
→ Find 'α'.





$$\tau_0 = I_0 \alpha$$

$$mgR = \left[\frac{mR^2}{2} + mR^2 + m(2R)^2\right]\alpha$$

$$g = \left[\frac{R}{2} + R + 4R\right]\alpha$$

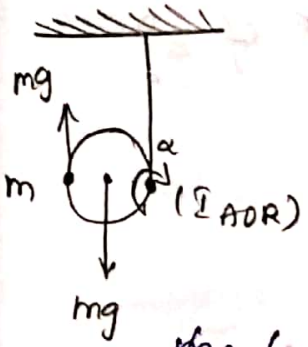
$$g = \alpha R\left(\frac{1}{2} + 5\right)$$

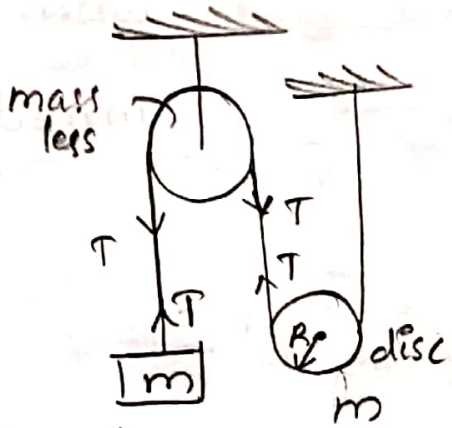
$$g = \frac{11}{2}\alpha R \Rightarrow \boxed{\alpha = \frac{2g}{11R}}$$

& $a_{cm} = R\alpha$

* $a_{block} = (2R)\alpha$

→ Find α of disc.





$$mg(2R) - mgR = \left[\frac{mR^2}{2} + mR^2 + m(2R)^2\right]\alpha$$

& $a_{cm} = R\alpha \ (\uparrow)$

$a_{block} = (2R)(\alpha)(\uparrow)$

$$2g - g = \alpha R\left(\frac{1}{2} + 1 + 4\right)$$

$$g = \frac{11\alpha R}{2}$$

$$\boxed{\alpha = \frac{2g}{11R}}$$

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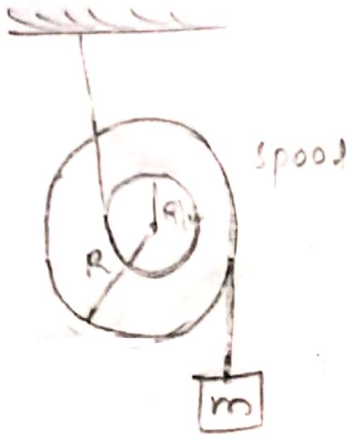
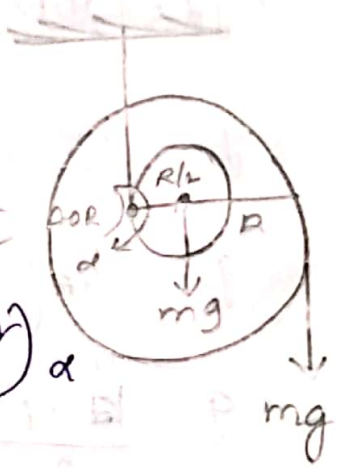


spool mass = m

$I = \frac{mR^2}{2}$

find α of spool.

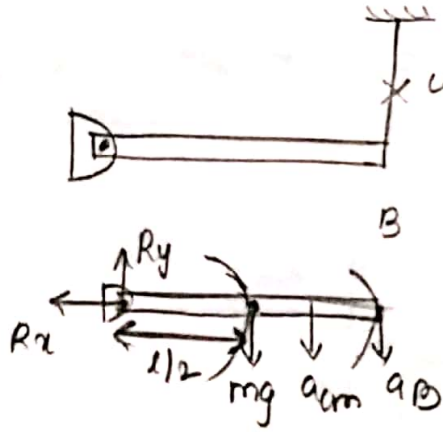


$mg\left(\frac{3R}{2}\right) + mg\left(\frac{R}{2}\right)$

$= \left(\left(I + m\left(\frac{R}{2}\right)^2\right) + m\left(R + \frac{R}{2}\right)^2\right)\alpha$

$a_{block} = \left(\frac{3R}{2}\right)\alpha$

$$\boxed{a_{block} = \frac{3R\alpha}{2}}$$



when string is cut, immediate angular of B?

$mg - R_y = ma_{cm}$ — (1)

$$\boxed{R_x = 0}$$ — (2)

$\tau_p = I_p \alpha$

$mg\frac{l}{2} = \frac{ml^2}{3}\alpha$ — (3)

$$\boxed{\alpha = \frac{3g}{2l}}$$

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* constraint eqn,

$$a_{cm} = \frac{l}{2}\alpha \quad -(4)$$

$$a_{cm} = \frac{\cancel{l}}{2} \times \frac{3g}{2\cancel{l}}$$

$$\boxed{a_{cm} = \frac{3g}{4}}$$

* acceleration of end 'B',

$$a_B = l\alpha = \cancel{l} \times \frac{3g}{2\cancel{l}}$$

$$\boxed{a_B = \frac{3g}{2}}$$

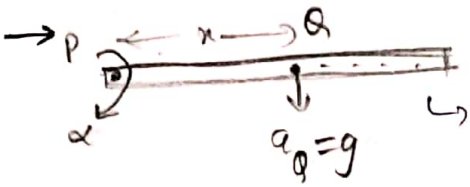
→

since $a_B = \frac{3g}{2} > g$.

since 'T' is a pulling force,

string slacks $\Rightarrow \boxed{T = 0}$

$$\boxed{\alpha = \frac{3g}{2l}}$$



↳ If block is suspended with string at the pts $x > \frac{2l}{3}$ then $\boxed{T = 0}$

$$a_Q = g$$

$$x\alpha = g$$

$$x \times \frac{3\cancel{g}}{2l} = \cancel{g} \Rightarrow \boxed{x = \frac{2l}{3}}$$

$\Rightarrow \alpha = \frac{2l}{3}$.

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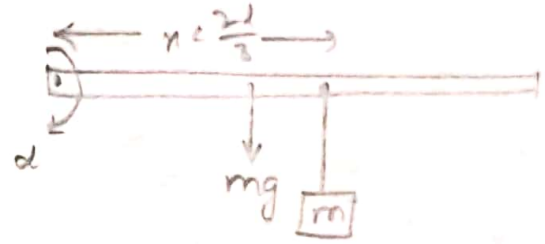
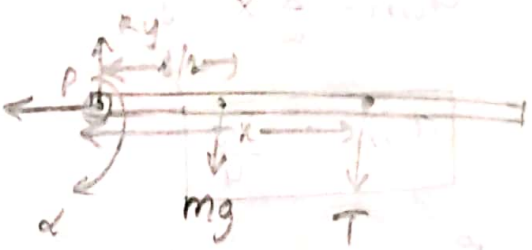
→



Sol * for rod,



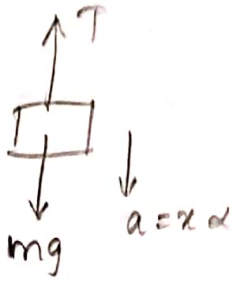
$T + mg - R_y = m a_{cm}$ — ①

$R_x = 0$

$\tau_p = I_p \alpha$

$\Rightarrow mg\frac{l}{2} + Tx = \frac{ml^2}{3}\alpha$ — ②

* for block,



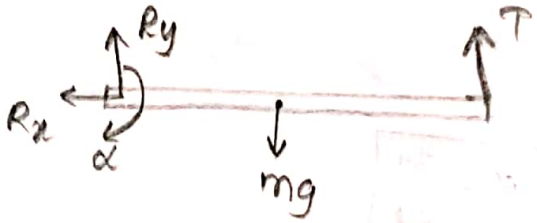
$mg - T = m x \alpha$ — ③

& $a_{cm} = \frac{l}{2}\alpha$ — ④

by solving we get — α, a_{cm}, T, R_y.

8. →

$mg - T - R_y = m a_{cm}$ — ①



$R_x = 0$ — ②

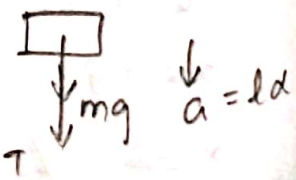
$\tau_p = I_p \alpha$

$\Rightarrow mg\frac{l}{2} - Tl = \frac{ml^2}{3}\alpha$ — ③

* for block,

$T + mg = ml\alpha$ — ④

by solving we get — α, a_{cm}, T, R_y.



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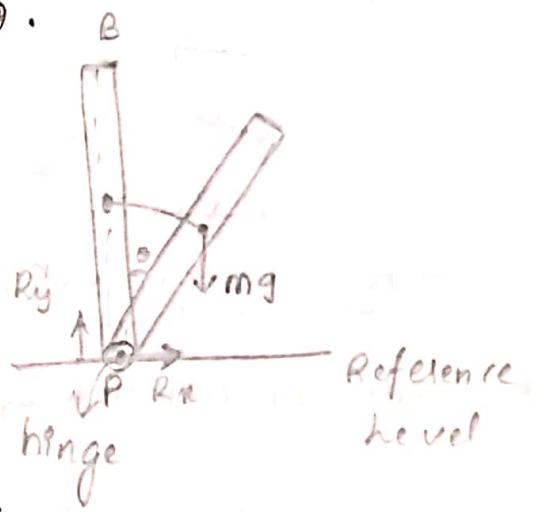
→ Rod released from vertical position, when it is turned through an angle θ.



To find α :

$$\tau_p = I_p \alpha$$

$$mg\frac{l}{2}\sin\theta = \frac{ml^2}{3}\alpha$$

$$\boxed{\alpha = \frac{3g}{2l}\sin\theta}$$ — standard eqn.

To find ω :

M-I : (calculus method)

$$\alpha = \frac{3g}{2l}\sin\theta$$

$$\omega \cdot \frac{d\omega}{d\theta} = \frac{3g}{2l}\sin\theta$$

$$\omega \cdot d\omega = \frac{3g}{2l}\sin\theta \cdot d\theta$$

$$\frac{\omega^2}{2} = \frac{3g}{2l}[\cos\theta]_0^\theta$$

$$\omega^2 = \frac{3g}{l}(1-\cos\theta)$$

$$\boxed{\omega = \sqrt{\frac{3g(1-\cos\theta)}{l}}}$$

M-II :

L.C.E,

$$U_1 + K_1 = U_2 + K_2$$

$$mg\frac{l}{2} + 0 = mg\frac{l}{2}\cos\theta + \frac{1}{2}I_p\omega^2$$

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→

$$mg\frac{l}{2}(1-\cos\theta) = \frac{1}{2} \times \frac{ml^2}{3} \times \omega^2$$

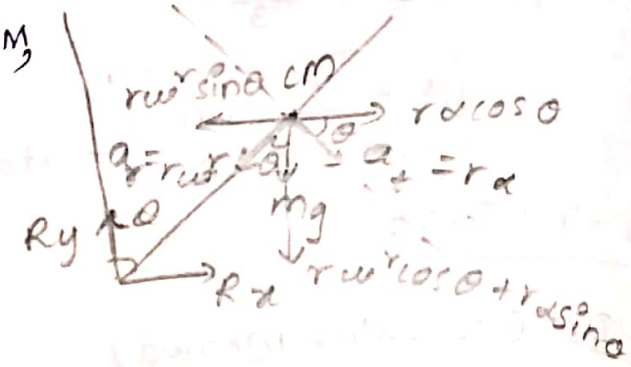
$$\boxed{\omega = \sqrt{\frac{3g}{l}(1-\cos\theta)}}$$

Sl

To find hinge rxn forces:

m

* Linear acceleration of CM,



$$a_t = r\alpha = \frac{l}{2}\left(\frac{3g}{2l}\sin\theta\right)$$

$$a_t = \frac{3g}{4}\sin\theta$$

$$a_r = r\omega^2 = \frac{l}{2} \times \frac{3g}{l}(1-\cos\theta)$$

$$a_r = \frac{3g}{2}(1-\cos\theta)$$

* Force eqns:

S

$$R_x = ma_x$$

$$R_x = m(r\alpha\cos\theta - r\omega^2\sin\theta) \quad -①$$

& $$mg - R_y = m(r\omega^2\cos\theta + r\alpha\sin\theta) \quad -②$$

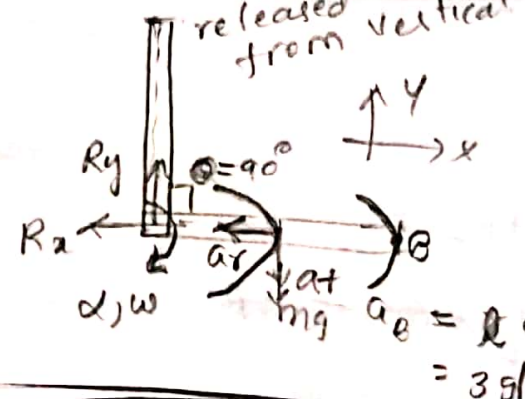
∴ net hinge rxn,

$$\boxed{R = \sqrt{R_x^2 + R_y^2}}$$

* Case:

*To find α:

released from vertical



$$mg\frac{l}{2} = \frac{ml^2}{3}\alpha$$

$$\boxed{\alpha = \frac{3g}{2l}}$$

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* To find ω :

I.C.E)

$$mg\frac{l}{2} + 0 = 0 + \frac{1}{2} m\frac{l^2}{3}\omega^2$$

$$\boxed{\omega = \sqrt{\frac{3g}{l}}}$$

* Linear accelerations of CM,

$$a_t = \frac{l}{2}\alpha = \frac{l}{2} \times \frac{3g}{2l} = \frac{3g}{4}$$

& $a_r = \left(\frac{l}{2}\right)\omega^2 = \frac{l}{2} \times \frac{3g}{l} = \frac{3g}{2}$

* net linear accelesation of CM,

$$a = \sqrt{\left(\frac{3g}{4}\right)^2 + \left(\frac{3g}{2}\right)^2}$$

* net acceleration of CM,

$$a = -\frac{3g}{2}\uparrow - \frac{3g}{4}\hat{j} - \alpha\hat{k}.$$

* Linear velocity of CM,

$$v_{cm} = r\omega = \frac{l}{2}\sqrt{\frac{3g}{l}}$$

$$\boxed{v_{cm} = \sqrt{\frac{3gl}{4}}}$$

& $v_B = l\omega = \sqrt{3gl}$

* Now, support is suddenly withdrawn,

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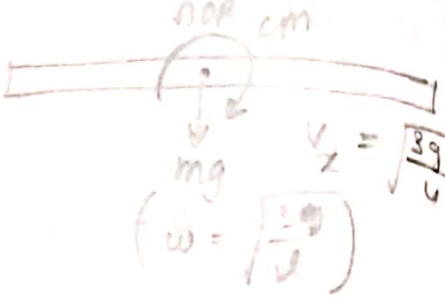
→ then $R_x = 0$, $R_y = 0$.



here $\boxed{\tau_{mg} = 0}$

$\Rightarrow \boxed{\alpha = 0}$

$\Rightarrow \boxed{\omega = \text{const}}$

* Velocity of rod at any time (t).

$$\boxed{V = V_0 + gt}$$

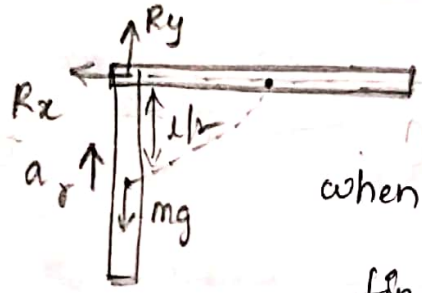
* K.E of rod at any time (t),

$$K.E = \frac{1}{2} m v^2 + \frac{1}{2} I_m \omega^2$$

$$K.E = \frac{1}{2} m (V_0 + gt)^2 + \frac{1}{12} \times \frac{m l^2}{12} \left(\sqrt{\frac{3g}{l}}\right)^2$$

$$\boxed{K.E = \frac{1}{2} m (V_0 + gt)^2 + \frac{mgl}{8}}$$

→



Rod released from height. when it reaches to vertical position, find α, ω.

$\tau_p = I_p \alpha$

$0 = I_p \alpha$

$\boxed{\alpha = 0}$

$a_t = a\alpha = 0$

$R_x = m a_t = (m)(0) = 0$

* we have,

$R_y - mg = m a_r$

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$$R_y - mg = m\frac{l}{2}\omega^2$$

$$R_y - mg = m\left(\frac{l}{2}\right)\left(\frac{3g}{l}\right)$$

$$R_y = mg + \frac{3mg}{2}$$

$$\boxed{R_y = \frac{5mg}{2}}$$

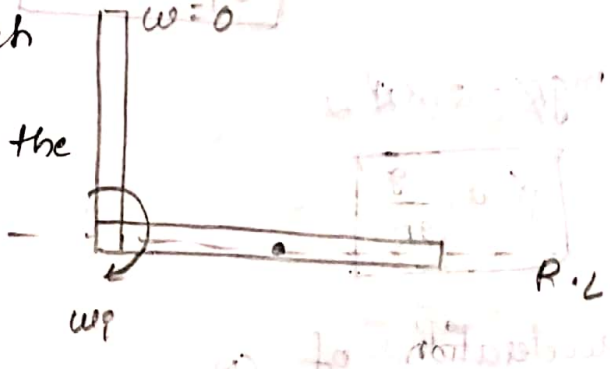
* To find ω :

L.C.E ,

$$0 + 0 = -m\frac{l}{2}g + \frac{1}{2}\frac{ml^2}{3}\omega^2$$

$$\boxed{\omega = \sqrt{\frac{3g}{l}}}$$

→ Find $(\omega_i)_{min}$ for which rod is able to reach the vertical position.



$$U_1 + K_1 = U_2 + K_2$$

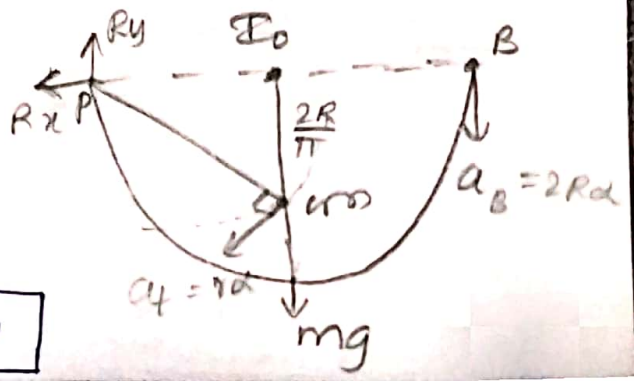
$$0 + \frac{1}{2}\frac{ml^2}{3}\omega_i^2 = mg\frac{l}{2} + 0$$

$$\boxed{\omega_i = \sqrt{\frac{3g}{l}}}$$

Semi-Circular Ring :

half ring is released from rest as shown.

$$\boxed{\omega = 0} \qquad \boxed{a_r = r\omega^2 = 0}$$



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→ To find α :

$$\tau_p = I_p \alpha$$

$$mgR = I_p \alpha \quad -(1)$$

To find I_p :

$$I_o = I_{cm} + m\left(\frac{2R}{\pi}\right)^2$$

$$mR^2 = I_{cm} + m\left(\frac{2R}{\pi}\right)^2 \quad -(2)$$

and

$$I_p = I_{cm} + mr^2$$

$$I_o = I_{cm} + m\left(R^2 + \left(\frac{2R}{\pi}\right)^2\right) \quad -(3)$$

eqn (3) - eqn (2) ,

$$\boxed{I_p = 2mR^2}$$

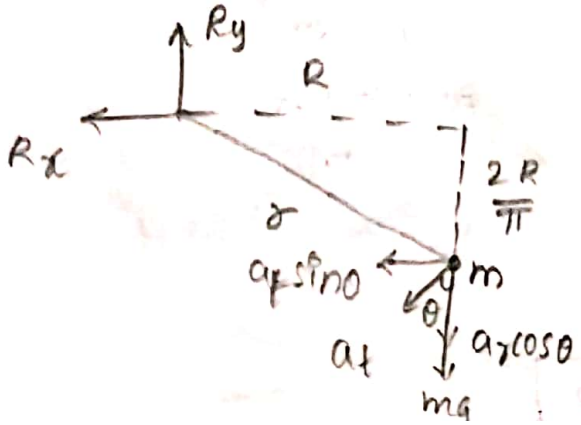
$$mgR = 2mR^2\alpha$$

$$\boxed{\alpha = \frac{g}{2R}}$$

* acceleration of CM,

$$a_t = r\alpha = \sqrt{\left(R^2 + \frac{4R^2}{\pi^2}\right)}\,\frac{g}{2R}$$

To find hinge rxns



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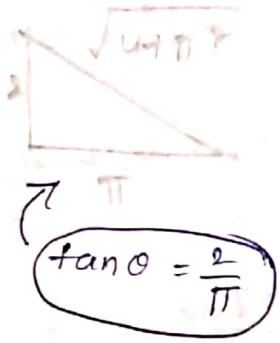
$R_x = ma_t \sin\theta$

& $mg - R_y = ma_t \cos\theta$

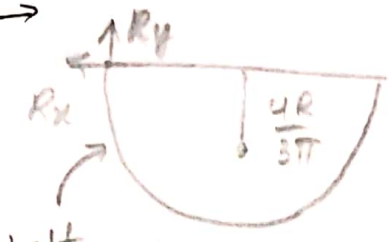
net hinge rx^n force,

$$R = \sqrt{R_x^2 + R_y^2}$$

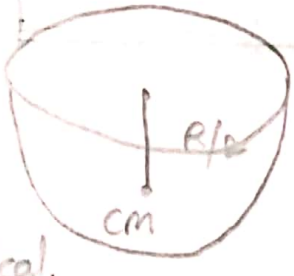


$$\tan\theta = \frac{2}{\pi}$$

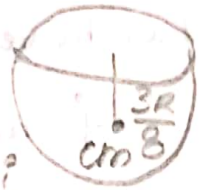
→



half disc



hemispherical shell



hemisphere

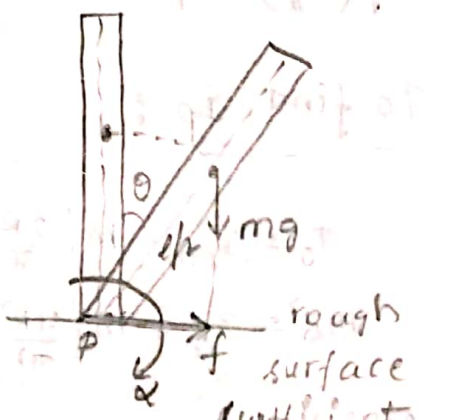
→ rod is released from vertical.

To find α :

$$\tau_P = I_P \alpha$$

$$mg\frac{l}{2}\sin\theta = \frac{ml^2}{3} \times \alpha$$

$$\alpha = \frac{3g}{2l}\sin\theta$$



To find ω :

(work energy theorem)

$$U_1 + K_1 = U_2 + K_2$$

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$$mg\frac{l}{2} + 0 = mg\frac{l}{2}\cos\theta + \frac{1}{2}I_p\omega^2$$

$$mgl(1-\cos\theta) = \frac{ml^2}{3}\omega^2$$

$$\frac{3g}{l}(1-\cos\theta) = \omega^2$$

$$\boxed{\omega = \sqrt{\frac{3g}{l}(1-\cos\theta)}}$$

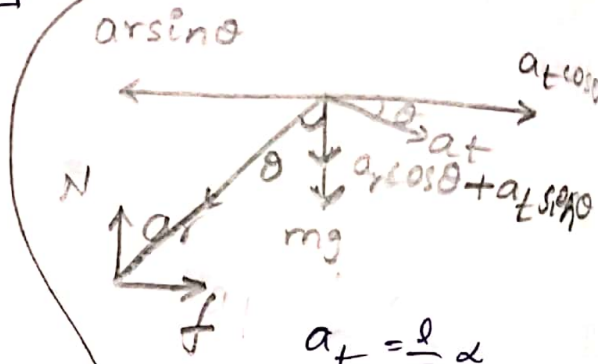
$a_t\cos\theta = a_r\sin\theta$

$\tan\theta = \frac{a_t}{a_r}$

$\theta = \tan^{-1}(a_t/a_r)$

by solving,

$\boxed{\theta = \cos^{-1}(2/3)}$



*we have,

$$f = ma_x$$

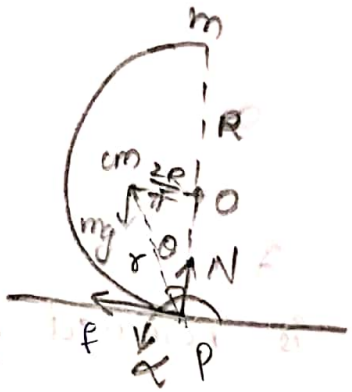
$$f = m(a_t\cos\theta - a_r\sin\theta)$$

$$mg - N = m(a_r\cos\theta + a_t\sin\theta)$$

$a_t = \frac{l}{2}\alpha$

$a_r = \frac{l}{2}\omega^2$

→ Find min coefficient of friction for no initial slipping of ring.



To find α :

$$\tau_p = I_p\alpha$$

$$mg\left(\frac{2R}{\pi}\right) = I_p\alpha$$

To find I_p :

$$I_O = I_{cm} + m\left(\frac{2R}{\pi}\right)^2$$

$$mR^2 = I_{cm} + m\left(\frac{4R^2}{\pi^2}\right) - (1)$$

& $I_p = I_{cm} + mr^2 - (2)$

eqn (2) − eqn (1)

$$\boxed{I_p = 2mR^2}$$

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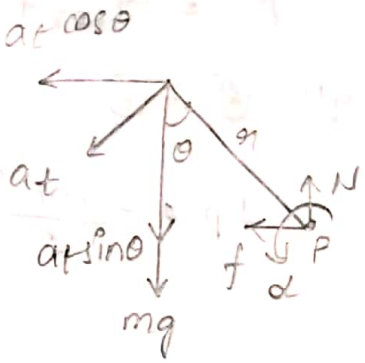
$$\therefore mg\left(\frac{2R}{\pi}\right) = \cancel{2}mR^2(\alpha)$$

$$\boxed{\alpha = \frac{g}{\pi R}}$$

& $\omega = 0$ (released from rest).



$$a_t = r\alpha = \left(\sqrt{R^2 + \frac{4R^2}{\pi^2}}\right)\alpha$$

$$\boxed{a_t = \frac{R\alpha}{\pi}\sqrt{\pi^2+4}}$$

$$\therefore f = ma_x$$

$$f = ma_t\cos\theta$$

& $mg - N = ma_t\sin\theta$

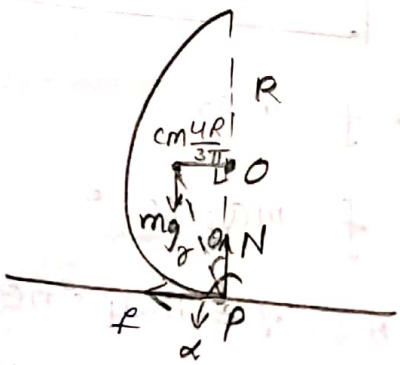
finally we have,

$$f \leq \mu N$$

$$\mu \geq \frac{f}{N}$$

$$\boxed{\mu_{min} = f/N} \Rightarrow \mu_{min} = \frac{a_t\cos\theta}{g - a_t\sin\theta}$$

→ H.W Find min coefficient of friction for no initial slipping of disc, [hollow & solid hemi sphere].



* disc:

To find α:

$$\tau_p = I_p\alpha$$

$$mg\left(\frac{4R}{3\pi}\right) = I_p\alpha$$

To find I_p:

$$r^2 = \left(\frac{4R}{3\pi}\right)^2 + R^2$$

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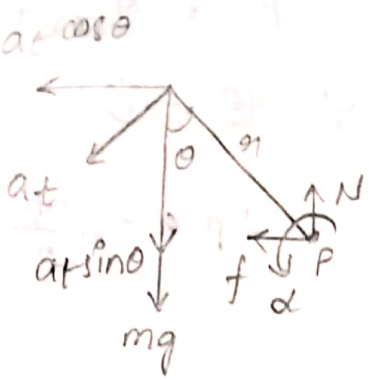
$$\frac{g}{\pi} = 2R\alpha$$

$$\boxed{\alpha = \frac{g}{\pi R}}$$

& $\omega = 0$ (released from rest).



$$a_t = R\alpha = \left(\sqrt{R^2 + \frac{4R^2}{\pi^2}}\right)\alpha$$

$$\boxed{a_t = \frac{R\alpha}{\pi}\sqrt{\pi^2+4}}$$

$\therefore f = ma_x$

$f = ma_t\cos\theta$

& $mg - N = ma_t\sin\theta$

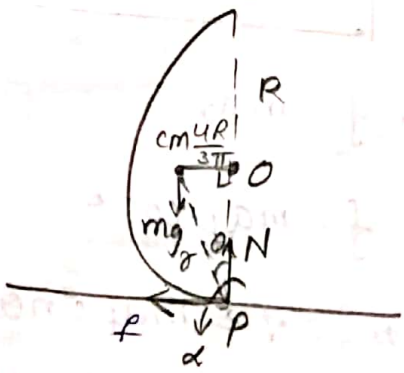
finally we have,

$f \leq \mu N$

$\mu \geq \frac{f}{N}$

$\boxed{\mu_{min} = f/N}$ $\Rightarrow$ $\mu_{min} = \dfrac{a_t\cos\theta}{g - a_t\sin\theta}$

→ H.W Find min coefficient of friction for ho initial slipping of disc, [hollow & solid hemisphere].



* disc:

To find α:

$\tau_p = I_p\alpha$

$mg\left(\frac{4R}{3\pi}\right) = I_p\alpha$

$r^2 = \left(\frac{4R}{3\pi}\right)^2 + R^2$

To find I_p:

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$$I_0 = I_{cm} + m\left(\frac{4R}{3\pi}\right)^2$$

$$\frac{mR^2}{2} = I_{cm} + \frac{m16R^2}{9\pi^2} \quad -(1)$$

& $I_p = I_{cm} + mr^2 \quad -(2)$

eqn (2) - (1)

$$I_p - \frac{mR^2}{2} = mr^2 - m\left(\frac{4R}{3\pi}\right)^2$$

$$= m\left(\frac{4R}{3\pi}\right)^2 + mR^2 - m\left(\frac{4R}{3\pi}\right)^2$$

$$I_p = mR^2 + \frac{mR^2}{2}$$

$$\boxed{I_p = \frac{3mR^2}{2}}$$

$$\therefore mg\left(\frac{4R}{3\pi}\right) = \frac{3mR^2}{2}(\alpha)$$

$$\boxed{\frac{8g}{9\pi R} = \alpha}$$

& $\omega = 0$ (released from rest)

$$a_t = r\alpha = \left(\sqrt{\frac{16R^2}{9\pi^2} + R^2}\right)\alpha$$

$$\boxed{a_t = \frac{R\alpha}{3\pi}\sqrt{16R^2 + 9\pi^2}}$$

$\therefore f = ma_x$

$f = ma_t\cos\theta$

$mg - N = ma_t\sin\theta$

finally we have,

$f \le \mu N \Rightarrow \boxed{\mu_{min} = f/N}$

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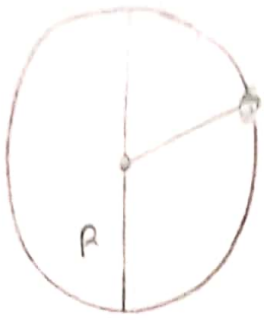
→ 

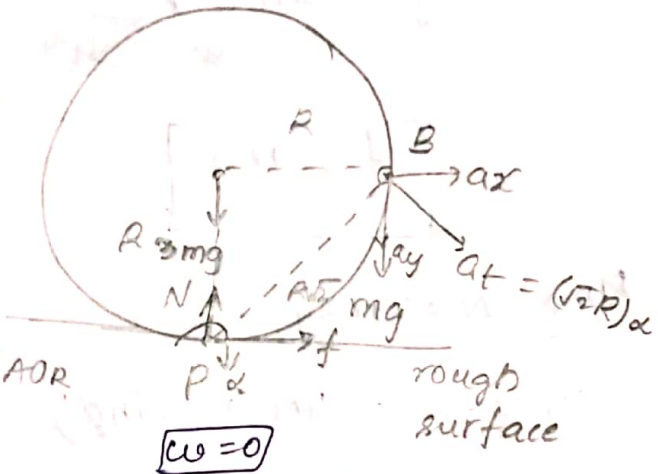
hoop or ring of mass 3m & radius R.

a bead of mass 'm' is attached to hoop.

The system is released from rest.

when $\theta = 90°$ there is no slipping b/w hoop & ground.

Find α, a, N & f, μ_{min}.



To find α:

$\tau_P = I_P \alpha$

$mgR = [(3mR^2 + 3mR^2) + m(\sqrt{2}R)^2]\alpha$

$3mgR = 8mR^2\alpha$

$\boxed{\alpha = g/8R}$

$\tan\phi = \frac{R}{R} = 1.$

$\boxed{\theta = 45°}$

* To find horizontal & vertical accelerations:

$a_B = (\sqrt{2}R)\alpha = \sqrt{2}R \times g/8R = \frac{g}{4\sqrt{2}}$

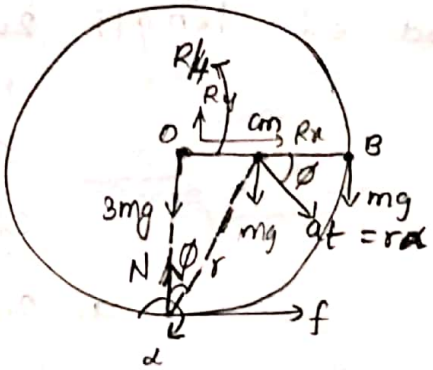
$a_x = a_B \cos 45° = g/8$

$a_y = a_B \sin 45° = g/8$

* To find N & f:

$r = \sqrt{R^2 + \frac{R^2}{16}} = \sqrt{\frac{17R^2}{16}}$

$\boxed{r = \frac{\sqrt{17}}{4}R}$



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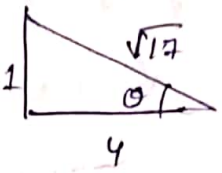
→ $a_x = r\alpha\cos\phi$

$a_y = r\alpha\sin\phi$



$\therefore f = \mu m a_x$

$f = \mu m (r\alpha\cos\phi)$

$f = \mu m \frac{\sqrt{17}R}{4} \times \frac{4}{\sqrt{17}} \times \frac{g}{8R}$

$$\boxed{f = \frac{mg}{2}}$$

$mg - N = m a_y$

$mg - N = m(r\alpha\sin\phi)$

$N = mg - \mu m\left(\frac{\sqrt{17}R}{4} \times \frac{g}{8R} \times \frac{1}{\sqrt{17}}\right)$

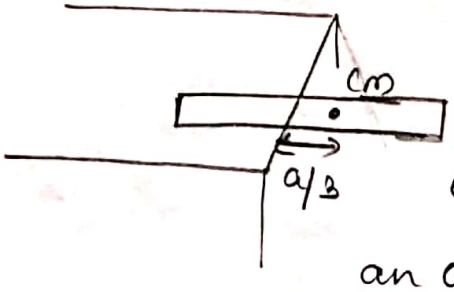
$= mg - \frac{mg}{8}$

$$\boxed{N = \frac{31mg}{8}}$$

$\mu_{min} = \frac{f}{N} = \frac{4}{31}$

$$\boxed{\mu_{min} = 4/31}$$

→ Rod of length '2a' & mass (m) released from rest as shown. Rod is about to slip from edge of table when it is turned thru an angle θ. Find μ blw rod & surface.



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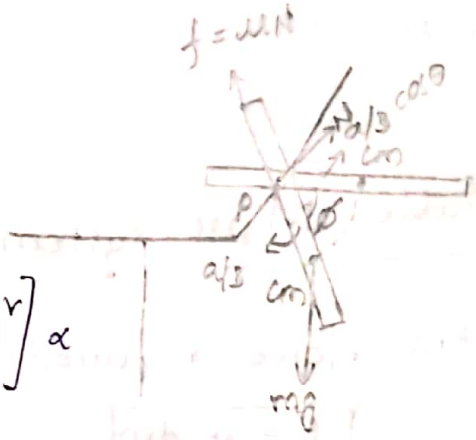
* To find α of rod :



$$\tau_p = I_p \alpha$$

$$mg \frac{a}{3}\cos\theta = \left[\frac{m(2a)^2}{12} + m\left(\frac{a}{3}\right)^2\right]\alpha$$

$$mg\frac{a}{3}\cos\theta = m\alpha a^2\left[\frac{1}{3} + \frac{1}{9}\right]$$

$$\frac{g\cos\theta}{a} = \alpha\,(4/3)$$

$$\boxed{\alpha = \frac{3g\cos\theta}{4a}}$$

* To find 'ω' of rod :

L.C.E,

$$U_1 + K_1 = U_2 + K_2$$

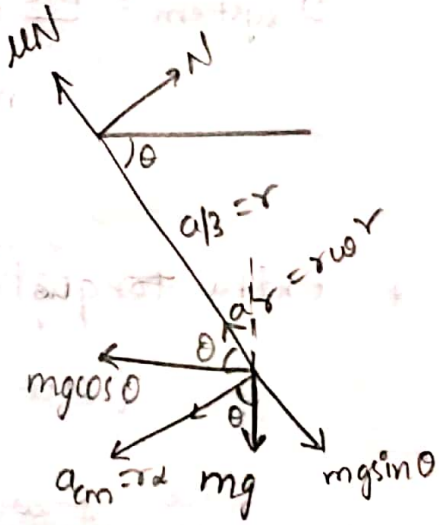
$$0 + 0 = -mg\frac{a}{3}\sin\theta + \frac{1}{2} I_p \omega^2$$

$$\omega^2 = \frac{2mga\sin\theta}{3I_p}$$

$$= \frac{2mga\sin\theta}{ma^2(4/3)}$$

$$\omega^2 = \frac{3g\sin\theta}{2a}$$

$$\boxed{\omega = \sqrt{\frac{3g\sin\theta}{2a}}}$$



* $mg\cos\theta - N = ma_t$ — (1)

$\mu N \sim mg\sin\theta = ma_r$ — (2)

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Ans: $\mu = \tan\theta$

Variable Mass system:

Extra force required

$$\boxed{F = v \cdot \frac{dm}{dt}}$$

Extra power required

$$\boxed{P = FV = v^2 \cdot \frac{dm}{dt}}$$

Similarly, For variable M.I system,

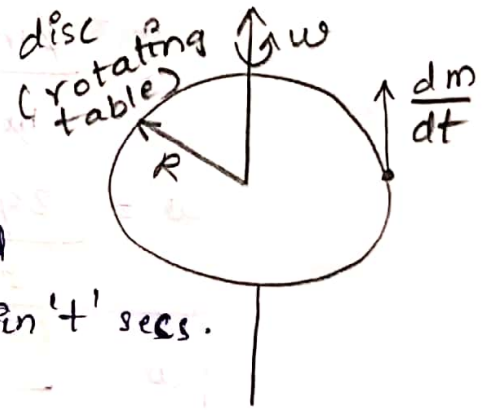
Extra torque required to move with const (ω),

$$\boxed{\tau = \omega \cdot \frac{dI}{dt}}$$

& extra power required,

$$P = \tau\omega = \omega^2 \frac{dI}{dt}$$

→ at any time 't'



$I_{system} = \frac{mR^2}{2} + mR^2$

mass of sand collected in 't' secs.

* Extra torque required,

$$\tau = \omega \cdot \frac{dI}{dt}$$

$$= \omega \cdot \frac{d}{dt}\left[\frac{MR^2}{2} + MR^2\right]$$

($\frac{MR^2}{2}$ → const)

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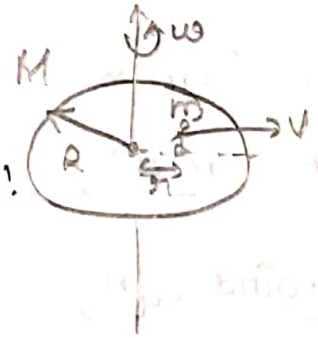
$$\boxed{\tau = \omega R^2 \cdot \frac{dm}{dt}}$$

Extra power required,

$$P = \tau \omega$$

$$\boxed{P = \omega^2 R^2 \cdot \frac{dm}{dt}}$$

→ man is walking radially at const speed v!



* $I = \frac{MR^2}{2} + mx^2$

* extra torque required,

$$\tau = \omega \cdot \frac{dI}{dt}$$

$$= \omega \cdot \frac{d}{dt}\left(\frac{MR^2}{2} + mx^2\right)$$

↳ const

$$= \omega m \left(2x \cdot \frac{dx}{dt}\right)$$

$$\boxed{\tau = 2m\omega x \cdot v}$$

$$\boxed{\tau \propto x}$$

* extra power, $P = \tau \omega$

$$= 2m\omega^2 x v$$

$$\boxed{P = 2m\omega^2 x v}$$

$$\boxed{P \propto x}$$

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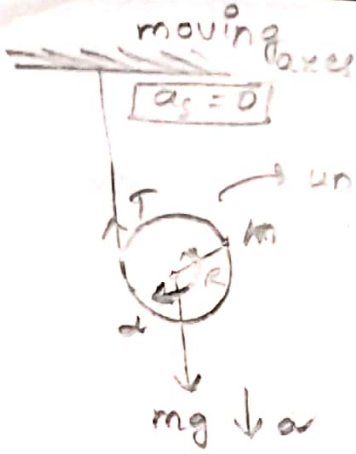


Unknowns:

a, α & T.

* $mg - T = ma$ — (1)

* $\tau_{cm} = I_{cm}\alpha$

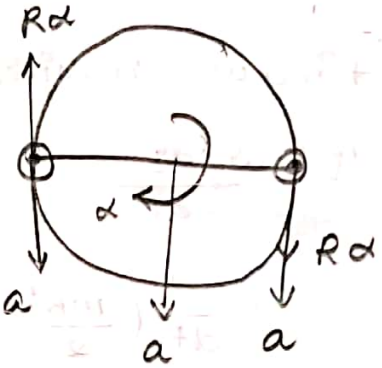
$TR = \frac{mR^2}{2}\alpha$ — (2)

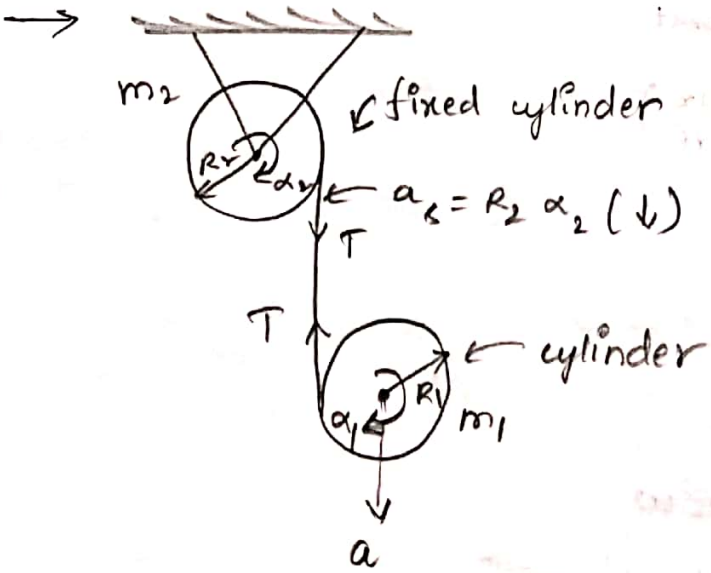
* Constraind eqn,

since there is no slipping blw string & cylinder,

$0 = a - R\alpha$

$\boxed{a = R\alpha}$ — (3)





Unknowns:

a, α_1, α_2, T.

* For lower cylinder,

$m_1 g - T = m_1 a$ — (1)

$TR_1 = \frac{m_1 R_1^2}{2}\alpha_1$ — (2)

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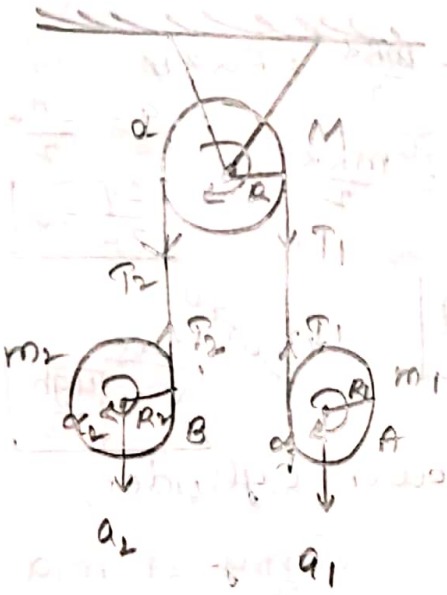
* for upper cylinder,

$$T R_2 = \frac{m_2 R_2^2}{2} \alpha_2 \quad -③$$

* constraint eqn,

$$R_2 \alpha_2 = a - R_1 \alpha_1 \quad -④$$

→



Unknowns:

$a_1, a_2, \alpha_1, \alpha_2, T_1, T_2, \alpha$.

* For cylinder A:

$$m_1 g - T_1 = m_1 a_1 \quad -①$$

$$T_1 R_1 = \frac{m_1 R_1^2}{2} \alpha_1 \quad -②$$

* For cylinder B:

$$m_2 g - T_2 = m_2 a_2 \quad -③$$

$$T_2 R_2 = \frac{m_2 R_2^2}{2} a_2 \quad -④$$

* for fixed cylinder:

$$T_1 R_1 - T_2 R = \frac{M R^2}{2} \alpha \quad -⑤$$

* constraint eqn:

$$R \alpha = a_1 - R_1 \alpha_1 \quad -⑥$$

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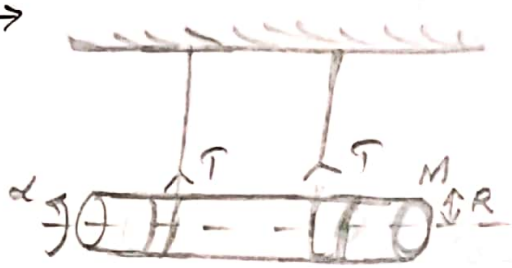
$R\alpha = R_2\alpha_2 - a_2$ — (7)

→



* $mg - 2T = ma$ — (1)

* $2TR' = \frac{mR^2}{2}\alpha$ — (2) $\Rightarrow 2T = \frac{mR\alpha}{2}$

$mg - \frac{mR\alpha}{2} = R\alpha \times m$

$g = \frac{3R\alpha}{2}$

$\boxed{\frac{2g}{3R} = \alpha}$

* constraint eqn,

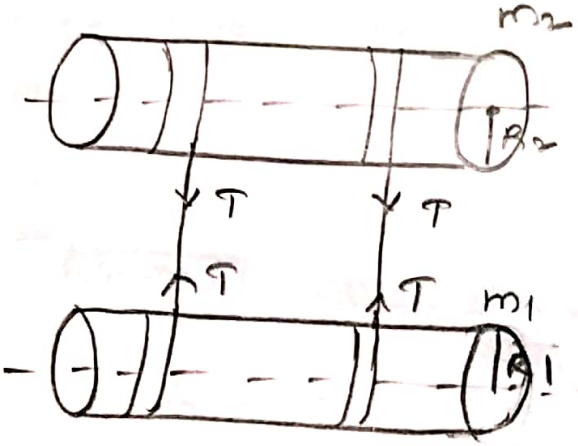
$a - R\alpha = 0$ — (3)

$\Rightarrow \boxed{a = \frac{2g}{3}}$

$v^2 - u^2 = 2as$

$\boxed{v = \sqrt{\frac{4gh}{3}}}$

→



lower cylinder,

$m_1 g - 2T = m_1 a$ — (1)

$2TR_1 = \frac{m_1 R_1^2}{2}\alpha_1$ — (2)

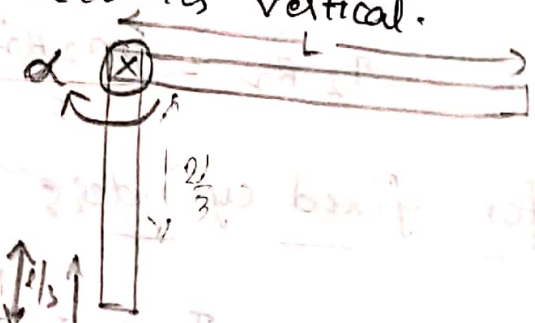
Upper cylinder,

$2TR_2 = \left(\frac{m_2 R_2^2}{2}\right)\alpha_2$

constraint eqn,

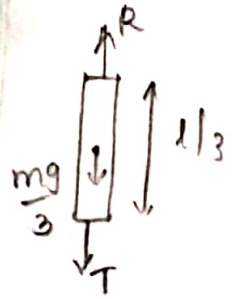
$R_2\alpha_2 = a - R_1\alpha$ — (4)

→ Find tension at a pt located at a distance $\frac{l}{3}$ from hinge pt when rod becomes vertical.



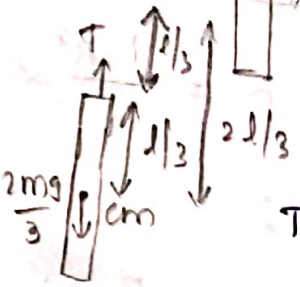
$\frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2 = mg\frac{l}{2}$

$\boxed{\omega = \sqrt{\frac{3g}{l}}}$



$T + \frac{mg}{3} = R$

$T = \frac{2mg}{3}$



$T - \frac{2mg}{3} = \frac{2m}{3}\left(\frac{2l}{3}\right)\omega^2$

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$$T - \frac{2mg}{3} = \frac{2m}{3} \times \frac{2\ell}{\ell} \times \frac{3g}{\ell}$$

$$T = \frac{4mg}{3} + \frac{2mg}{3}$$

$$T = \frac{6mg}{3}$$

$$\boxed{T = 2mg}$$

→ Rod is released from horizontal position. Find rxn at the hinge when K.E of rod is maximum.

$$mg\frac{\ell}{4} = \frac{1}{2} I_p \omega^2$$

$$mg\frac{\ell}{4} = \frac{1}{2} m\ell^2 \left(\frac{1}{12} + \frac{1}{16}\right)\omega^2$$

$$\frac{g}{2} = \ell\left(\frac{16+12}{12\times 16}\right)\omega^2$$

$$\frac{g}{2} = \frac{\ell \times 28}{6 \times 16_4}\omega^2$$

$$\boxed{\omega^2 = \frac{24g}{7\ell}}$$

$$R_y - mg = m\left(\frac{\ell}{4}\right)\omega^2$$

$$R_y - mg = m \times \frac{\ell}{4} \times \frac{24g}{7\ell}$$

$$R_y = \frac{6mg}{7} + mg$$

$$\boxed{R_y = \frac{13mg}{7}}$$

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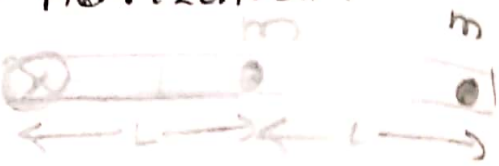
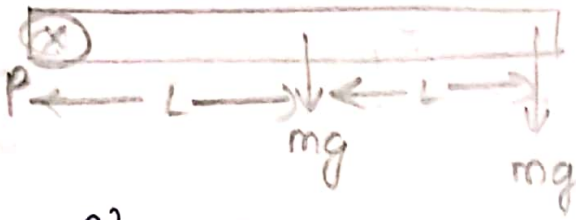
→ Find initial angular acceleration & hinge rxn force when light rod is released from horizontal.



i)

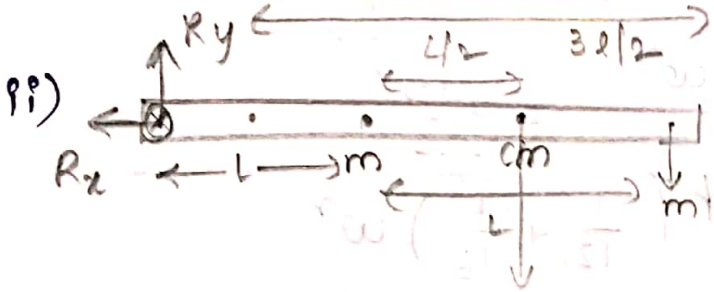


$$\tau_P = I_P \alpha$$

$$mg\ell + mg(2\ell) = 5m\ell^2 \times \alpha$$

$$\boxed{\alpha = \frac{3g}{5\ell}}$$

ii)



$\omega = 0$

$\Rightarrow R_x = 0$

$$a_t = \frac{3\ell}{2} \times \alpha = \frac{3\ell}{2} \times \frac{3g}{5\ell} = \frac{9g}{10}$$

$$\overline{F}_{ext} = m\overline{a}_{cm}$$

$$2mg - R_y = (2m)\left(\frac{9g}{10}\right)$$

$$2mg - \frac{9gm}{5} = R_y$$

$$\frac{10mg - 9mg}{5} = R_y$$

$$\boxed{R_y = \frac{mg}{5}}$$

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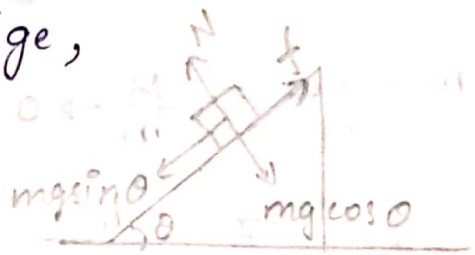
Toppling of wedge:

There is no slipping b/w wedge & ground. Find the mass of block so that toppling of wedge will not occur.

case-(i):

Block is at rest on wedge,



$mg\sin\theta = f$

$mg\cos\theta = N$

we have,

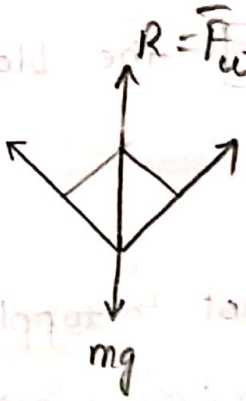
$f \leq \mu N$

$mg\sin\theta \leq \mu \times mg\cos\theta$

$$\boxed{\mu \geq \tan\theta}$$

since block is at rest,

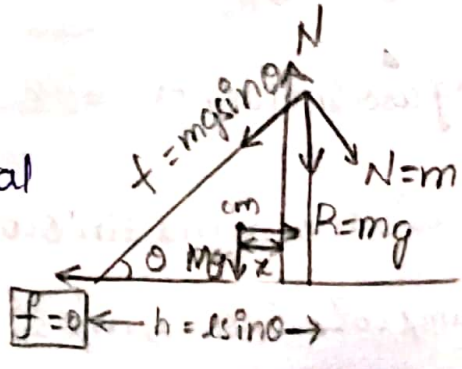
$\bar{F}_g + \bar{F}_{wedge} = 0$



$$\boxed{F_{wedge} = R = mg\ (\uparrow)}$$

for wedge,

since net horizontal force on wedge is zero $\Rightarrow \boxed{f = 0}$



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$\tau_o = 0$

$Mg(x) = R\left(\frac{h}{3} - x\right)$ $\qquad Mgx = mg\left(\frac{h}{3} - x\right)$

~~$Mgx = mg\left(\frac{h}{3} - x\right)$~~ $\qquad Mx + mx = \frac{mh}{3}$

~~$Mx = \frac{h}{3}$~~ ✗

$$\boxed{x = \frac{mh}{3(M+m)}}$$

If $m \to 0 \Rightarrow x \to 0$

if $m \to \infty \Rightarrow \frac{M}{m} \to 0$

$$x = \frac{h}{3\left(\frac{M}{m}+1\right)}$$

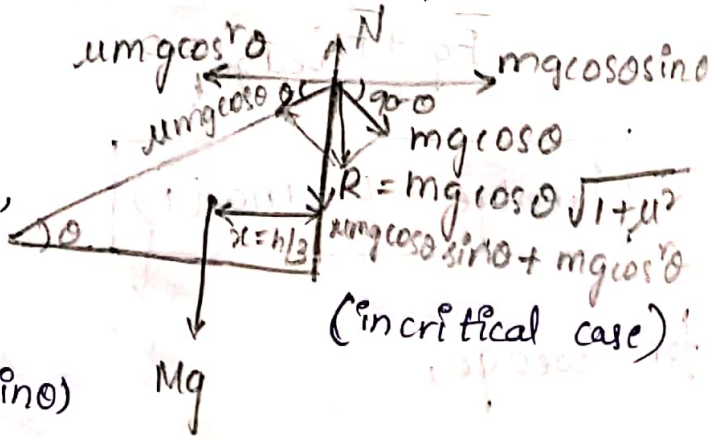
$$\Rightarrow \boxed{x = \frac{h}{3}}$$

Topple never occurs for any mass 'm' of block.

Case-(ii):

If $\boxed{\mu = \tan\theta}$ the block slides on wedge.

On wedge:



for wedge not to topple,

$Mg\frac{h}{3} + umg\cos^2\theta(d\sin\theta) \geq mg\cos\theta\sin\theta(d\sin\theta)$

(in critical case)

$M\frac{h}{3} + dum\cos^2\theta\sin\theta \geq md\sin^2\theta\cos\theta$

$M\frac{(d\cos\theta)}{3} + umd\cos^2\theta\sin\theta \geq md\sin^2\theta\cos\theta$

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$$\frac{M}{3} + \mu m \cos\theta \sin\theta \geq m \sin^2\theta$$

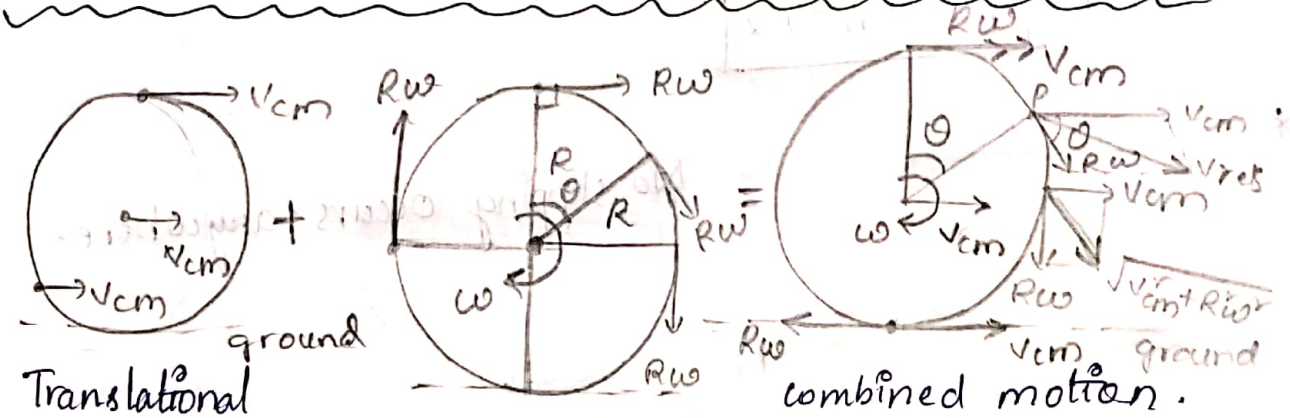
$$\frac{M}{3} \geq m\sin\theta(\sin\theta - \mu\cos\theta)$$

$$\boxed{m \leq \frac{M}{3\sin\theta(\sin\theta - \mu\cos\theta)}}$$

* if $m > \dfrac{M}{3\sin\theta(\sin\theta - \mu\cos\theta)}$

Toppling of wedge occurs.

Combined Motion (both Translation & Rotation):



* In general, $V_{res} = V_p = \sqrt{V_{cm}^2 + R^2\omega^2 + 2V_{cm}R\omega\cos\theta}$

* If there is no slipping b/w rolling body & ground (pure rolling), then

$$V_{ground} = V_{body}$$

$$0 = V_{cm} - R\omega$$

$$\boxed{V_{cm} = R\omega}$$

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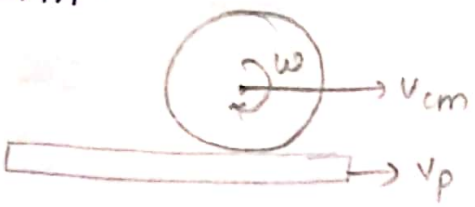
$\Rightarrow$ $\boxed{a_{cm} = R\alpha}$

↓

tangential acceleration

* If body is rolling without slipping on moving platform.



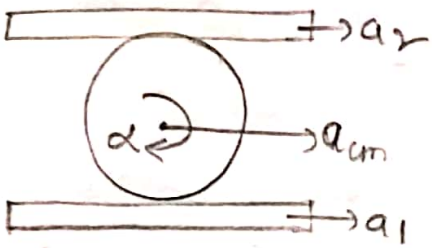
at pt of contact,

$$v_p = v_{body}$$

$$\boxed{v_p = v_{cm} - R\omega}$$

& $\boxed{a_p = a_{cm} - R\alpha}$

*



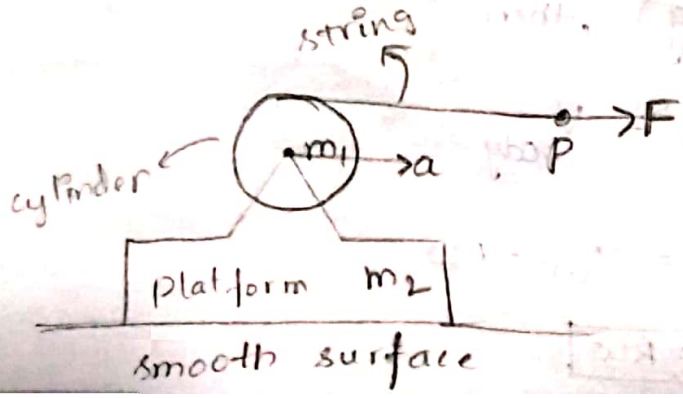
No slipping occurs anywhere.

$a_1 = a_{cm} - R\alpha$ — ①

$a_2 = a_{cm} + R\alpha$ — ②

① + ② $\Rightarrow$ $\boxed{a_{cm} = \frac{a_1 + a_2}{2}}$ ② − ① $\Rightarrow$ $\boxed{\alpha = \frac{a_2 - a_1}{2R}}$

→



There is no slipping blw string & pulley.

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* $a = \frac{F}{m_1+m_2}$ — ①

now, $\tau_{cm} = FR = I_{cm}\alpha$

$\frac{m_1R^2}{2}\alpha = FR$

$$\boxed{\alpha = \frac{2F}{m_1R}}$$

* $a_p = a + R\alpha$

$= \frac{F}{m_1+m_2} + \frac{2F}{m_1}$

$= \frac{F(m_1+2m_1+2m_2)}{m_1(m_1+m_2)}$

$$\boxed{a_p = \frac{F(3m_1+2m_2)}{m_1(m_1+m_2)}}$$

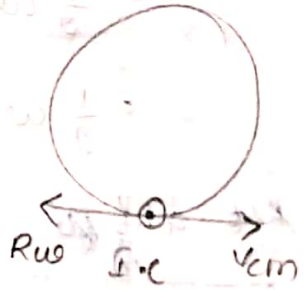
* displacement of string particle in t-sec,

$$\boxed{S_p = \frac{1}{2}a_pt^2}$$

* Instantaneous Centre* (I.C) & (net velocity = 0).

$v_{cm} - R\omega = 0$

$$\boxed{v_{cm} = R\omega}$$



* Instantaneous Axis of Rotation:

The axis of rotation is passing through I.C & ller

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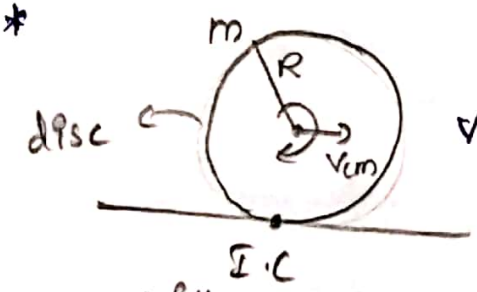
to plane of rotation.

* I.C may lie inside or outside the body.
* I.C is used to find the velocity of particles of the body.
* I.C will not be used to find acceleration of the bodies bcoz inst. centre may be accelerating.
* Combined motion of the body will become pure rotation about instantaneous AOR.
* Angular velocity of the body about the AOR passing through COM and I.C is same.

*



$v_{cm} = R\omega$;

without slipping

* K.E of body = K.E of CM + K.E about C.M

$$= \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} m R^2 \omega^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} \omega^2 (mR^2 + I_{cm})$$

$$= \frac{1}{2} \omega^2 (I_{IAOR})$$

$$\boxed{K.E_{body} = \frac{1}{2} I_{IAOR} \omega^2}$$

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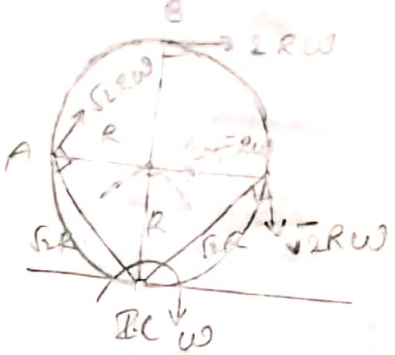
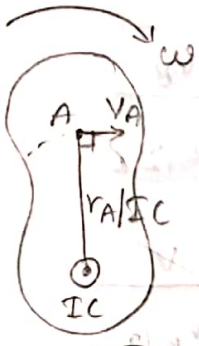
*



To locate I.C :

①

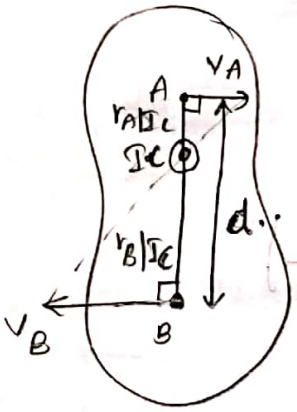


$$r_{A/IC} = \frac{V_A}{\omega}$$

②

$$\omega = \frac{V_A}{r_{A/IC}} = \frac{V_B}{r_{B/IC}}$$

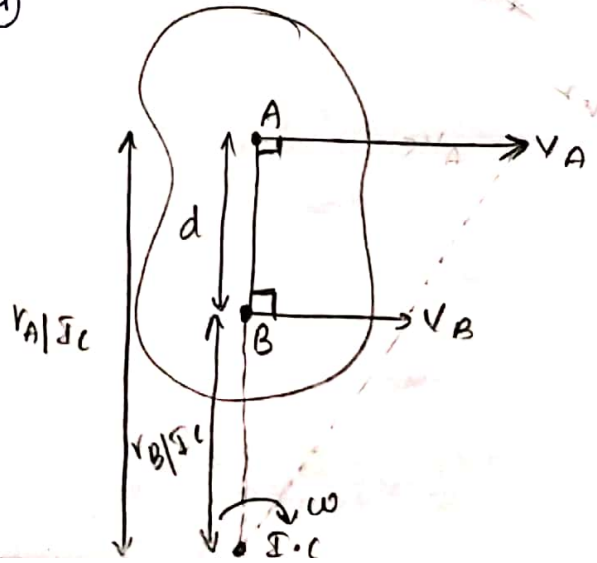
③



$$\omega = \frac{V_A}{r_{A/IC}} = \frac{V_B}{r_{B/IC}}$$

$$r_{A/IC} + r_{B/IC} = d$$

④



$$\omega = \frac{V_A}{r_{A/IC}} = \frac{V_B}{r_{B/IC}}$$

$$r_{A/IC} - r_{B/IC} = d$$

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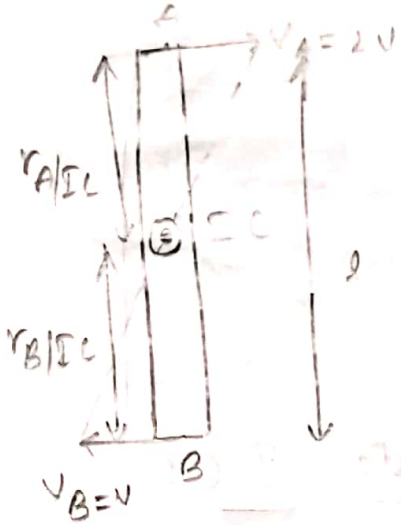
→ Find ① Angular velocity of rod & velocity of COM

② K.E of rod.



① $r_{A/IC} + r_{B/IC} = l$

$3r_{B/IC} = l$

$r_{B/IC} = l/3$

$\omega = \frac{V_B}{r_{B/IC}} = \frac{V}{l/3} = \frac{3V}{l}$

$\frac{V_A}{r_{A/IC}} = \frac{V_B}{r_{B/IC}}$

$\frac{2V}{r_{A/IC}} = \frac{V}{r_{B/IC}}$

$2 r_{B/IC} = r_{A/IC}$

$V_{cm} = r\omega = \left(\frac{l}{3} - \frac{l}{2}\right)\omega$

$= \frac{l}{6} \times \frac{3V}{l}$

$\boxed{V_{cm} = \frac{V}{2}}$

② $K.E = \frac{1}{2} \times m \times \frac{V^2}{4} + \frac{1}{2} \times \frac{ml^2}{12} \times \left(\frac{3V}{l}\right)^2$

$= \frac{1}{8} mv^2 + \frac{1}{2} \times \frac{ml^2}{12} \times \frac{9V^2}{l^2}$

$= \frac{4mv^2}{8} = \frac{1}{2} mv^2$

$\boxed{K.E = \frac{1}{2} mv^2}$

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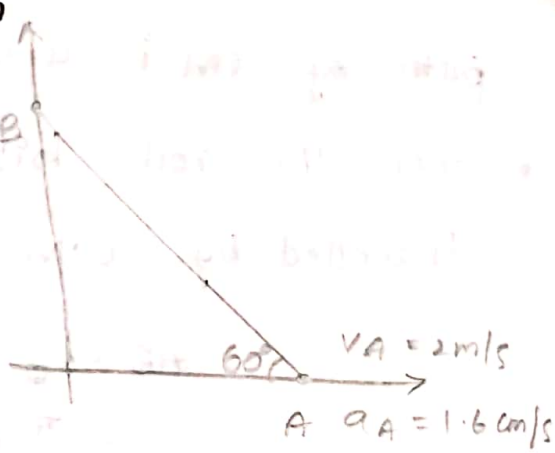
→ Find velocity & acceleration of end 'B'.

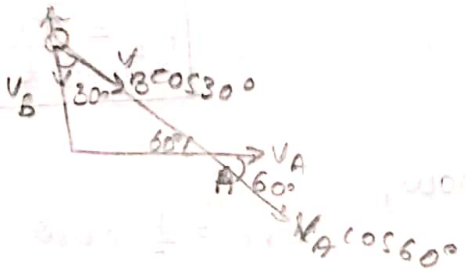


Velocity of end B:

M-I: velocity component along the rods are equal.

$$V_B \cos 30° = V_A \cos 60°$$

$$\boxed{V_B = \frac{2}{\sqrt{3}} \text{ m/s}} \quad (\downarrow)$$

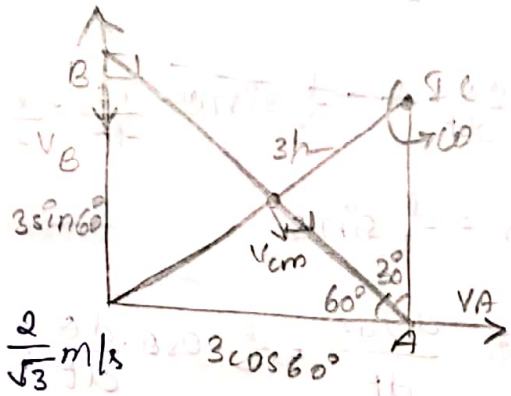


M-II:

Velocity of CM

$$V_{cm} = \frac{3}{2}\omega$$

$$= \frac{\cancel{3}}{2} \times \frac{2}{\cancel{3}\sqrt{3}} = \frac{2}{\sqrt{3}} \text{ m/s}$$



$$\omega = \frac{V_A}{3\sin 60°} = \frac{V_B}{3\cos 60°}$$

$$V_B \cos 30° = V_A \cos 60°$$

$$\boxed{V_B = \frac{2}{\sqrt{3}} \text{ m/s}}$$

$$\omega = \frac{V_A}{3\sin 60°}$$

$$= \frac{2}{3 \times \frac{\sqrt{3}}{2}}$$

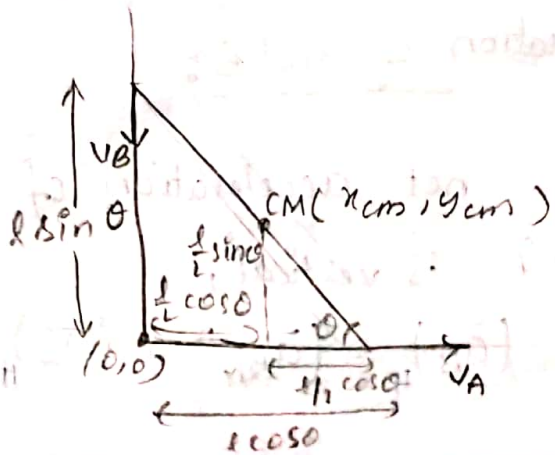
$$\boxed{\omega = \frac{4}{3\sqrt{3}} \text{ rad/s}}$$

* $x_{cm} = \frac{l}{2}\cos\theta$

$y_{cm} = \frac{l}{2}\sin\theta$

$$\frac{x_{cm}^2}{\left(\frac{l}{2}\right)^2} + \frac{y_{cm}^2}{\left(\frac{l}{2}\right)^2} = 1$$

$$\boxed{x_{cm}^2 + y_{cm}^2 = \frac{l^2}{4}}$$



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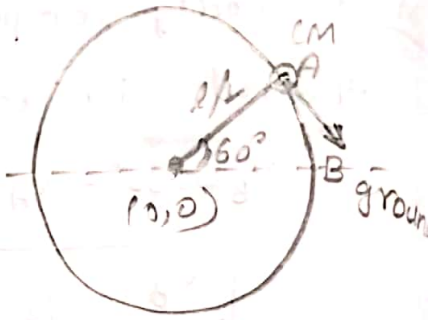
- path of CM is a circle of radius $\frac{l}{2}$.

* when the rod hits the ground, distance travelled by C.M of rod is

$$AB = r\theta$$

$$= \frac{\pi}{3} \times \frac{l}{2}$$

$$\boxed{AB = \frac{\pi l}{6} m}$$



* now,

$$x_{cm} = \frac{l}{2}\cos\theta$$

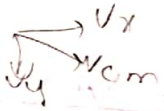
$$v_x = \frac{dx}{dt} = -\frac{l}{2}\sin\theta \cdot \frac{d\theta}{dt} = \frac{l}{2}\cos\sin\theta$$

$$y_{cm} = \frac{l}{2}\sin\theta$$

$$v_y = \frac{dy_{cm}}{dt} = \frac{l}{2}\cos\theta \cdot \frac{d\theta}{dt} = \frac{l}{2}\omega\cos\theta$$

now, $v_{cm} = \sqrt{v_x^2 + v_y^2}$

$$= \sqrt{\left(\frac{l}{2}\omega\sin\theta\right)^2 + \left(\frac{l}{2}\omega\cos\theta\right)^2}$$



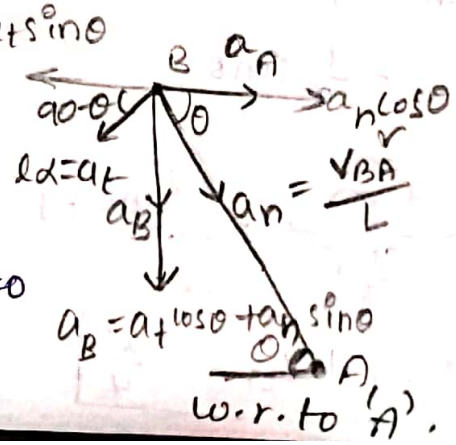
$$v_{cm} = \frac{l}{2}\omega = \frac{2}{2} \times \frac{4^2}{2\sqrt{3}}$$

$$\boxed{v_{cm} = 2/\sqrt{3}\ m/s}$$

*Acceleration of end B?

Since net acceleration of end 'B' is vertical,

$$(\bar{a}_B)_H = (\bar{a}_{B/A})_H + (\bar{a}_A)_H = 0$$



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$$a_A + a_n\cos\theta = a_t\sin\theta$$

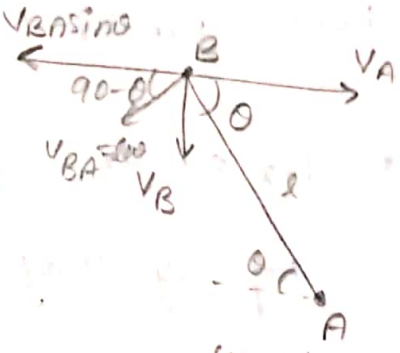
$$a_A + \frac{V_{BA}^2}{l}\cos\theta = l\alpha\sin\theta \quad -(1)$$

$$a_A + \frac{V_A^2}{l\sin\theta}\cos\theta = l\alpha\sin\theta$$

$$1.6 + \frac{2^2}{3\times\left(\frac{8}{4}\right)_2}\times\frac{1}{2} = 3\times\alpha\times\frac{\sqrt{3}}{2}$$

$$1.6 + \frac{8}{9} = 3*\alpha*\frac{\sqrt{3}}{2}$$

$$\alpha = ____$$



w.r.to A

$$\bar{V}_B = \bar{V}_{B|A} + \bar{V}_A$$

$$(\bar{V}_B)_H = (\bar{V}_{B|A})_H + (\bar{V}_A)_H$$

$$0 = -V_{BA}\sin\theta + V_A$$

$$\boxed{V_{BA} = \frac{V_A}{\sin\theta}}$$

finally,

$$a_B = a_t\cos\theta + a_n\sin\theta$$

$$= l\alpha\cos\theta + \frac{V_{BA}^2}{l}\sin\theta \quad (\downarrow)$$

Acceleration of CM:

$$V_x = \frac{l}{2}\omega\sin\theta$$

$$a_x = \frac{dV_x}{dt} = \frac{l}{2}\times\frac{d}{dt}(\omega\cdot\sin\theta) = \frac{l}{2}\omega\cos\theta\cdot\frac{d\theta}{dt} + \frac{l}{2}\sin\theta\cdot\frac{d\omega}{dt}$$

$$\boxed{a_x = \frac{l}{2}\omega^2\cos\theta + \frac{l}{2}\alpha\sin\theta}$$

**Similarly,

$$a_y = \frac{dV_y}{dt} = \frac{d}{dt}\left(-\frac{l}{2}\omega\cos\theta\right)$$

$$= \frac{l}{2}\omega\sin\theta\cdot\frac{d\theta}{dt} - \frac{1}{2}\cos\theta\cdot\frac{d\omega}{dt}$$

$$\boxed{a_y = \frac{l}{2}\omega^2\sin\theta - \frac{l}{2}\cos\theta\cdot\frac{d\omega}{dt}}$$

$$a_y = \frac{l}{2}\omega^2\sin\theta - \frac{l}{2}\cos\theta\cdot\alpha$$

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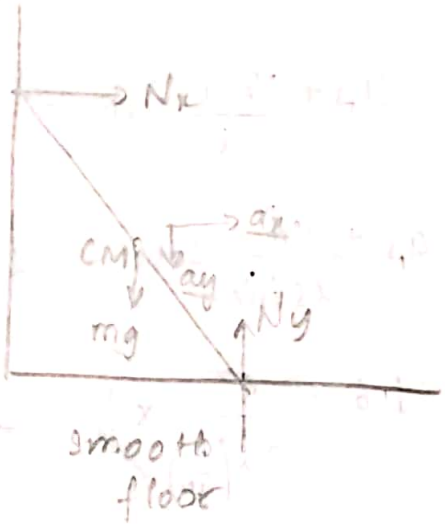
now, $a_{cm} = \sqrt{a_x^2 + a_y^2}$

* $N_x = ma_x$

$mg - N_y = ma_y$

** rod looses contact with vertical floor,

$\therefore N_x = 0, a_x = 0$



$$-\frac{l}{2}\omega^2\cos\theta + \frac{l}{2}\alpha\sin\theta = 0$$

$$\boxed{\tan\theta = \frac{\omega^2}{\alpha}}$$

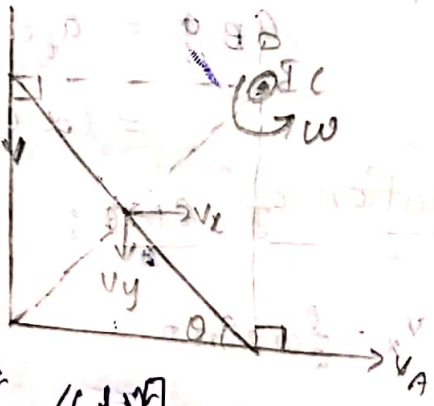
→ rod is slightly disturbed from vertical position.

M-2

* To find ω's

• L.C.E,

$$mg\frac{l}{2} + 0 = mg\frac{l}{2}\sin\theta + \frac{1}{2}I_p\omega^2$$



$$mg\frac{l}{2} = mg\frac{l}{2}\sin\theta + \frac{1}{2}\left[\frac{ml^2}{12} + m\left(\frac{l}{2}\right)^2\right]\omega^2$$

$$\frac{g}{l} = \frac{g}{l}\sin\theta + \frac{1}{2}\left[\frac{1}{12} + \frac{1}{4}\right]\omega^2$$

$$\frac{g}{l}(1-\sin\theta) = \frac{1}{3}\omega^2$$

$$\boxed{\omega = \sqrt{\frac{3g(1-\sin\theta)}{l}}}$$

now,

$$v_A = l\sin\theta(\omega) = l\sin\theta\sqrt{\frac{3g(1-\sin\theta)}{l}}$$

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$$V_B = l\cos\theta\,\omega$$

$$= l\cos\theta\sqrt{\frac{3g(1-\sin\theta)}{l}}$$

$$v_{cm} = \frac{l}{2}\omega = \frac{l}{2}\sqrt{\frac{3g(1-\sin\theta)}{l}}$$

II - Methods

* To find ω:

2 · C · E)

$$mg\frac{l}{2} + 0 = mg\frac{l}{2}\sin\theta + \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$$

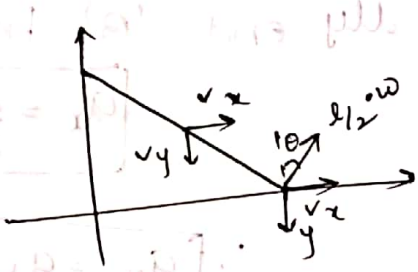
$$mg\frac{l}{2}(1-\sin\theta) = \frac{1}{2}m(v_x^2+v_y^2) + \frac{1}{2}\frac{ml^2}{12}\times\omega^2 \quad \text{---(1)}$$

$$gl(1-\sin\theta) = v_x^2 + v_y^2 + \frac{l^2\omega^2}{12} \quad \text{---(1)}$$

Since net velocity of end A along vertical = 0

$$\boxed{\frac{l}{2}\omega\cos\theta = v_y}$$

$$\boxed{v_A = v_x + \frac{l}{2}\omega\sin\theta}$$



Ally net velocity of end 'B' is vertical,

$$\boxed{v_x = \frac{l}{2}\omega\sin\theta}$$

∴ net velocity of end B is,

$$\boxed{v_B = v_y + \frac{l}{2}\omega\cos\theta}$$

$$\Rightarrow v_A = l\omega\sin\theta, \quad v_B = l\omega\cos\theta$$

$$\boxed{\omega = \sqrt{\frac{3g(1-\sin\theta)}{l}}}$$

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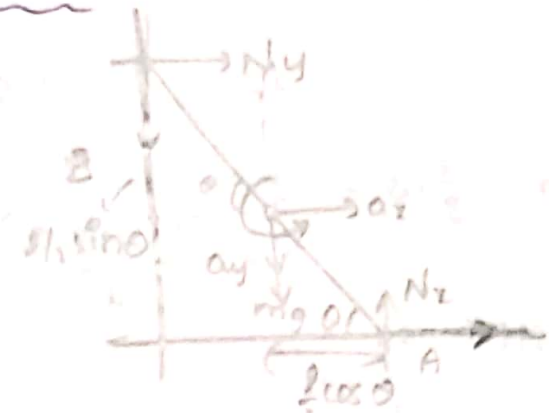
* To find accelerations:



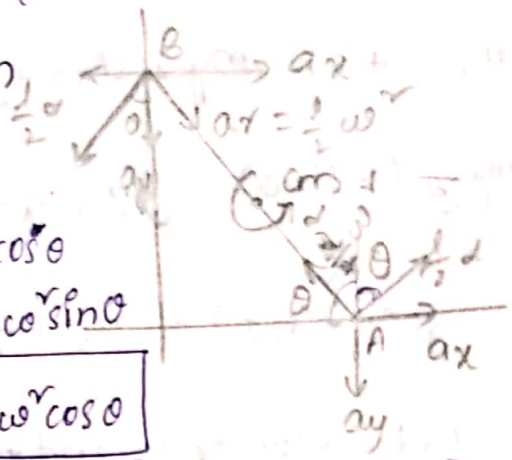
$N_x = ma_x$ — ①

$mg - N_y = ma_y$ — ②

and $\tau_{cm} = I_{cm}\alpha$

$$N_y \frac{l}{2}\cos\theta - N_x \frac{l}{2}\sin\theta = \frac{ml^2}{12}(\alpha) \text{ — ③}$$

* end A has no acceleration in vertical direction,



$a_y = \frac{l}{2}\alpha\cos\theta + \frac{l}{2}\omega^2\sin\theta$

$$\therefore a_A = a_x + \frac{l}{2}\alpha\sin\theta - \frac{l}{2}\omega^2\cos\theta$$

My end 'B' has no acceleration along horizontal.

$a_x = \frac{l}{2}\sin\theta$ $\quad a_x = \frac{l}{2}\alpha\sin\theta - \frac{l}{2}\omega^2\cos\theta$

$$\therefore a_B = a_y + \frac{l}{2}\alpha\cos\theta + \frac{l}{2}\omega^2\sin\theta$$

③ $\Rightarrow (mg - ma_y)\frac{l}{2}\cos\theta - (ma_x)\frac{l}{2}\sin\theta = \frac{ml^2}{6}\alpha$

$$(g - a_y)\cos\theta - a_x\sin\theta = \frac{l\alpha}{6}$$

$$g\cos\theta - (a_y\cos\theta + a_x\sin\theta) = \frac{l\alpha}{6}$$

$$g\cos\theta - \left(\frac{l}{2}\alpha\right) = \frac{l\alpha}{6}$$

$$g\cos\theta = \frac{l\alpha}{6} + \frac{3l\alpha}{6}$$

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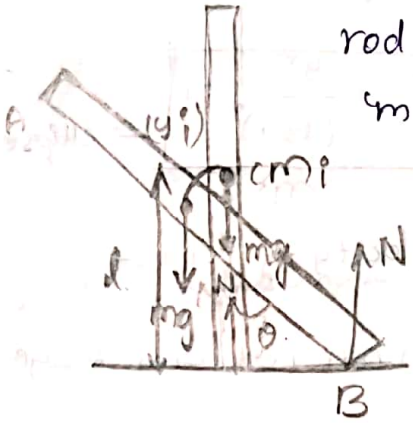
$$g\cos\theta = \frac{\gamma_2 l\alpha}{k^2}$$

$$\boxed{\alpha = \frac{3g}{2l}\cos\theta}$$

Rod released on smooth floor from vertical position



rod of mass 'm' & length '2l'.

since, there $F_H = 0$

$$\frac{dP_H}{dt} = 0$$

$$\boxed{P_H = \text{const}}$$

$$m(v_{cm})_H = \text{const}$$

i.e, $\boxed{(v_{cm})_H = 0}$

∴ There is no displacement of COM of rod along horizontal.

*CM of rod moves along vertical direction only. [along st. line path].

* $(y_{cm})_i = l$

$(y_{cm})_f = l\cos\theta$

* displacement of COM,

$$\Delta y_{cm} = (y_i)_{cm} + (y_f)_{cm}$$

$$\boxed{\Delta y_{cm} = l(1-\cos\theta)}$$

* Torque on the rod about a point on the ground just below end B.

$$\tau_B = mgl\sin\theta$$

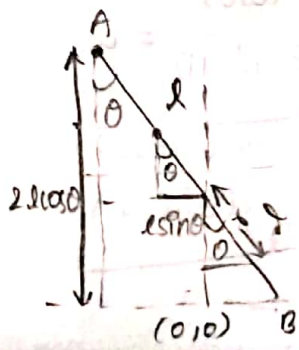
$$\boxed{\tau_B \propto \sin\theta}$$

* path of end A of the rod :

Co-ordinates of end 'A' :

$$x = -l\sin\theta \quad \text{—①}$$

& $y = 2l\cos\theta$ —②



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now, $\boxed{\frac{x^2}{l^2} + \frac{y^2}{4l^2} = 1}$

∴ path of end 'A' is ellipse.

In general,

$$x = \pm r\sin\theta$$

$$y = (l \pm r)\cos\theta$$

$$\boxed{\frac{x^2}{r^2} + \frac{y^2}{(l \pm r)^2} = 1} \rightarrow \text{ellipse.}$$

* To find velocity of CM :-

$$y = l\cos\theta \text{ (at any stage)}$$

$$v_{cm} = \frac{dy}{dt} = l(-\sin\theta)\cdot\frac{d\theta}{dt}$$

$$\boxed{v_{cm} = -l\omega\sin\theta}$$

$$\therefore |v_{cm}| = l\omega\sin\theta.$$

* To find ω :-

use $\Delta \cdot C \cdot E$,

$$mgl = mgl\cos\theta + \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$$

$$mgl = mgl\cos\theta + \frac{1}{2}m(l\omega\sin\theta)^2 + \frac{1}{2}\frac{m(2l)^2}{12}\omega^2$$

$$g = g\cos\theta + \frac{1}{2}l\omega^2\sin^2\theta + \frac{1}{24}\times 4l\omega^2$$

$$2g(1-\cos\theta) = \left(l\sin^2\theta + \frac{l}{3}\right)\omega^2$$

$$\frac{6g(1-\cos\theta)}{l(3\sin^2\theta+1)} = \omega^2$$

$$\boxed{\omega = \frac{6g(1-\cos\theta)}{l(3\sin^2\theta+1)}}$$

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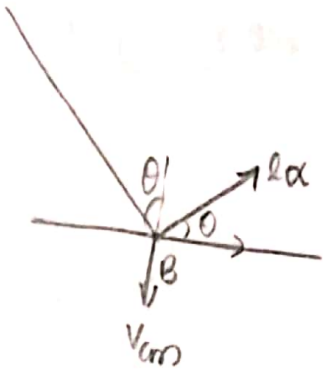
* we have,



$$(V_B)_V = 0$$

$$\boxed{l\omega\sin\theta = V_{cm}}$$

& $\boxed{V_B = l\omega\cos\theta}$

* acceleration of CM :-

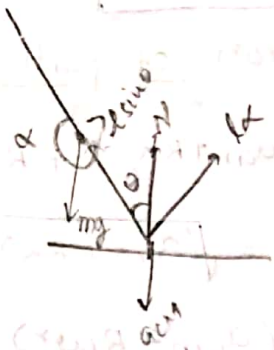
$$a_{cm} = \frac{dV_{cm}}{dt} = -l\cdot\frac{d(\omega\sin\theta)}{dt}$$

$$= -l\left(\cos\theta\cdot\frac{d\theta}{dt}\right)\omega - l\sin\theta\frac{d\omega}{dt}$$

$$\boxed{a_{cm} = -l\omega^2\cos\theta - l\alpha\sin\theta}$$

$\tau_{cm} = I_{cm}\alpha$

$$\Rightarrow N l\sin\theta = \frac{m(2l)^2}{12}(\alpha)$$



$$N = \frac{ml^2}{3}\times\frac{\alpha}{l\sin\theta}$$

$$\boxed{N = \frac{ml\alpha}{3\sin\theta}}$$

$\Rightarrow$ from constraint eqn,

$$\boxed{l\alpha\sin\theta = a_{cm}}$$

$$a_{cm} = -a_{cm} - l\omega^2\cos\theta$$

$$a_{cm} = -\frac{l\omega^2\cos\theta}{2}$$

$$|a_{cm}| = \frac{l\omega^2\cos\theta}{2}$$

$$l\alpha\sin\theta = \frac{l\omega^2\cos\theta}{2}$$

$$\boxed{\alpha = \frac{\omega^2\cot\theta}{2}}$$

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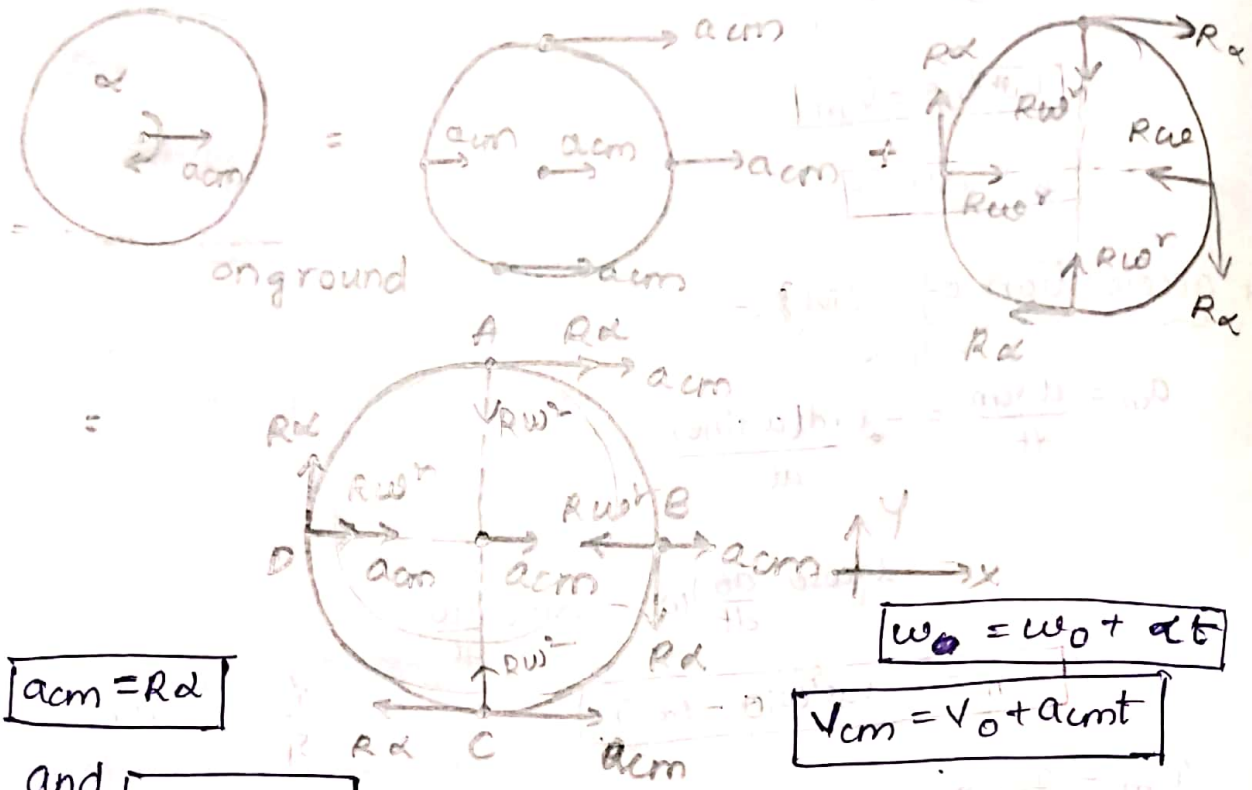
- Pure rolling in accelerated motion:



$\omega = \omega_0 + \alpha t$

$V_{cm} = V_0 + a_{cm}t$

* $\boxed{a_{cm} = R\alpha}$

and $\boxed{V_{cm} = R\omega}$

* Acceleration of particles of body:

* $\bar{a}_A = (a_{cm} + R\alpha)\uparrow + R\omega^2(-\uparrow)$

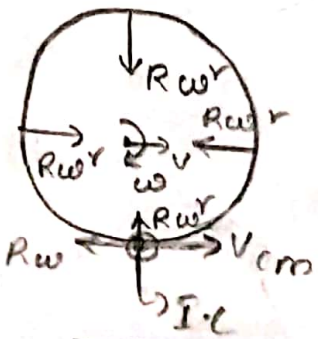
$$|\bar{a}_A| = \sqrt{(a_{cm} + R\alpha)^2 + (R\omega^2)^2}$$

* $\bar{a}_B = (a_{cm} - R\omega^2)\uparrow + R\alpha(-\uparrow)$

* $\bar{a}_C = (a_{cm} - R\alpha)\uparrow + R\omega^2(\uparrow)$

* $\bar{a}_D = (R\omega^2 + a_{cm})\uparrow + R\alpha\uparrow$

* In uniform pure rolling motion:



ω = const & V = const

$\Rightarrow \alpha = 0 \Rightarrow \boxed{a = 0}$

$\boxed{V_{cm} = R\omega}$

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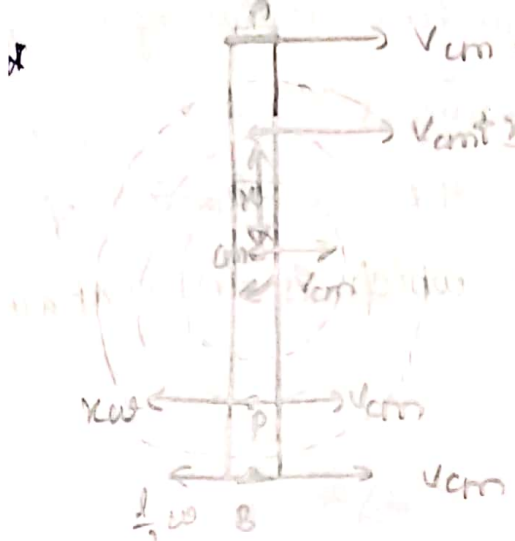


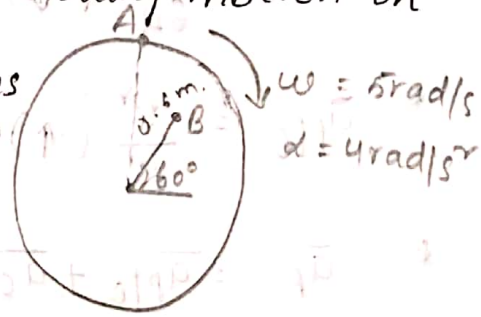
At P $V_{net} = 0$ i.e, instantaneous centre

$V_{cm} = xw$

$$x = \frac{V_{cm}}{w}$$

→ disc of radius 1m have pure rolling motion on ground. Find ① velocity of particles A & B.

② acceleration of particles A & B.



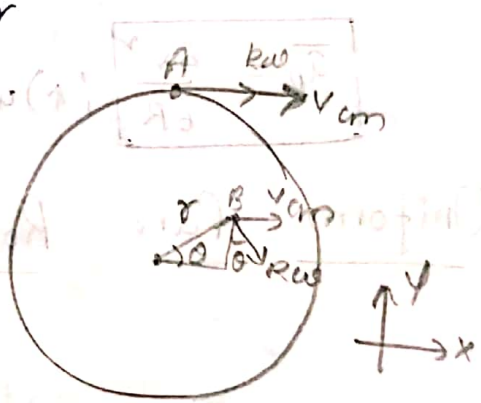
Since there is no slipping,

$V_{cm} = Rw = 1 \times 5 = 5\ m/s$

& $a_{cm} = R\alpha = 1 \times 4 = 4\ m/s^2$

i) $V_A = Rw + V_{cm} = 2Rw = 2 \times 1 \times 5$

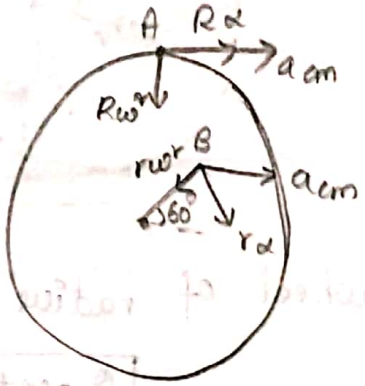
$$\boxed{V_A = 10\ m/s}$$



$V_B = (V_{cm} + rw\sin 60°)\hat{i} + (rw\cos 60°)\hat{j}$

$= \left(5 + 5 \times \frac{\sqrt{3}}{2}\right)\hat{i} + \left(5 \times \frac{1}{4}\right)\hat{j}$

$$\boxed{V_B = \frac{2(10+5\sqrt{3})\hat{i} + 5\hat{j}}{2}}$$



ii) $a_A = R\alpha + a_{cm} = 4 + 1 \times 4 = 8$

$$\boxed{a_A = 8\ m/s^2}$$

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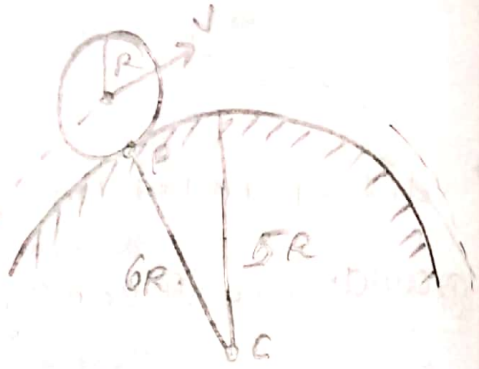
→ A convex surface has uniform radius of curvature 5R. A wheel of radius R is rolling without sliding on it with a const speed V. Find the acceleration of point P of the ~~which~~ wheel which is in contact with convex surface.



From ground,

$$\bar{a}_c = \frac{v^2}{6R} \ (\downarrow) \text{down}$$

* $\bar{a}_{p/c} = \frac{v^2}{R} \ (\uparrow) \text{up}$

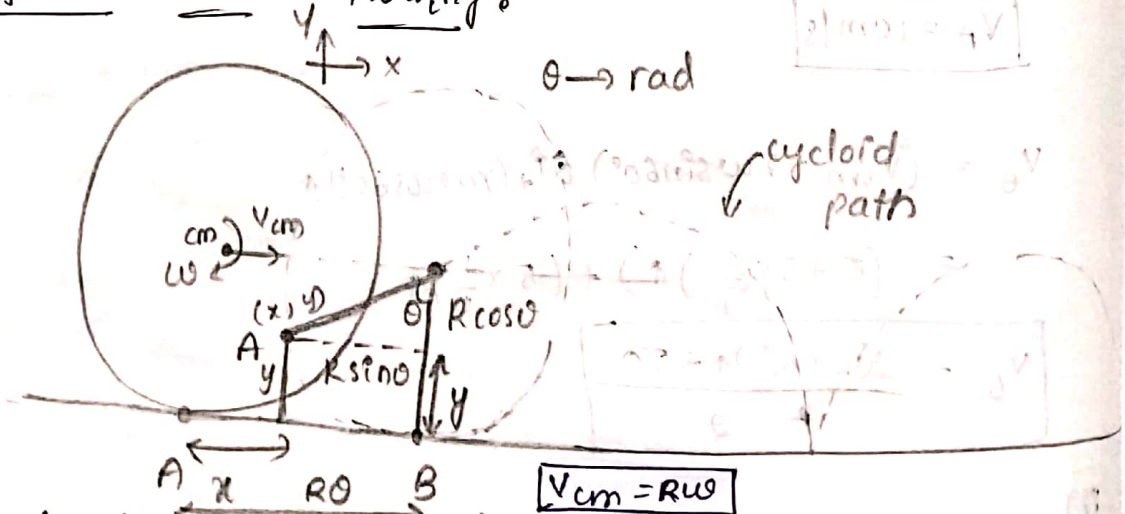
* $\bar{a}_p = \bar{a}_{p/c} + \bar{a}_c$

$$= \frac{v^2}{R} + \left(-\frac{v^2}{6R}\right)$$

$$= \frac{v^2}{R} - \frac{v^2}{6R} = v^2\left(\frac{6-1}{6R}\right) = \frac{5v^2}{6R}$$

$$\boxed{\bar{a}_p = \frac{5v^2}{6R}} \ (\uparrow) \text{up}.$$

Uniform Pure Rolling:



$\boxed{v_{cm} = R\omega}$

wheel of radius R is in uniform pure rolling.

$\boxed{\theta = \omega t}$

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* $\omega = \frac{d\theta}{dt}$

$\int d\theta = \int \omega \cdot dt$

$\boxed{\theta = \omega t}$

* from figure,

horizontal & vertical displacements of particle 'A' on the rim are,

$x + R\sin\theta = R\theta$

$\boxed{x = R\theta - R\sin\theta}$

and $y + R\cos\theta = R$

$\boxed{y = R - R\cos\theta}$

∴ net displacement = $\sqrt{x^2 + y^2} = |\bar{S}|$

→ For half revolution,

$\theta = \pi$ rad

$x = \pi R$, $y = 2R$.

$\bar{S} = \pi R\uparrow + 2R\uparrow \Rightarrow |\bar{S}| = R\sqrt{\pi^2 + 4}$

$\boxed{|\bar{S}| = R\sqrt{\pi^2 + 4}}$

* velocity of the particle on the rim,

$v_x = \frac{dx}{dt} = \frac{d}{dt}(R\theta - R\sin\theta)$

$= R\omega - R\omega\cos\theta$

$\boxed{v_x = R\omega(1 - \cos\theta)}$

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& $V_y = \frac{dy}{dt} = \frac{d}{dt}(R - R\cos\theta)$

$$\boxed{V_y = R\omega\sin\theta}$$

$\therefore$ velocity, $\bar{V} = V_x\uparrow + V_y\uparrow$

* speed of the particle $= |\bar{V}| = \sqrt{V_x^2 + V_y^2}$

$$|\bar{V}| = R\omega\sqrt{1 + \cos^2\theta - 2\cos\theta + \sin^2\theta}$$

$$= R\omega\sqrt{2 - 2\cos\theta}$$

$$|\bar{V}| = 2R\omega\sin\theta/2$$

$$\boxed{V = 2R\omega\sin\left(\frac{\omega t}{2}\right)}$$

* distance travelled by any particle on the rim of wheel in one complete revolution,

$$d = \int_{t=0}^{t=T=\frac{2\pi}{\omega}} V\,dt$$

$$d = \int_0^{\frac{2\pi}{\omega}} 2R\sin\left(\frac{\omega t}{2}\right)\cdot dt$$

$$= \left\{2R\sin\left(\frac{\omega}{2}\times\frac{2\pi}{\omega}\right) - 2R\cos 0\right\}$$

$$d = \left\{2R(-1) - 2R\right\}$$

$$= 4R \times 2 = 8R$$

$$\boxed{d = 8R}$$

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* distance travelled by C.M of wheel in one complete revolution,

$$S_{cm} = V_{cm} \times t$$

$$= V_{cm} \times T$$

$$= R\omega \times \frac{2\pi}{\omega}$$

$$\boxed{S_{cm} = 2\pi R}$$

① If $V_{cm} = R\omega$ (or) $S_{cm} = 2\pi R$

⇒ wheel is in pure rolling.

② If $V_{cm} > R\omega$ (or) $S_{cm} > 2\pi R$.

⇒ body is having "forward slipping".

friction → kinetic,

which opposes V_{cm} & supports 'ω'.

③ If $V_{cm} < R\omega$ (or) $S_{cm} < 2\pi R$

⇒ body is having "backward slipping".

friction → kinetic,

which opposes 'ω' & supports 'V_{cm}'.

* In pure rolling,

friction $f = 0$ (horizontal surface)

(or) f = static (Inclined plane).

Kinetic Energy of Body in Pure Rolling:

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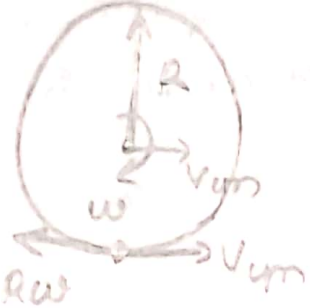

K.E is of two parts.

1) Translational K.E :

$$K.E_T = \frac{1}{2} m v_{cm}^2$$

2) Rotational K.E :

$$K.E_R = \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} m k^2 \left(\frac{v_{cm}}{R}\right)^2$$

$$= \frac{1}{2} m v_{cm}^2 \left(\frac{k^2}{R^2}\right)$$

$$\boxed{K.E_R = \frac{1}{2} m v_{cm}^2 \beta}$$

$\beta = \frac{k^2}{R}$ — dimensionless quantity.

* for ex,

(a) ring / hollow cylinder ;

$$I_{cm} = mR^2$$

$$\beta = \frac{k^2}{R^2} = \frac{R^2}{R^2} = 1 \qquad \boxed{\beta = 1}$$

(b) for disc / solid cylinder ;

$$I_{cm} = \frac{mR^2}{2}$$

$$\beta = \frac{R^2}{2 \times R^2} = \frac{1}{2} . \qquad \boxed{\beta = 1/2}$$

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c) for solid sphere,

$$I_{cm} = \frac{2}{5} mR^2$$

$$\boxed{\beta = \frac{2}{5}}$$

d) for hollow sphere,

$$I_{cm} = \frac{2}{3} mR^2$$

$$\boxed{\beta = 2/3}$$

* for sliding bodies,

$$\boxed{\beta = 0}$$

* now, $K.E_R = (\beta) K.E_T$

$$\boxed{\frac{K.E_R}{K.E_T} = \beta}$$

* Total K.E of rolling body,

$$K.E_{roll} = K.E_T + K.E_{rot}$$

$$= K.E_T + \beta K.E_T$$

$$K.E_{roll} = K.E_T (1+\beta)$$

$$\boxed{K.E_{roll} = \frac{1}{2} m v_{cm}^2 (1+\beta)}$$

* fractional total K.E of rolling body converts into translation K.E is

$$\boxed{\frac{K.E_T}{K.E_{roll}} = \frac{1}{1+\beta}}$$

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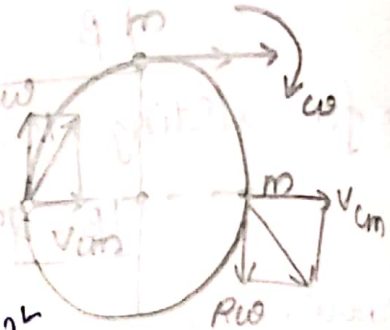
* $$\frac{K.E_{roll} - K.E_{rot}}{K.E_{roll}} = 1 - \frac{1}{1+\beta} = \frac{1+\beta-1}{1+\beta} = \frac{\beta}{1+\beta}$$

$$\boxed{\frac{K.E_{rot}}{K.E_{roll}} = \frac{\beta}{1+\beta}}$$

→ Ring of mass 'm' and radius 'R' with particles is moving without slipping. Find K.E of system.



$$K.E_{sys} = \frac{1}{2} m v_{cm}^2 (1+\beta) + \frac{1}{2}(2m)(\sqrt{2}R\omega)^2$$

$$+\frac{1}{2} m(2R\omega)^2 + \frac{1}{2} m(\sqrt{2}R\omega)^2$$

$$\boxed{v_{cm} = R\omega}$$

$$= \frac{1}{2} m v_{cm}^2 (1+\beta) + \frac{1}{2}(2m)(2R^2\omega^2)$$

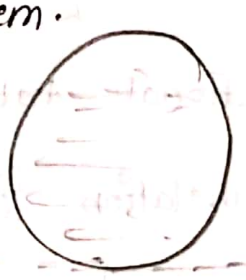
$$+\frac{1}{2} m(2R^2\omega^2) + \frac{1}{2} m(2R^2\omega^2)$$

$$= \frac{1}{2} m(R^2\omega^2)(2) + 2mR^2\omega^2 + 2mR^2\omega^2 + mR^2\omega^2$$

$$\boxed{K.E = 6mR^2\omega^2}$$

$$\boxed{K.E_{sys} = 6mv_{cm}^2}$$

→ Hollow sphere of mass (m_1) and radius 'R' is filled completely with Non-viscous liquid of mass (m_2) is rolling without slipping. Find K.E of system.



$$K.E_{sys} = \frac{1}{2} m_1 v_{cm}^2 (1+\beta) + \frac{1}{2} m_2 v_{cm}^2$$

(hollow sphere) (liquid)

$$= \frac{1}{2} m_1 v_{cm}^2 \left(\frac{5}{3}\right) + \frac{1}{2} m_2 v_{cm}^2 = \frac{1}{2} v_{cm}^2 \left(\frac{5}{3} m_1 + m_2\right).$$

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→ If the sphere is filled with thick liquid (viscous liq) then ~~k.E of~~ system is

$$K.E_{sys} = \frac{1}{2} m_1 v_{cm}^2 (1+\beta_1) + \frac{1}{2} m_2 v_{cm}^2 (1+\beta_2)$$

$$= \frac{1}{2} m_1 v_{cm}^2 \left(\frac{5}{3}\right) + \frac{1}{2} m_2 v_{cm}^2 \left(\frac{7}{5}\right)$$

$$\boxed{K.E_{sys} = \frac{1}{2} v_{cm}^2 \left(\frac{5}{3} m_1 + \frac{7}{5} m_2\right)}$$

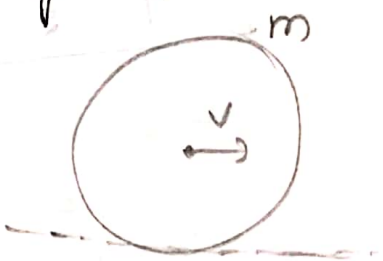
→ ring is rolling without slipping.

M-I 6

* K.E of ring $= \frac{1}{2} m v^2 (1+\beta)$

$= \frac{1}{2} m v^2 (1+1)$

$$\boxed{K.E = mv^2}$$



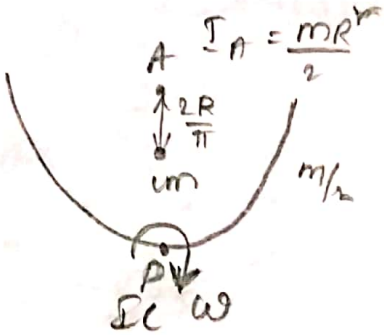
* K.E of lower half of ring:

$$K.E_{I_{AOR}} = \frac{1}{2} I_p \omega^2$$

we have,

$$I_A = I_{cm} + \frac{m}{2}\left(\frac{2R}{\pi}\right)^2 \quad -(1)$$

$$I_p = I_{cm} + \frac{m}{2}\left(R - \frac{2R}{\pi}\right)^2 \quad -(2)$$



eq(2) − eq(1)

$$I_p - I_A = \frac{m}{2}\left(R - \frac{2R}{\pi}\right)^2 - \frac{m}{2}\left(\frac{2R}{\pi}\right)^2$$

$$I_p = \frac{mR^2}{2} + \frac{m}{2}\left(\left(R - \frac{4R}{\pi}\right)(R)\right) = \frac{mR^2}{2}\left[1 + 1 - \frac{4}{\pi}\right]$$

$$\boxed{I_p = \frac{mR^2}{2}\left[2 - \frac{4}{\pi}\right]}$$

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$\therefore$ K.E lower half $= \frac{1}{2} I_p \omega^2$

$= \frac{1}{2} mv^2 - \frac{mv^2}{\pi}$

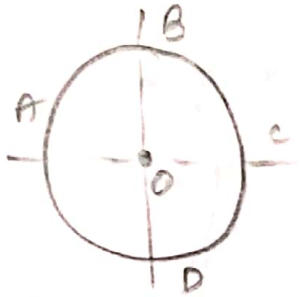
$= \frac{mv^2}{2} - \frac{mv^2}{\pi}$

* K.E of upper half of ring,

$K.E_{up} = mv^2 - mv^2\left(\frac{1}{2} - \frac{1}{\pi}\right)$

$= mv^2\left(1 - \frac{1}{2} + \frac{1}{\pi}\right)$

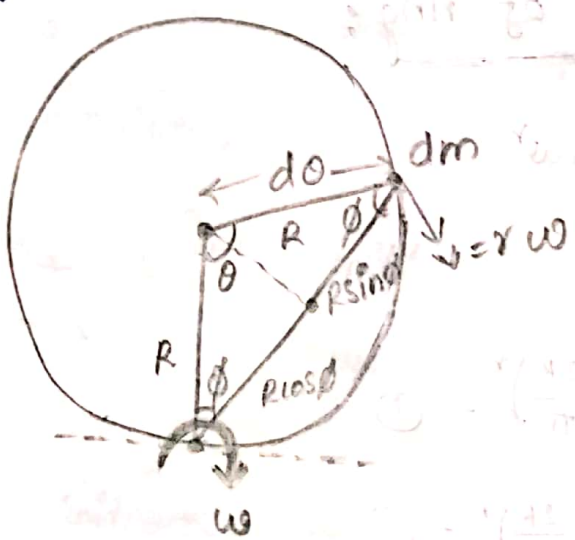
$$\boxed{K.E_{up} = \frac{mv^2}{2} + \frac{mv^2}{\pi}}$$



① $K.E_{ABC} > K.E_{ADC}$

② $K.E_{BCD} = K.E_{DAB}$.

M-II:



$dK.E = \frac{1}{2}(dm)v^2$

$= \frac{1}{2}\left(\frac{m}{2\pi R} \times R(\cos\theta)\right)(r\omega)^2$

$= \frac{m}{4\pi}(2R\cos\theta)^2 \omega^2 . d\theta$

$= \frac{m}{\pi}\omega^2 R^2 \cos^2(90 - \theta/2)\, d\theta$

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$$dK.E = \frac{m\omega^2 R^2}{\pi} \sin^2(\theta/2) \cdot d\theta$$

$$K.E = \int_{\theta=-\pi/2}^{\theta=\pi/2} dKE = \frac{m\omega^2 R^2}{\pi} \int_{\theta=-\pi/2}^{\theta=\pi/2} \sin^2(\theta/2) \cdot d\theta$$

$$= \frac{m\omega^2 R^2}{2\pi} \int_{\theta=-\pi/2}^{\theta=\pi/2} (1-\cos\theta) \cdot d\theta$$

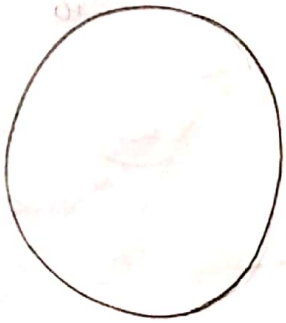
$$= \frac{m\omega^2 R^2}{2\pi} \left\{ [\theta]_{-\pi/2}^{\pi/2} - [\sin\theta]_{-\pi/2}^{\pi/2} \right\}$$

$$= \frac{m\omega^2 R^2}{2\pi} \left\{ \frac{\pi}{2} + \frac{\pi}{2} - [1+1] \right\}$$

$$= \frac{mv^2}{2\pi} \{\pi - 2\}$$

$$K.E = \frac{mv^2}{2} - \frac{mv^2}{\pi}$$

→ A ring of radius R rolls without slidding with a const velocity. The radius of curvature of the path followed by any particle of the ring at the highest pt of it's path will be,



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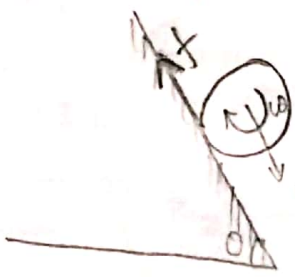
Problems:

→ AB → rough, BC → smooth, AB = BC. A solid cylinder is in pure rolling in path AB. Find ratio of K.E$_T$ & K.E$_R$ at 'C'.

In pure rolling,

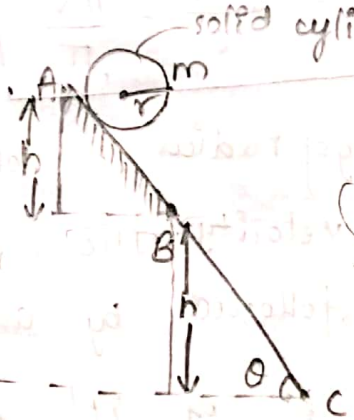
① $V = R\omega$

② $f = 0$



① $V = R\omega$

② f : static



$\frac{(KE)_{TC}}{(KE)_{RC}}$

$K_T = \frac{1}{2} mv^2 (1+\beta)$

K_T

$\frac{(KE)_T}{(KE)_R} = \beta$

L.E.E,

$mgh = K_T + K_R$

we have, $K_R = \beta K_T = \frac{K_T}{2}$.

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$mgh = K_T + \frac{K_T}{2}$

$mgh = \frac{3K_T}{2}$

$\boxed{K_T = \frac{2mgh}{3}}$ at B

& $K_R = \frac{K_T}{2} = \frac{mgh}{3}$

$\boxed{K_R = \frac{mgh}{3}}$ at B.

for BC (smooth plane):

$\Rightarrow f = 0 \Rightarrow \tau_f = 0 \Rightarrow \alpha = 0 \Rightarrow \boxed{\omega = \text{const}}$

$\Rightarrow \boxed{K.E_R = \text{const}}$

$\therefore$ L.C.E)

$mgh = \Delta K_T$

$mgh = (KE)_{fi}$

$mgh = (KE_T)_f - (KE_T)_i$

$mgh = (K_T)_C - (K_T)_B$

$(K_T)_{atC} = (K_T)_B + \Delta K_T$

$(K_T)_{atC} = \frac{2mgh}{3} + mgh$

$(K_T)_{atC} = \frac{5mgh}{3}$

& $(K.R)_{atC} = (K_R)_{atB} = \frac{mgh}{3}$

$\therefore \left(\frac{K_T}{K_R}\right)_{atC} = \frac{\frac{5mgh}{3}}{\frac{mgh}{3}} = 5$

ratio = 5 : 1.

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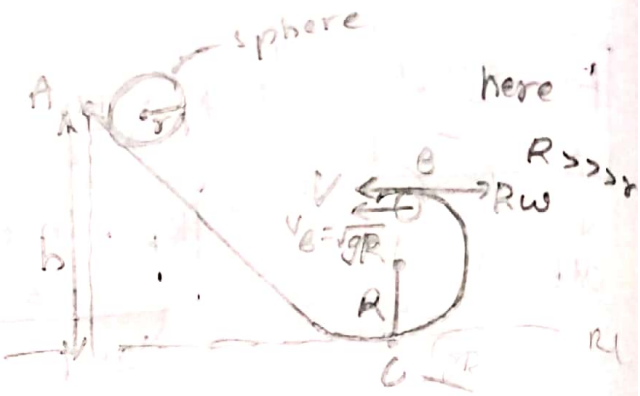
→ sphere is released from rest & rolling without slipping. Find min height 'h' for which sphere completes the loop.

$$mg + N^{0} = \frac{mv^2}{R}$$

$$\boxed{V = \sqrt{gR}}$$



To complete the loop,

velocity at top-most pt $= \sqrt{gR}$

use L.C.E b/w A & B.

$$mgh = mg(2R) + \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$mgh = mg(2R) + \frac{1}{2}mv^2 + \frac{1}{2} \times \frac{2}{5} \times mr^2 \times \frac{v^2}{r^2}$$

$$\boxed{h = 2.7R}$$

$$K.E_T = \frac{1}{2}mv^2 = \frac{1}{2}mgR \quad -(1)$$

$$K.E_R = \beta K_T = \frac{2}{5} \times \frac{mgR}{2} = \frac{mgR}{5}$$

$$\therefore K.E_B = \frac{7mgR}{10}$$

At C:

$$K_T + K_R = \frac{7mgR}{10} + mg(2R)$$

$$K_T(1+\beta) = \frac{27mgR}{10}$$

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$$\frac{7}{5} K_T = \frac{27mgR}{10}$$

$$K_T = \frac{27mgR}{14}$$

$$\frac{1}{2}mv^2 = \frac{27mgR}{14}$$

$$\boxed{v = \sqrt{\frac{27gR}{7}}}$$ — at bottom most pt.

* L.C.E b/w A & C is

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$= \frac{1}{2}m \times \frac{27gR}{7} + \frac{1}{2} \times \frac{2}{5} m v^2 \times \frac{v^2}{r^2}$$

$$= \frac{27mgR}{14} + \frac{mv^2}{5}$$

$$= \frac{27mgR}{14} + \frac{27mgR}{7 \times 5}$$

$$= \frac{27mgR}{7}\left(\frac{1}{2} + \frac{1}{5}\right)$$

$$= \frac{27mgR}{7} \times \left(\frac{7}{10}\right)$$

$$mgh = \frac{27mgR}{10}$$

$$\boxed{h = 2.7R}$$

→ ball of mass 'm' and radius 'r' is released from 'A'. It rolls with out slipping. when it reaches

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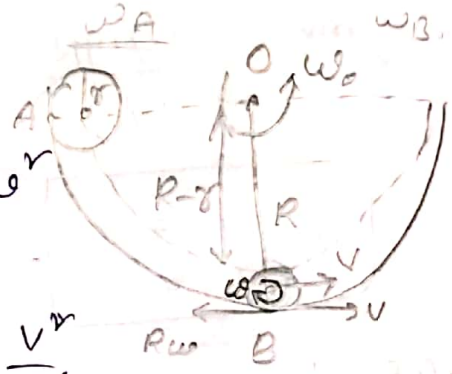
to bottom-most pt 'B'. Find

(1) Angular velocity of centre of ball about centre of shell.

(2) Normal force exerted by track on the ball.



1) L.C.E,

$$mg(R-r) = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$mg(R-r) = \frac{1}{2}mv^2 + \frac{1}{2}\times\frac{2mr^2}{5}\times\frac{v^2}{r^2}$$

$$g(R-r) = v^2\left(\frac{5+2}{10}\right)$$

$$v^2 = \frac{10g(R-r)}{7}$$

$$\boxed{v = \sqrt{\frac{10g(R-r)}{7}}}$$

$$\therefore \omega_0 = \frac{v}{R-r} = \sqrt{\frac{10g(R-r)}{7(R-r)^2}} = \sqrt{\frac{10g}{7(R-r)}}$$

$$\boxed{\omega_0 = \sqrt{\frac{10g}{7(R-r)}}}$$

2) $$N - mg = \frac{mv^2}{R-r} = \frac{10g(m)}{7}$$

$$N = \frac{10mg}{7} + mg$$

$$\Rightarrow \boxed{N = \frac{17mg}{7}}$$

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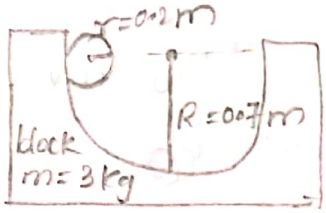
→ Solid cylinder of mass $m = 2kg$ and radius $r = 0.2m$ is released from rest as shown. when ball reaches to bottom of track (rolls without slipping). Find

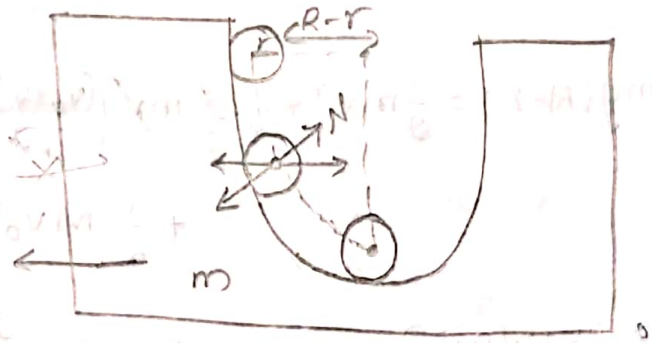
(1) Distance travelled by the block.

(2) speed of cylinder.

(3) speed of pt of contact of cylinder and block w.r.to ground.



* since net force on the system along horizontal = 0.



along horizonal,

$$(v_{cm})_f = (v_{cm})_i = 0$$

$$(\Delta s_{cm})_H = 0$$

$$m_1 \overline{\Delta x_1} + m_2 \overline{\Delta x_2} = 0$$

$$-Mx + m(R-r-x) = 0$$

$$Mx = m(R-r-x)$$

$$Mx + mx = m(R-r)$$

$$x = \frac{m(R-r)}{M+m} = \frac{2 \times 0.5}{10} = 0.2$$

displacement of block.

& displacement of cylinder w.r.to ground = $R-r-x$.

$= 0.7 - 0.2 - 0.2$

$= 0.3$

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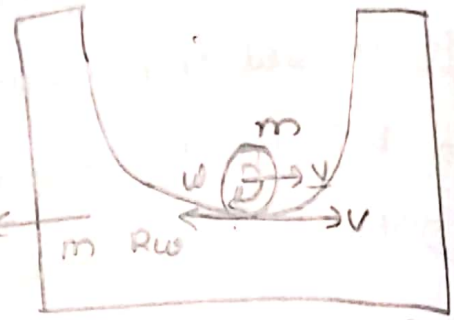
* Since, $\bar{P}_H$ = const



$$0 = -MV_0 + mv$$

$$MV_0 = mv \quad \text{— (1)}$$

* L.C.E, $V_0 = \frac{2v}{3}$

$$mg(R-r) = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}MV_0^2$$

constraint eqn:

$$r\omega - v = V_0$$

$$\omega = \frac{V_0 + v}{r} \quad \text{— (1)}$$

$$mg(R-r) = \frac{1}{2}mv^2 + \frac{1}{2}\times\frac{2}{5}mr^2\left(\frac{(V_0+v)^2}{r^2}\right) + \frac{1}{2}MV_0^2$$

$$2\times10(0.5) = \frac{1}{2}\times2\times v^2 + \frac{1}{5}\times2\times(V_0+v)^2 + \frac{1}{2}\times3\times V_0^2$$

$$100 = v^2 + \frac{2}{5}\times\left(\frac{5}{3}v\right)^2 + \frac{1}{2}\times3\times\left(\frac{2v}{3}\right)^2$$

$$100 = v^2\left[1 + \frac{10}{9} + \frac{2}{3}\right]$$

$$100 = v^2\left[\frac{9+10+6}{9}\right]$$

$$100 = v^2\left[\frac{25}{9}\right]$$

$$v = 6 \text{ m/s}$$

$$V_0 = \frac{2v}{3} = \frac{2}{3}(6) = 4$$

$$V_0 = \frac{4}{\sqrt{11}} \text{ m/s}$$

$$mg(R-r) = \frac{1}{2}\times mv^2 + \frac{1}{2}\times\frac{mr^2}{2}\times\frac{(V_0+v)^2}{r^2} + \frac{1}{2}\times M\left(\frac{V_0}{1}\right)^2$$

$$2\times10\times0.5 = \frac{1}{2}\times2\times v^2 + \frac{1}{2}\times\frac{2}{2}\times\left(\frac{5}{3}v\right)^2 + \frac{1}{2}\times3\left(\frac{2v}{3}\right)^2$$

$$100 = v^2\left(1 + \frac{25}{18} + \frac{2}{3}\right)$$

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$$V^2\left(\frac{18+25+12}{18}\right) = 100$$

$$V^2\left(\frac{55}{18}\right) = 100$$

$$V^2 = \frac{36}{11}$$

$$\boxed{V = \frac{6}{\sqrt{11}}\ m/s}$$

→ The path AB of the track is a semicircle of Radius 'R' of vertical plane the disc rolls without slipping and leaves contact with track at the highest pt flying through the air, It strikes the ground at pt 'C'. The velocity of COM of disc makes 30° below horizontal at the time of strike. At the same instant velocity of topmost pt of disc is found to be 6 m/s. Find

i) value of R.

ii) Velocity of COM of disc when it strikes the ground.

iii) distance AC. (Ans in next page)

N + g

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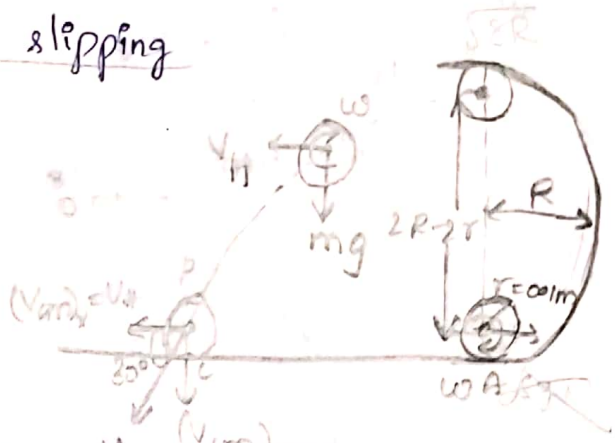
since, there is no slipping b/w disc & track.

$$0 = v_H - r\omega$$

$$\boxed{v_H = r\omega}$$



. when it is flying in air,

horizontal component of velocity of every pt is not changing & 'ω' is also not changing.

Clearly $v_p = (v_{cm})_y = v_y = 6\ m/s$.

$$\tan 30^\circ = \frac{(v_{cm})_y}{v_H} \Rightarrow \boxed{v_H = 6\sqrt{3}\ m/s}$$

$$v_{cm} = \sqrt{v_H^2 + (v_{cm})_y^2}$$

$$= \sqrt{(6\sqrt{3})^2 + (6)^2}$$

$$= \sqrt{6^2 \times 2^2}$$

$$\boxed{v_{cm} = 12\ m/s}$$

② L.C.E b/w B & C

$$mg\,2(R-r) = \frac{1}{2}mv_p^2$$

$$g \times 10(R - 0.1) = 36 \times \frac{1}{2}$$

$$R - 0.1 = 0.9$$

$$\boxed{R = 1m}$$

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③ distance travelled by (AC)

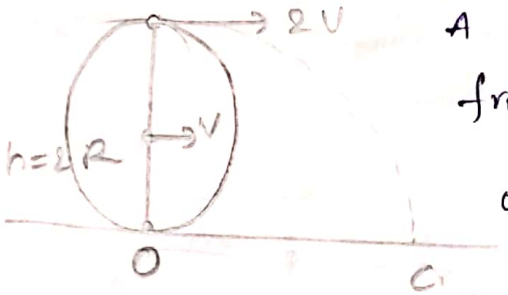
$$AC = V_H(t) = V_H \sqrt{\frac{2(2(R-r))}{g}}$$

$$= 6 \times \sqrt{3} \times 2\sqrt{0.09}$$

$$= 6 \times \sqrt{3} \times 2 \times 0.3$$

$$\boxed{AC = 3.6\sqrt{3}\ m}$$

*



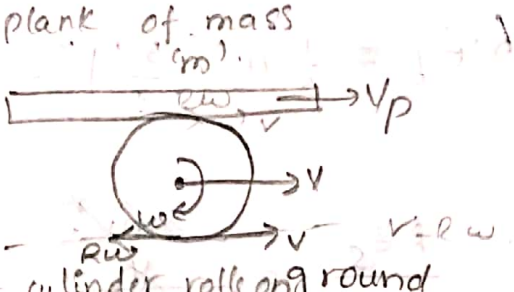
A mud particle released from top of wheel.

wheel rolling on ground with out slipping.

$$OC = (2V)t = 2V\sqrt{\frac{2(2R)}{g}} = 4V\sqrt{\frac{R}{g}}$$

→ there is no slipping b/w any two contact surface.

Find $\frac{K.E_{cy}}{K.E_{plank}} =$ _____.



$$K.E_{cyl} = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$= \frac{1}{2} \times mv^2 + \frac{1}{2} \times \frac{mR^2}{2} \times \frac{v^2}{R^2}$$

$$= \frac{1}{2}mv^2 + \frac{1}{4}mv^2$$

$$\boxed{K.E_{cy} = \frac{3mv^2}{4}}$$

$V_D = V + R\omega$

$(\because V = R\omega)$

↓

no slipping

$V_D = 2V$

$$K.E_{plank} = \frac{1}{2}m(2V)^2 = \frac{1}{2} \times m \times 4V^2 = 2mv^2$$

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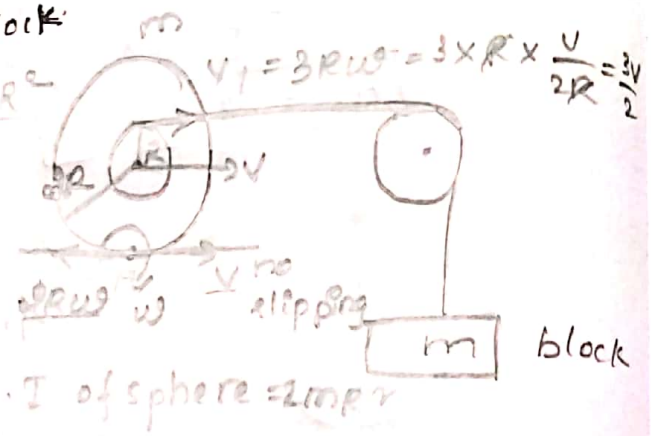
$$\frac{K.E_{cy}}{K.E_{pl}} = \frac{\frac{3mv^2}{4}}{\frac{2mv^2}{2}} = \frac{3}{8}.$$

→ Find $K.E_{spher}$ & $K.E_{block}$



we have,

$V = 2R\omega$

$$\boxed{\omega = \frac{V}{2R}}$$

$$\therefore K.E_{sphere} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2} \times mv^2 + \frac{1}{2} \times \frac{2}{5}mR^2 \times \frac{v^2}{4R^2}$$

$$= \frac{1}{2}mv^2\left(1 + \frac{1}{5}\right) = \frac{3mv^2}{4}$$

$$K.E_b = \frac{1}{2}m\left(\frac{3V}{2}\right)^2 = \frac{9mv^2}{8}$$

$$\frac{K.E_s}{K.E_b} = \frac{\frac{3mv^2}{4}}{3\frac{9mv^2}{8}\frac{1}{2}} = \frac{2}{3}.$$

→ A small sphere of mass 1kg is rolling without slipping with linear speed $v = \sqrt{\frac{200}{7}}$ m/s. It leaves the inclined plane at 'C'. Find

① Linear speed at pt 'C'.

② Find the ratio of rotational & translational

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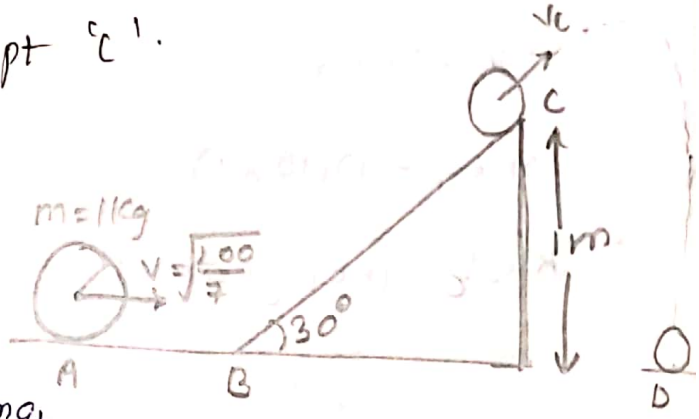
K.E of sphere when it strikes the ground after leaving from pt 'C'.



L.C.E,

$$T.E_B = T.E_C$$

$$\frac{1}{2}mv^2(1+\beta) = K.E_T + K.E_R + mgh$$

$$\frac{1}{2} \times 1 \times \frac{100}{7} \times \frac{7}{5} = K.E_T + \beta K.E_T + 1(10)(1)$$

$$(\beta+1)K_T = 10$$

$$\frac{7}{5}K_T = 10$$

$$K.E_T = \frac{50}{7}$$

$$\frac{1}{2}mv_c^2 = \frac{50}{7}$$

$$mv_c^2 = \frac{100}{7}$$

$$v_c = \sqrt{\frac{100}{7}}$$

now,

$$(K.E_R)_{at\,C} = \beta (K.E_T)_C$$

$$= \frac{2}{5} \times \frac{50}{7}$$

$$= \frac{20}{7} J.$$

when it is flying in air,

$$\tau_g = 0 \Rightarrow \alpha = 0 \Rightarrow \omega = \text{const}$$

$$K.E_R = \frac{20}{7} J = \text{const}$$

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L·C·E b/w C & D:

$mgh = \Delta K.E_T$

$\Delta K.E_T = (1)(10)(1) = 10\ J$

$\therefore (K.E_T)_C = (KE_T)_C + \Delta K.E_T$

$= \frac{50}{7} + 10$

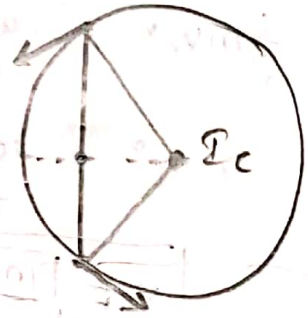
$(K.E_T)_C = \frac{120}{7}\ J$

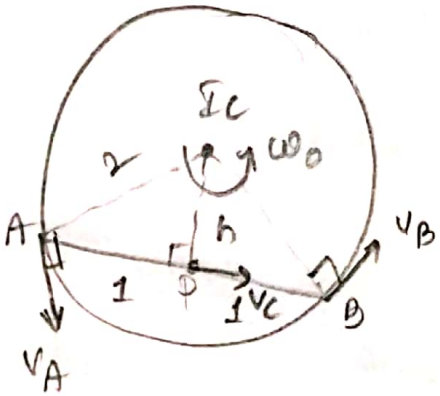
& $(K.E_R)_D = (K.E_R)_C = \frac{20}{7}\ J$

→ rod of mass 'm' & length $\ell = 2m$ is released from vertical position as shown in a fixed spherical shell of radius 2m. Find

① Angular velocity of rod when it becomes horizontal.

② Find velocity of centre of rod at this instant.





$h^2 + 1^2 = 2^2$

$h = \sqrt{3}$

from L·C·E,

$mgh = \frac{1}{2} I_P \omega^2$

$mg(\sqrt{3}) = \frac{1}{2}\left(\frac{m\ell^2}{12} + mh^2\right)\omega^2$

$2\sqrt{3}g = \left(\frac{\ell^2 + 12h^2}{12}\right)\omega^2$

$$\sqrt{\frac{24\sqrt{3}g}{\ell^2 + 12h^2}} = \omega$$

$\omega = \sqrt{\frac{24\sqrt{3}\times 10}{4+36}} = \sqrt{\frac{6\cancel{24}\times\sqrt{3}\times\cancel{10}}{\cancel{40}\ 4}} = \sqrt{6\sqrt{3}}$

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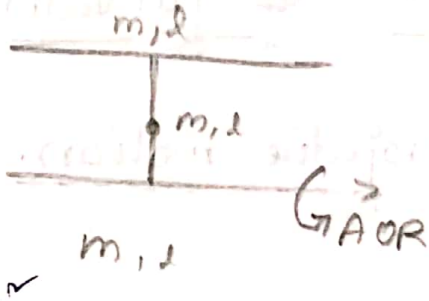
now, velocity of centre of rod,

$$v_c = h\omega$$

& $v_A = R\omega$, $v_B = R\omega$

→ Body is released from vertical position as shown. Find angular velocity of body when it reaches to the vertical position below AOR.



L.C.E,

$$mg\left(\frac{l}{2}+\frac{l}{2}\right) + mg(l+l) = \frac{1}{2}\left(\frac{m(2l)^2}{3} + m(2l)^2\right)\omega^2$$

$$2g + 2g = \frac{1}{2}\left(4l\left(\frac{4l^2}{3}\right)\right)\omega^2$$

$$3g = \frac{2l\omega^2}{3}$$

$$\boxed{\omega = 3\sqrt{\frac{g}{2l}}}$$

K.E of body at this moment,

$$K.E = \frac{1}{2} I_{AOR}\,\omega^2$$

$$= \frac{1}{2} \times \left(\frac{ml^2}{3} + ml^2\right)\omega^2$$

$$= \frac{1}{2} \times \frac{4ml^2}{3} \times \frac{3}{2}\frac{g}{l}$$

$$\boxed{K.E = 3mgl}$$

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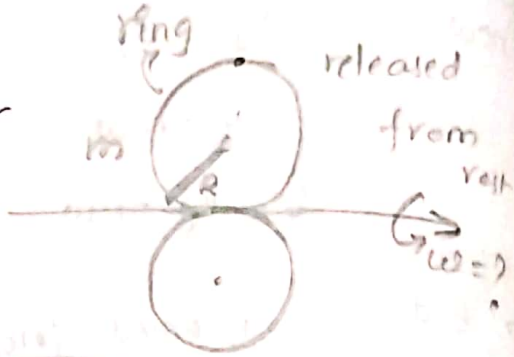
→

$$mg(2R) = \frac{1}{2}\left(\frac{mR^2}{2} + mR^2\right)\omega^2$$

$$4g = \frac{3R}{2}\omega^2$$

$$\boxed{\omega = \sqrt{\frac{8g}{3R}}}$$

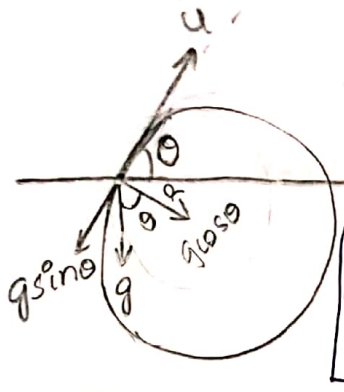


$\omega = 2\sqrt{\frac{2g}{3R}}$

Radius of Curvature of a pt in rolling body:

* projectile motion.

$$\boxed{R_1 = \frac{(u\cos\theta)^2}{g}}$$



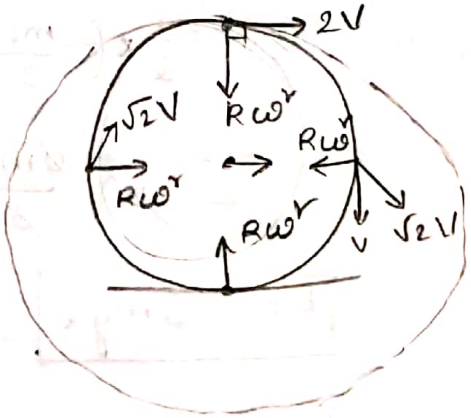
$$\boxed{R = \frac{u^2}{g\cos\theta}}$$

In case of rolling body:

* for pure rolling,

Radius of curvature at highest pt,

$$R_A = \frac{(2v)^2}{R\omega^2} = \frac{4v^2}{R\frac{v^2}{R^2}} = 4R$$



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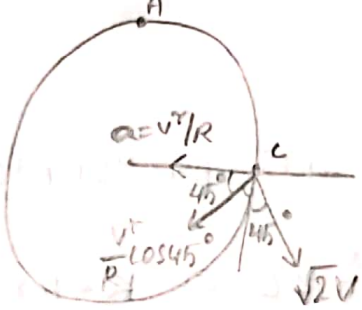
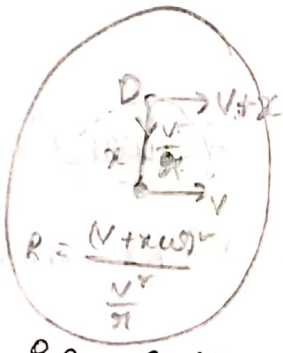
$$R_c = \frac{(\sqrt{2}v)^2}{\frac{v^2}{R}\cos 45^\circ}$$

$$= \frac{2v^2}{\frac{v^2}{R} \times \frac{1}{\sqrt{2}}}$$

$$\boxed{R_c = 2\sqrt{2}R}$$

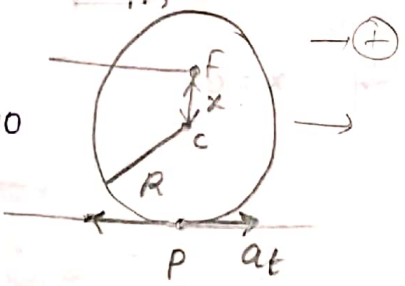




Direction of friction in case of rolling motion:

**

Here, Bottom-most pt has two accelerations.



① linear acceleration,

$$\boxed{a_t = \frac{F}{M}} (\rightarrow)$$

$a_p = a_t - R\alpha$
↓
a_2

② tangential acceleration,

$$a_2 = R\alpha = R\frac{a}{\alpha}$$

$$\boxed{a_2 = \frac{RFx}{Mk^2}} (\leftarrow)$$

∴ net acceleration of bottom-most pt

$$a_p = \frac{F}{M} - \frac{RFx}{Mk^2} = \frac{F}{M}\left(1 - \frac{Rx}{k^2}\right)$$

$$\boxed{a_p = \frac{F}{M}\left(1 - \frac{Rx}{k^2}\right)}$$

points:

① If $1 - \frac{Rx}{k^2} > 0 \Rightarrow k^2 > Rx$ then $a_p = +ve$ i.e, slipping is

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to right.

∴ friction acts "backwards".

② If $1-\frac{Rx}{k^2} < 0 \Rightarrow k^2 < Rx$ then a_p = -ve. i.e, slipping is to left.

∴ friction acts "forward".

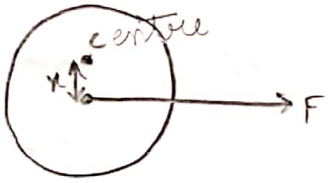
③ If $1-\frac{Rx}{k^2} = 0 \Rightarrow k^2 = Rx$ then $a_p = 0$, i.e, no slipping occurs. $\boxed{f=0}$

④ If $x = 0$

$a_p = \frac{F}{m}$ = +ve

∴ friction acts backwards.

⑤

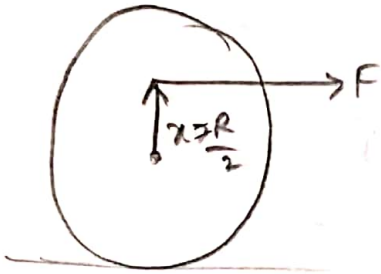


Here $(x = -ve)$

$a_p = \frac{F}{m}\left[1+\frac{Rx}{k^2}\right]$ = +ve

∴ friction acts backwards.

→



① ring: $k^2 = R^2$ & $Rx = R\left(\frac{R}{2}\right) = \frac{R^2}{2}$;

$\beta = 1$, $\frac{k^2}{R^2} = 1$ $\quad k^2 > Rx \quad R^2 > \frac{R^2}{2}$

∴ friction acts backwards.

② disc: $k^2 = \frac{R^2}{2}$; $Rx = R\left(\frac{R}{2}\right) = \frac{R^2}{2}$;

$\beta = \frac{1}{2}$ $\quad k^2 = Rx$

$\therefore \boxed{f=0}$

③ hollow sphere:

$\beta = \frac{2}{3}$ $\quad k^2 = \frac{2R^2}{3}$; $Rx = \frac{R^2}{2}$; $k^2 = Rx$

∴ friction acts backwards.

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④ solid sphere :

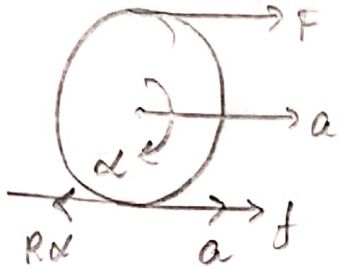
$\beta = \frac{2}{5}$, $k^2 = \frac{2}{5}R^2$, $Rx = R^2$

$k^2 < Rx$

∴ friction acts forward.

→

A solid sphere of mass 'm' and radius 'r'. A force 'f' is applied tangentially at top-most pt, body rolls without slipping (pure rolling). Find acc of CM of sphere.



$F + f = ma$ — ①

$\tau_{cm} = I_{cm}\alpha$

$FR - fR = \frac{2}{5}mR^2(\alpha)$

$F - f = \frac{2}{5}ma$

$F + f = ma$

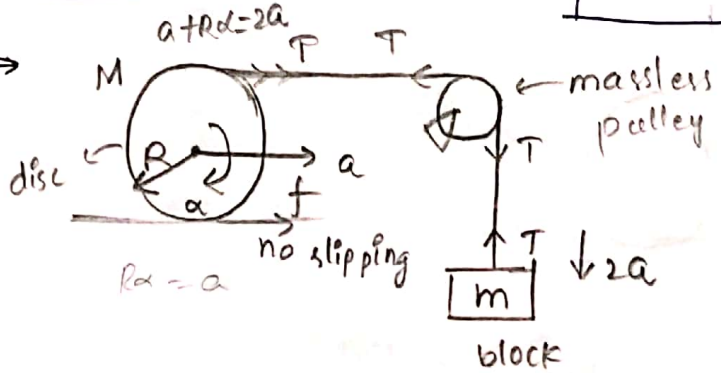
$2F = \frac{7ma}{5}$

$\boxed{F = \frac{7ma}{10}}$

Constraint eqn

$a = R\alpha$

$\boxed{f = \frac{3ma}{10}}$

→ 

Let us assume friction f_s to right.

For discs :

$T + f = Ma$ — ①

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$$\tau_{cm} = I_{cm}\alpha$$

$$TR - fR = \frac{MR^2}{2} \times \alpha \quad \text{— (2)}$$

Constraint eqn, $a = R\alpha$

$$T - f = \frac{Ma}{2}$$

$$T + f = Ma$$

$$\boxed{T = \frac{3Ma}{4}} \qquad \boxed{f = \frac{Ma}{4}}$$

For block

$$mg - T = m(2a)$$

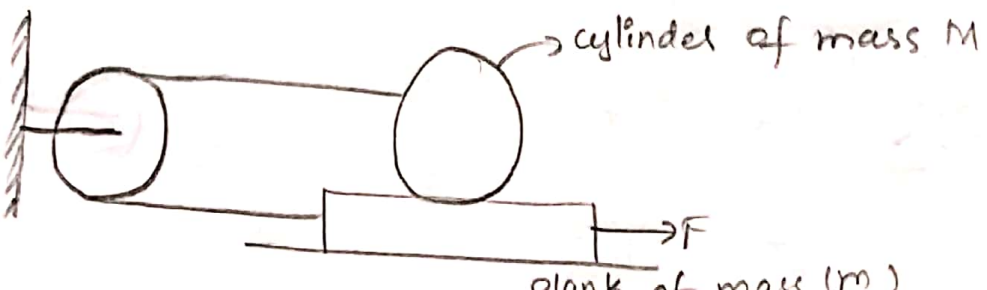
$$mg - \frac{3Ma}{4} = 2ma$$

$$mg = a\left(\frac{8m+3M}{4}\right)$$

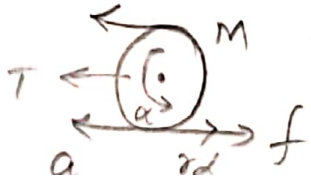
$$\boxed{a = \frac{4mg}{8m+3M}}$$

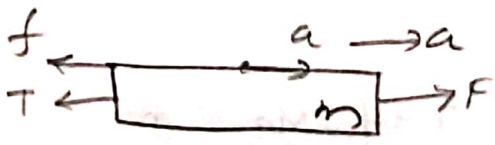
$$\boxed{\alpha = \frac{4mg}{(8m+3M)R}}$$

→



There is no slipping b/w cylinder & plank.





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For cylinder :

$$T - f = Ma \quad -(1)$$

$$\tau_{cm} = I_{cm}\alpha$$

$$fR = \frac{MR^2}{2}\alpha \quad -(2)$$

$\Rightarrow \boxed{T = 2Ma}$, $\boxed{f = Ma}$

for plank :

$$F - f - T = ma \quad -(3)$$

$$F = ma + 3Ma$$

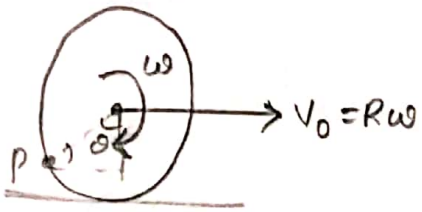
$$\boxed{a = \frac{F}{m+3M}}$$

constraint eqn :

$$a = R\alpha - a \Rightarrow \boxed{a = \frac{R\alpha}{2}} \Rightarrow \alpha = \frac{2F}{(m+3M)R}$$

Here $\boxed{f \le \mu_s N}$

*



P is a particle on the rim of wheel moving

net speed $V = 2R\omega \sin\left(\frac{\omega t}{2}\right)$ (or) $V = 2R\omega\cos\left(\frac{\omega t}{2}\right)$

$V^2 = 4V_0^2\cos^2\left(\frac{\omega t}{2}\right)$

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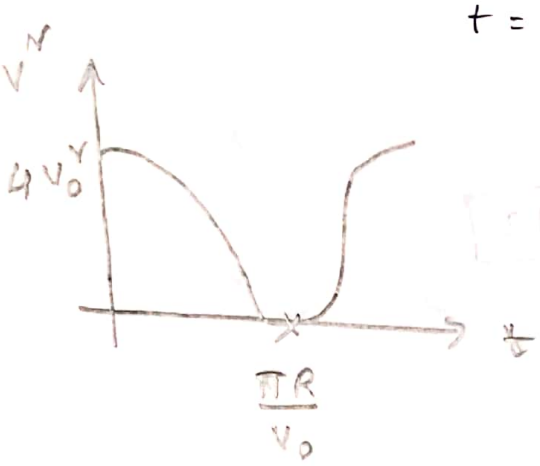
when $t=0 \Rightarrow \cos^2\left(\frac{\omega t}{2}\right)=1 \Rightarrow v^2=4v_0^2$ Ans

$v=0$ when $\cos\left(\frac{\omega t}{2}\right)=0$

$$\frac{\omega t}{2}=\frac{\pi}{2}$$

$$t=\frac{\pi}{\omega}=\frac{\pi R}{v_0}$$

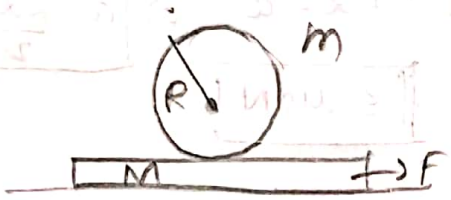


Pure rolling on accelerated frame:

→ plank of mass M is placed on smooth surface. A cylinder of mass (m) & radius (R) is kept on the plank. Sufficient friction exists b/w cylinder & plank for pure rolling If force F is applied on plank then find

① acceleration of planck

② acceleration of cylinder.



M-I:

from ground:

* on cylinder:-

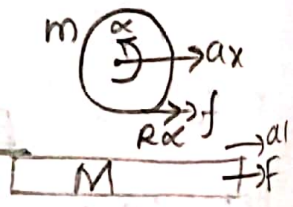
$f=ma_2$ — ①

$\tau_{cm}=I_{cm}\alpha$

$fR=\frac{mR^2}{2}\alpha$ ②

* on plank:-

$F-f=ma_1$ — ③



* constraint eqn:-

$a_2+R\alpha=a_1$ — ④

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unknowns

a_1, a_2, α & f — (4)

& $f \leq \mu_s N$

$\Rightarrow \boxed{\mu_s \geq \frac{f}{N}}$ cohere $N = mg$

$$\boxed{(\mu_s)_{min} = \frac{f}{N}}$$

* If $\mu < (\mu_s)_{min}$

then slipping occurs b/w cylinder & plank.

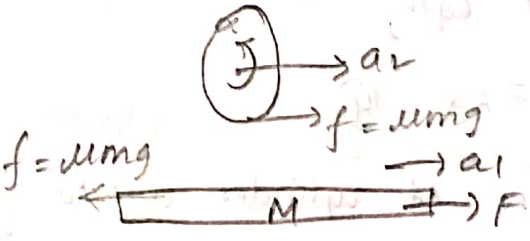
∴ In this case,

$f = \mu N = \mu mg$ (Kinetic)

For cylinder :-

$\mu mg = m a_2$

$\boxed{a_2 = \mu g}$ — (1)



& $fR = I_{cm}\alpha$

$\mu mg R = \frac{mR^2}{2}\alpha$

$\boxed{\alpha = \frac{2\mu g}{R}}$ — (2)

For plank :-

$F - \mu mg = Ma_1$

$\boxed{a_1 = \frac{F - \mu mg}{m}}$ — (3)

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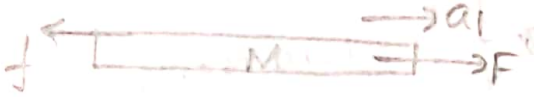
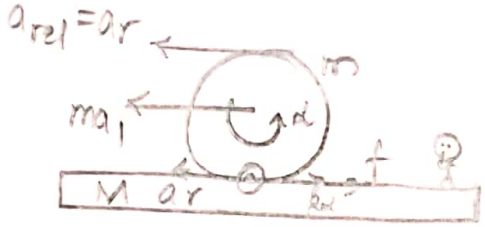
here we should not use constraint eqn.

M-II:

from plank frame:

* $F - f = Ma_1$ — ①





w.r.to plank,

$ma_1 - f = ma_r$ — ②

$\tau_{cm} = I_{cm}\alpha$

$fR = \frac{mR^2}{2}\alpha$

$f = \frac{mR\alpha}{2}$ — ③

constraint eqn,

$a_r = R\alpha$ — ④

Accelesation of cylinder w.r.to ground,

$\bar{a}_2 = \bar{a}_r + \bar{a}_1$

$\therefore \boxed{a_2 = a_1 - a_r}$

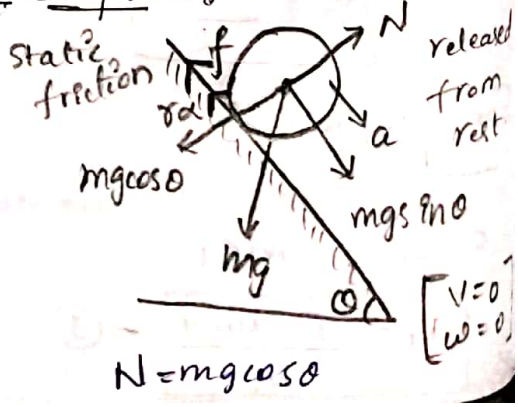
~~Acceleration~~ Pure rolling on inclined plane:

* forces on the body,

N, mg, f

$\therefore mg\sin\theta - f = ma$ — ①

$\tau_{cm} = I_{cm}\alpha$



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$$f R = I_{cm}\alpha$$

$$f = \frac{I_{cm}\alpha}{r}$$

$$\boxed{f = \frac{mk^2\alpha}{r}} \quad -(2)$$

constraint eqn, $a = r\alpha$

$$f = \frac{mk^2 a}{r^2}$$

$$\boxed{f = ma\beta} \quad -(3)$$

eqn (1) becomes

$$mg\sin\theta - ma\beta = ma$$

$$\boxed{a = \frac{g\sin\theta}{1+\beta}}$$ — acceleration of (pure) rolling body motion

* static friction b/w body & surface,

$$f = ma\beta$$

$$\boxed{f = \frac{mg\sin\theta\beta}{1+\beta}}$$

* In general,

$$f \le \mu_s N$$

$$\Rightarrow mg\sin\theta\left(\frac{\beta}{1+\beta}\right) \le \mu_s mg\cos\theta$$

$$\boxed{\tan\theta \le \mu_s\left(\frac{1+\beta}{\beta}\right)} \quad -(4)$$

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∴ max value of θ for pure rolling is,

$$\boxed{\theta = \tan^{-1}\left(\mu_s\left(\frac{1+\beta}{\beta}\right)\right)}$$

* If $\theta > \tan^{-1}\left[\mu_s\left(\frac{1+\beta}{\beta}\right)\right]$

then "slipping occurs" while rolling on plane & friction becomes $\boxed{f = \mu mg\cos\theta,\ \text{kinetic}}$

** eqn (4), $\boxed{\mu_s \geq \left(\frac{\beta}{1+\beta}\right)\tan\theta}$

∴ min coeff of friction for rolling (pure),

$$\boxed{\mu_{min} = \left(\frac{\beta}{1+\beta}\right)\tan\theta}$$

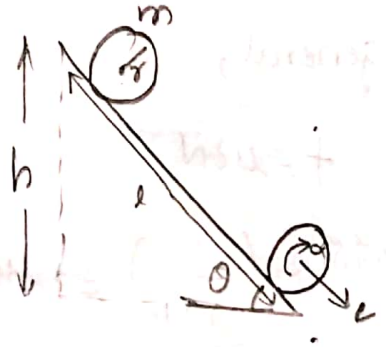
* If $\mu < \left(\frac{\beta}{1+\beta}\right)\tan\theta$

then slipping occurs while rolling,

∴ friction becomes kinetic.

$$\boxed{f = \mu mg\cos\theta}$$

Velocity of body on reaching bottom of incline (in pure rolling):



* use C.E.,

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$mgh = \frac{1}{2}mv^2(1+\beta)$$

$$\boxed{v = \sqrt{\frac{2gh}{1+\beta}}}$$

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here, $v \propto \frac{1}{\sqrt{1+\beta}}$ · for $\beta(\uparrow) \Rightarrow v(\downarrow)$

* $\beta_{solid\ sphere} < \beta_{disc} < \beta_{hollow\ sphere} < \beta_{ring}$.

Time taken by the body to reach the bottom of incline:

$$d = \cancel{ut}^{0} + \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2d}{a}}$$

$$t = \sqrt{\frac{2l(1+\beta)}{g\sin\theta}}$$

$$t = \frac{2h(1+\beta)}{g\sin^2\theta}$$

$$t = \frac{1}{\sin\theta}\sqrt{\frac{2h(1+\beta)}{g}}$$

$$\boxed{t \propto \sqrt{1+\beta}}$$

points:

① For sliding bodies, $\boxed{\beta = 0}$

② $a_{rolling} = \frac{g\sin\theta}{1+\beta}$ (min)

In general,

$$\boxed{\frac{g\sin\theta}{1+\beta} \leq a \leq g\sin\theta}$$

③ for rolling body,

$$a = \frac{g\sin\theta}{1+\beta} \Rightarrow a = \frac{g\sin\theta}{1+\frac{K^2}{r^2}}$$

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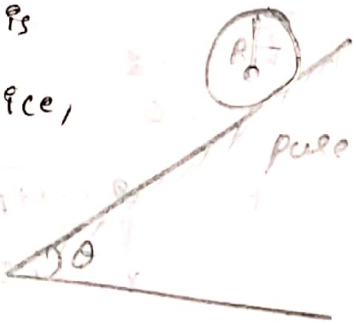
$$a = \frac{mg\sin\theta}{m + \frac{mk^2}{r^2}}$$

$$\boxed{a = \frac{mg\sin\theta}{m + \frac{I_{cm}}{r^2}}}$$

→ shell is filled with liquid of mass 'm' . mass of shell is m_1 is released on inclined plane as shown,

i) Then acceleration of body is

ii) If liquid freezes into ice, then acceleration is



i) $$a = \frac{(m_1+m_2)g\sin\theta}{(m_1+m_2) + \left(\frac{2}{3}m_1\right)}$$

$$\boxed{a = \frac{3(m_1+m_2)g\sin\theta}{5m_1+3m_2}}$$

ii) $$a = \frac{(m_1+m_2)g\sin\theta}{(m_1+m_2) + \left(\frac{2}{3}m_1r^2 + \frac{2}{5}m_2r^2\right)}$$

$$\boxed{a = \frac{15(m_1+m_2)g\sin\theta}{25m_1+21m_2}}$$

(4) Bodies will not perform pure rolling motion on smooth inclined planes.

* but if that inclined plane is accelerating with $\boxed{a = g\tan\theta}$ **, bodies will perform "pure rolling motion".

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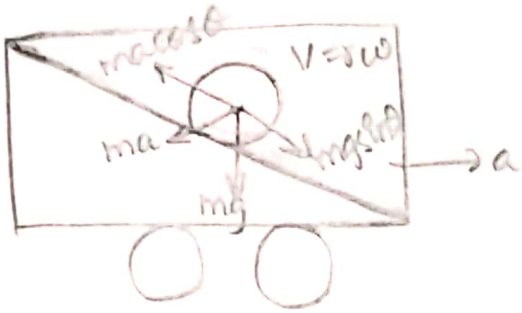
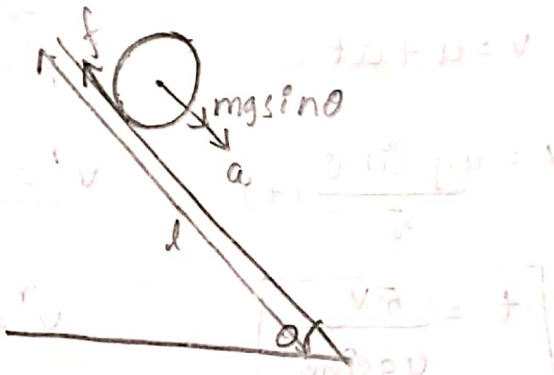


$ma\cos\theta = mg\cos\theta$

$\Rightarrow \boxed{a = g\tan\theta}$

→ A shell is released from top of the inclined plane with inclination θ. Calculate μ_{min} b/w shell & plane to prevent sliding. Assuming friction coeff b/w shell & the plane to be half the calculated μ_{min} value. Find kinetic energy of the shell as it moves a length l down the inclined plane.



$$\mu_{min} = \tan\theta\left(\frac{\beta}{1+\beta}\right)$$

$$= \tan\theta\left(\frac{2/5}{1+2/5}\right)$$

$$\boxed{\mu_{min} = \frac{2}{5}\tan\theta}$$

If $\mu = \frac{1}{2}\mu_{min}$

$$\boxed{\mu = \frac{1}{5}\tan\theta}$$

slipping occurs

$$f = \mu mg\cos\theta = \frac{1}{5}\tan\theta \times mg\cos\theta$$

$$\boxed{f = \frac{1}{5}mg\sin\theta}$$ (kinetic friction)

$$mg\sin\theta - f = ma$$

$$mg\sin\theta - \frac{1}{5}mg\sin\theta = ma$$

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$$\boxed{a = \frac{4g\sin\theta}{5}}$$

and

$$\tau_{cm} = I_{cm}\alpha$$

$$f r = \frac{2}{5} m r^2 \alpha$$

$$\frac{mg\sin\theta}{5} = \frac{2}{3} m r \alpha$$

$$\boxed{\alpha = \frac{3g\sin\theta}{10r}}$$

* we have,

$$v = u + at \qquad v^2 - u^2 = 2al$$

$$v = \frac{4g\sin\theta}{5}(t) \qquad v^2 = 2\left(\frac{4g\sin\theta}{5}\right) l$$

$$\boxed{t = \frac{5v}{4g\sin\theta}} \qquad v^2 = \frac{8g\sin\theta l}{5}$$

& $\omega = \omega_0 + \alpha t$

$$\omega = \left(\frac{3g\sin\theta}{10r}\right)\left(\frac{5v}{4g\sin\theta}\right)$$

$$\omega = \frac{3v}{8r}$$

$$K.E = K.E_T + K.E_R = \frac{1}{2} m v^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} \times m \times \frac{8g\sin\theta(l)}{5} + \frac{1}{2} \times \frac{2}{3} m r^2 \times \frac{9v^2}{64r^2}$$

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$$= \frac{4mgl\sin\theta}{5} + \frac{3}{8} \times \frac{8g\sin\theta\, lm}{5}$$

$$= \frac{32mgl\sin\theta + 3\times mgl\sin\theta}{5\times 8}$$

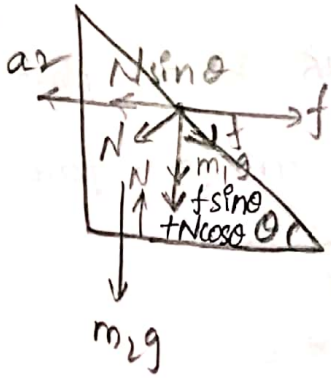
$$= \frac{35}{5\times 8} mgl\sin\theta$$

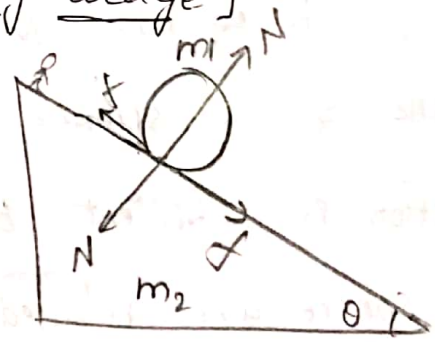
$$\boxed{K.E = \frac{7}{8} mgl\sin\theta}$$

→ A sphere of mass 10kg is placed on inclined surface of rough wedge of inclination 30°. If the mass of wedge is also 10kg. Calculate acceleration of wedge when the sphere is allowed to roll down with out slipping. take $g = 10 m/s^2$ & wedge is placed on smooth surface.

[Pure Rolling Motion on moving wedge]

on wedge:

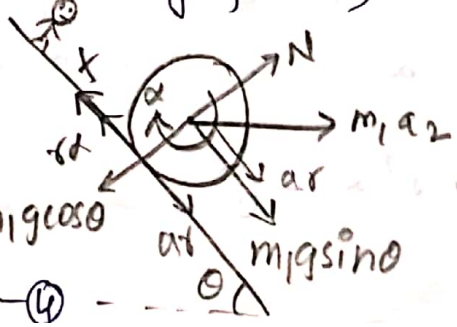




$$N\sin\theta - f\cos\theta = m_2 a_2 \quad -(1)$$

& $$f\sin\theta + N\cos\theta + m_2 g = N_1 \quad -(2)$$

* On spheres (w.r.to wedge frame)



$$m_1 g\sin\theta + m_1 a_2\cos\theta - f = m_1 a_r \quad -(3)$$

and

$$N + m_1 a_2 \sin\theta = m_1 g\cos\theta \quad -(4)$$

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$$\tau_{cm} = I_{cm}\alpha$$

$$f r = \frac{2}{5} m r^2 (\alpha)$$

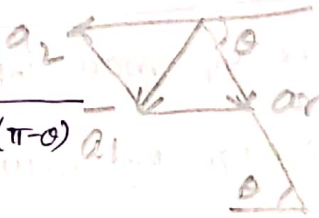
$$f = \frac{2mr\alpha}{5} \quad -(5)$$

constraint eqn,

$$\boxed{a_r = r\alpha} \quad -(6)$$

* finally, net acceleration of sphere

$$\vec{a_1} = \vec{a_r} + \vec{a_2}$$



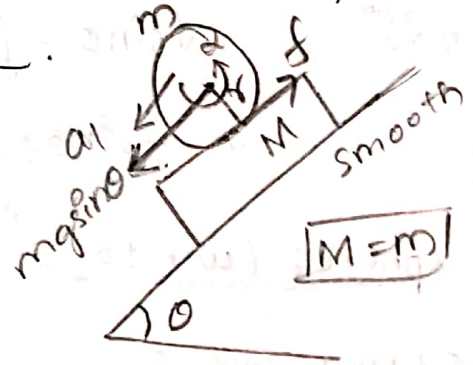
$$a_1 = \sqrt{a_r^2 + a_2^2 + 2(a_r)(a_2)\cos(\pi-\theta)}$$

we have,

$$f \leq \mu_s N$$

$$\boxed{(\mu_s)_{min} = \frac{f}{N}}$$

→ A plank of mass M is placed on a smooth inclined plane & a sphere is also placed over the plank. Friction is sufficient b/w sphere & plank if the plank & sphere are released from rest. Find frictional force on the sphere is ____.

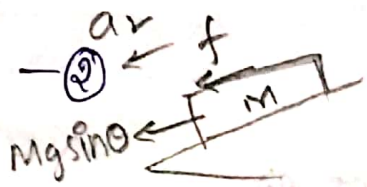


for sphere,

$$mg\sin\theta - f = ma_1 \quad -(1)$$

& $f r = I_{cm}\alpha$

$$fr = \frac{2}{5} m r^2 \alpha \quad -(2)$$



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$$Mg\sin\theta + f = Ma_2 \quad -(3)$$

constraint eqn,

$$a_1 - r\alpha = a_2 \quad -(4)$$

$$\frac{mg\sin\theta - f}{m} - r\left(\frac{5f}{2mr}\right) = \frac{mg\sin\theta + f}{m}$$

$$-f - \frac{5f}{2} = f$$

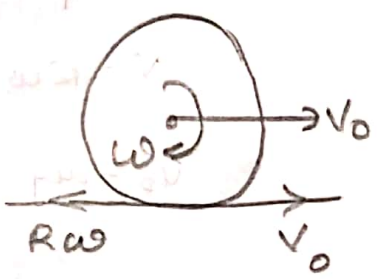
$$\boxed{f = 0}$$

now, $\boxed{\begin{aligned} a_1 &= g\sin\theta \\ a_2 &= g\sin\theta \end{aligned}}$

Rolling with slipping:

Case-(i): forward slipping ($V_0 > R\omega$)

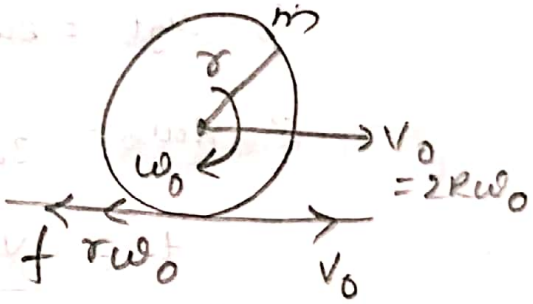
pt of contact slips forward & kinetic friction acts backwards.



→ kinetic friction

$$f = \mu mg \ (\leftarrow)$$

f=0 T=t (pure rolling)



t=0

acceleration of disc,

$$a = \frac{-\mu mg}{m} = -\mu g$$

Angular acceleration of disc,

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$$\alpha = \frac{\tau}{I} = \frac{\mu mgr}{\frac{mr^2}{2}} = \frac{2\mu g}{r}$$

Velocity of disc at any time 't',

$$v = v_0 + at$$

$$\boxed{v = v_0 - \mu g t} \quad -(1)$$

Time for velocity becomes zero

$$v_0 - \mu g t = 0 \Rightarrow \boxed{t = \frac{v_0}{\mu g}}$$

angular of disc at any time 't',

$$\omega = \omega_0 + \alpha t$$

$$\boxed{\omega = \omega_0 + \frac{2\mu g}{r} t} \quad -(2)$$

Cond^n for pure rolling,

$$v = R\omega$$

$$v_0 - \mu g t = R\left(\omega_0 + \frac{2\mu g}{R} t\right)$$

$$v_0 - \mu g t = R\omega_0 + 2\mu g t$$

$$v - R\omega_0 = 3\mu g t \qquad (\because v_0 = 2r\omega_0)$$

$$t = \frac{v_0 - v_0/2}{3\mu g}$$

$$\boxed{t = \frac{v_0}{6\mu g}} \text{ or } \boxed{t = \frac{r\omega_0}{3\mu g}}$$

Velocity of body in pure rolling,

$$v = v_0 + at$$

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$$= V_0 - \mu g \times \frac{V_0}{6\mu g}$$

$$\boxed{v = \frac{5V_0}{6}}$$

$$\omega = \frac{v}{r} = \frac{5V_0}{6r}$$

K.E of the rolling body without slipping,

$$K.E = \frac{1}{2} m v^2 (1+\beta)$$

$$= \frac{1}{2} m \left(\frac{5V_0}{6}\right)^2 \left(1+\frac{1}{2}\right)$$

$$\boxed{(K.E)_f = \frac{25 m V_0^2}{48}}$$

Work done by kinetic friction,

$$W_f = (\Delta K)_{system}$$

$$W_f = (K.E_f)_{system} - (K.E_i)_{sys}$$

$$= \frac{25 m V_0^2}{48} - \left[\frac{1}{2} m V_0^2 + \frac{1}{2} I_{cm} \omega_0^2\right]$$

$$= \frac{25 m V_0^2}{48} - \left[\frac{1}{2} m V_0^2 + \frac{1}{2} \times \frac{m r^2}{2} \times \frac{V_0^2}{4r^2}\right]$$

$$= \frac{25 m V_0^2}{48} - \frac{9 m V_0^2}{16}$$

$$= \frac{25 m V_0^2 - 27 m V_0^2}{48} = \frac{-m V_0^2}{24}$$

distance travelled by body before coming to pure rolling,

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$$S = v_0 t + \frac{1}{2} a t^2 \qquad a = -\mu g$$

$$t = \frac{v_0}{6\mu g}$$

&

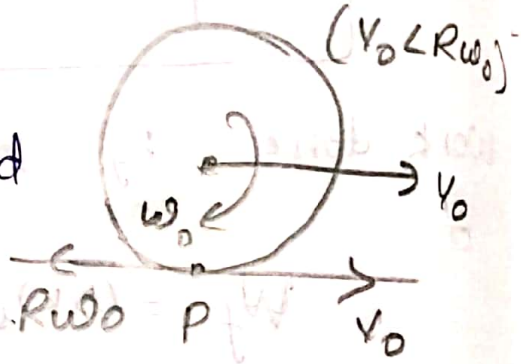
total angle formed by the body before coming to pure rolling,

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

Number of revolutions made by disc before coming to pure rolling,

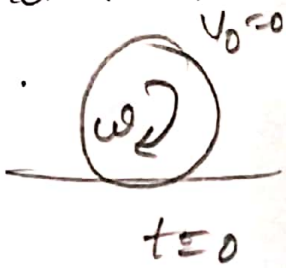
$$\theta = 2\pi N \Rightarrow \boxed{N = \frac{\theta}{2\pi}}$$

Case (ii): Backward slipping

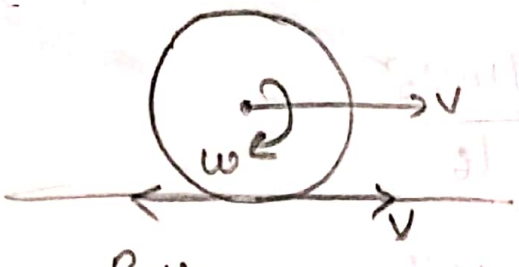


pt of contact slips backward & kinetic friction on body acts forwards.

→ Sphere of mass (m) & radius R is spinned about it's own axis upto 'ω_0' and then gently placed on rough surface.



$v = R\omega$ (pure rolling)



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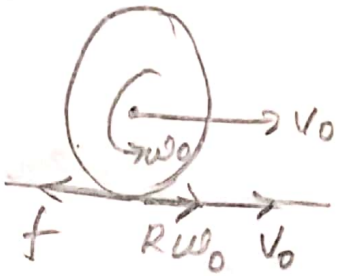
Case-(iii): Reverse spin problem

disc of mass 'm' and radius 'R'

$$f = \mu mg$$

$$a = -\frac{\mu mg}{m}$$

$$\boxed{a = -\mu g}$$



$$\alpha = -\frac{\tau}{I} = -\frac{\mu mgR}{\frac{mR^2}{2}} = \frac{-2\mu g}{R}$$

velocity of disc at any time 't'

$$v = v_0 - \mu g t$$

$$\omega = \omega_0 - \frac{2\mu g}{R} t$$

} both v, ω decreases with time.

when $t = t_1$ $(v = 0)$

$$v_0 - \mu g t_1 = 0$$

$$\boxed{t_1 = \frac{v_0}{\mu g}} \quad -(1)$$

when $t_2 = t_2$ $(\omega = 0)$

$$\omega_0 - \frac{2\mu g}{R} t_2 = 0$$

$$t_2 = \frac{\omega_0 R}{2\mu g}$$

$$= \frac{2v_0}{2\mu g}$$

$$\boxed{t_2 = \frac{2v_0}{\mu g}} \quad -(2)$$

Clearly, $\boxed{t_1 < t_2}$

i.e, v becomes zero first than ω.

Angular velocity of disc at $t = t_1$

$$\omega = \omega_0 - \frac{2\mu g}{R} \times \frac{v_0}{\mu g}$$

$$= \omega_0 - \frac{2}{R} \times \frac{\omega_0 R}{2} \Rightarrow \boxed{\frac{\omega_0}{2} = \omega}$$

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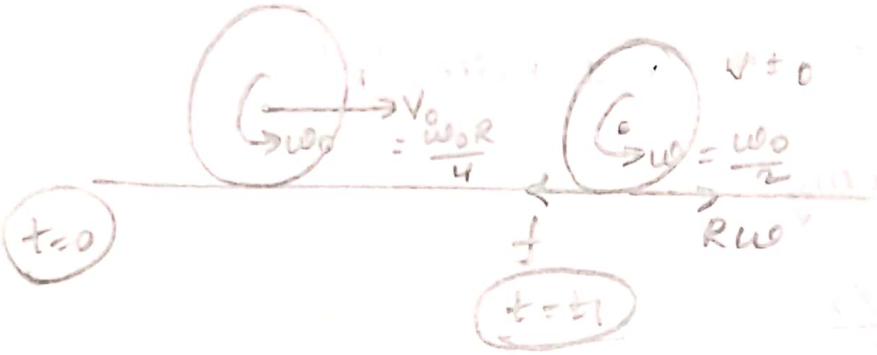

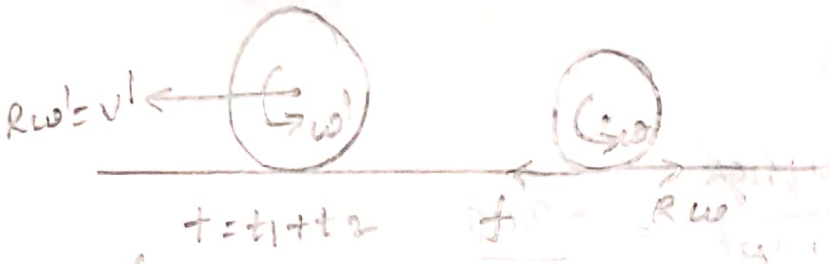

(pure rolling)

for next part,

condn for pure rolling,

$$v' = R\omega'$$

$$(0 + at) = R(\omega + \alpha t)$$

$$\mu g \times t_3 = R\left(\frac{\omega_0}{2} - \frac{2\mu g}{R} t_3\right)$$

$$\mu g t_3 = \frac{R\omega_0}{2} - 2\mu g t_3$$

$$\frac{R\omega_0}{2} = 3\mu g t_3$$

$$\boxed{t_3 = \frac{R\omega_0}{6\mu g}}$$

Total time taken for pure rolling,

$$t = t_1 + t_3 = \frac{v_0}{\mu g} + \frac{R\omega_0}{6\mu g}$$

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$$= \frac{R\omega_0}{4\mu g} + \frac{R\omega_0}{6\mu g}$$

$$\boxed{t = \frac{5R\omega_0}{12\mu g}}$$

velocity of body in pure rolling,

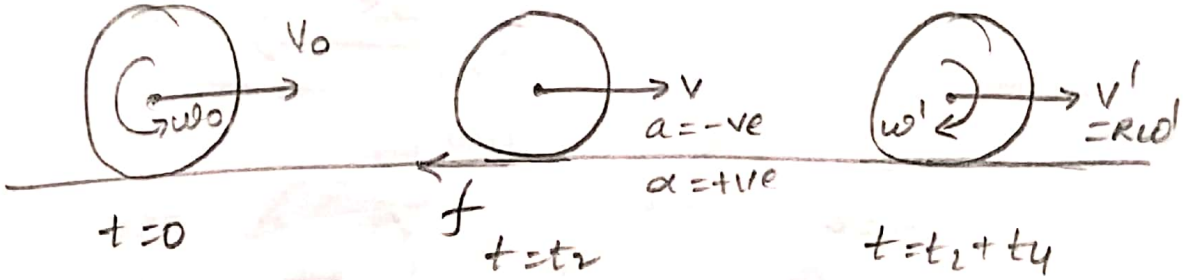
$$v' = a t_3$$

$$\omega' = \omega + \alpha t_3$$

$$\omega' = \frac{\omega_0}{2} - \frac{2\mu g}{R} t_3$$

points:

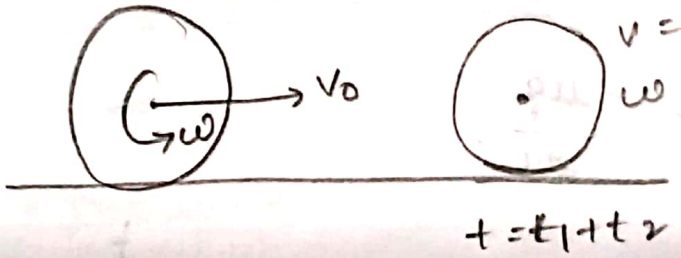
If 'ω' becomes zero first earlier to 'v'



$$v = v_0 + a t_2$$

$\Rightarrow t_2$ = time when $\omega = 0$;

If both 'ω', 'v' becomes zero simultaneously,



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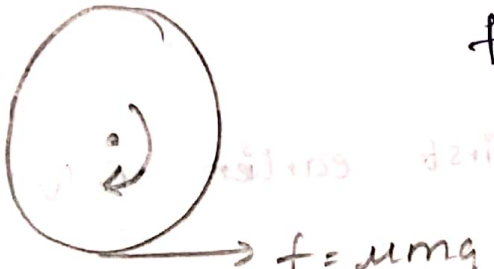
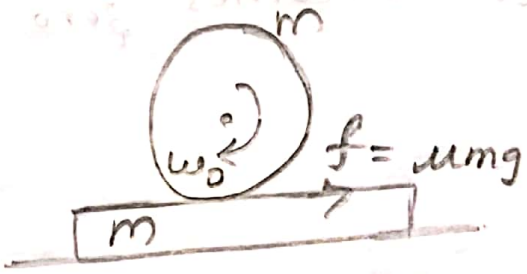
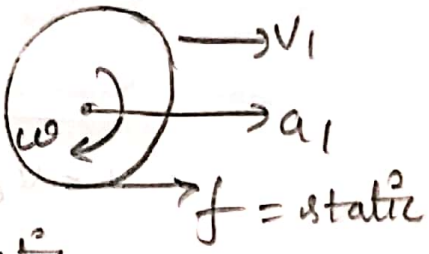
Rolling Motion On Moving Platform:

→ plank of mass m is placed on smooth surface, a sphere of same mass m & radius R is spinned about its axis about ω_0 & then gently placed on plank. If μ is the coeff of friction b/w sphere & plank the time taken by sphere to come to "pure rolling on plank" is



from ground frame,



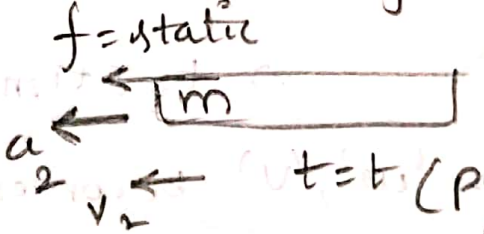


for sphere,

$$a_1 = \frac{\mu m g}{m}$$

$$\boxed{a_1 = \mu g}$$

$$\alpha = \frac{-\mu m g r}{\frac{2}{5} m r^2} = \frac{-\mu g}{\frac{2}{5} r}$$



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for plane,

$$a_2 = \frac{\mu m g}{m} = \mu g$$

constraint eqn,

for pure rolling,

$$\boxed{V_2 = r\omega - V_1}$$

$$a_2 t = r(\omega_0 + \alpha t) - a_1 t$$

$$\mu g t = r\left(\omega_0 - \frac{5\mu g}{2r} t\right) - \mu g t$$

$$2\mu g t = \omega_0 r - \frac{5\mu g t}{2}$$

$$\frac{9\mu g t}{2} = \omega_0 r \Rightarrow \boxed{t = \frac{2\omega_0 r}{9\mu g}}$$

Velocities,

$V_2 = a_2 t$ (plank)

$V_1 = a_1 t$ (sphere)

$$\omega = \frac{V_1 + V_2}{2} \quad \text{(sphere)}$$

$$= \frac{(a_1 + a_2)t}{2}$$

$$= \frac{2\mu g}{2}\left(\frac{2\omega_0 r}{9\mu g}\right)$$

$$\boxed{\omega = \frac{2\omega_0 r}{9}}$$

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displacement of plank,

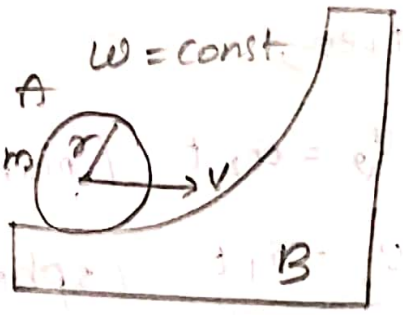
$$S_p = \frac{1}{2} a_2 t^2$$

work done by friction,

$$W_f = (\Delta K)_{system}$$

$$= (K_f)_{sys} - (K_i)_{sys}$$

→ Ring 'A' is initially rolling without slipping with a velocity 'v' on the horizontal surface. The body 'B' having same mass all surfaces are smooth & 'B' have no initial velocity, what will be the max height reached by 'A' on 'B'.



points: ① At highest pt

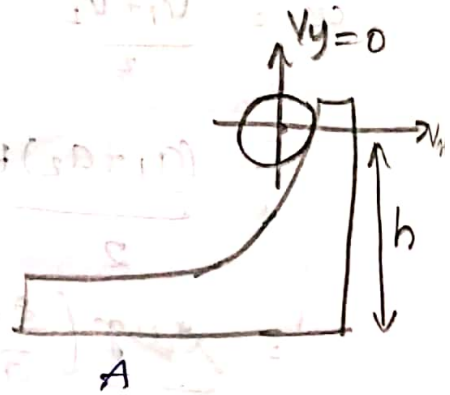
$$V_y = 0$$

② along common normal velocities equal.

Sol: L.C.L.M, (along horizontal)

$$mv = mv_x + mv_x$$

$$\boxed{v_x = v/2}$$



L.C.E,

$$\frac{1}{2} m v^2 + \frac{1}{2} I_{cm} \omega^2 = \frac{1}{2} m v_x^2 + \frac{1}{2} I_{cm} \omega^2 + mgh + \frac{1}{2} m v_x^2$$

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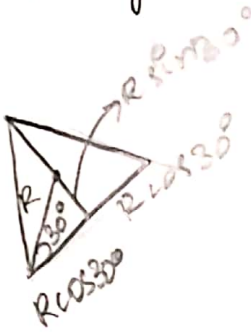
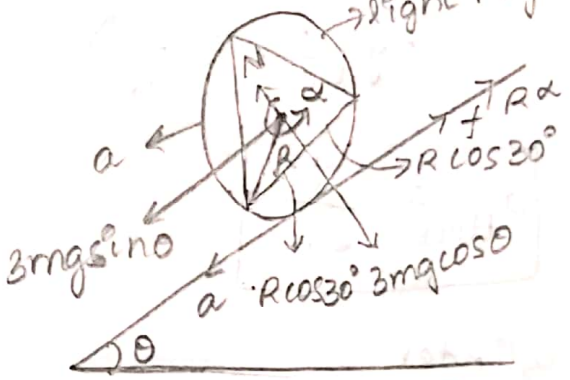
$$\boxed{h = \frac{v^2}{4g}}$$

→ Three identical rods each of mass 'm' forms an equilateral Δle. Rigid body released from rest. Find μ_{min} of static friction that will allow pure rolling motion.





$$3mg\sin\theta - f = 3ma \quad -(1)$$

$$3mg\cos\theta = N \quad -(2)$$

$$\tau_{cm} = I_{cm}\alpha$$

$$fR = \left[\frac{m(2R\cos30°)^2}{12} + m(R\cos45°)^2\right]3\alpha \quad -(3)$$

constraint eqn

$$R\alpha = a \quad -(4)$$

unknowns

a, α, f, N

finally $f \le \mu_s N$

$$\frac{f}{N} \le \mu_s$$

$$(\mu_s)_{min} = \frac{f}{N}$$

$$\boxed{(\mu_s)_{min} = \frac{1}{3}\tan\theta}$$

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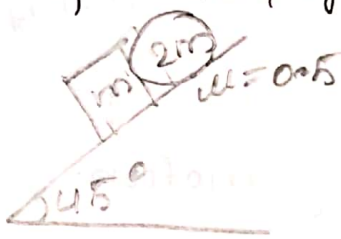
→ A block of mass 'm' & cylinder of mass 'm' are released on an rough inclined plane of $\theta = 45°$. The coefficient of friction for all contact surfaces is 0.5. Find the accelerations of blocks & cylinder.



$$a_{block} = g(\sin\theta - \mu\cos\theta)$$

$$= g\left(\frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{1}}\right)$$

$$\boxed{a_{block} = \frac{g}{2\sqrt{2}}}$$

for cylinder

μ_{min} for pure rolling,

$$\mu_{min} = \tan\theta\left(\frac{\beta}{1+\beta}\right)$$

$$= (1)\left(\frac{1/2}{3/2}\right)$$

$$= \frac{1}{3}$$

$$\boxed{\mu_{min} = 0.33}$$

but given $\mu = 0.5$ $\quad \mu_{min} < \mu$

∴ cylinder is in pure rolling.

now, $a_{cylinder} = \dfrac{g\sin\theta}{1+\beta} = \dfrac{g}{\sqrt{2}\,(3/2)} = \dfrac{2g}{3\sqrt{2}} = \dfrac{\sqrt{2}g}{3}$.

clearly, $\boxed{a_{cy} > a_{block}}$

∴ both the bodies move with common ~~accel~~ acceleration 'a'.

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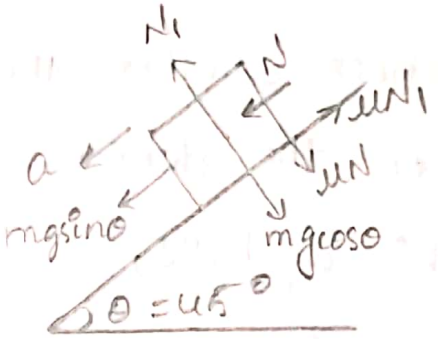
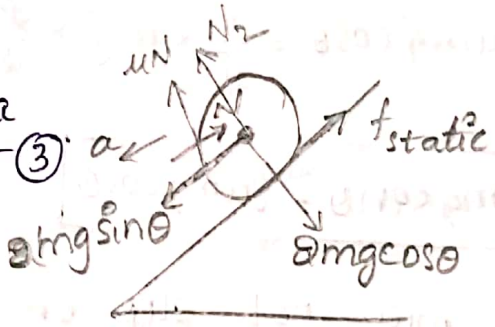
for block



$mg\sin\theta + N - \mu N_1 = ma$ — (1)

$N_1 = mg\cos\theta + \mu N$ — (2)

for cylinder



$2mg\sin\theta - N - f = 2ma$ — (3)

$N_2 + \mu N = 2mg\cos\theta$ — (4)

$\tau_{cm} = I_{cm}\alpha$

$fR - \mu N R = \frac{(2m)R^2}{2}\alpha$

$f - \mu N = mR\alpha$ — (5)

constraint eqn,

$\boxed{a = R\alpha}$ — (6)

unknowns

$a, N_1, N_2, N, \alpha, f$ — (6)

$a = \frac{3g}{5\sqrt{2}}$, $N = \frac{\sqrt{2}mg}{15}$, $f = \frac{\sqrt{2}mg}{3}$, $N_2 = \frac{29\sqrt{2}mg}{30}$

→ A spinning cylinder of mass m and radius r is lowered on a rough inclined plane of $\theta = 30^\circ$ and $\mu = 1/\sqrt{3}$. The cylinder is released at a height of 3R from the horizontal. Find the total time taken by the cylinder to reach the bottom of the cylinder.

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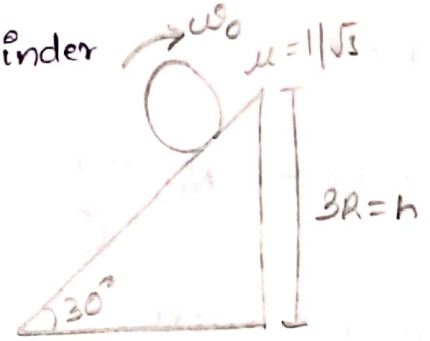
first, check whether the cylinder slipps on the plane.



for this we have,

$$mg\sin\theta = \frac{mg}{2}$$

& $\mu mg\cos\theta = \frac{1}{\sqrt{3}} \times mg \times \frac{\sqrt{3}}{2}$

$$\boxed{mg\sin\theta = \mu mg\cos\theta}$$

cylinder will not slip on the plane till its angular velocity becomes zero.

now, $\tau_{cm} = I_{cm}\alpha$

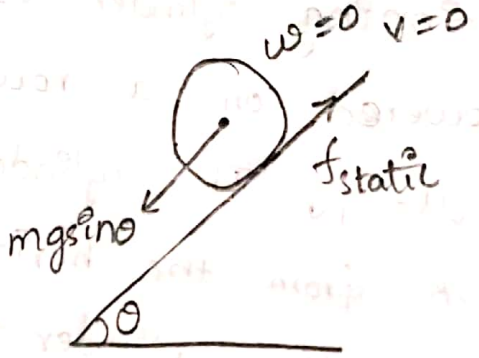
$$\alpha = -\frac{\tau}{I}$$

$$= -\frac{\mu mg\cos\theta R}{\frac{mR^2}{2}}$$

$$= -\frac{\mu g\sqrt{3}}{2 \times \frac{1}{2} \times R}$$

$$\boxed{\alpha = -\frac{\mu g\sqrt{3}}{R}}$$

After $t = t_1$



cylinder starts rolling down the plane without slipping. at $t = t_1$

* time taken by the cylinder to reach bottom

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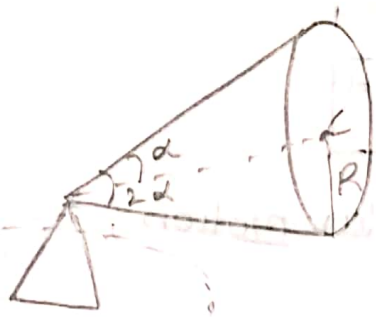
of plane.

$$t_2 = \frac{1}{\sin\theta}\sqrt{\frac{2h(1+\beta)}{g}}$$

∴ total time taken to reach the bottom of plane. $t = t_1 + t_2$

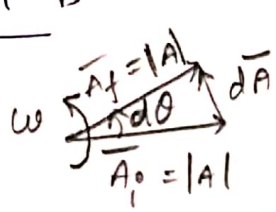
In this case, cylinder will never slides up the plane.

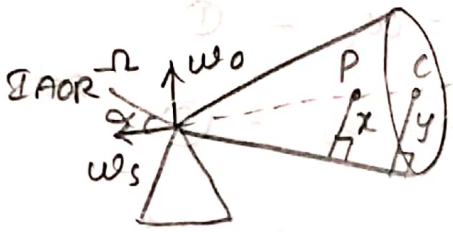


A round cone rolls uniformly without slipping over a horizontal plane as shown. The cone apex is hinged at the pt 'O' which is on the same level with the pt 'C'. The cone base centre. The velocity of 'C' is 10cm/s. Find the magnitude of i) the vector of the angular velocity of cone and the angle it makes with vertical.
ii) The vector of the angular acceleration of cone.

Ω = net angular velocity of cone.

Concept B





for small angle 'dθ' dA becomes arc length

$$dA = (d\theta)A$$

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$$\frac{dA}{dt} = A \cdot \frac{d\theta}{dt}$$

$$\boxed{\frac{dA}{dt} = \omega A}$$

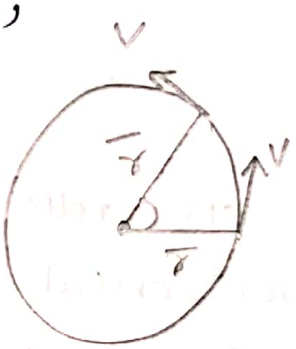
In vector form,

$$\boxed{\frac{d\bar{A}}{dt} = \bar{\omega} \times \bar{A}}$$

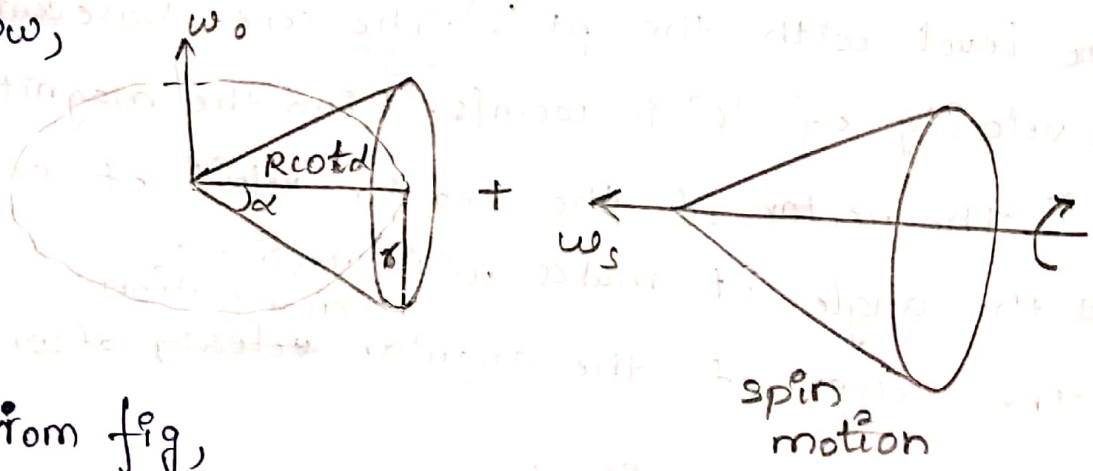
In uniform circular motion,

$$\frac{d\bar{r}}{dt} = \bar{\omega} \times \bar{r}$$

$$\boxed{\frac{d\bar{v}}{dt} = \bar{\omega} \times \bar{v}}$$



now,



from fig,

$$\Omega \cos\alpha = \omega_s \quad - (1)$$

$$\Omega \sin\alpha = \omega_0 \quad - (2)$$

and $$v_c = R \cot\alpha \, \omega_0 \quad - (3)$$

from eqn (2)

$$\Omega \sin\alpha = \frac{v_{oc}}{R \cot\alpha}$$

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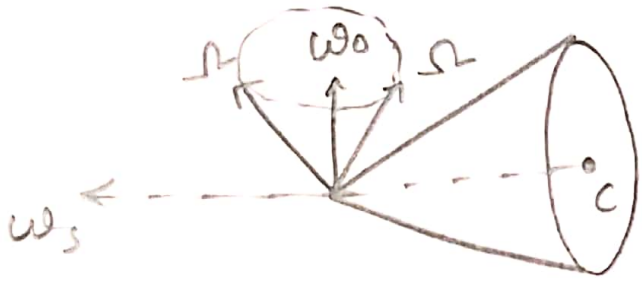
$$\Omega = \frac{V_c}{R\cos\alpha} = \frac{10 \times 2}{5 \times \sqrt{3}}$$

$$\boxed{\Omega = 4/\sqrt{3} \text{ rad/s}}$$ 60° with vertical.

now,



Here, ω_0 is not changing but cone is rotating at 'ω_0'.

now, $\bar{\alpha} = \frac{d\bar{\Omega}}{dt}$

$= \frac{d}{dt}(\bar{\omega_0} + \bar{\omega_s})$

$= \frac{d\bar{\omega_0}}{dt}^{0} + \frac{d}{dt}\bar{\omega_s}$

$$\boxed{\bar{\alpha} = \bar{\omega_0} \times \bar{\omega_s}}$$ Here $\bar{\omega_0} \perp \bar{\omega_s}$

$\bar{\alpha} = (\omega_0)(\omega_s)$

$= \frac{2}{\sqrt{3}} \times 2$

$$\boxed{\bar{\alpha} = 4/\sqrt{3} \text{ rad/sec}^2}$$

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Angular Momentum:

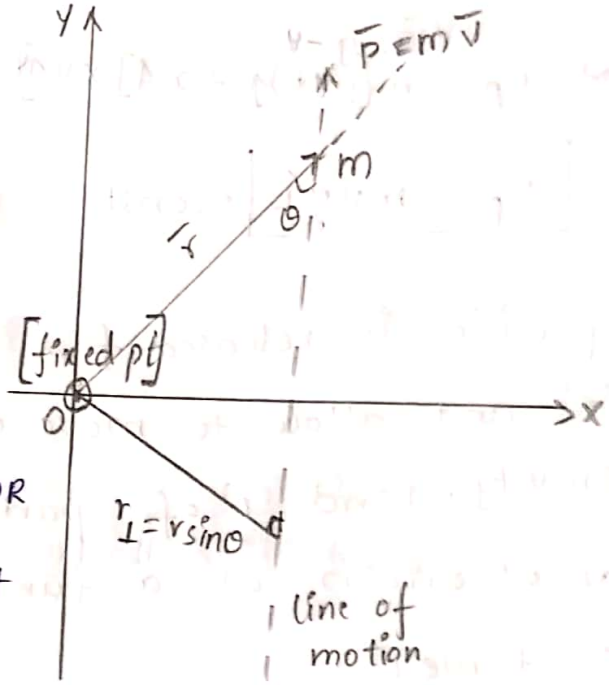
It is the moment of linear momentum of the body about the AOR is called angular momentum about AOR.

i.e, L = (Arm moment) x linear moment

$L = (r_\perp)mv$

$L = mvr_\perp$

$\boxed{L = mvr\sin\theta}$



It is an "axial vector" whose direction is along AOR and is decided by right hand thumb rule.

$\boxed{L_p = \bar{r} \times \bar{P}}$

$= \bar{r} \times m\bar{v}$

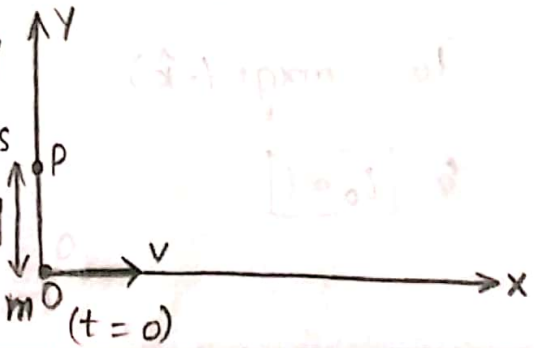
$= m(\bar{r} \times \bar{v})$

Units: kgm^2/s

Cases:-

① In case of translational motion of particle:

→ A particle of mass (m) moving with const speed (v) along x-axis as shown. Find Angular momentum of particle about 'p' as a



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function of time 't'?

Sol: we have,

$$\bar{L}_p = m(\bar{r} \times \bar{v})$$

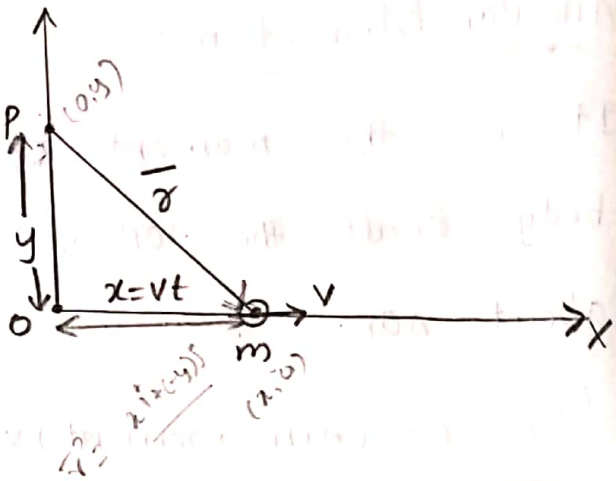
where,

$$\bar{r} = y(-\hat{\jmath}) + x\hat{\imath}$$

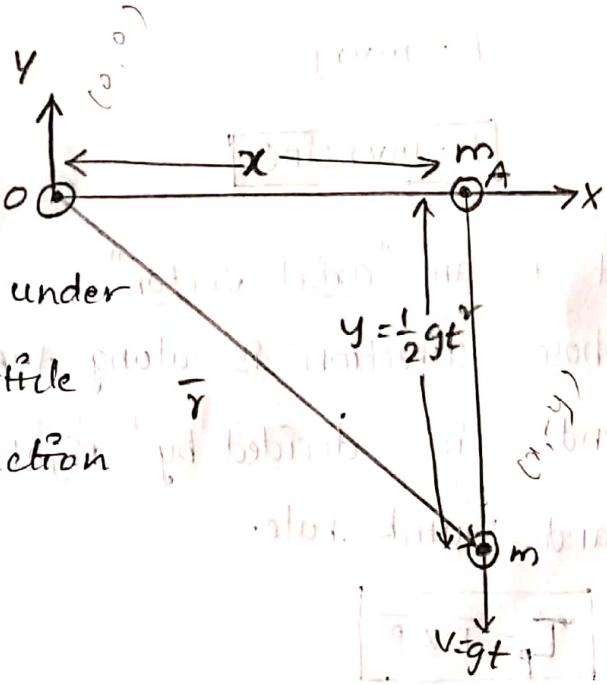
$$\bar{v} = v\hat{\imath}$$

now $L_p = m[[(-y)\hat{\jmath} + x\hat{\imath}] \times v\hat{\imath}]$

$$\boxed{\bar{L}_p = mvy\hat{k}} = \text{const}$$



(2) → particle is released from 'A' and allow to move under gravity. Find 'L' of particle about origin as a function of time?



$$\bar{L}_o = m(\bar{r} \times \bar{v})$$

where

$$\bar{r} = x\hat{\imath} + y(-\hat{\jmath})$$

$$\bar{v} = gt(-\hat{\jmath})$$

$$\bar{L}_o = m[(x\hat{\imath} - y\hat{\jmath}) \times gt(-\hat{\jmath})]$$

$$\boxed{\bar{L}_o = -mxgt\hat{k}}$$

$$\bar{L}_o = mxgt(-\hat{k})$$

& $\boxed{L_o \propto t}$

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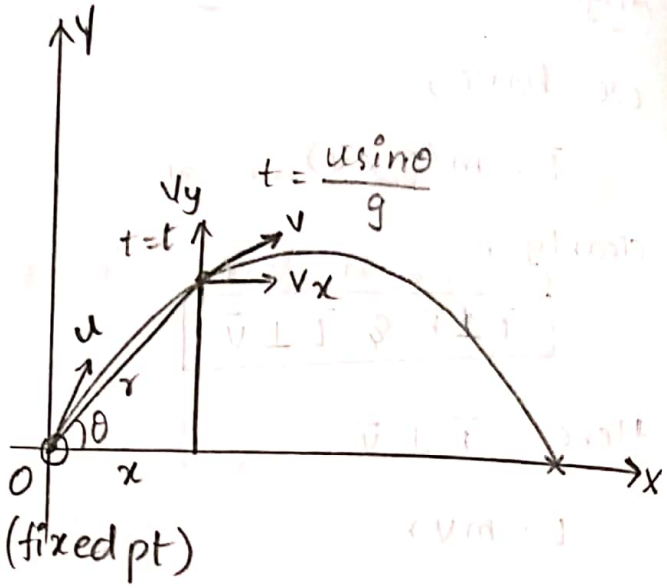
③ In case of projectile Motion :



$\bar{L}_0 = m(\bar{r} \times \bar{v})$

here $\bar{r} = x\uparrow + y\uparrow$

where $x = u\cos\theta t$

& $y = u\sin\theta t - \frac{1}{2}gt^2$

& $\bar{v} = v_x\uparrow + v_y\uparrow$

where $v_x = u\cos\theta$, $v_y = u\sin\theta - gt$

now, $$\bar{r} \times \bar{v} = \begin{vmatrix} \uparrow & \uparrow & \hat{k} \\ x & y & 0 \\ v_x & v_y & 0 \end{vmatrix}$$

$$= \uparrow(0-0) - \uparrow(0-0) + \hat{k}(xv_y - yv_x)$$

now, $\bar{L}_0 = m(\bar{r} \times \bar{v})$

$= m(xv_y - yv_x)\hat{k}$

$= m\left[(u\cos\theta t)(u\sin\theta - gt) - (u\sin\theta t - \frac{1}{2}gt^2)(u\cos\theta)\right]\hat{k}$

$= m\left[u^2\sin\theta\cos\theta t - ugt^2\cos\theta - u^2\sin\theta\cos\theta t + \frac{1}{2}ugt^2\cos\theta\right]\hat{k}$

$$\boxed{\bar{L}_0 = -\frac{1}{2}mugt^2\cos\theta(\hat{k})}$$

$\bar{L}_0 = mugt^2\cos\theta(-\hat{k})$

└ along (-z-axis)

& $\boxed{L_0 \propto t^2}$

① At highest pt — $t = \frac{u\sin\theta}{g}$

② At the time striking the ground, $t = \frac{2u\sin\theta}{g}$.

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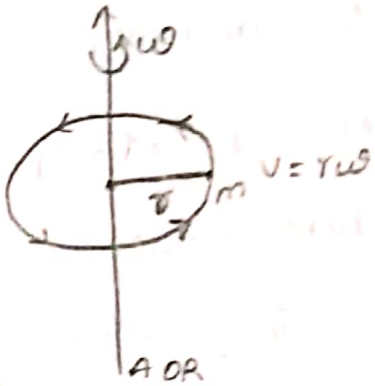
Angular Momentum of particle in Circular Motion:



we have,

$$\bar{L} = m(\bar{r} \times \bar{v})$$

Clearly,

$$\boxed{\bar{L} \perp \bar{r} \text{ & } \bar{L} \perp \bar{v}}$$

Here $\bar{r} \perp \bar{v}$

$$L = mvr$$

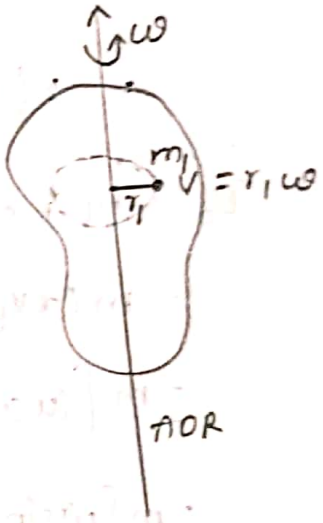
$$L = m(r\omega)r$$

$$L = mr^2\omega$$

$$L = I\omega$$

$$\boxed{\bar{L} = I\bar{\omega}}$$

For a rotating rigid body:



Angular momentum of i^{th} particle about AOR,

$$d\bar{L}_i = m_i(\bar{r}_i \times \bar{v}_i)$$

$$dL_i = m_i r_i v_i$$

$$dL_i = m_i r_i^2 \omega$$

∴ Total angular momentum of body,

$$\bar{L} = \sum_{i=1}^{n} d\bar{L}_i = \sum_{i=1}^{n} m_i r_i^2 \omega$$

$$\boxed{\bar{L} = I\bar{\omega}}$$

I = Moment of inertia of body about AOR.

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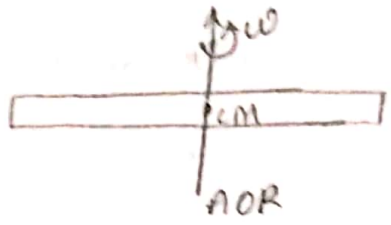
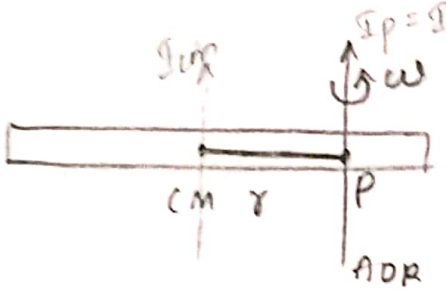
For centroidal axis of rotation,



$$\bar{L} = I_{cm}\,\bar{\omega}$$

For non-centroidal rotation,

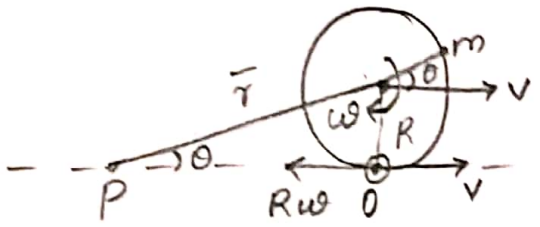


$$L_p = I_p\,\omega$$

$$L_p = [I_{cm} + mr^2]\,\omega$$

where $I_p = I_{cm} + mr^2$.

In case of combined Motion (both translational & rotational) :



$$\sin\theta = \frac{R}{r}$$

$$R = r\sin\theta$$

Angular momentum of body about 'P' is,

$$L_p = L_{\text{of CM}} + L_{\text{about CM}}$$

(translational) (pure rotation)

It depends on pt of observation

It doesn't depends on pt of observation.

$$\bar{L}_p = m(\bar{r}\times\bar{v}) + I_{cm}\,\bar{\omega}$$

$$\bar{L}_p = mrv\sin\theta\,(\odot W) + I_{cm}\,\omega\,(\odot W)$$

$$\bar{L}_p = (mrv\sin\theta + I_{cm}\omega)\,(\odot W)$$

For pure rolling, $V = R\omega$

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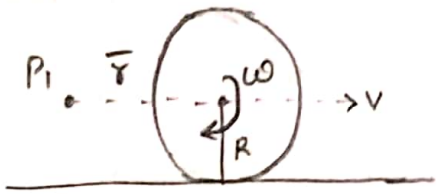
$$\bar{L}_p = (mR^2 + I_{cm})\omega \quad (\odot \cdot W)$$

$$\boxed{\bar{L}_p = I_o \omega \ (\odot \cdot W)} \Rightarrow M \cdot I \text{ of body about an axis passing through I.C.}$$

Clearly, about I.C body is in pure rotation,

'L' of body about 'P_1' is

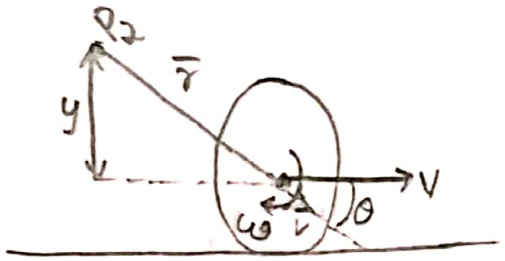


$$\bar{L}_{P_1} = m(\bar{r} \times \bar{v}) + I_{cm}(\bar{\omega})$$

Here $\bar{r} \parallel \bar{v}$

$$\bar{L}_{P_1} = I_{cm}\,\bar{\omega}$$

*



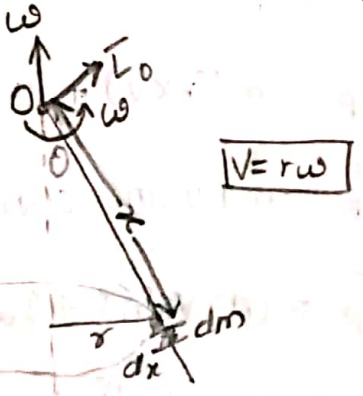
$$\bar{L}_{P_2} = m(\bar{r} \times \bar{v}) + I_{cm}\,\bar{\omega}$$

$$\bar{L}_{P_2} = mrv\sin\theta \ (A \cdot C \cdot W) + I_{cm}\omega \ (\odot \cdot W)$$

$$= mVy \ (A \cdot C \cdot W) - I_{cm}\omega \ (A \cdot C \cdot W)$$

$$= (mVy - I_{cm}\omega)(A \cdot C \cdot W)$$

→ Find Angular momentum of rod of mass 'm' & length 'l' inclined at angle θ with vertical as shown.



$\sin\theta = \frac{r}{x}$ $\quad \boxed{V = r\omega}$

now, Angular momentum of small element about 'O' is

$$dL_0 = dm\,(\bar{r} \times \bar{v})$$

$$= \left(\frac{m}{l}dx\right)(x v)$$

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$$= \left(\frac{m}{l} x\,dx\right) r\omega$$

$$= \left(\frac{m}{l} x\,dx\right)(x\sin\theta)\,\omega$$

$$dL_o = \frac{m}{l}\sin\theta\,\omega x^2\,dx$$

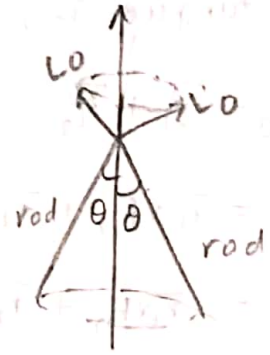
∴ Total angular momentum,

$$L_o = \int dL_o = \int_{x=0}^{x=l} \frac{m}{l}\sin\theta\cdot\omega x^2\,dx$$

$$L_o = \frac{m}{l}\omega\sin\theta\,\frac{l^3}{3}$$

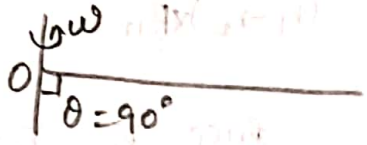
$$\boxed{L_o = \frac{ml^2}{3}\omega\sin\theta} \quad *$$

$(L_o = I_o\omega)$



If rod is rotating in horizontal position

$$\boxed{\bar{L}_o = \frac{ml^2}{3}\omega}$$



If AOR of rod is passing through CM then L, ω may or may not be in same direction. But if it is planar motion then L, ω be in same direction.

Relation b/w Torque and angular momentum:

we have,

$$\bar{\tau} = I\bar{\alpha} \quad \text{(for rotating body)}$$

$$\bar{\tau} = I\cdot\frac{d\bar{\omega}}{dt}$$

$$\bar{\tau} = \frac{d}{dt}(I\bar{\omega}) \quad [I = \text{const}]$$

$$\boxed{\bar{\tau} = \frac{d\bar{L}}{dt}} \quad [\bar{L} = I\bar{\omega}]$$

ext torque Internal torque

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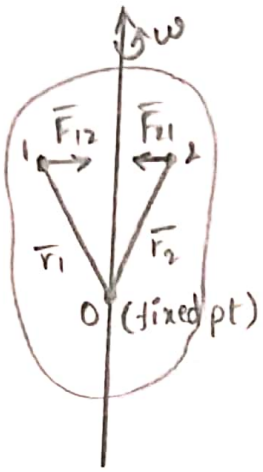


we have,

$$\bar{F}_{12} = -\bar{F}_{21}$$

Torque due to internal forces about 'O' are

① $\bar{\tau}_1 = \bar{r}_1 \times \bar{F}_{12}$

② $\bar{\tau}_2 = \bar{r}_2 \times \bar{F}_{21}$

* net torque is,

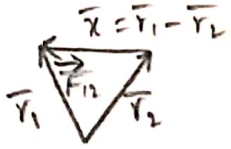
$$\bar{\tau}_0 = \bar{\tau}_1 + \bar{\tau}_2$$

$$= \bar{r}_1 \times \bar{F}_{12} + \bar{r}_2 \times \bar{F}_{21}$$

$$= \bar{r}_1 \times \bar{F}_{12} + \bar{r}_2 \times (-\bar{F}_{12})$$

$$= \bar{r}_1 \times \bar{F}_{12} - \bar{r}_2 \times \bar{F}_{12}$$

$$\bar{\tau}_0 = (\bar{r}_1 - \bar{r}_2) \times \bar{F}_{12}$$



since $\bar{r}_2 = \bar{r}_1 - \bar{r}_2$ & $\bar{F}_{12}$ are anti parallel,

$$(\bar{r}_1 - \bar{r}_2) \times \bar{F}_{12} = 0$$

$\bar{r}_2 + \bar{x} = \bar{r}_1$

$$\boxed{\bar{x} = \bar{r}_1 - \bar{r}_2}$$

$$\Rightarrow \boxed{\bar{\tau}_0 = 0}$$

i.e, net torque due to internal forces is always "zero".

Finally $\boxed{\bar{\tau}_{net} = \frac{d\bar{L}}{dt}}$ – Newton's 2nd law of rotation.

Statement –

Net external torque of a body is ~~calle~~ equal to rate of change of angular momentum.

where as for translation motion,

$$\boxed{\bar{F}_{net} = \frac{d\bar{P}}{dt}}$$

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Angular Impulse:

Angular Impulse (A·I)

$$= \int_{t_1}^{t_r} \bar{\tau} \cdot dt = \Delta \bar{L}$$

Angular Impulse - Momentum theorem,

$$\bar{L}_i \pm \int_{t_1}^{t_r} \bar{\tau} \cdot dt = \bar{L}_f$$

For translation motion:

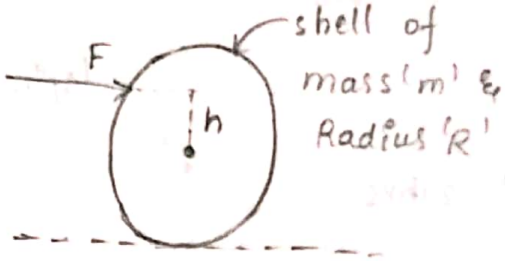
linear impulse,

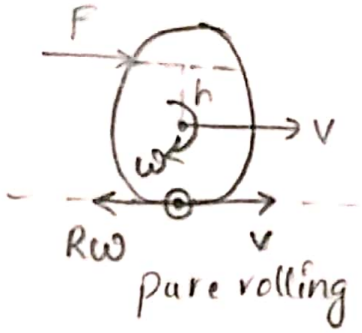
$$J = \int_{t_1}^{t_r} \bar{F} \cdot dt = \Delta \bar{P}$$

Impulse-momentum theorem,

$$\bar{P}_i \pm \int_{t_1}^{t_r} \bar{F} \cdot dt = \bar{P}_f$$

→ Immediately, shell comes to "pure rolling motion". Find 'h'?





Impulse eqns,

$$0 + \int F \cdot dt = mv \quad -(1)$$

& $$0 + \int \tau \cdot dt = I_{cm}\,\omega$$

$$\int Fh \cdot dt = \frac{2}{3} mR^2 \omega \quad -(2)$$

constraint eqn,

$$V = R\omega \quad -(3)$$

L·C·A·M

$$mvh = \frac{2}{3} mR^2 \left(\frac{v}{R}\right)$$

$$\boxed{h = \frac{2}{3}R}$$

'ω' changes when pt of application of force is changes.

when $h > \frac{2}{3}R$, V doesn't change but ω increases.

$\therefore V < R\omega$ (backward slipping takes place).

friction will be in forward direction.

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Law of conservation of angular momentum:

we have,

$$\bar{\tau}_{ext} = \frac{d\bar{L}}{dt}$$

When $\bar{\tau}_{ext} = 0$

$$\frac{d\bar{L}}{dt} = 0$$

$$\Rightarrow \boxed{\bar{L} = \text{const}}$$

i.e, $\boxed{\bar{L}_f = \bar{L}_i}$

In case of translation,

$$\bar{F}_{ext} = \frac{d\bar{P}}{dt}$$

If $\bar{F}_{ext} = 0$

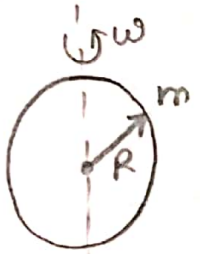
then $\frac{d\bar{P}}{dt} = 0$

$$\Rightarrow \boxed{\bar{P} = \text{const}}$$

i.e, $\boxed{\bar{P}_i = \bar{P}_f}$

Law of conservation of linear momentum

Applications:

① 

$L_{spin} = I\omega = \frac{2}{5} mR^2\omega$

$T_1 = 24\text{hrs}$

a) Keeping mass of earth const:-

If radius of earth is changing,

then $I_1\omega_1 = I_2\omega_2$

$$\frac{2}{5} m R_1^2 \frac{2\pi}{T_1} = \frac{2}{5} m R_2^2 \frac{2\pi}{T_2}$$

$$\boxed{\frac{R_1^2}{T_1} = \frac{R_2^2}{T_2}}$$

b) Keeping density of earth const:-

$$m = \frac{4}{3}\pi R^3 \rho$$

If radius of earth is changing,

then $I_1\omega_1 = I_2\omega_2$

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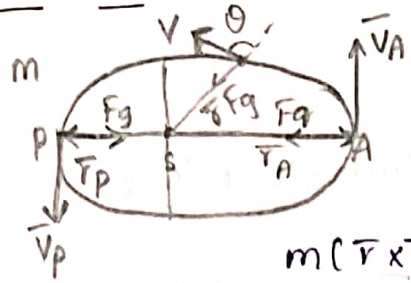
$$\frac{2}{5} m_1 R_1^2 \omega_1 = \frac{2}{5} m_2 R_2^2 \omega_2$$

$$\frac{4}{3}\pi R_1^3 \rho \cdot R_1^2 \cdot \frac{2\pi}{T_1} = \frac{4}{3}\pi R_2^3 \rho \cdot R_2^2 \cdot \frac{2\pi}{T_2}$$

$$\boxed{\frac{R_1^5}{T_1} = \frac{R_2^5}{T_2}}$$

② Kepler's Law :-



$\bar{\tau}_g = 0$

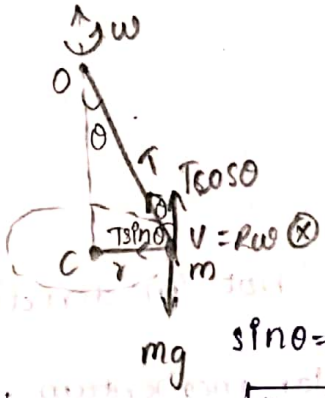
$\boxed{\bar{L} = \text{const}}$

$m(\bar{r} \times \bar{v}) = \text{const}$

$\boxed{m r_A V_A = m r_B V_B = MrV\sin\theta}$

law of conservation of angular momentum.

③ Angular Momentum of a conical pendulum :-



$T\cos\theta = mg$

$T\sin\theta = \frac{mv^2}{r}$

$\boxed{\tan\theta = \frac{v^2}{rg}}$

$\sin\theta = \frac{r}{l}$

$\boxed{r = l\sin\theta}$

and $T\cos\theta = mg$

$T\sin\theta = \frac{m(r\omega)^2}{r}$

$= \frac{m r^2 \omega^2}{r}$

$T\sin\theta = mr\omega^2$

$T\sin\theta = m(l\sin\theta)\omega^2$

$\boxed{T = ml\omega^2}$

$ml\omega^2\cos\theta = mg \Rightarrow \boxed{\cos\theta = \frac{g}{l\omega^2}}$

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As ω increases, $\cos\theta$ decreases, θ increases.

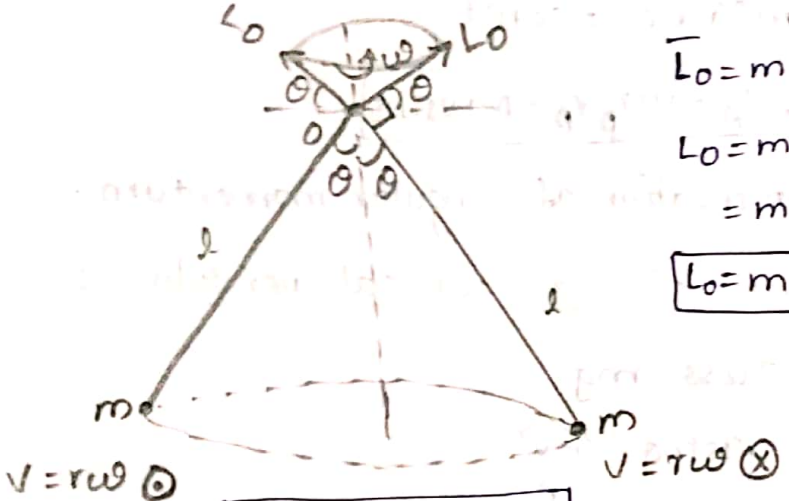
$$\therefore \omega^2 = \frac{g}{l\cos\theta}$$

$$\boxed{\omega = \sqrt{\frac{g}{l\cos\theta}}}$$

$$T = \frac{2\pi}{\omega}$$

$$\boxed{T = 2\pi\sqrt{\frac{l\cos\theta}{g}}}$$

④ Angular momentum of pendulum about 'O':-



$$\bar{L}_O = m(\bar{r} \times \bar{v})$$

$$L_O = mlv$$

$$= ml r\omega$$

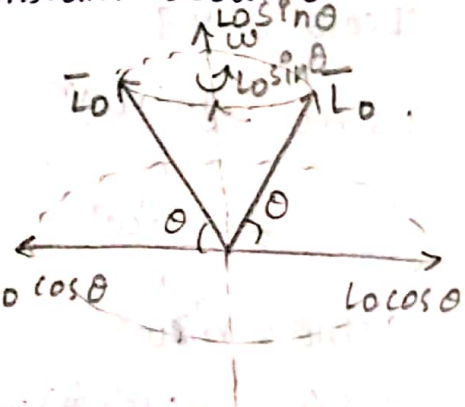
$$\boxed{L_O = ml^2 \sin\theta\, \omega}$$

$$\boxed{\bar{L}_O \perp \bar{I} \;\&\; \bar{L}_O \perp \bar{V}}$$

$\therefore$ 'L_O' is const in magnitude but not in direction.

$\therefore$ $\bar{L}_O$ is not a const vector i.e, angular momentum of conical pendulum is not constant about 'O'.

* Clearly, horizontal component "$L_O\cos\theta$" is rotating in horizontal plane with angular velocity 'ω'.



whose vertical component "$L_O\sin\theta$" is const both in magnitude & direction.

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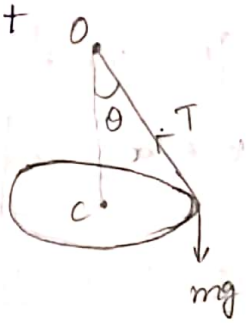
* Torque acting on bob about O



'O' $\Rightarrow \tau_O = mgr = mgl\sin\theta \neq 0$

$\frac{dL_O}{dt} \neq 0 \Rightarrow \bar{L}_O \neq \text{const}$

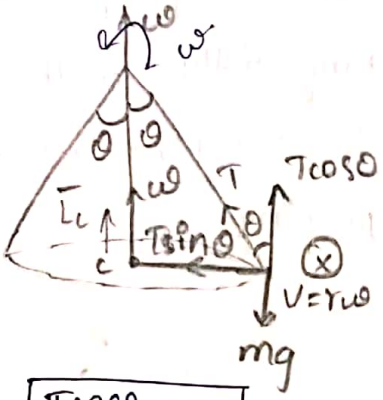
Angular momentum of pendulum about C :-

$\bar{L}_C = m(\bar{r} \times \bar{v})$

$= mrv$

$= m(l\sin\theta)(l\sin\theta\omega)$

$\boxed{L_C = ml^2\sin^2\theta\,\omega}$ parallel to $\bar{\omega}$



Clearly, $\bar{\tau}_C = rmg - T\cos\theta r$ $\quad \boxed{T\cos\theta = mg}$

$= 0$

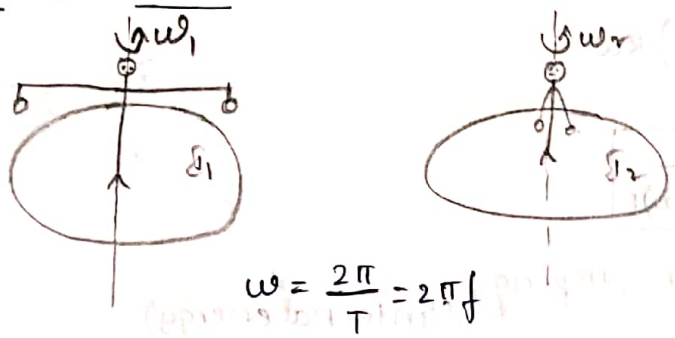
$\boxed{\tau_C = 0} \Rightarrow \frac{dL_C}{dt} = 0$

$\Rightarrow \boxed{L_C = \text{const}}$

∴ Angular momentum of conical pendulum is const about 'C'.

Turntable Problems :

①



$\omega = \frac{2\pi}{T} = 2\pi f$

Here,

$I_1\omega_1 = I_2\omega_2$

$I_1 > I_2 \Rightarrow \boxed{\omega_1 < \omega_2}$

and Rotation K.E,

$K.E = \frac{1}{2} I\omega^2 = \frac{1}{2}\frac{I^2\omega^2}{I} = \frac{L^2}{2I}$

$\Rightarrow \boxed{K.E = \frac{L^2}{2I}}$

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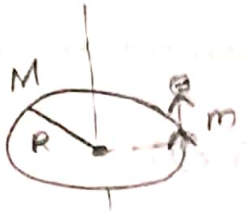
when $\boxed{L = \text{const}}$

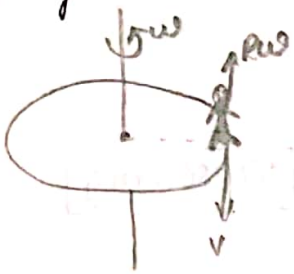
$$\boxed{K.E \propto \frac{1}{I} \propto \omega}$$

② Jumping from turn table :-

Man is jumping from turntable tangentially with speed 'v' relative to it. Find ω of table just after jumping.



Initially system is at rest.



Here linear momentum is not conserved as external force is acting on disc due to axle, where Angular momentum of system is conserved about Axle.

$\bar{L}_i = \bar{L}_f$ $\quad [\bar{V}_m = \bar{V}_{mT} + \bar{V}_T]$

$$0 = m(v - R\omega)R - \frac{MR^2}{2}\omega$$

$$m(v - R\omega) = \frac{MR}{2}\omega$$

$$mv = \left(\frac{M}{2} + m\right)R\omega$$

$$\boxed{\omega = \frac{2mv}{(M+2m)R}}$$

Energy required for jumping, (internal energy)

$$E = \frac{1}{2}m(v - R\omega)^2 + \frac{1}{2}\left(\frac{MR^2}{2}\right)\omega^2$$

* If system is rotating with ω_0 initially,

Then

$$\bar{L}_i = \bar{L}_f$$

$$\left(\frac{MR^2}{2} + mR^2\right)\omega_0 (A.C.W) = m(v - R\omega)R(C.W) + \frac{mR^2}{2}\omega(A.C.W)$$

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$$\frac{(M+2m)R\omega_0}{2} = mR\omega - mv + \frac{MR\omega}{2}$$

$$(M+2m)R\omega_0 = m(3R\omega - 2v)$$

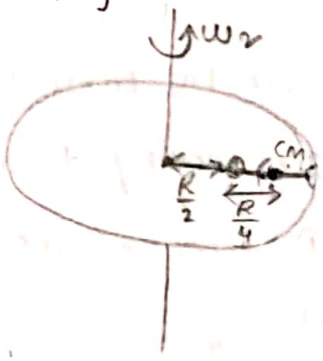
$$\boxed{\omega_0 = \frac{m(3R\omega - 2v)}{(M+2m)R}}$$

→ man of mass $\frac{M}{2}$ & height $\frac{R}{2}$ is standing at edge of table initially. Now, man gradually lay down on the table with his head at $\frac{R}{2}$ from centre of table. Assume, man as rod. Find new angular velocity of system.

Sol: $\bar{L}_i = \bar{L}_f$

$$\left[\frac{MR^2}{2} + \left(\frac{M}{2}\right)R^2\right]\omega_1 = \left[\frac{MR^2}{2} + \underbrace{\frac{M}{2}\left(\frac{R}{2}\right)^2}_{12} + \frac{M}{2}\left(\frac{3R}{4}\right)^2\right]\omega_2$$

↓ disc ↓ man ↓ disc — man



Man Walking on the turn table:

→ along the edge:

Initially system is at rest. Man starts walking with angular velocity (ω) relative to table. Find

① Angle turned by table by the time man reaches to same position on table.

② Angle travelled by man w.r.to ground in this time

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①

$\theta_{mT} = 2\pi$ — ①

and $\theta_{mT} = \omega t$

$t = \frac{2\pi}{\omega}$

here $\bar{L}_i = \bar{L}_f$.

$\theta_T = \frac{mR^2(2\pi)}{mR^2 + \frac{MR^2}{2}}$

$\boxed{\theta_T = \frac{4m\pi}{M+2m}}$ (A.C.W)

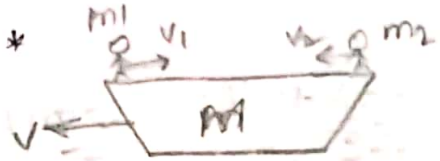
②

$\bar{\theta}_m = \bar{\theta}_{mT} + \bar{\theta}_T$

$= \omega t(\circlearrowleft) + \frac{4m\pi}{M+2m}(A.C.W)$

$\bar{\theta}_m = \left(\omega t - \frac{4m\pi}{M+2m}\right)(C.W)$

*



Here $\boxed{V_{cm} = 0}$

$0 = -MV + m_1(v_1 - V) - m_2(V_2 + V)$

$MV + m_2V_2 + m_2V = m_1v_1 - m_1V$

$(m_1 + m_2 + M)V = m_1v_1 - m_2V_2$

$\boxed{V = \frac{m_1v_1 - m_2V_2}{m_1 + m_2 + M}}$

* Similar to this,

* $\boxed{\theta_T = \frac{m_1R_1^2\theta_{AT} + m_2R_2^2\theta_{BT}}{m_1R_1^2 + m_2R_2^2 + \frac{MR^2}{2}}}$

$v_1, v_2 \rightarrow$ relative velocities.

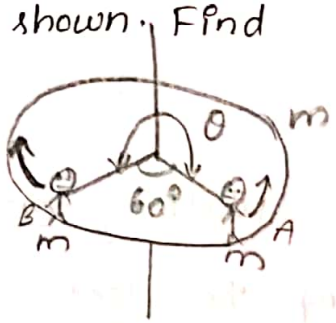
→ w.r.to turn table $\omega_A = \pi$ rad/sec, $\omega_B = 3\pi$ rad/sec. Both are moving towards each other as shown. Find

① time of meeting

② θ_T

③ θ_B

④ θ_A } w.r.to ground } at the time of meeting.



Sol:

$\theta = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

& $\theta = \theta_{AT} + \theta_{BT} = \frac{5\pi}{3}$

$\omega_A t + \omega_B t = \frac{5\pi}{3}$

& $\frac{\theta_{AT}}{\theta_{BT}} = \frac{\omega_A t}{\omega_B t} = \frac{\pi}{3\pi} = \frac{1}{3}$

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and $\theta_{AT} + 3\theta_{AT} = \frac{5\pi}{3}$

$$\boxed{\theta_{AT} = \frac{5\pi}{12}}$$

& $$\boxed{\theta_{BT} = \frac{5\pi}{4}}$$

$\theta_{AT} = \omega_A t$ $\qquad$ $\theta_{BT} = \omega_{BT}(t)$

$\frac{5\pi}{12} = \pi(t)$ $\qquad$ $\frac{5\pi}{4} = 3\pi(t)$

$\boxed{t = \frac{5}{12}\text{ sec}}$ $\qquad$ $\boxed{t = \frac{5}{12}\text{ sec}}$

time of meeting.

② Angle turned by table is

$$\theta_T = \frac{\pi R^2\left(\frac{5\pi}{12}\right)(\text{C}\cdot\omega) + \pi R^2\left(\frac{5\pi}{4}\right)(\text{A}\cdot\text{C}\cdot\omega)}{\frac{\pi R^2}{1} + \frac{\pi R^2}{1} + \frac{\pi R^2}{2}}$$

$$= \frac{-\frac{5\pi}{12} + \frac{5\pi}{4}}{\frac{5}{2}}$$

$$= \frac{\frac{\pi}{12}\left(1 - \frac{1}{3}\right)}{1/2}$$

$$= \frac{\pi}{12}(2)\left(\frac{1}{3}\right) = \frac{\pi}{3}$$

$$\boxed{\theta_T = \frac{\pi}{3}(\text{A}\cdot\text{C}\cdot\omega)}$$

$(60°)$.

③ $\overline{\theta_A} = \overline{\theta_{AT}} + \overline{\theta_T}$

$$= \frac{5\pi}{12}(\text{C}\cdot\omega) + \frac{\pi}{3}(\text{A}\cdot\text{C}\cdot\omega)$$

$$= \frac{5\pi}{12} - \frac{\pi}{3} = \frac{\pi}{12}(\text{C}\cdot\omega)$$

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$$\vec{\theta}_B = \vec{\theta}_{BT} + \vec{\theta}_T$$

$$= \frac{5\pi}{4}(A\cdot C\cdot\omega) + \frac{\pi}{3}(A\cdot C\cdot\omega)$$

$$= \frac{15\pi + 4\pi}{12} = \frac{19\pi}{12}(A\cdot C\cdot\omega)$$

Man walking towards Centre :-

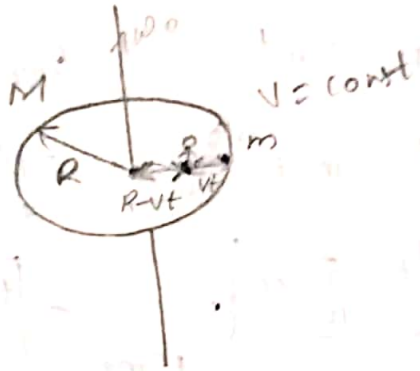
$L_i = L_f$

$$\left(\frac{MR^2}{2} + mR^2\right)\omega_0$$

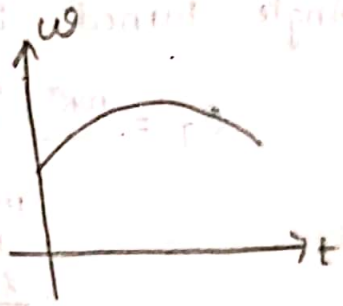
↓ at $t=0$

$$= \left[\frac{MR^2}{2} + m(R-Vt)^2\right]\omega$$

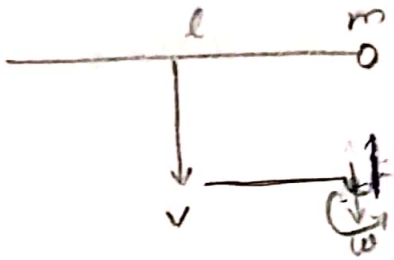
↓ at $t=t$



$$\Rightarrow \boxed{\omega = \frac{\left(\frac{MR^2}{2} + mR^2\right)\omega_0}{\frac{MR^2}{2} + m(R-Vt)^2}}$$



→



$$\frac{mV^2}{2} = \frac{ml^2}{3}\omega$$

$$\boxed{\omega = \frac{3V}{2l}}$$

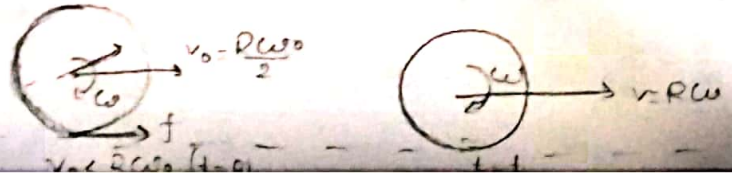
$$K.E = \frac{1}{2}I\omega^2$$

$$= \frac{1}{2} \times \frac{ml^2}{3}\omega^2$$

$$= \frac{1}{2} \times \frac{ml^2}{3} \times \frac{3}{4}\frac{V^2}{l^2}$$

$$\boxed{K.E = \frac{3mV^2}{8}}$$

* In case of rolling motion,



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* $\vec{L_i} = \vec{L_f}$

$\Rightarrow mv_0R(\odot) + I_{cm}\omega_0(\odot) = mvR(\odot) + I_{cm}\omega(\odot)$

$\Rightarrow mv_0R + \frac{mR^2}{2} \times \frac{2v_0}{R} = mvR + \frac{mR^2}{2}\left(\frac{v}{R}\right)$

$\Rightarrow 2mv_0R = \frac{3}{2}mvR$

$\boxed{v = \frac{4v_0}{3}}$ & $\boxed{\omega = \frac{4v_0}{3R}}$

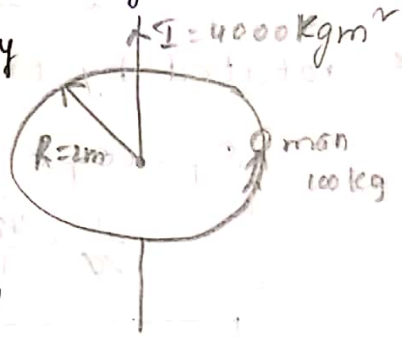
→ time for coming pure rolling,

$v = v_0 + at$

$\frac{v_0}{3} = \mu g t \Rightarrow \boxed{v_0 = 3\mu g t} \Rightarrow \frac{4v_0}{3} = v_0 + \mu g t$

$\boxed{t = \frac{v_0}{3\mu g}}$

→ The whole system is initially at rest. If the man walks along the outer edge of disc with a velocity 1 m/s relative to earth.



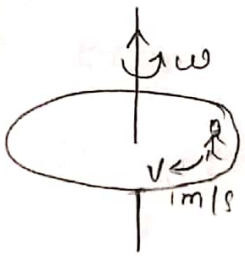
i) At what Angular velocity & in what direction will the turn-table rotate,

ii) Through what angle will it have rotated when the man reaches his initial position on the turn table.

iii) Through what angle will it have rotated when the man reaches his initial position relative to ground.

Sol:

i)



Let the man walks in clockwise direction on the table

$L_i = L_f$

$0 = mvR - I_t\omega_t$

$\omega_t = \frac{mvR}{I_t} = \frac{100 \times 1 \times 2}{4000}$

$\omega_t = \frac{1}{20}$ rad/sec,

anti clockwise.

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(ii) $\theta_{mT} = 2\pi$

$(\omega_{mT})t = 2\pi$ $\qquad \omega_m = \frac{V}{R} = \frac{1}{2}$ rad/sec

$(\omega_m - \omega_T)t = 2\pi$

$\left(\frac{1}{2} - \left(-\frac{1}{20}\right)\right)t = 2\pi$

$\left(\frac{10+1}{20}\right)t = 2\pi$

$$\boxed{t = \frac{40\pi}{11} \text{ sec}}$$

Angle rotated by table,

$\theta_T = \omega_T t$

$= \frac{1}{20} \times \frac{40\pi}{11}$

$$\boxed{\theta = \frac{2\pi}{11} \text{ rad}} \quad (A \cdot C \cdot \omega)$$

(iii) $\theta_m = 2\pi$

$(\omega_m)t = 2\pi$

$\frac{1}{2} \times t = 2\pi$

$$\boxed{t = 4\pi \text{ sec}}$$

$\theta_T = \omega_T t$

$= \frac{1}{20} \times 4\pi$

$$\boxed{\theta_T = \frac{\pi}{5} \text{ rad}} \quad (A \cdot C \cdot \omega)$$

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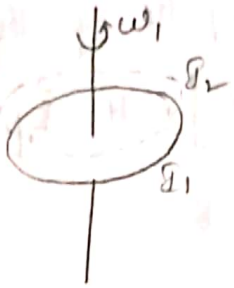
→ Initially disc-1 is rotating with Angular velocity 'ω'. Another disc of moment of Inertia I_2 is placed gently & co-axially on disc-1. Sufficient friction exist blw 2 discs. Find



① Common angular velocity of discs.

② Loss in K.E of disc.

③ Time for getting common velocity.

① friction is internal force of system.

∴ net torque due to friction on system = 0.

⇒ Angular momentum of system about AOR is constant.

$$\vec{L}_i = \vec{L}_f$$

$$I_1\omega_1 = (I_1+I_2)\omega$$

$$\boxed{\omega = \frac{I_1\omega_1}{I_1+I_2}}$$

* After getting common angular velocity friction blw the discs will becomes static & τ_{net} will be '0'

② loss in K.E of system,

$$\Delta K = K.E_i - K.E_f$$

$$= \frac{1}{2} I_1\omega_1^2 - \frac{1}{2}(I_1+I_2)\omega^2$$

$$\boxed{\Delta K = \frac{1}{2}\frac{I_1 I_2}{I_1+I_2}\omega_1^2}$$

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Angular impulse of disc-1,

$$L_i \pm \int \tau \cdot dt = L_f$$

on disc-1 &-

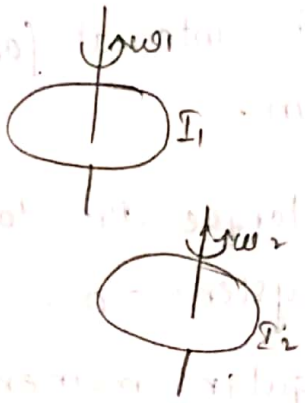
$$I_1\omega_1 - \int \tau \cdot dt = I_2\omega_2$$

$$\therefore \int \tau \cdot dt = I_1\omega_1 - I_1\omega \quad -①$$

on disc-2 :-

$$0 + \int \tau \cdot dt = I_2\omega$$

$$\int \tau \cdot dt = I_2\omega \quad -②$$



* In general:

L.C.A.M,

$$I_1\bar{\omega}_1 + I_2\bar{\omega}_2 = (I_1+I_2)\bar{\omega}$$

$$\boxed{\bar{\omega} = \frac{I_1\bar{\omega}_1 + I_2\bar{\omega}_2}{I_1+I_2}}$$

Loss in k.E,

$$\Delta K = \frac{1}{2}\frac{I_1I_2}{I_1+I_2}|\bar{\omega}_1 - \bar{\omega}_2|^2$$

Angular impulse eqns,

$$I_1\bar{\omega}_1 \pm \int_{t_1}^{t_2} \bar{\tau} \cdot dt = I_1\bar{\omega}$$

& $$I_2\bar{\omega}_2 \pm \int_{t_1}^{t_2} \tau \cdot dt = I_2\bar{\omega}$$

This case is similar to perfectly inelastic collision.

$$\boxed{\bar{V} = \frac{m_1\bar{v}_1 + m_2\bar{v}_2}{m_1+m_2}}$$

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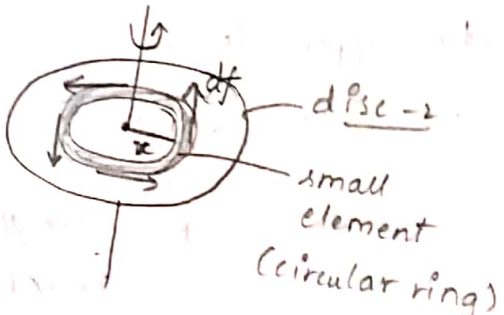
& $\Delta K = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}|\vec{v}_1 - \vec{v}_2|^2$

(3) Torque due to friction on small element about AOR,



$$d\tau = x\,df$$

$$d\tau = x(\mu)\cdot dN$$

$$= x\mu(dm)g$$

$$= x\mu g \times \frac{m}{\pi R^2} \times 2\pi x\cdot dx$$

$$= \frac{2\mu mg}{R^2}\cdot x^2\cdot dx$$

∴ Total torque,

$$\tau = \int_{x=0}^{x=R} d\tau = \frac{2\mu mg}{R^2} \times \frac{R^3}{3}$$

$$\boxed{\tau = \frac{2\mu mgR}{3}}$$

on disc-1,

$$\alpha_1 = \frac{-\tau}{I_1} = \frac{-2\mu mgR}{3I_1}$$

& on disc-2, $\alpha_2 = \frac{\tau}{I_2} = \frac{2\mu mg}{3I_2}$

∴ we have,

$$\boxed{\omega = \omega_0 + \alpha t}$$

for lower disc,

$$\omega = \omega_0 + \alpha_1 t$$

$$\frac{\omega_0}{2} = \omega_0 - \frac{2\mu mgR}{3I_1}t \Rightarrow \frac{\omega_0}{2} = \frac{2\mu mgRt}{3\times\frac{mR^2}{2}} = \frac{4\mu g}{3R}t$$

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$$\boxed{t = \frac{3R\omega_0}{8\mu g}}$$

given

$$I_1 = I_2 = I = \frac{mR^2}{2}$$

$$\omega = \frac{I_1\omega_0}{I_1 + I_2}$$

$$\boxed{\omega = \frac{\omega_0}{2}}$$

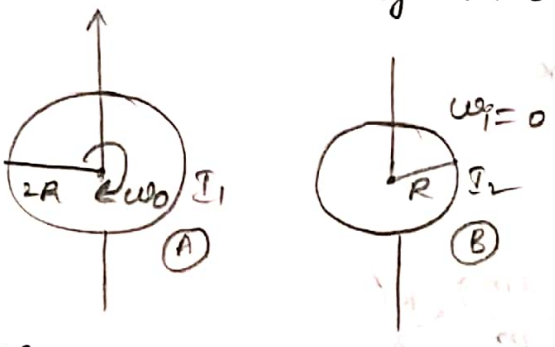
for upper disc,

$$\omega = \omega_0 + \alpha_2 t$$

$$\frac{\omega_0}{2} = 0 + \frac{2\mu mgR\, t}{3 \times \frac{mR^2}{2}}$$

$$\boxed{t = \frac{3R\omega_0}{8\mu g}}$$

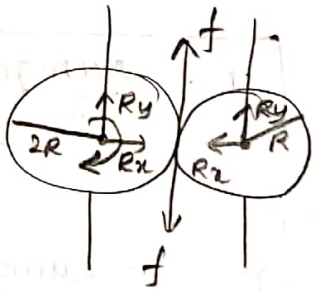
→ Two discs are brought in contact to their surfaces..



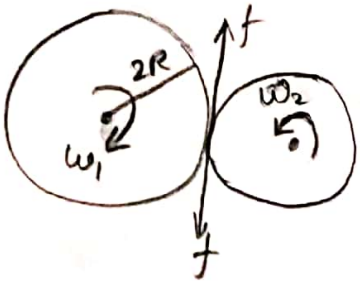
Here,

$\tau_{sys} \neq 0$

∴ L.C.A.M. is not applicable as there is slipping b/w discs at their contact surfaces.



Impulse eqns,



when slipping ceases

$$I_1\omega_0 - \int f(2R)\cdot dt = I_1\omega_1 \quad \text{—②}$$

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$$0 + \int fR \cdot dt = I_2 \omega_2 \quad -(2)$$

constraint eqn,

unknowns,

ω_1, ω_2.

$$V_1 = V_2$$

$$(2R\omega_1) = R\omega_2$$

$$\omega_1 = \frac{\omega_2}{2} \quad -(3)$$

$$I_1\omega_0 - 2I_2\omega_2 = I_1\omega_1$$

$$I_1\omega_0 - 2I_2\omega_2 = I_1\omega_1$$

$$\underline{I_1\omega_0 = (4I_2 + I_1)\omega_1}$$

$$\boxed{\omega_1 = \frac{I_1\omega_0}{4I_2 + I_1}}$$

&

$$\boxed{\omega_2 = \frac{2I_1\omega_0}{4I_2 + I_1}}$$

now, Angular impulse on disc A

$$\int f(2R) \cdot dt = I_1\omega_0 - I_1\omega_1$$

on disc B :

$$\int fR \cdot dt = I_2\omega_2$$

* loss in K.E of system,

$$= K.E_i - K.E_f$$

$$= \frac{1}{2}I_1\omega_0^2 - \left(\frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2\right)$$

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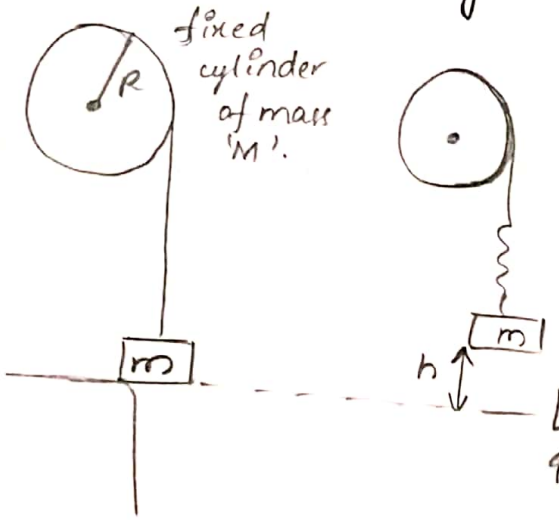
→ Just before the string is taut. Evaluate the

① angular velocity ω_0 of cylinder & speed v_0 of falling body.

② K.E K_0 of the system

③ Evaluate the corresponding quantities ω_1, v_1, K_1 just after string becomes taut.



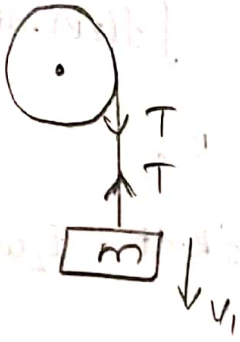
* Just before & Just after string taut, we can not use L.C.E but we can use L.C.A.M.

Just before string taut,

$v_0 = \sqrt{2gh}$ & $\omega_0 = 0$ as $T = 0$,

$K_0 = \frac{1}{2} m v_0^2 = mgh$.

Just after string taut,



T → impulsive

& mg → non-impulsive

& hence T is internal force

∴ net torque about C is zero.

∴ we can use L.C.A.M about C.

⇒ $L_i = L_f$

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$$mv_0\ell = mv_1\ell + \frac{MR^2}{2}\omega_1 \quad -(1)$$

for no slipping,

$$v_1 = R\omega_1 \quad -(2)$$

$$mv_0\ell = mR\omega_1\ell + \frac{MR}{2}\omega_1$$

$$\boxed{\omega_1 = \frac{2mv_0}{(M+2m)R}}$$

& $$v_1 = R\omega_1 = \frac{2mv_0}{M+2m}$$

finally, $$K.E_1 = K_1 = \frac{1}{2}mv_1^2 + \frac{1}{2}\frac{MR^2}{2}\omega_1^2$$

II-Method:

Impulse eqns:

on blocks:-

$$mv_0 - \int T\,dt = mv_1 \quad -(1)$$

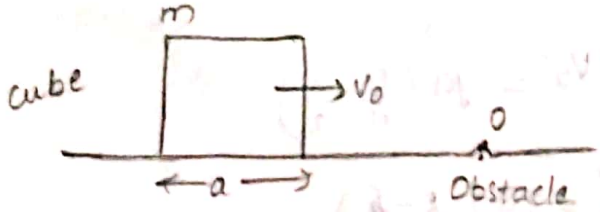
on cylinder:-

$$0 + \int TR\,dt = \frac{MR^2}{2}\omega_1 \quad -(2)$$

for no slipping,

$$v_1 = R\omega_1 \quad -(3)$$

→



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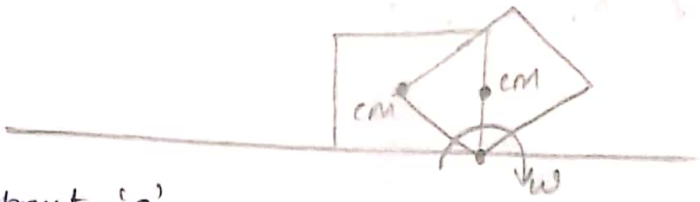

About 'O'

* $\bar{L_i} = \bar{L_f}$

$$mV_0\left(\frac{a}{2}\right) = I_0\omega$$

$$mV_0\left(\frac{a}{2}\right) = \left[\frac{ma^2}{6} + m\left(\frac{a}{\sqrt{2}}\right)^2\right]\omega$$

$$\frac{V_0}{2} = \left(\frac{a}{6} + \frac{a}{2}\right)\omega$$

$$V_0 = \frac{4a}{3}\omega$$

$$\boxed{\omega = \frac{3V_0}{4a}}$$

$$K.E = \frac{1}{2} I_0 \omega^2$$

Min velocity (V_0) required for block to rise the C.M to vertical position.

After impact,

C.C.E,

$$\frac{1}{2} I_0\omega^2 + mg\frac{a}{2} = mg\frac{a}{\sqrt{2}}$$

$$\frac{1}{2}\left[\frac{ma^2}{6} + m\left(\frac{a}{\sqrt{2}}\right)^2\right]\left(\frac{3V_0}{4a}\right)^2 + mg\frac{a}{2} = mga/\sqrt{2}$$

$$\frac{1}{2}\left(m a^2\left(\frac{1}{6} + \frac{1}{2}\right)\right)\frac{9V_0^2}{16a^2} + mg\frac{a}{2} = mga/\sqrt{2}$$

$$\frac{8}{4} \times \frac{1}{32} \times \frac{3}{4} 9V_0^2 = ga\left(\frac{1}{\sqrt{2}} - \frac{1}{2}\right)$$

$$\frac{3V_0^2}{16} = ga\left(\frac{2-\sqrt{2}}{2\sqrt{2}}\right)$$

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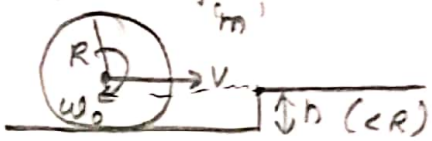
$$\frac{3}{8}V_0^2 = \left(\frac{2-\sqrt{2}}{\sqrt{2}}\right)ga$$

$$V_0^2 = \frac{8ga(2-\sqrt{2})}{\sqrt{2}\times 3}$$

$$V_0^2 = \frac{8ga(\sqrt{2}-1)}{3}$$

$$\boxed{V_0 = \sqrt{\frac{8ga(\sqrt{2}-1)}{3}}}$$

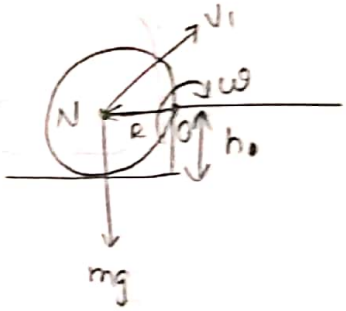
→ cylinder of mass 'm'



L·C·A·M about 'O'

$$\bar{L}_i = \bar{L}_f$$

$$\boxed{mV_0(R-h) + \frac{MR^2}{2}\omega_0 = \frac{3MR^2}{2}\omega}$$



during ~~collision~~ climbing,

$$-N + mg\cos\theta = \frac{mv_1^2}{R}$$

$$\boxed{N = mg\cos\theta - \frac{mv_1^2}{R}}$$

Here θ↓ & V(↓)
⇒ N(↑)

⇒ Velocity of C·M of sphere just after impact

$V_C = R\omega$.

⇒ K·E of body ~~xxxx~~ just after impact,

$$K.E = \frac{1}{2}I_0\omega_0^2.$$

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* about 'O' body is in pure rotation, i.e, all particles are moving in circular path with centre at O.

$$mg\cos\theta - N = \frac{mv_c^2}{R}$$

$$\boxed{N = mg\cos\theta - \frac{mv_c^2}{R}}$$

As sphere climbs, $\theta(\downarrow)$ $\cos\theta(\uparrow)$, and also $v_c(\downarrow)$,

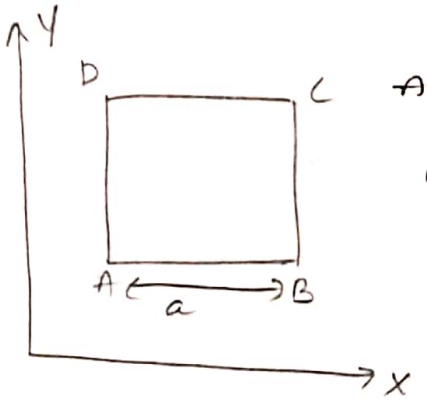
$\Rightarrow$ (N) Increases.

for example,

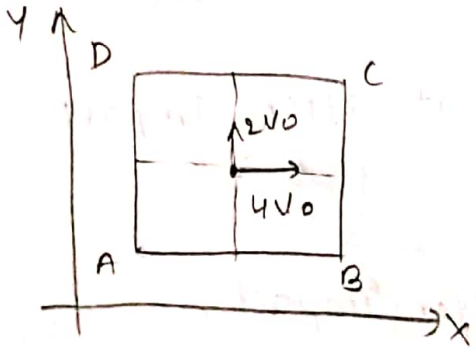
$h = R/2$

$\Rightarrow \theta = 60^\circ$.

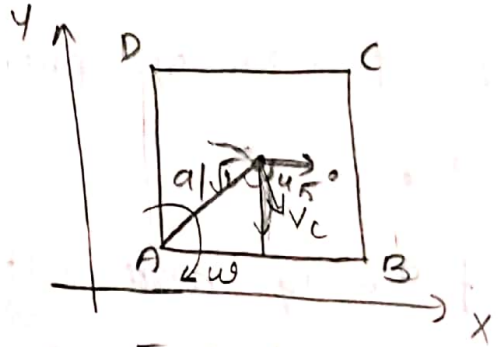
→



ABCD is a uniform square plate of mass (m) & side length (a) moving on smooth horizontal table with velocity $\overline{V} = V_0(4\uparrow + 2\uparrow)$ with out rotation. Now end (A) is suddenly fixed with nail & Find velocity of C.M of plate just after fixed.



Just before fixed



Just after fixed

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using L·C·A·M about A.

$$\bar{L}_i = \bar{L}_f$$

$$m(2v_0)\left(\frac{a}{2}\right)(\odot) + m(2v_0)\left(\frac{a}{2}\right)(\curvearrowright) = I_A\,\bar{\omega}$$

$$I_A\,\bar{\omega} = mv_0 a\,(\odot)$$

$$\left[\frac{ma^2}{6} + m\left(\frac{a}{\sqrt{2}}\right)^2\right]\omega = mv_0$$

$$\left(\frac{a}{6} + \frac{a}{2}\right)\omega = v_0$$

$$\frac{a+3a}{6}\,\omega = v_0$$

$$\omega = \frac{6v_0}{4a}$$

$$\boxed{\omega = \frac{3v_0}{2a}}$$

now, $v_c = \frac{a}{\sqrt{2}}\omega$

$$= \frac{3v_0}{2\sqrt{2}}$$

$$\boxed{v_c = \frac{3v_0}{2\sqrt{2}}}$$

$$\bar{v}_c = v_{0c}\cos 45^\circ \uparrow + v_{0c}\sin 45^\circ (-\hat{j})$$

$$= \frac{3v_0}{4}\uparrow + \frac{3v_0}{2\sqrt{2}}(-\hat{j})$$

$$\boxed{\bar{v}_c = \frac{3v_0}{4}\uparrow - \frac{3v_0}{4}\hat{j}}$$

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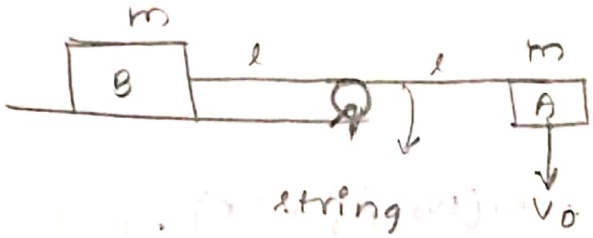
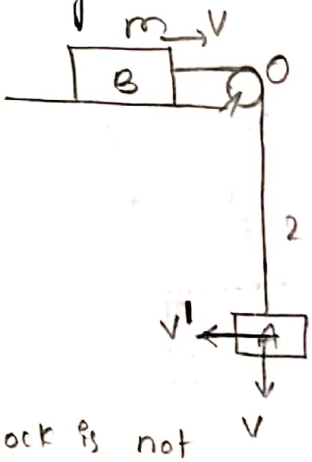


Two identical blocks each of mass m are placed on smooth horizontal surface (fixed). Block B comes to rest after hitting pulley. Then find,

① Velocities of blocks A & B just before block 'B' hits pulley

② velocities of blocks A & B just after block 'B' hits pulley.



Here $(\tau_{tension})_{about\ O} = 0$

A·M of sys about 'O' is conserved. i.e,

$\bar{L_i} = \bar{L_f}$

$mv_0 \ell = mv'(2\ell)$

$v' = v_0/2$

block is not hanging. Everything is happened on ground.

and since system is smooth, use 'L.C.E'.

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}mv'^2$$

$$mv_0^2 = 2mv^2 + m\frac{v_0^2}{4}$$

$$\frac{3mv_0^2}{4} = 2mv^2$$

$$v = \sqrt{\frac{3}{8}}\,v_0$$

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∴ speed of blocks just before hitting,

$$v_B = v = v_0\sqrt{\frac{3}{8}}$$

& $v_A = \sqrt{v'^2 + v^2} = \sqrt{\frac{v_0^2}{4} + \frac{3}{8}v_0^2}$

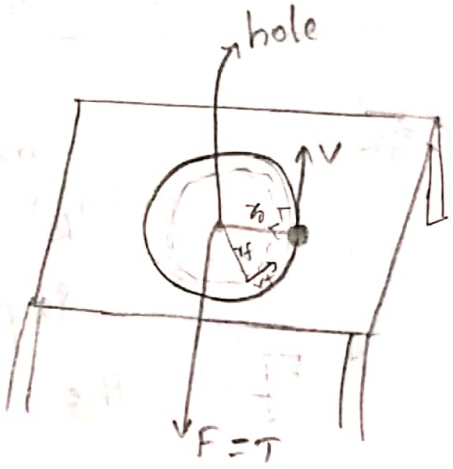
$$\boxed{v_A = \sqrt{\frac{5}{8}}\, v_0}$$

Just after hitting,

$$v_B = v = 0$$

& $v_A = v' = \frac{v_0}{2}$

→ In this process of pulling the string, Torque due to Tension(F) about 'O' = 0.



∴ A·M of particle is conserved about 'O'. i.e,

$$l_0 = mv_0 r = mv_f r_f$$

$$\boxed{v_f = \frac{v_0 r}{r_f}}$$

* K.E of particle at radius 'r',

$$K = \frac{1}{2}mv^2$$

we have,

$$l_0 = mvr$$

$$v = \frac{l_0}{mr}$$

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$$K = \frac{1}{2} m \times \frac{L_0^2}{m^2 r^2}$$

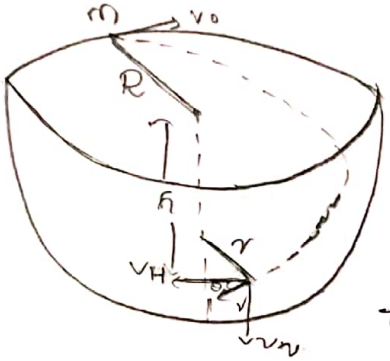
$$\boxed{K = \frac{L_0^2}{2mr^2}}$$

* work done by applied force during r_0 to r

$$W = K_f - K_i$$

$$\boxed{W = \frac{L_0^2}{2m}\left[\frac{1}{r^2} - \frac{1}{r_0^2}\right]}$$

→



Smooth hollow sphere of radius 'R'. A particle of mass 'm' is projected tangentially to the surface with speed v_0.

In this process,

Torque due to gravity about A.O.R = 0.

∴ A.M about AOR is const.

i.e, $\overline{L_i} = \overline{L_f}$

$$mv_0 R = mv_H r$$

$$\boxed{v_H = \frac{v_0 R}{r}} \quad -(1)$$

where $\boxed{v_H = v\cos\theta}$

Since, surface of bowl is smooth.

Energy is conserved.

$$\frac{1}{2}mv_0^2 + mgh = \frac{1}{2}m\left(v_H^2 + v_v^2\right) \quad -(2)$$

$\boxed{v_v = v\sin\theta}$

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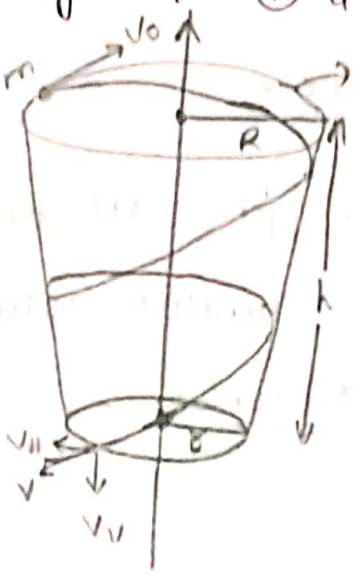
Solving eqns ① & ② we get v & θ.

→



L·C·A·M, about AOR,

$$mv_0R = mv_Hr \quad \text{—— ①}$$

L·C·E,

$$\frac{1}{2}mv_0^2 + mgh = \frac{1}{2}m(v_H^2 + v_V^2)$$

$$v_0^2 + 2gh = (v_H^2 + v_V^2)$$

$$v_0^2 + 2gh = \frac{v_0^2R^2}{r^2} + v_V^2$$

* For $r = \frac{R}{2}$. Find min 'h' so that particles will never comes out of friction from base?

Here vertical component of velocity becomes "zero" before coming out of bottom.

$$2gh + v_0^2\left(1 - \frac{R^2}{\left(\frac{R}{2}\right)^2}\right) = 0$$

$$\boxed{h = \frac{3v_0^2}{2g}}$$

→

Two identical beads are released symmetrically along smooth wires.

In this process,

L.C.E & L·C·A·M are used & are applicable.

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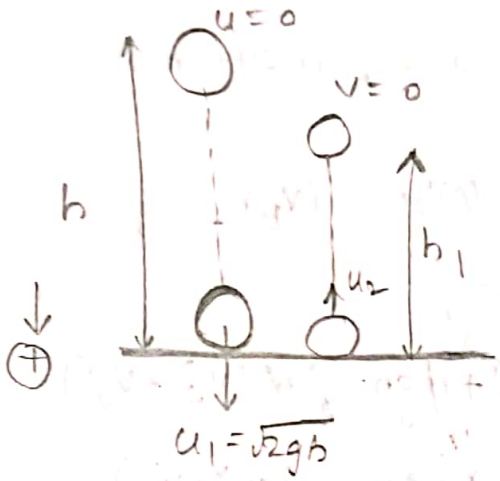
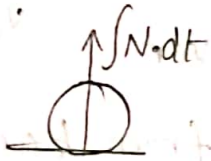
Collision in Rotational dynamics:-

① Collision of body with surface:



Here large forces are developed in short duration called impulsive forces.

Impulse eqns:



$$mu_1 - \int N\,dt = -mv_1 \quad \text{—①}$$

$$\boxed{\int N\,dt = m(u_1 + v_1)}$$

and $\frac{1}{2}mv_1^2 = mgh_1$

$$v_1 = \sqrt{2gh_1}$$

$$\boxed{\int N\,dt = m\left(\sqrt{2gh} + \sqrt{2gh_1}\right)}$$

* If collision is elastic,

$$e = \frac{\bar{v}_2 - \bar{v}_1}{\bar{u}_1 - \bar{u}_2} = \frac{-\bar{v}_1}{\bar{u}_1}$$

$$\bar{v}_1 = -e\bar{u}_1$$

$$\boxed{v_1 = eu_1}$$

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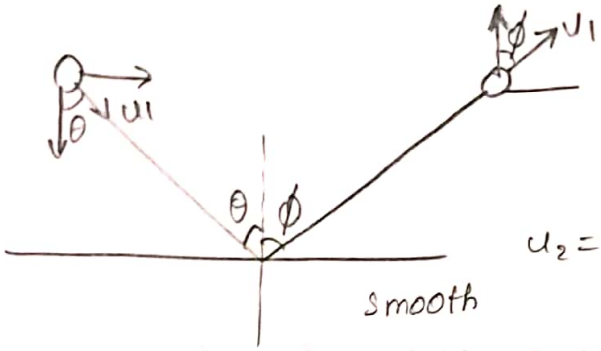
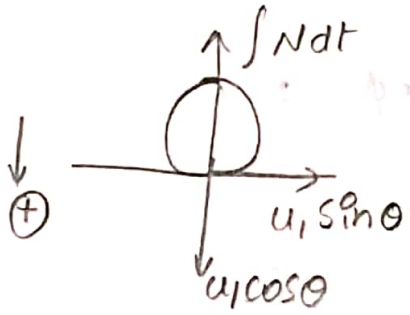
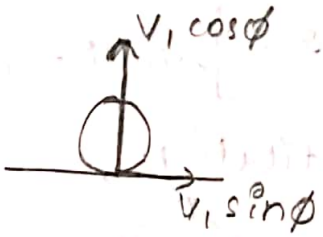
②







$$m u_1 \cos\theta - \int N \cdot dt = -m v_1 \cos\phi \quad -(1)$$

$$u_1 \sin\theta = v_1 \sin\phi \quad -(2)$$

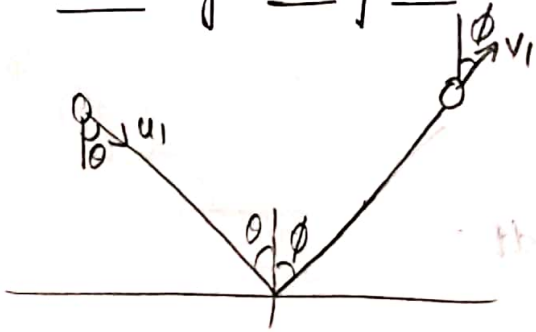
i.e, velocity component of each body along C.T is same if colliding surfaces are smooth.

③ Coefficient of restitution, (along L.O.I)

$$e = \frac{\overrightarrow{v_2}^{\,0} - \overrightarrow{v_1}}{\overrightarrow{u_1}^{\,0} - \overrightarrow{u_2}} \Rightarrow \quad e = \frac{+(+v_1 \cos\phi)}{u_1 \cos\theta}$$

$$\boxed{e = \frac{v_1 \cos\phi}{u_1 \cos\theta}}$$

* If colliding surfaces are rough:



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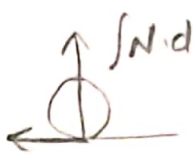


$\int f \cdot dt = \int \mu N \cdot dt$

Impulse eqns :-

$$mu_1\cos\theta - \int N\cdot dt = -mv_1\cos\phi \quad -(1)$$

$$mu_1\sin\theta - \int \mu N\cdot dt = mv_1\sin\phi \quad -(2)$$

coeff of restitution,

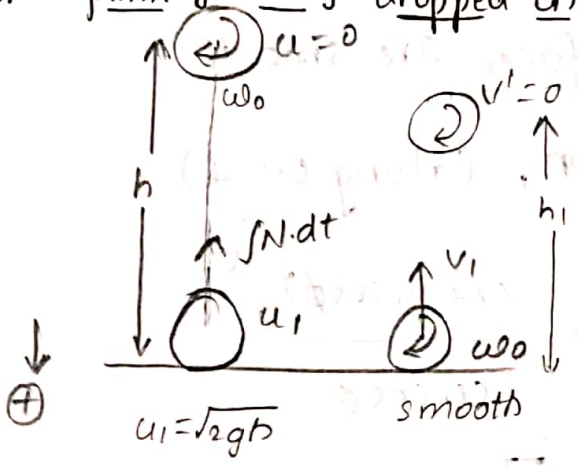
(along NOR)

$$e = \frac{\cancel{\overrightarrow{v_2}^{\,0}} - \overline{v_1}}{\overline{u_1} - \cancel{\overrightarrow{u_2}^{\,0}}} = \frac{-\overline{v_1}}{\overline{u_1}}$$

unknowns, v_1, ϕ, μ

net impulse $\Rightarrow J\sqrt{1+\mu^2}$

* Spinning body dropped on smooth surface :



* colliding surfaces are smooth hence 'ω' doesn't change.

$$mu_1 - \int N\cdot dt = -mv_1 \quad -(1)$$

and $$e = \frac{\cancel{\overrightarrow{v_2}^{\,0}} - \overline{v_1}}{\overline{u_1} - \cancel{\overrightarrow{u_2}^{\,0}}}$$

$$e = \frac{-\overline{v_1}}{\overline{u_1}} \quad -(2)$$

unknowns, v_1 & $\int N\cdot dt$.

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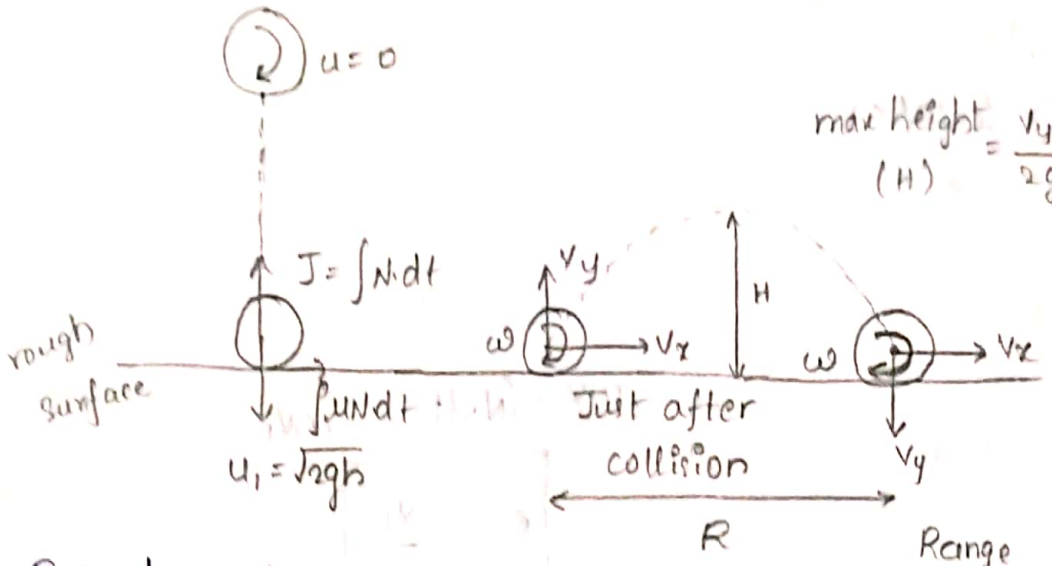
→ spinning body dropped on rough surface:



Impulse eqn:

① $mu_1 - \int N\cdot dt = -mv_y$ — ①

② $0 + \int \mu N\cdot dt = mv_x$ — ②

③ $I\omega_0 - \int \mu NR\cdot dt = I\omega$ — ③

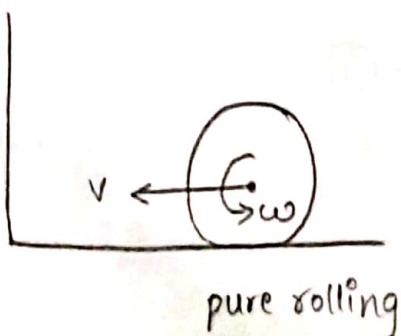
④ coefficient of restitution,

$$e = \frac{\overline{v_2} - \overline{v_1}}{\overline{u_2} - \overline{u_1}} = \frac{-(-v_y)}{u_1} \quad \text{— ④}$$

unknowns, $v_x, v_y, \int N\cdot dt, \omega$ — ④

Net linear impulse, $\Rightarrow J\sqrt{1+\mu^2}$.

→



pure rolling

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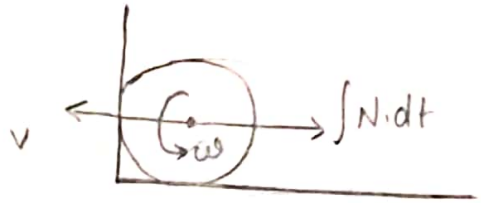
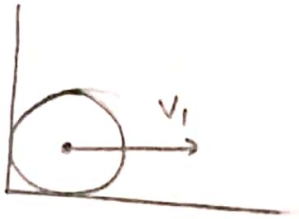
If wall is smooth,

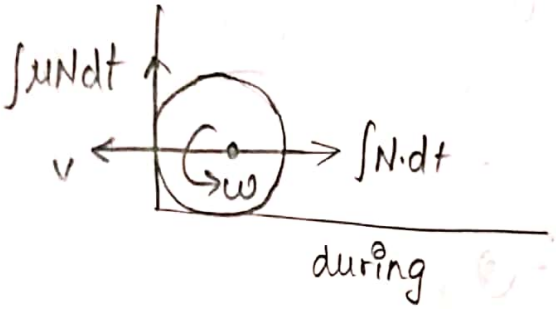


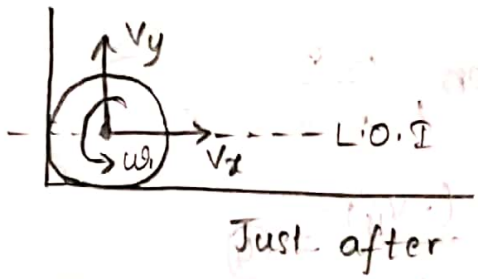


$$mv - \int N\cdot dt = -mv_1$$

$$e = \frac{-\overline{v_1}}{\overline{v}}$$

If wall is rough,





$$mv - \int N\cdot dt = -mv_x \quad -(1)$$

$$\int \mu N\cdot dt = mv_y \quad -(2)$$

$$I\omega - \int \mu NR\,dt = I\omega_1 \quad -(3)$$

$$e = \frac{-\overline{v_x}}{\overline{v_y}}$$

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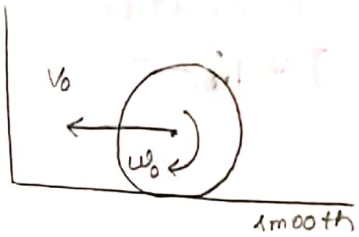
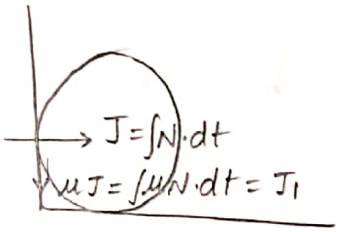
→ Solid sphere of radius 1m & mass 2kg, $V_0 = 4$ m/s, ω_0 = 9 rad/sec. It collides elastically with rough wall of coefficient of friction μ. Find

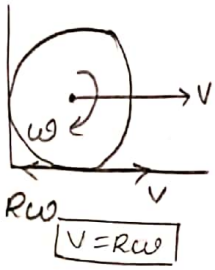


① If the sphere after colliding with the wall rolls without slipping in opp direction. Find 'μ' value.

② Find net linear impulse imparted by the wall on sphere during Impact.

Ans:
① $\mu = \frac{1}{4}$
② $4\sqrt{17}$ N-s





Impulse eqns

$mV_0 - J = -mv$ — ①

$I\omega_0 - J_1 R = I\omega$ — ②

$V = R\omega$ — ③

$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{v_1}{u_1}$

$v_1 = u_1$ ($\because$ elastic $\Rightarrow e = 1$)

$v = V_0 = 4$

$\Rightarrow V = R\omega \Rightarrow V = \omega = 4$

solving eqn ①, ②, ③ we get V, ω, J & J_1

$I(\omega_0 - \omega) = J_1 R$

$\frac{2}{5} m R^2 (9 - \omega) = \mu J R$

$\frac{2}{5} R(9 - \omega) = \mu(4 + R\omega)$

$\frac{2}{5}(9 - \omega) = \mu(4 + \omega)$

$\frac{2}{5}(5) = \mu(5) \Rightarrow \boxed{\mu = 1/4}$

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net linear impulse $= \sqrt{J^2 + J_1^2} = J\sqrt{1+u^2}$

$J = m(v_0 + v)$

$= 2(4+4)$

$J = 16$

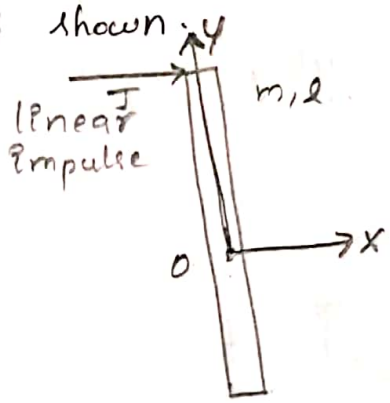
net impulse

$= 16\sqrt{1+\frac{1}{16}}$

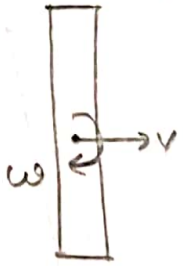
$= 16\sqrt{\frac{16+1}{16}} = 4\sqrt{16+1}$

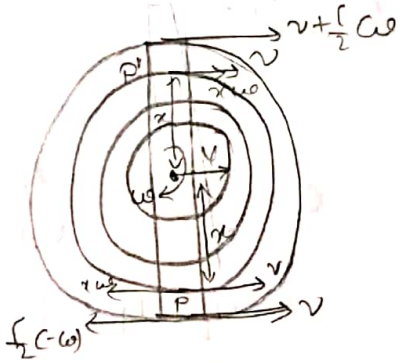
$= 4\sqrt{17}$ N-S.

→ A uniform rod of mass 'm' & length 'l' is placed on smooth horizontal surface. An impulse is given at one end of the rod as shown.



Just after impulse,





* Impulse eqns,

$J = mv \Rightarrow \boxed{v = \frac{J}{m}}$

* About C.M of rod,

$J\left(\frac{l}{2}\right) = I_{cm}\omega$

$J\frac{l}{2} = \frac{ml^2}{12}\omega$

$J = \frac{ml}{6}\omega$

$\boxed{\omega = \frac{6J}{ml}}$

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* K.E of rod after impulse,

$$K.E = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

* If net velocity of 'P' is zero,

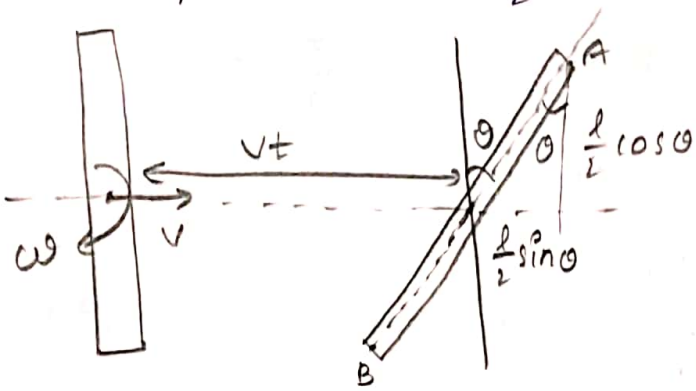
then $v = x\omega$

$$x = \frac{v}{\omega} = \frac{\frac{I}{m}}{\frac{6I}{ml}}$$

$$\boxed{x = \frac{l}{6}}$$

* Velocity of P':

$$V_{P'} = V + x\omega = V + \frac{l}{2}\omega$$



*position coordinates of end 'A'

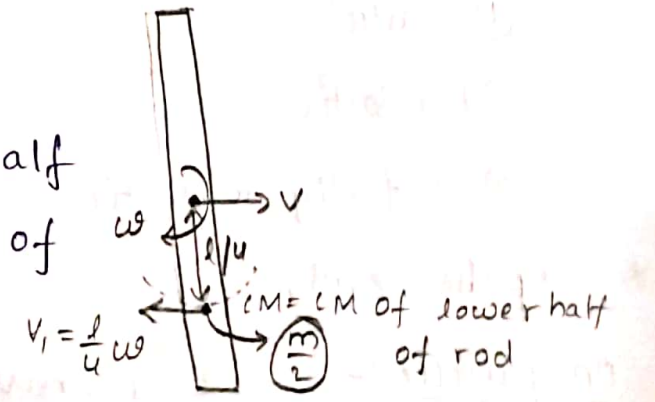
$$= \left(V + \frac{l}{2}\sin\theta, \frac{l}{2}\cos\theta\right)$$

*position co-ordinates of end 'B'

$$= \left(V - \frac{l}{2}\sin\theta, -\frac{l}{2}\cos\theta\right)$$

* force exerted by one half of the rod due to another half:

velocity of C.M of lowerhalf of rod w.r.to centre of rod.



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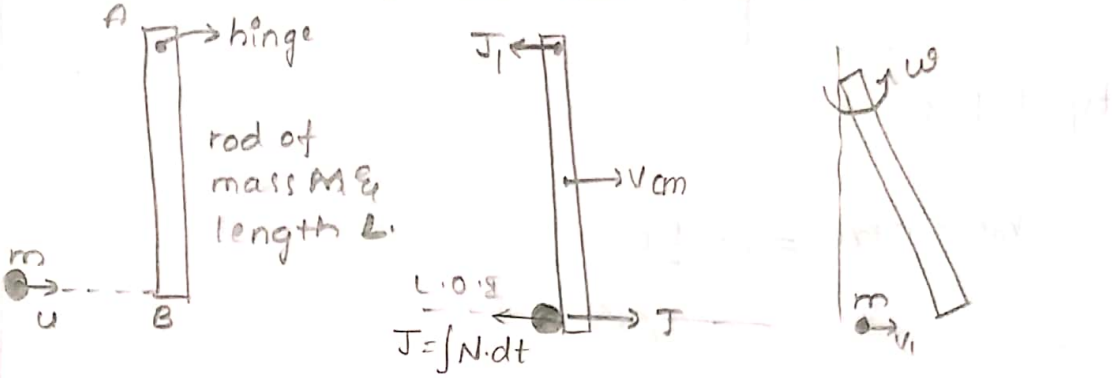
$$V_1 = \frac{l}{4}\omega$$

$$F = \frac{\frac{m}{2}v_1^2}{r} = \frac{\frac{m}{2}\left(\frac{l}{4}\omega\right)^2}{l/4}$$

$$\boxed{F = \frac{m l \omega^2}{8}}$$

Collision blw Particle & Body :-

① Body is hinged at some pt :



After collision,

* particle may ~~move~~
  1. continue to move forward
  2. move backwards
  3. Comes to rest.

* Here collision blw particle & rod may be
  1. elastic
  2. Inelastic
  3. perfectly inelastic

* Impulse eqns,

on particle :- $mu - J = mV_1 - ①$

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about hinge :

$$mud - J\ell = m_0 v_1 \ell \quad -②$$

on rod :-

$$0 + (J - J_1) = MV_{cm} - ③$$

$$\boxed{V_{cm} = \frac{\ell}{2}\omega}$$

⇒ about hinge,

$$0 + J\ell = I_{hinge}\,\omega$$

$$J\ell = \frac{M\ell^2}{3}\omega \quad -④$$

* coefficient of restitution :

$$e = \frac{\overline{v_2} - \overline{v_1}}{\overline{u_1} - \overline{u_2}}$$

$$e = \frac{\ell\omega - v_1}{u} \quad -⑤$$

If collision is elastic → $(e = 1)$

we use L.C.E just before & just after collision.

* work done by the hinge force on system = 0

because pt of application of force is at rest.

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}I_{hinge}\omega^2 \quad -⑥$$

combining eqn ① & ③,

$$\boxed{mu - J_1 = mv_1 + MV_{cm}}$$

i.e, L.C.L.M is not applicable.

combining eqns ② & ③,

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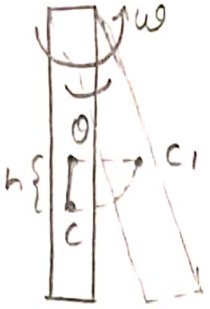
$$mud = mv_1 l + \frac{Ml^2}{3}\omega$$

i.e, L.C.A.M is valid (about hinge)

* After collision,

$$h = \frac{l}{2}(1-\cos\theta)$$



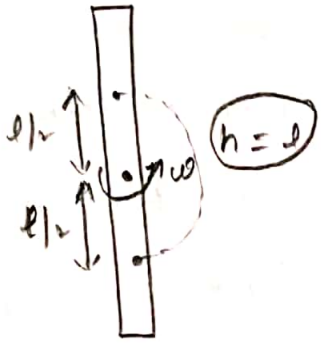
height raised by CM of rod = h

using L.C.E for rod,

$$\frac{1}{2} I_{hinge}\omega^2 = mgh$$

→ Find initial angular velocity, when the rod completes one revolution,

If the rod completes one revolution,

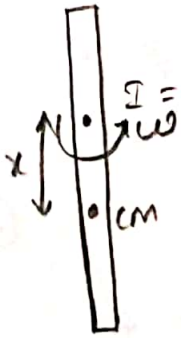


$$\frac{1}{2} I_{hinge}\omega^2 = mgh$$

$$\frac{1}{2} \times \frac{Ml^2}{3}\omega^2 = mgl$$

$$\omega_{min} = \sqrt{\frac{6g}{l}}$$

* If rod is hinged at some other pt,



$$I = I_{cm} + Mx^2$$

Rod completes one revolution.

for this,

$$\frac{1}{2} I_{hinge}\omega_{min}^2 = mgh$$

here

$$h = \frac{l}{2} + \frac{l}{2}$$

$$= l$$

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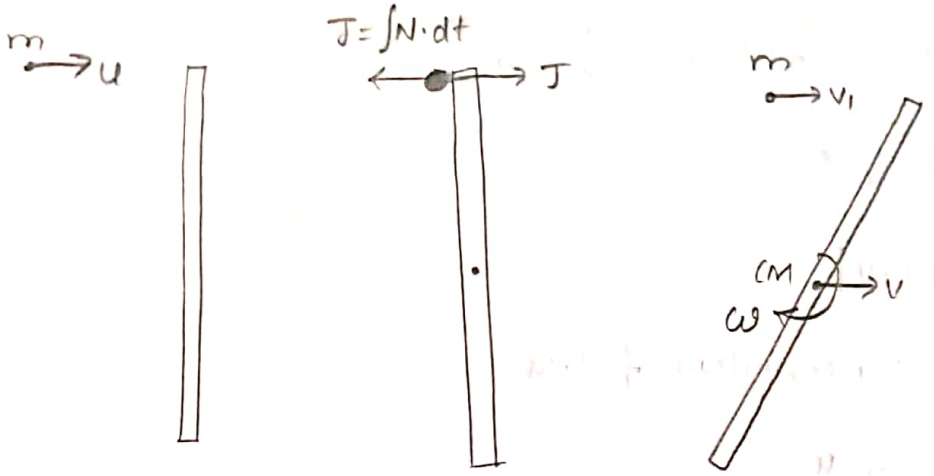
② Rod is placed on smooth horizontal surface-



Impulse eqns,

on particle,

$$mu - J = mv_1 \quad -①$$

About CM of rod,

$$mu\frac{l}{2} - J\frac{l}{2} = mv_1\frac{l}{2} \quad -②$$

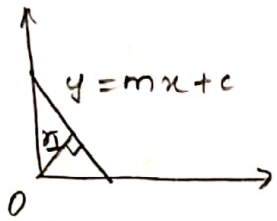
on rod,

$$0 + J = Mv \quad -③$$

About CM of rod,

$$0 + J\left(\frac{l}{2}\right) = I_{cm}\omega \quad -④$$

→ eq^n module



$$L_0 = mvr_\perp$$

$$r_\perp = \frac{|c|}{\sqrt{a^2+b^2}}$$

---

coefficient of restitution,

$$e = \frac{\overline{v_2} - \overline{v_1}}{\overline{u_1} - \overline{u_2}} = \frac{(v + \frac{l}{2}\omega) - v_1}{u} \quad -⑤$$

If collision is elastic,

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L.C.E is valid,

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2 \quad -(6)$$

from eqns (1) & (3),

$$\boxed{mu = mv_1 + Mv}$$

(Law of conservation of L.M.

from eqns (2) & (4),

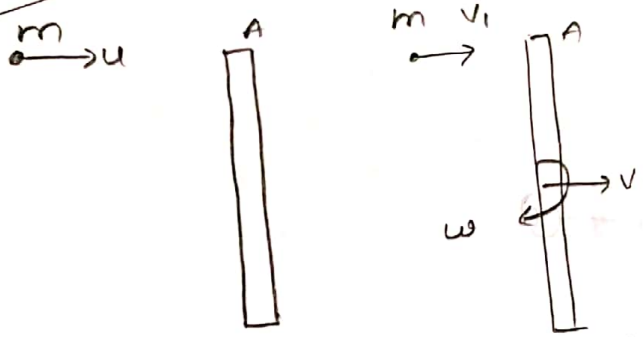
$$\boxed{mu\frac{l}{2} = mv_1\frac{l}{2} + \frac{Ml^2}{12}\omega}$$

(L.C.A.M about a pt on the ground just below the centre of rod. or we can write about the hinged pt.

* K.E of rod just after the collision,

$$K.E = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$$

I-Method



L.C.L.M,

$$mu = mv_1 + Mv_2 \quad -(1)$$

L.C.A.M, about a pt on the ground just below end 'A'.

$$0 = 0 + m\frac{l}{2}v\,(\uparrow) + I_{cm}\omega\,(\circlearrowleft) \quad -(2)$$

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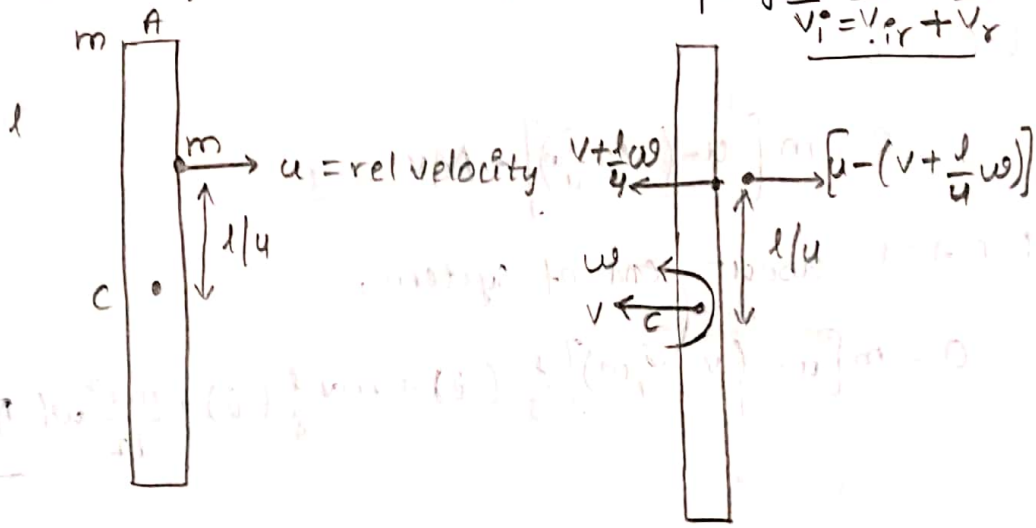
→ A uniform rod AB of mass m & length l is placed on smooth horizontal surface. An insect of same mass which is at rest on rod initially, jumps with velocity 'u' relative to rod. Find

① v & ② ω of rod just after Jumping.

$\bar{v}_i = \bar{v}_{ir} + \bar{v}_r$



L.C.L.M,

$$0 = m\left[u - \left(v + \frac{l}{4}\omega\right)\right] - mv \longrightarrow ①$$

L.C.A.M, about a pt on ground just below centre of rod.

$$0 = m\left[u - \left(v + \frac{l}{4}\omega\right)\right]\frac{l}{4}\left(\frac{l}{2}\right) + \frac{ml^2}{12}\omega\left(\frac{l}{2}\right) - ②$$

Solving eqns ① & ② we get v & ω.

* Energy required for Jumping;

$$E = \frac{1}{2}m\left(u - \left(v + \frac{l}{4}\omega\right)\right)^2 + \frac{1}{2}Mv^2 + \frac{1}{2}\times\frac{ml^2}{12}(\omega^2)$$

(first term: Insect; remaining terms: rod)

M-II:

We can do this about centre of mass of system.

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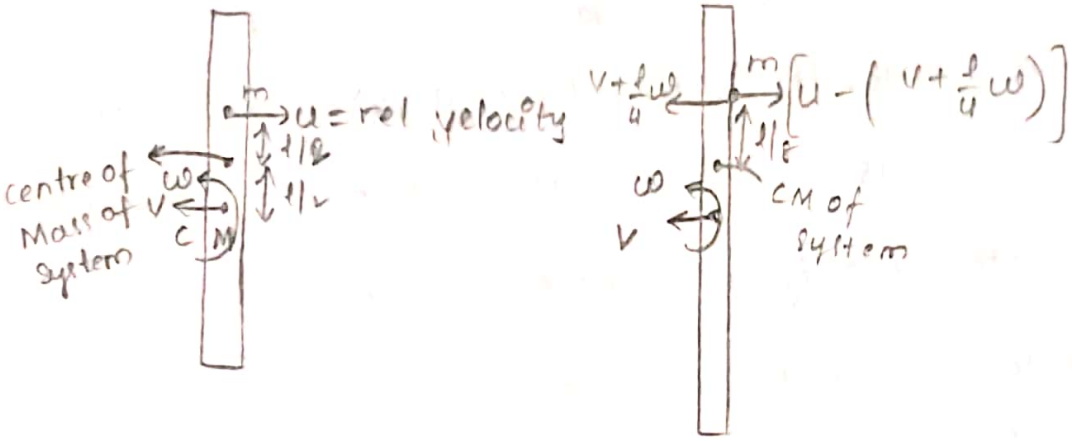

L·C·L·M,

$$0 = m\left[u-\left(v+\frac{l}{4}\omega\right)\right] - MV \quad -(1)$$

L·C·A·M about CM of system,

$$0 = m\left[u-\left(v+\frac{l}{4}\omega\right)\right]\frac{l}{8}\left(\frac{l}{2}\right) + mv\frac{l}{8}\left(\frac{l}{2}\right) + \frac{ml^2}{12}\omega\left(\uparrow\right) \quad -(2)$$

(1) $\Rightarrow$ $MV = m\left[u-\left(v+\frac{l}{4}\omega\right)\right]$

(1) $\Rightarrow$ $MV + mv = mu - \frac{ml}{4}\omega$

(2) $\Rightarrow$ $MV\frac{l}{8} + mv\frac{l}{8} - \frac{ml^2}{12}\omega = 0$

$$\frac{MV+mv}{8_2} = \frac{ml\omega}{12_3}$$

$$(M+m)V = \frac{2}{3}ml\omega$$

$$mu - \frac{ml}{4}\omega = \frac{2}{3}ml\omega$$

$$u = \left(\frac{2}{3}+\frac{1}{4}\right)l\omega$$

$$\frac{12u}{11l} = \omega \Rightarrow \boxed{\omega = \frac{12u}{11l}}$$

$$(M+m)V = mu - \frac{ml}{4}\times\frac{12u}{11l}^{3}$$

$$(M+m)V = mu - \frac{3mu}{11}$$

$$V = mu\left(\frac{8}{11}\right)\times\frac{1}{M+m}$$

$$\boxed{V = \frac{8mu}{11(M+m)}}$$

$m = M$

$$\Rightarrow V = \frac{4}{\cancel{2}m\cancel{u}}{11\times\cancel{2}m}$$

$$\boxed{V = \frac{4u}{11}}$$

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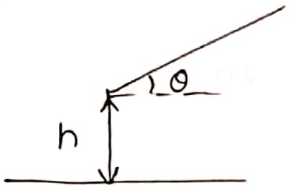
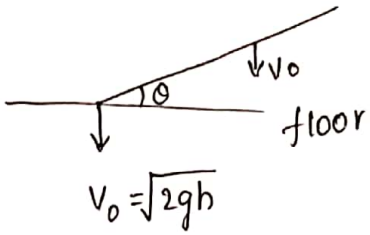
Rod falling on smooth Surface :

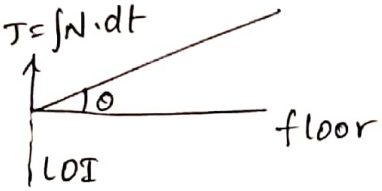
A rod of mass 'm' & length 'l' falling from height 'h'. It collides with floor elastically.

Just before collision,

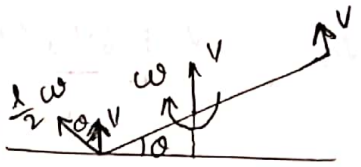




During collision,



Just after collision,



Impulse eqns,

$$mv_0 - J = -mv \quad \text{— (1)}$$

Angular impulse about CM of rod,

$$\frac{J\cos\theta}{2} = \frac{ml^2}{12}\omega \quad \text{— (2)}$$

$$J = \frac{ml\omega}{6\cos\theta} \quad \text{— (3)}$$

Coefficient of restitution,

$$e = \frac{\bar{v}_2 - \bar{v}_1}{\bar{u}_1 - \bar{u}_2} \Rightarrow 1 = \frac{v + \frac{l}{2}\omega\cos\theta}{v_0}$$

$$v_0 = v + \frac{l}{2}\omega\cos\theta \quad \text{— (4)}$$

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$$V_0 = \frac{mV_0 - J}{-m} + \frac{l}{2}\omega\cos\theta$$

$$V_0 = -V_0 + \frac{l\omega}{6\cos\theta} + \frac{l\omega}{2}\cos\theta$$

$$2V_0 = l\omega\left(\frac{1+3\cos^2\theta}{6\cos\theta}\right)$$

$$\boxed{\omega = \frac{12V_0\cos\theta}{l(1+3\cos^2\theta)}}$$

&

$$V_0 = V + \frac{l}{2}\cos\theta\left(\frac{12V_0\cos\theta}{l(1+3\cos^2\theta)}\right)$$

$$V_0 = V + \frac{6V_0\cos^2\theta}{1+3\cos^2\theta}$$

$$V = \left(\frac{1+3\cos^2\theta - 6\cos^2\theta}{1+3\cos^2\theta}\right)V_0$$

$$\boxed{V = \left(\frac{1-3\cos^2\theta}{1+3\cos^2\theta}\right)V_0}$$

Clearly 'ω' depends on 'θ' (for a given 'h')

$\therefore$ for maximum ω,

$$\frac{d\omega}{d\theta} = 0$$

$$\frac{d}{d\theta}\left(\frac{12V_0\cos\theta}{l(1+3\cos^2\theta)}\right) = 0$$

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$$\frac{d}{d\theta}\left(\frac{\cos\theta}{1+3\cos^2\theta}\right) = 0$$

$$\frac{(1+3\cos^2\theta)(-\sin\theta) - \cos\theta(-6\cos\theta\sin\theta)}{(1+3\cos^2\theta)^2} = 0$$

$$(1+3\cos^2\theta)\sin\theta = 6\cos^2\theta\sin\theta$$

$$1 = 3\cos^2\theta$$

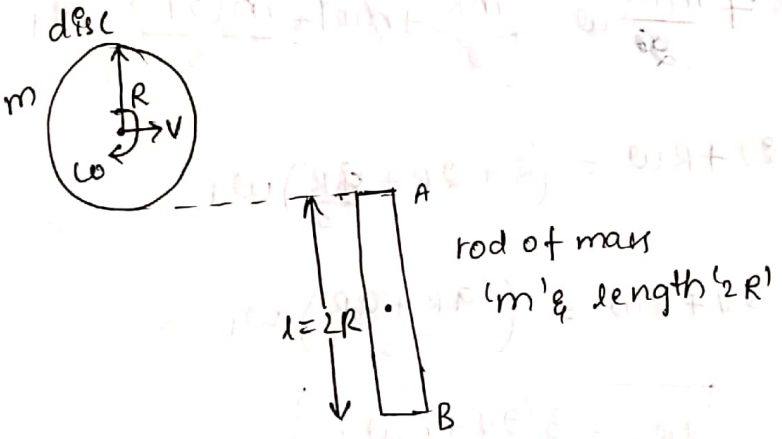
$$\cos\theta = 1/\sqrt{3} \Rightarrow \boxed{\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)}$$

$$\therefore \omega_{max} = \frac{12V_0 \times 1/\sqrt{3}}{(1+3(1/3))l} = \frac{4\sqrt{3}V_0}{2l} = \frac{2\sqrt{3}V_0}{l}$$

$$\boxed{\omega_{max} = \frac{2\sqrt{3}V_0}{l}}$$

collision b/w bodies :-

Eg:-



Disc is moving on smooth surface with the flat circular face on the surface as shown. It just manages to hit the rod AB at it's end A & sticks to the disc. Find the velocity & angular velocity of system after collision.

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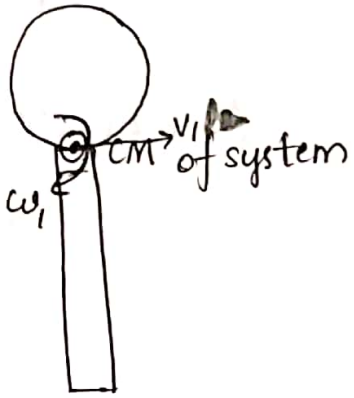
Sol^n - from CM of system,

position of CM from centre of disc

$$= \frac{m(2R)}{m+m} = \frac{2R}{2}$$

$$= R$$



L.C.L.M,

$$mv = (2m)v_1$$

$$\boxed{v_1 = v/2}$$

L.C.A.M about CM of system,

$$\overline{L_i} = \overline{L_f}$$

$$\Rightarrow mvR + \frac{mR^2}{2}\omega = \left(I_{\substack{\text{system} \\ \text{about CM}}}\right)\omega_1$$

$$mvR + \frac{mR^2}{2}\omega = \left[\frac{mR^2}{2} + mR^2 + \frac{m(2R)^2}{3}\right]\omega_1$$

$$2v + R\omega = \left(R + 2R + \frac{8R}{3}\right)\omega_1$$

$$2v + R\omega = \left(\frac{9R + 8R}{3}\right)\omega_1$$

$$\boxed{\omega_1 = \frac{3(2v + R\omega)}{17R}}$$

↳ angular velocity of system after hitting.

now, velocity of CM of disc,

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$$V_{disc} = V_1 + R\omega_1 = \frac{V}{2} + \frac{3(2V+R\omega)}{17} = \frac{17V+12V+6R\omega}{17}$$

Velocity of CM of rod,

$$= \frac{29V+6R\omega}{17}$$

$$V_{rod} = V_1 - R\omega_1 = \frac{V}{2} - \frac{3(2V+R\omega)}{17}$$

$$= \frac{17V-12V-6R\omega}{17} = \frac{5V-6R\omega}{17}.$$

Case 8

If $V = R\omega$

$$\omega_1 = \frac{3(3V)}{17R} = \frac{9}{17}\frac{V}{R}$$

$$\boxed{\omega_1 = \frac{9}{17}\omega}$$

loss in K.E,

$$\Delta E = K.E_i - K.E_f$$

$$= \left(\frac{1}{2}mv^2 + \frac{1}{2}\frac{mR^2}{2}\omega^2\right) - \left(\frac{1}{2}(2m)V_1^2 + \frac{1}{2}\,I_{sys\ about\ CM}\,\omega_1^2\right)$$

put $V_1 = V/2$ & $\omega_1 = \frac{3(2V+R\omega)}{17R}$

After simplyfing,

$$\Delta E = \frac{5}{68}mv^2 + \frac{7}{34}mR^2\omega^2 - \frac{3}{17}mvR\omega$$

Here ΔE depends on 'V'. (for a given ω)

* loss in K.E is min,

when $\frac{d(\Delta E)}{dV} = 0$

$$\frac{5}{68}m(2v) + 0 - \frac{3}{17}mR\omega = 0$$

$$\frac{5V}{2} = 3R\omega \Rightarrow \boxed{V = \frac{6R\omega}{5}}$$

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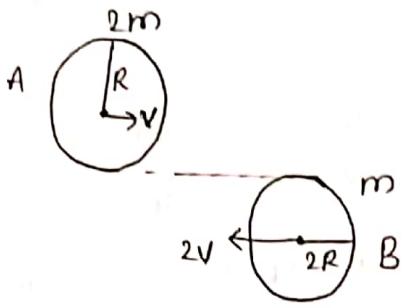
at $v = \frac{6}{5}R\omega$, loss in K.E is minimum.

Eg:



They are just managed to collide & stick to each other. Find

① Angular velocity of composite system after collision.

② Loss in K.E due to collision.

Sol: From C.M of system:

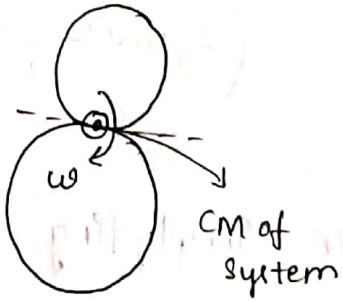
position of centre of mass from centre of disc 'A'

$$= \frac{(2R)m}{3m} = R$$

L.C.L.M

$$(2m)v - m(2v) = 3m\,V_{cm}$$

$$\Rightarrow \boxed{V_{cm} = 0}$$



L.C.A.M about system's CM,

$$(2m)vR + m(2v)(2R) = (I_{sys})\omega'$$

$$6mvR = \left[\frac{2mR^2}{2} + 2mR^2 + \frac{4mR^2}{2} + m(2R)^2\right]\omega'$$

$$v = [R + 2R + 2R + 4R]\omega'$$

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$$\omega' = \frac{2V}{3R}$$

$$\boxed{\omega' = \frac{2V}{3R}}$$

Loss in K.E,

$$= \frac{1}{2}(3m)v^2 + \frac{1}{2}m(2v)^2 - \frac{1}{2}(I_{sys})\omega^2$$

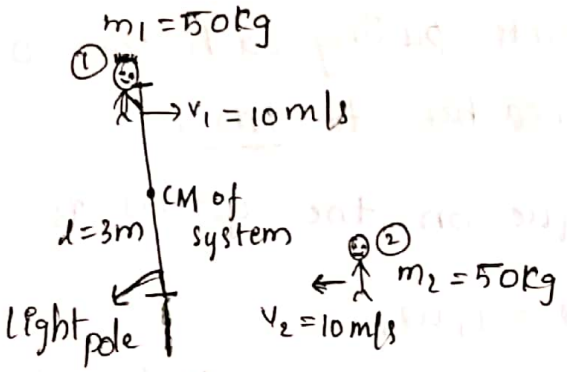
$$= \frac{1}{2}(3mv^2 + 4mv^2) - \frac{1}{2}\left(mR^2 + 2mR^2 + 2mR^2 + 4mR^2\right)\left(\frac{2V}{3R}\right)^2$$

$$= 3mv^2 - \frac{1}{2}(m + 2m + 2m + 4m)\left(\frac{4v^2}{9}\right)$$

$$= 3mv^2 - \frac{9mv^2}{9}(2)$$

$$= mv^2.$$

Eg:



Initial seperation b/w them is 3m. 1st person is skating with help of pole & now 2nd person grabs the pole, after grabing, Find 'ω' of system.

Sol: from C.M of system:

L.C.L.M is applicable as there is no ext force acting on system.

$$m_1v_1 - m_2v_2 = (m_1 + m_2)v_{cm}$$

$$\boxed{v_{cm} = 0}$$

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L.C.A.M about CM of system,

$$m_1 v_1 \frac{l}{2} + m_2 v_2 \frac{l}{2} = \left[m_1\left(\frac{l}{2}\right)^2 + m_2\left(\frac{l}{2}\right)^2\right]\omega$$

$$50 \times 10 \times 8 = \left(50 \times \frac{4}{4}^3 + 50 \times \frac{3}{4}\right)\omega$$

$$10 = \frac{6^3}{4_2}\omega$$

$$\boxed{\omega = 20/3 \text{ rad/s}}$$

K.E after grabing,

$$K.E = \frac{1}{2} I_{cm} \times \omega^2$$

$$= \frac{1}{2} \times 50 \times \frac{18^9}{4_2} \times \frac{100}{2} \times \frac{400}{2}$$

$$\boxed{K.E = 5000 \text{ J}}$$

* If the person starts pulling each other so that their seperation reaches to 1m.

Here net torque on the system is zero.

$$I\omega = I_1 \omega_1$$

$$\frac{20}{3} \times \left(m_1\left(\frac{l}{2}\right)^2 + m_2\left(\frac{l}{2}\right)^2\right) = \left(m_1\left(\frac{1}{2}\right)^2 + m_2\left(\frac{1}{2}\right)^2\right)\omega_1$$

$$\frac{20}{3} \times 25 \times 8^3 = 50\left(\frac{1}{2}\right)\omega_1$$

$$\boxed{\omega_1 = 60 \text{ rad/s}}$$

now, $K.E_1 = \frac{L^2}{2I_1}$

$$\frac{K.E}{K.E_1} = \frac{I_1}{I} = \frac{25}{225} = \frac{1}{9}$$

$$\boxed{K.E_1 = 9K.E}$$

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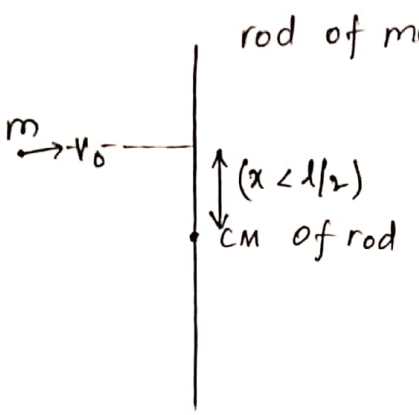
here, K.E of system increases.

bcoz work done on each person is positive.

Eg:

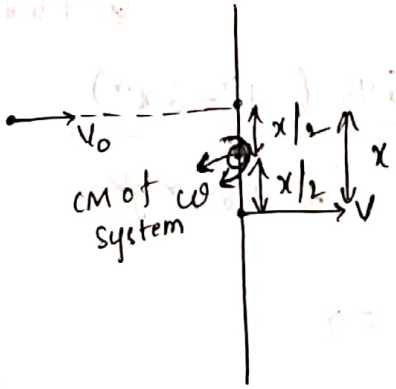


Rod is placed on smooth horizontal surface. A particle of same mass 'm' moving with velocity v_0 strikes the rod perpendicular to it as shown. After collision if the particle sticks to the rod. Find angular velocity of system after collision.

Sol:

Method-1:

from centre of mass of system:



LCLM,

$$mv_0 = 2mv$$

$$\boxed{v = v_0/2}$$

LCAM,

$$mv_0\left(\frac{x}{2}\right) = I_{system}\,\omega$$

$$mv_0\left(\frac{x}{2}\right) = \left[\underbrace{m\left(\frac{x}{2}\right)^2}_{\text{particle}} + \underbrace{\frac{ml^2}{12} + m\left(\frac{x}{2}\right)^2}_{\text{rod}}\right]\omega$$

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$$mv_0\frac{x}{2} = \left(\frac{mx^2}{2} + \frac{md^2}{12}\right)\omega \qquad mv_0\left(\frac{x}{2}\right)$$

$$6$$

$$\omega = \frac{6v_0}{7x}$$

$$mv_0\left(\frac{x}{2}\right) = \left(\frac{mx^2}{2} + \frac{md^2}{12}\right)\omega$$

$$6$$

$$\boxed{\omega = \frac{6v_0 x}{6x^2 + d^2}}$$

* ω depends on 'x'

when $x = 0$ (centre of rod)

$$\Rightarrow \boxed{\omega = 0}$$

*To find minimum | max value of ω:

① $$\frac{d\omega}{dx} = \frac{d}{dx}\left(\frac{6v_0 x}{d^2 + 6x^2}\right)$$

$$= 6v_0\left(\frac{d^2 + 6x^2 - x(12x)}{(d^2 + 6x^2)^2}\right)$$

$$\frac{d\omega}{dx} = \frac{6v_0\,(d^2 - 6x^2)}{(d^2 + 6x^2)^2}$$

now, $$\frac{d\omega}{dx} = 0$$

$$\Rightarrow d^2 = 6x^2$$

$$\boxed{x = d/\sqrt{6}}$$

and $$\left.\frac{d^2\omega}{dx^2}\right|_{x = \frac{d}{\sqrt{6}}} = \frac{6v_0\left[(d^2 + 6x^2)(-12x) - 2(d^2 - 6x^2)(12x)(d^2 + \right.}{(d^2 + 6x^2)^4}$$

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$$\therefore \quad \frac{d^2\omega}{dx^2} = -ve \ .$$

i.e, $x = l/\sqrt{6}$, ω is maximum.

Conclusion

As 'x' increases from $x = 0$ to $x = \frac{l}{2}$.

'ω' increases first and then decreases.

linear impulse on rod,

$$J = |\bar{P}_f - \bar{P}_i|$$

$$J = \bar{P}_i - \bar{P}_f$$

Impulse in maximum when $\omega = 0$

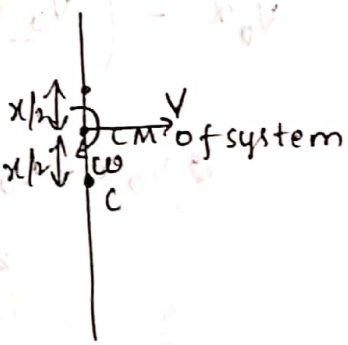
i.e, when particle hits at centre of rod.

$\therefore$ Max impulse $= mv_0 - \frac{mv_0}{2} = \frac{mv_0}{2}$

$$\boxed{J_{max} = \frac{mv_0}{2}}$$

* velocity of the particle after collision,

$$= v + \frac{x}{2}\omega$$



* Velocity of centre of rod,

$$= v - \frac{x}{2}\omega$$

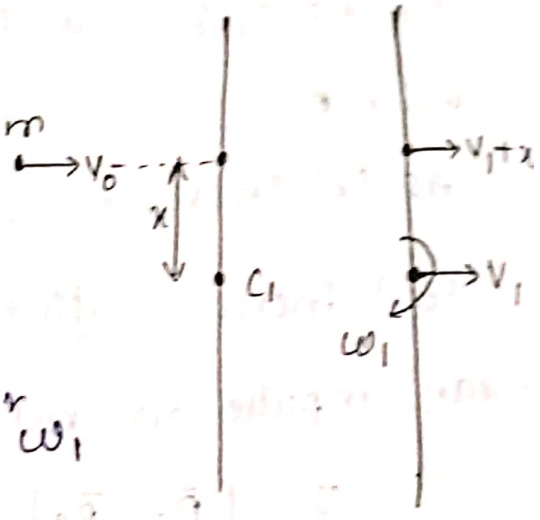
* loss in K.E of system,

$$= \frac{1}{2} m v_0^2 - \left[\frac{1}{2}(2m)(v)^2 + \frac{1}{2} I_{sys\ about\ CM}\, \omega^2\right]$$

*

Method-2:

from a pt on the ground bflow centre of rod:



L.C.L.M,

$$mv_0 = m(v_1 + x\omega_1) + mv_1 \quad \text{——①}$$

L.C.A.M,

$$mv_0(x) = m(v_1 + x\omega_1)x + \frac{md^2}{12}\omega_1$$

(particle, rod)

$$v_0 x = (v_1 + x\omega_1)x + \frac{d^2}{12}\omega_1$$

$$v_0 x = \left(\frac{v_0 - x\omega_1}{2} + x\omega_1\right)x + \frac{d^2}{12}\omega_1$$

$$v_0 x = \left(\frac{v_0 + x\omega_1}{2}\right)x + \frac{d^2}{12}\omega_1$$

$$v_0 x - \left(\frac{v_0 + x\omega_1}{2}\right)x = \frac{d^2}{12}\omega_1$$

$$\frac{2v_0 x - v_0 x - x^2\omega_1}{2} = \frac{d^2}{12}\omega_1$$

$$v_0 x - x\omega_1 = \frac{d^2}{6}\omega_1$$

$$6v_0 x = (d^2 + 6x^2)\omega_1$$

$$\boxed{\omega_1 = \frac{6v_0 x}{d^2 + 6x^2}}$$

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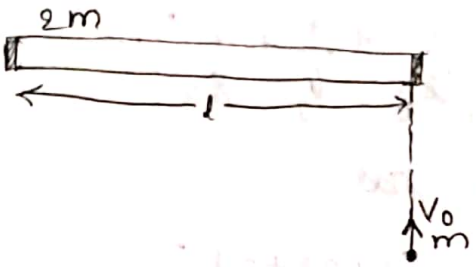
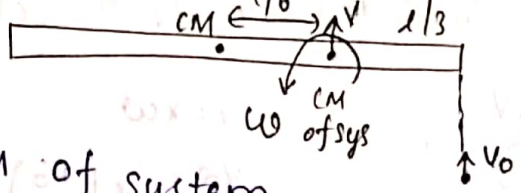
Eg:



rod of mass '2m' & length 'l' is placed on smooth surface. A particle of mass 'm' moving with speed 'v_0' collides one end of rod & sticks to it. Find angular velocity of system after collision

Sol:



position of CM of system

$$= \frac{m(l/2)}{3m} = l/6 .$$

L.C.L.M,

$$mv_0 = 3mv$$

$$\boxed{v = v_0/3}$$

L.C.A.M,

$$mv_0\left(\frac{l}{3}\right) = (I_{sys})\omega$$

$$mv_0\left(\frac{l}{3}\right) = \left[2m\left(\frac{l}{6}\right)^2 + \frac{2m\cdot l^2}{12} + m\left(\frac{l}{3}\right)^2\right]\omega$$

$$\frac{v_0}{3} = \left(\frac{2l}{18} + \frac{l}{12} + \frac{l}{9}\right)\omega$$

$$v_0(12) = (8l + 3l + 4l)\omega$$

$$\omega = \frac{12v_0}{15l}$$

$$v_0(6) = (4l + 3l + 2l)$$

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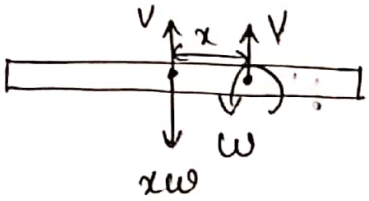
* Position of $\frac{v_0}{3} l = \left(\frac{l}{18_6} + \frac{l}{6_2} + \frac{l}{9_3}\right)\omega$

$$v_0 = \left(\frac{l+3l+2l}{6}\right)\omega$$

$$\boxed{\frac{v_0}{l} = \omega}$$

* Position of pt on the rod which remains stationary immediately after collision is.



$V = x\omega$

$$\frac{v_0}{3} = x\left(\frac{v_0}{l}\right)$$

$$\boxed{x = \frac{l}{3}}$$

from end A,

$$= \frac{l}{3} + x$$

$$= \frac{l}{3} + \frac{l}{3} = \frac{2l}{3}$$

velocity of centre of rod $= V - \frac{l}{6}\omega = \frac{v_0}{3} - \frac{l}{6} \times \frac{v_0}{l}$

$$= \frac{v_0}{6}.$$

& velocity of end 'A' of rod,

$$V = V + \frac{l}{2}\omega$$

$$= \frac{v_0}{3} + \frac{l}{3}\left(\frac{v_0}{l}\right)$$

$$\boxed{V = \frac{2v_0}{3}}$$

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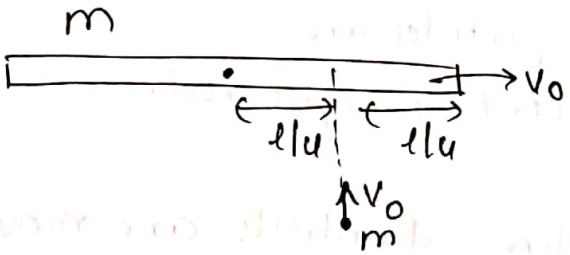
Eg: A uniform rod of mass 'm' & length l is translating on a smooth horizontal plane with a velocity V_0. when a particle of same mass 'm' travelling with same velocity 'V_0' as shown strikes normally & becomes embedded in the bar. Determine angular velocity of system after collision, V_{cm} of rod, velocity of the particle.



$$\omega = \frac{12V_0}{11l}$$

$$\overline{V}_{rod} = \frac{V_0}{2}\uparrow + \frac{4V_0}{11}\uparrow$$

$$\overline{V}_{particle} = \frac{V_0}{2}\uparrow + \frac{7V_0}{11}\uparrow$$

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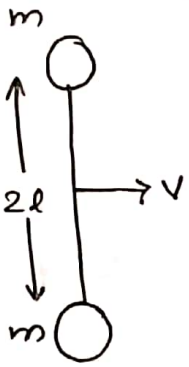
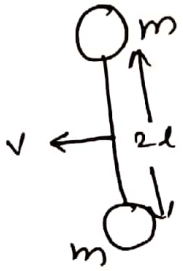
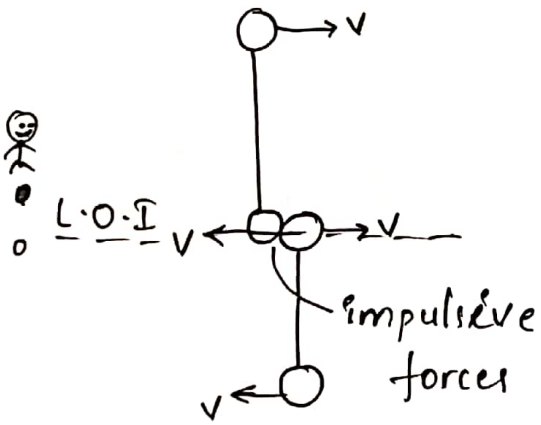
Collision b|w dumbells:

↳ two particles are connected by light rod.

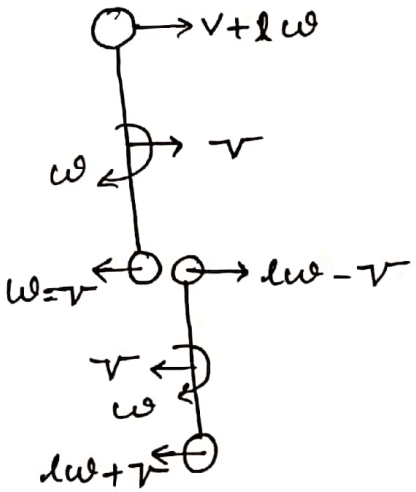


Here dumbells are moving on smooth horizontal surface and make elasti collision.





Just before collision



Just after collision.

* Impulse Eqn:

$$2mV - J = 2mv$$

Angular impulse about CM of rod,

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$$0 + Jd = I_{cm}\omega \Rightarrow J\cancel{d} = (m\cancel{d}^2 + m\ell^2)\omega \quad -(2)$$

coefficient of restitution,

$$e = \frac{\overline{v_2} - \overline{v_1}}{\overline{u_1} - \overline{u_2}}$$

$$e = \frac{(\ell\omega - v) - [-(\ell\omega - v)]}{v - (-v)}$$

$$e = \frac{2\ell\omega - 2v}{2v}$$

$$e = \frac{\ell\omega - v}{v} \quad -(3)$$

If collision is elastic,

L.C.E,

$$\frac{1}{2}(m)v^2 + \frac{1}{2}(2m)v^2 = 2\left[\frac{1}{2}(2m)v^2 + \frac{1}{2}(m\ell^2 + m\ell^2)\omega^2\right] \quad -(4)$$

L.C.A.M about pt 'O' on the ground,

$$mv\cancel{\ell} + mv\cancel{\ell} = \cancel{2}\left[2mv\cancel{\ell} + \cancel{2}(m\ell^2 + m\ell^2)\omega\right] \quad -(5)$$

$$2v = \cancel{2}v + \cancel{2}\ell\omega$$

$$\boxed{v = v + \ell\omega} \quad -(6)$$

If collision is elastic

$$e = 1 \Rightarrow \ell\omega - v = v \quad -(7)$$

adding eqns (6) & (7),

$$\cancel{2}\ell\omega = \cancel{2}v$$

$$\boxed{\omega = v/\ell}$$

angular velocity of rod after collision.

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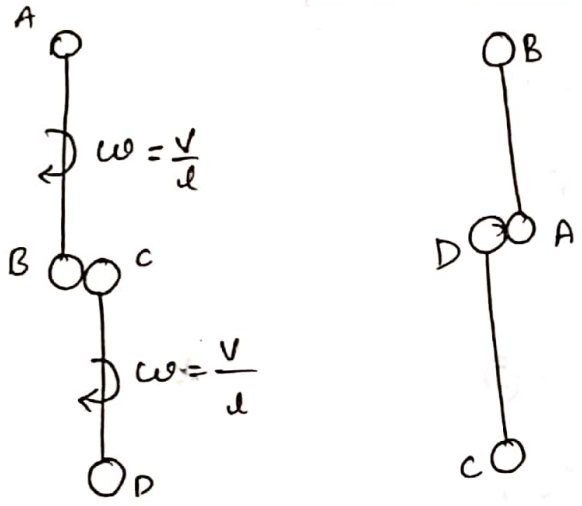
and $\ell\left(\frac{V}{\ell}\right) = V = V$

$$\boxed{V = 0}$$

i.e, linear velocity of rod after collision is zero.

* Time for next collision :



Here $\theta = \omega t$

$\pi = \omega t$

$\pi = \frac{V}{\ell} t$

$$\boxed{t = \frac{\pi \ell}{V}}$$

Eg: A disc of mass 'm' and radius 'R' is spinning about it's own axis with angular velocity 'ω' on smooth horizontal surface. A particle of same mass 'm' moving with speed 2Rω towards centre of disc as shown. If the collision is perfectly inelastic, Find

① Angular velocity of disc after collision

② loss in K.E of system.

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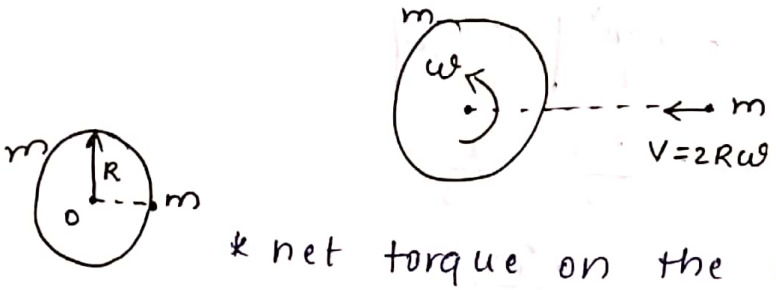
③ Impulse on the particle due to disc.

④ Impulse on disc due to hinge.

Sol:



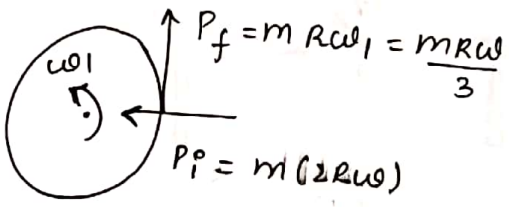
* net torque on the system about CM of disc = 0.

* $\bar{L_i} = \bar{L_f}$

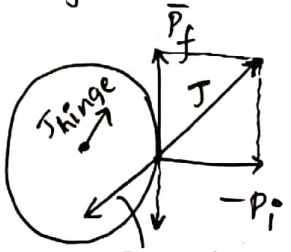
$$\frac{mR^2}{2}\omega = \left(\frac{mR^2}{2} + mR^2\right)\omega_1$$

$$\boxed{\omega_1 = \frac{\omega}{3}}$$

* Impulse on the particle,



$$\bar{J} = \bar{P}_f - \bar{P}_i$$



Impulse on disc due to particle = J

$$J = \sqrt{(m2R\omega)^2 + \left(\frac{mR\omega}{3}\right)^2} = \frac{\sqrt{37}\, mR\omega}{3}$$

* disc has no linear (momentum) motion,

$$\bar{J}_{net} = 0$$

$$\boxed{J_H = \frac{\sqrt{37}\, mR\omega}{3}}$$

Angular impulse on disc,

$$= \bar{L}_f - \bar{L}_i$$

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$$= \left| \frac{mR^2}{2}\omega_1 - \frac{mR^2}{2}\omega \right|$$

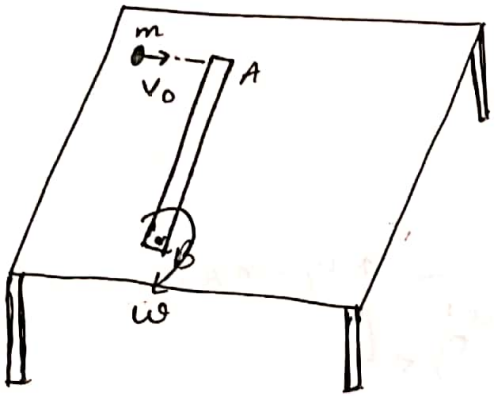
$$= \frac{mR^2}{2}\left| \frac{\omega}{3} - \omega \right|$$

$$= \frac{mR^2}{2} \times \frac{2\omega}{3}$$

$$= \frac{mR^2\omega}{3}$$

Eg:

Metal rod of mass 'M' & length l is kept on smooth table is pivoted at one end.



After hitting,

① L.C.A.M, about hinge,

$$mv_0 l = mv_1 l + \frac{Ml^2}{3}\omega \quad -①$$

② coeff of restitution,

$$e = \frac{\bar{v}_2 - \bar{v}_1}{\bar{u}_1 - \bar{u}_2}$$

$$e = \frac{l\omega - v_1}{v_0} \quad -②$$

③ If collision is elastic,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}\left(\frac{Ml^2}{3}\right)\omega^2 \quad -③$$

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(4) Initial momentum,

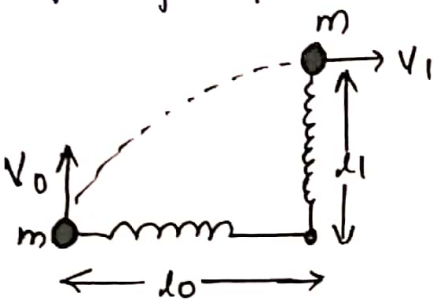
$$P_i = mV_0$$

Final momentum,

$$P_f = mV_1 + MV_{cm} \quad \left(V_{cm} = \frac{l}{2}\omega\right)$$

clearly $P_f > P_i$.

Eg.



$x = l_1 - l_0$

L.C.A.M,

$$mV_0 l_0 = mV_1 l_1$$

$$V_0 l_0 = V_1 l_1 \quad \text{—— (1)}$$

L.C.E,

$$\frac{1}{2} m V_0^2 = \frac{1}{2} m V_1^2 + \frac{1}{2} k x^2$$

$$mV_0^2 = mV_1^2 + k(l_1 - l_0)^2 \quad \text{—— (2)}$$

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